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Artificial Intelligence for Military Applications with Blockchain

EDITED BY

Gururaj H L, Gowtham M, Ajay A V,
and Pramod H B

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Artificial Intelligence for Military Applications with Blockchain

In an era where advanced technology plays a critical role in maintaining national security, *Artificial Intelligence for Military Applications with Blockchain* investigates how combining AI and blockchain could transform military operations. This comprehensive guide offers creative answers for contemporary military problems while addressing the most important defense-related concerns, from data security to decision-making. It explores constrained networking middleware for defense applications, guaranteeing smooth communication under critical circumstances. This book starts with an in-depth examination of blockchain's potential to improve document management across defense departments, then moves to a detailed discussion of security and privacy in military applications. The integration of AI and blockchain in military context is then the main topic of discussion, along with its advantages, disadvantages, and real-time applications.

The potential of blockchain and AI to protect data and streamline operations is also explored, providing readers with insights into the military and healthcare sectors. This book offers a thorough examination of the military's current and future use of AI, as well as a breakdown of cybersecurity issues and how blockchain technology is being used to improve military cybersecurity. A dedicated chapter examines the ways in which blockchain technology is being used by computational intelligence to transform the defense environment.

Key features:

- Examines privacy and security issues in military blockchain applications
- Investigates military operations using constrained networking middleware
- Discusses integrating AI and blockchain technology for military applications
- Includes case studies of blockchain and AI uses in the military and healthcare
- Thoroughly examines cybersecurity issues and how blockchain technology can help

This book is essential for military personnel, defense academics, and cybersecurity specialists interested in the use of AI and blockchain for defense. It presents real-world examples and case studies together with an outlook on how these technologies will influence future military operations.



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Edited by
Gururaj H L, Gowtham M, Ajay A V, and
Pramod H B



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Preface

The book serves as a descriptive summary of *Artificial Intelligence for Military Applications with Blockchain*, offering a collection of case studies and analyses contributed by diverse authors from across the globe. In today's rapidly evolving digital landscape, the integration of *Artificial Intelligence (AI) and Blockchain technology* is fundamentally reshaping the future of defense and military operations. The contributions in this book reflect the growing importance of these technologies in enhancing security, efficiency, and decision-making within the military domain.

The book begins with Chapter 1, where the authors elaborate on the *foundations of security and privacy* in military applications using blockchain. By exploring blockchain's potential to secure military systems, they highlight how it can serve as a critical tool in protecting sensitive data from unauthorized access and breaches.

In Chapter 2, the focus shifts to *document management in defense*. Here, blockchain technology is presented as a solution for ensuring *secure, transparent, and tamper-resistant* data sharing across various military departments, a necessity for seamless and efficient communication in defense environments.

Chapter 3 provides an in-depth analysis of *networking middleware for military operations*, emphasizing the significance of constrained environments where bandwidth and resources are limited. The authors demonstrate how blockchain can support reliable, secure communication, even under challenging conditions.

Chapters 4, 5, and 6 delve into the broader topic of *AI and Blockchain for military applications*, providing an overview of the *opportunities, challenges, and practical applications* of these technologies in defense contexts. These chapters underscore the transformative power of AI and blockchain, from automating decision-making to enhancing logistical coordination and resource management in military operations.

Chapter 7 further explores the *intersection of AI and blockchain*, highlighting case studies of AI in military applications. The authors showcase real-world examples where these technologies have been deployed, offering insights into their effectiveness and potential for future use in defense strategies.

Chapter 8 presents a comprehensive *case study of AI in military applications*, offering practical illustrations of how AI is being utilized in real-time military environments. This chapter bridges theoretical concepts with on-the-ground applications, making it an invaluable resource for defense professionals.

Chapters 9 and 10 focus on the *critical importance of cybersecurity* in military contexts. The authors analyze various *cybersecurity challenges, issues, and countermeasures*, providing an in-depth look at the threats faced by modern defense systems. Additionally, these chapters explore how blockchain can be harnessed to *fortify military cybersecurity* by providing a robust, decentralized defense against increasingly sophisticated cyber threats.

The book concludes with Chapters 11 and 12, which explore *computational intelligence techniques* that combine AI with blockchain. These chapters demonstrate the diverse possibilities for enhancing military operations through these technologies, from *autonomous decision-making* systems to *advanced data analytics* for battlefield intelligence.

This book brings together a wide array of perspectives, providing a *holistic view* of how AI and blockchain are not only transforming the military but also paving the way for a more secure and efficient future.

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Security and privacy in military application using blockchain

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1.1 INTRODUCTION

Military operations require the highest levels of security and privacy to protect sensitive information and ensure mission success. Traditional centralized systems are vulnerable to cyber threats, necessitating the use of innovative technologies such as blockchain to strengthen military applications. Blockchain is a decentralized, distributed ledger technology that uses cryptography to ensure transparency, immutability, and security. Decentralization, consensus mechanisms, cryptographic hashing, and smart contracts all contribute to the strength of blockchain. The blockchain's distributed ledger ensures that once added, data cannot be changed or deleted, resulting in a tamper-proof record of military operations and communications. Blockchain uses advanced cryptographic techniques to protect data, communications, and transactions from unauthorized access. Blockchain allows users to conduct transactions and communicate using cryptographic keys, which improves the confidentiality of military personnel and operations. Advanced blockchain implementations support private transactions, allowing information to be selectively disclosed to only authorize parties. Blockchain can be used in the military supply chain to increase transparency and traceability, lowering the risk of counterfeit goods or compromised equipment entering the system. Automated smart contracts can be used to ensure that suppliers and vendors follow predefined security protocols, reducing vulnerabilities.

Blockchain enables decentralized communication, reducing reliance on centralized communication infrastructure, which may be vulnerable to attack. In the event of a network outage, decentralized communication via blockchain ensures continuous connectivity among military units. Blockchain strengthens resilience against cyberattacks by dispersing critical military information across a decentralized network, making it difficult for adversaries to compromise a centralized point of control. Blockchain enables secure, real-time updates to security protocols, ensuring the military's ability to respond to evolving cyber threats. The use of blockchain technology in military applications represents a paradigm shift in security and privacy,

providing a strong, transparent, and decentralized framework for protecting sensitive data. Blockchain technology continues to be researched and developed to optimize its application in military contexts, promising even greater advancements in security and privacy.

The incorporation of blockchain technology into military applications marks a significant shift in security and privacy. Military organizations can significantly improve sensitive information protection by leveraging blockchain's decentralized, transparent, and tamper-resistant nature, thereby contributing to a more secure and resilient defense infrastructure. Ongoing blockchain research and development are likely to result in additional innovations in this critical area.

1.2 BACKGROUND

Improving security and privacy in military applications is critical, given the sensitive nature of military operations and the potential consequences of security breaches. Blockchain technology provides a compelling solution to these challenges. Security and privacy mechanisms in blockchain are critical for maintaining data integrity, confidentiality, and resilience in a decentralized and distributed environment. Block chain technology is located at the heart of the peer-to-peer network, or internet. This block chain technology enables fundamental communication between businesses and the government. This era of technology has undoubtedly laid the groundwork for economic and social systems. The block chain's main strength is that it makes our files more durable through damage resistance and modification [Suma (2019); Angin (2020); Joshi et al. (2018)]. Blockchain technology can significantly improve data security in military autonomous systems (Sharma et al., 2023). Blockchain's decentralized, transparent, and tamper-resistant nature can help to solve several data security challenges in autonomous military applications (Zhong & Huang, 2020). While blockchain offers promising data security solutions, extensive testing, and simulations, as well as adherence to ethical and regulatory considerations, are required to ensure the technology's effectiveness in military applications (Zhang et al., 2019).

A literature review on blockchain technology's security and privacy issues reveals a complex landscape characterized by both challenges and innovative solutions, as well as an overview of key research findings and perspectives on security and privacy concerns in the blockchain domain (Wrona & Jarosz, 2019). The literature reveals a complex landscape of security and privacy issues in blockchain technology. Researchers and practitioners continue to investigate novel solutions, and the ever-changing nature of this field emphasizes the importance of ongoing research to address emerging challenges and ensure the secure and private adoption of blockchain technology (Krishna et al., 2021; Abualsauod, 2022; Reisman, 2019)).

The integration of blockchain technology with the Internet of Things (IoT) (Lv et al., 2021) has received a lot of attention as a promising solution to security and privacy issues in IoT ecosystems. The inherent characteristics of blockchain, such as decentralization, immutability, and cryptographic security, have the potential to improve the overall security and privacy of IoT devices and data (Lee & Kim, 2021). Blockchain technology has significant potential for improving the security of the Internet of Drones (IoD). The IoD is a network of connected drones, and it is critical to ensure the security of data, communications, and transactions within that network (Wu et al., 2021; Ahmad, et al., 2021). The incorporation of blockchain technology into the IoD creates a comprehensive framework for addressing security concerns while also cultivating a trustworthy and efficient drone ecosystem (Golam et al., 2020). To ensure that blockchain is successfully implemented in IoD applications, extensive testing, compliance checks, and collaboration with regulatory bodies are required. Creating a secure and confidential blockchain solution tailored for communication within the BoT ecosystem is critical (Krichen et al., 2022; Alladi et al., 2020). Furthermore, extensive research is underway to investigate the benefits of incorporating blockchain technology and cybersecurity measures into the development of BoT applications (Al-madani & Gaikwad, 2020).

1.3 TYPES OF MILITARY APPLICATIONS

Military applications include a wide range of technologies and systems designed to meet a variety of defense and security requirements (Yazdinejad et al., 2020; Zhu et al., 2020). These applications frequently use cutting-edge technology to improve the capabilities of armed forces and ensure national security. Here are some common military applications:

- **Communications Systems:**
Advanced communication systems enable secure and dependable information exchange between military units. This includes encrypted radio and satellite communication, as well as secure data networks.
- **Surveillance and Reconnaissance:**
Military surveillance employs unmanned aerial vehicles (UAVs or drones) (Alladi et al., 2020), satellites, and other sensors to gather intelligence, monitor activities, and conduct reconnaissance in both strategic and tactical contexts.
- **Unmanned Aerial Vehicles (UAVs):**
Drones are used extensively in military applications for surveillance, reconnaissance, intelligence gathering, and even offensive operations. They can be operated remotely or autonomously.

- **Cybersecurity and Information Warfare:**
Military forces make significant investments in cybersecurity to protect critical infrastructure, networks, and information systems (Yahouza et al., 2021). Information warfare is the use of electronic and cyber tools to gain a strategic advantage.
- **Missile Defense Systems:**
These systems are designed to detect, track, and intercept incoming missiles, thereby protecting against ballistic missile threats. Examples include anti-ballistic missile systems.
- **Electronic Warfare (EW):**
EW refers to the use of electromagnetic spectrum control to disrupt or deny enemy communications and radar systems (Tariq et al., 2019). It encompasses electronic countermeasures, electronic countermeasures, and electronic support.
- **Navigation and GPS Systems:**
Military navigation systems ensure precise positioning and navigation for troops, vehicles, and aircraft. These systems frequently use encrypted GPS signals to prevent spoofing or interference.
- **Strategic Defense Systems:**
Systems designed to defend against strategic threats, such as advanced anti-aircraft and anti-missile defense systems. Examples include the Patriot missile system.
- **Biometric and Identification Systems:**
Biometric technologies, such as fingerprint and facial recognition, are used in military installations to ensure secure access and identification.

These military applications demonstrate the diverse technological landscape that contributes to the defense and security capabilities of nations worldwide (Demertzis et al., 2022). The development and deployment of these technologies frequently involve collaboration among government agencies, defense contractors, and research institutions.

1.4 BLOCKCHAIN TECHNOLOGY FOR MILITARY APPLICATIONS

Blockchain technology has great potential for improving many aspects of military applications. Its decentralized, transparent, and tamper-proof design can address the security and privacy concerns inherent in military operations. Here's how blockchain can be used for military applications.

- **Secure Communication:**
Implementing blockchain in military communication systems creates a decentralized and secure channel for transmitting sensitive data. Smart contracts can be used to automate secure communication protocols, allowing only authorized entities to access specific information.

- **Data Integrity and Immutability:**
Blockchain's immutability makes it an ideal solution for ensuring the integrity of military data (Mohril et al., 2021). Once information is recorded on the blockchain, it cannot be changed, resulting in a tamper-resistant ledger. This is critical for ensuring the accuracy of mission-critical data like operational plans, intelligence reports, and logistics information.
- **Supply Chain Management:**
Blockchain can be used in military supply chains to improve transparency, traceability, and security. Using blockchain in the supply chain ensures that every step, from manufacturing to delivery, is documented and auditable, lowering the risk of counterfeit goods, sabotage, and unauthorized access to military equipment (Harbi et al., 2023).
- **Identity Management:**
Blockchain-based identity management systems can improve the security of military access controls. Each military personnel member can have a unique cryptographic key stored on the blockchain, ensuring secure access to sensitive information and facilities.
- **Logistics and Asset Tracking:**
Integrating blockchain into logistics operations enables real-time tracking of military assets. Smart contracts can automate the verification of asset conditions, ownership transfers, and adherence to transportation protocols, ensuring the security of military resources.
- **Smart Contracts for Compliance:**
Smart contracts can help to automate and enforce compliance with military regulations and protocols. Contracts can be programmed to carry out predefined actions based on specific triggers, streamlining administrative processes and lowering the risk of human error.
- **Decentralized Command and Control:**
Blockchain can enable decentralized command and control structures, reducing the vulnerability of centralized systems to cyberattacks. Peer-to-peer communication via blockchain ensures that critical information is distributed securely and without a single point of failure.
- **Cybersecurity Resilience**
The distribution of critical military information across a blockchain network improves resilience to cyber-attacks. Blockchain's decentralization of storage and communication makes it more difficult for adversaries to compromise a centralized point, improving overall cybersecurity (Sharma et al., 2023).
- **International Collaboration and Interoperability:**
Blockchain technology can enable secure data sharing and collaboration among international military entities. Smart contracts can be used to ensure that shared information meets predefined privacy and security standards, thereby promoting interoperability.

- **Auditing and Accountability:**
The transparent and auditable nature of blockchain enables accountability in military operations. Every transaction, communication, or update can be tracked and audited, resulting in a complete record for accountability and compliance.
- **Humanitarian Aid and Disaster Relief:**
Blockchain technology can be used in military-led humanitarian efforts to ensure the transparent and accountable distribution of aid. Smart contracts can automate the distribution process, lowering the risk of corruption and ensuring that aid reaches those in need quickly.
- **Resilient Communication in Remote Areas:**
In remote or hostile areas with limited communication infrastructure, blockchain can serve as a decentralized communication network. This ensures that military units can communicate even in difficult conditions, increasing operational effectiveness.

1.5 PROPERTIES OF BLOCKCHAIN-BASED MILITARY SYSTEM

The properties of the blockchain based military system is given in the below Table 1.1.

Table 1.1 Properties of blockchain based military system

Security properties	Mechanism
Integrity	Integrity has been achieved through digital signatures and cryptographic hash functions
Authentication	Data authentication is done through digital signature and public key cryptography
Availability	Transformation of messages to the P2P network
Confidentiality	Data encryption with session keys

1.5.1 Confidentiality of the data in military system can be enhanced using double encryption (case study)

The following points are to be considered from the military system of view:

1. The military system or even the key center derives private key and shared key from master key with the help of polynomials and prime numbers. Private Key is stored on the data owner site where secret key is shared between the data owner and the cloud service provider (CSP).

2. Initially military system encrypts all data files composed of keywords using private key.
3. Before storing data files in the server or data center, the military system encrypts data files that are already encrypted using shared key. Noted that specified keywords and the data files are also encrypted.
4. When the military system searches the specified keywords from the server, he or she encrypts the specified keywords using private key and sends them to the server system.
5. After the server system has completed its search operation, the data owner receives encrypted keywords from the server and decrypts keywords using private key and performs computing on the original keywords.

From the data server point of view, the following points are to be considered:

1. Server receives encrypted keywords to be searched by the military system and encrypts keywords that are already encrypted using shared key. Server uses double encrypted keywords, searches the keywords in the cloud storage, and extracts the keywords.
2. Military Server decrypts double encrypted keywords using shared key and sends them to the data owner. The proposed Scheme is based on the symmetric cryptosystem used to search encrypted keywords over encrypted data.

1.5.1.1 Design requirements

The requirement of the proposed framework for addressing confidentiality in addition to the privacy preservation of data processing is set out below:

1. To improve the efficiency of computation for solving time and efforts, only sensitive keywords are collected from relevant data.
2. Allows the CSP to search and retrieve the relevant document without knowing what the CSP is searching and retrieving.
3. Protection of irrelevant data from exposure while coordinating with the data owner at the time of sensitive data collection. This means that the privacy of irrelevant users is not disclosed during data processing.

1.5.1.2 Proposed scheme

Initially the proposed scheme is designed to allow a single keyword and it can be extended easily for multiple keywords. The Military server specifies the single keyword denoted as x^* , which is l -bit long. The data which are stored on the server is considered as a set of documents, which is denoted as $\{X_1, X_2, X_3 \dots X_L\}$. A document $X \in \{X_1, X_2, X_3 \dots, X_L\}$ has included a sequence of words is represented as $X = \{x_1, x_2, x_3, \dots, x_n\}$ where x_i is l -bit

length. The keyword x^* and the document W is assumed that they come from the same domain. The encryption of w^* and W is represented as $E(x^*)$ and $E(X) = \{E(x_1), E(x_2), E(x_3) \dots E(x_n)\}$ where E is the encryption.

The proposed scheme is designed to ensure the confidentiality and privacy of data outsourced to the military environment. Confidentiality is provided by private key (PR_k), which is derived from master key (MS_k). Algorithm 1 is designed to generate private key in the military system. The master key is concatenated with a polynomial value and applies the hash function on concatenated result. After that verifies the hash function which is invertible or not by computing gcd on hash value hf and prime number p . Once the military system sends a request to server to outsource the data after the registration has been successful, generates a private key (PR_k) by invoking key generation algorithm. The private key is retained at military system for further processing.

Algorithm 1	Generate Private key (PR _k) at the Military System site
Input:	Polynomial f , Master Key (MS _k) and prime number p
Output:	PR _k
1:	generate 128 bits MS _k from standard Algorithm (Blowfish)
2:	Compute $hf = \text{HashFunc}(f \parallel \text{MS}_k)$
3:	Verify hf is invertible $\text{gcd}(hf, p) = 1$ Increment q and repeat Step-3 if the condition fail
4:	Invoke key generation algorithm $\text{PR}_k = \text{KeyGen}(hf, p)$
5:	record PR _k for the next processing

Algorithm 2 is designed to derive shared key (SH_k) at the Military system site. Master key value is concatenated with a generated polynomial value g . Hash function is applied on concatenated result of master key and polynomial value g . The hash code generated by the hash function has been verified. It means that gcd of hash code and polynomial value q is 1 or not. If it is true, then invoke the key generation algorithm to obtain the shared key otherwise repeat the step 3 by incrementing the q value.

Algorithm 2	Generate shared key (SH _k) at the military system site
Input:	Polynomial g , Master Key (MS _k) and prime number q
Output:	SH _k
1:	receive MS _k from Algorithm 1
2:	Compute $hg = \text{HashFunc}(g \parallel \text{MS}_k)$
3:	Verify hg is invertible $\text{gcd}(hg, q) = 1$ Increment q and repeat Step-3 if the condition fail
4:	invoke key generation algorithm $\text{SH}_k = \text{KeyGen}(hg, q)$
5:	share SH _k to server for further processing

1.5.1.3 Confidentiality by first encryption

Initially military system encrypts all data files composed of keywords using private key PR_k . Before storing data files in the server, the military system encrypts data files that are already encrypted using shared key SH_k . After some time, when the data owner wants to search the specified keyword from the cloud, the data owner encrypts his identified keyword (x^*) using his PR_k and the encrypted keyword $E(PR_k(x^*))$ is sent to the server. Noted that specified keyword and the documents are also encrypted. In this context, the server is not able to know which keyword the data owner wants and this leads to the requirement of confidentiality which will prevent the server from looking plain text data. There is a requirement that the server does not build the correspondence between original and encrypted documents. After search operation has completed by the server, the military system receives encrypted keywords from the server and decrypt the keywords using PR_k and perform the computation on original keywords.

1.5.1.4 Privacy preservation using double encryption

The military server uses encrypted keywords to search encrypted data files, retrieve those keywords, and send only appropriate data to the military system. This means that the encrypted files and their corresponding plain text files consisting of the specified keyword(s). The military system decrypts the relevant data with his PR_k and the data owner of military system can identify the data only on the relevant data. The irrelevant data, i.e. those without the keyword(s), will not be sent to the military system. To search encrypted keyword(s) over encrypted data, the server has no idea what the data owner is looking for. The important point is that the documents without having the keyword w^* will never send to the military system, so privacy of other documents can also be protected completely.

The double encryption has the following properties to make data confidentiality and privacy preservation:

1. To achieve less computational cost, only relevant keyword is encrypted by the server instead of all data has stored on the server is known as selective encryption.
2. The server encrypts the keyword which is already encrypted blindly by computing $E(SH_k(R1))$ is known as blind encryption.
3. The server verifies that if the relevant data is really belonging to the data owner and ensures that the data owner does not retrieve other irrelevant data from the server. The data owner of military system needs the server cooperation to prove that without alteration the relevant data has come from the server. It is possible that the server can delegate the key to a third party and let the relevant data is retrieved by third party.

In this case study, the underlying assumption is that the data owner of military system may have trust with server to extract all the searching results when the searching criteria is satisfied. This trust is same as trust in our postman, that is, believes the postman will carry the letters without he opens the letters secretly. The encryption process, document searching and retrieving is carried out on the server side.

1.5.1.5 Implementation

The proposed study based on double encryption is implemented using symmetric cryptosystem:

Compute $E(PR_k(X))$: The data owner of military system encrypts each document consisting of keywords with his private (PR_k) such that $R1=E(PR_k(X))$, where $E()$ is an encryption function.

Compute $E(K2(R1))$: Prior to store the documents in a cloud, the data owner of military system encrypts each document which is already encrypted with his shared key (SH_k).

Compute $E(PR_k(w^*))$: After some time, data owner of military system going to extract the specified keyword from the cloud. Prior to send the keyword to the server, data owner encrypts the keyword using private key PR_k such that $R2=E(PR_k(x^*))$.

Compute $E(SH_k(R2))$: The server receives the keyword $R2=E(PR_k(x^*))$ encrypt the keyword which is already encrypted using shared key (SH_k) such that $R3=E(SH_k(R2))$.

Search $R3=E(SH_k(R2))$: The server uses the encrypted keyword $R3=E(SH_k(R2))$ to search on encrypted documents and extract $R3=E(SH_k(R2))$ if W hold x^* (that is, $x^* \in X$).

Decrypt $R3=E(SH_k(R2))$: The server decrypt the encrypted keyword $R3=E(SH_k(R2))$ using shared key (SH_k), extract $R2=E(PR_k(x^*))$ and send it the data owner.

Decrypt $R2=E(PR_k(x^*))$: The data owner of military system receives encrypted keyword $R2=E(PR_k(x^*))$, decrypt it using private key (PR_k) and extract the original keyword for computation.

1.5.1.6 Security analysis

The security analysis of the proposed study is evaluated based on the polynomial values f , g and master key which is generated from the standard symmetric algorithm. In addition to this selection of large prime numbers p and q also influence the enhanced security of the proposed method. Here, the two different kinds of secret keys are generated such as private key as well as shared key. The private key is always kept at data owner site whereas the shared key is relatively shared between the data owner of military system and server. The shared key is delivered to the server via secure channel

so that entry of unauthorized access is successfully handled. Also, secure keyword searching is achieved to a greater extent so that only server can be able to search the required keywords. Even some situation the attacker will capture the master key then it is impossible that attacker derive the private key also shared key in extent because large prime numbers and polynomial values are incorporated during key generation. In addition to that the hash functions are used for key generation process.

1.5.1.7 Summary of the study

The proposed scheme can ensure the unauthorized disclosing of data from the server using double encryption. It is considerably secure in case that the server cannot know anything from the computation and the data owner of military system cannot know anything other than the searching results. The proposed study is designed to use double-encrypted symmetric cryptography to ensure confidentiality and privacy preservation of sensitive data. The scheme may help the server to check whether the data owner cheats on extracting irrelevant data from the server and whether there is a change in the sensitive data. Confidentiality of irrelevant data can thus be achieved in a practical way. In the future, the scheme will be extended to multiple keywords from several documents and the proposed technique can be applied in the field of Blockchain technology.

1.6 DESIGN APPROACH FOR MILITARY APPLICATIONS THROUGH BLOCKCHAIN

Designing a system to improve security and privacy in military applications using blockchain technology necessitates careful consideration of numerous factors. The following is a comprehensive design approach:

- **Requirement Analysis**
Determine specific security and privacy requirements for military operations, such as data sensitivity, communication needs, and supply chain security. Understand the current military infrastructure, protocols, and systems to ensure smooth integration with blockchain technology.
- **Use Case Identification**
Define specific military use cases where blockchain can provide the greatest security and privacy benefits. This could include secure communication, supply chain tracking, and data integrity.
- **Blockchain Platform Selection**
Choose a blockchain platform that meets your needs. Consider consensus mechanisms, scalability, interoperability, and smart contract functionality. Permissioned blockchains may be preferred for controlling access and ensuring that only authorized parties participate (Janardhana Swamy et al., 2022).

- **Data Encryption and Hashing**
Implement strong cryptographic algorithms for data encryption and hashing to protect the confidentiality and integrity of military data stored on blockchain. Use advanced encryption standards to safeguard sensitive communications and transactions.
- **Identity Management**
Integrate a secure identity management system to ensure that only authorized users can access the blockchain. Use cryptographic keys for user authentication and authorization. Consider solutions such as zero-knowledge proofs to improve user privacy.
- **Smart Contracts for Compliance**
Smart contracts can be created to automate and enforce compliance with military regulations and security standards. They can also be used to validate and authorize supply chain transactions, ensuring that only verified and secure components enter the military infrastructure.
- **Private Transactions**
Implement mechanisms for private transactions to protect sensitive data. This may entail incorporating privacy-enhancing features such as ring signatures or zk-SNARKs. Ensure that only authorized parties have access to specific transaction details while preserving the blockchain's overall transparency (Rangappa et al., 2023).
- **Supply Chain Integration**
Integrate blockchain into the military supply chain to improve traceability and reduce the risk of counterfeiting. Use RFID technology or IoT devices to connect physical assets to the blockchain, ensuring real-time tracking and visibility.
- **Decentralized Communication**
Create a decentralized communication system with blockchain to reduce reliance on centralized networks. Implement peer-to-peer messaging protocols protected by blockchain encryption to ensure the confidentiality of military communications.
- **Testing and Simulation:**
Conduct extensive testing and simulations to evaluate the system's security and privacy features in different scenarios. Simulate cyberattacks and assess the blockchain's resilience to ensure it meets military-grade security requirements.
- **Continuous Monitoring and Updates**
Set up a continuous monitoring system to identify and respond to potential security threats. Regularly update the blockchain system to address new vulnerabilities and incorporate the most recent advancements in blockchain security.

1.7 BLOCKCHAIN ARCHITECTURE FOR MILITARY APPLICATIONS

Designing the architecture of military applications using blockchain involves creating a robust and secure framework that addresses the unique requirements and challenges of military operations. Figure 1.1 shows the architectural framework for military applications leveraging blockchain technology.

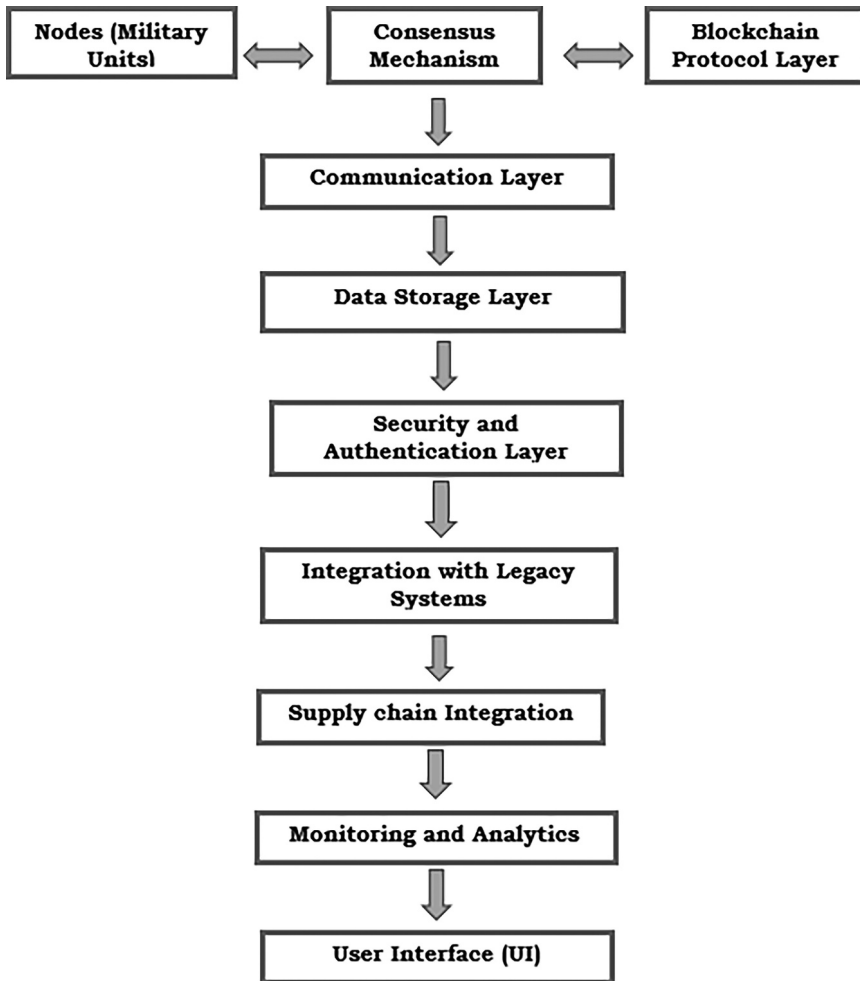


Figure 1.1 Blockchain architecture for military application.

- **Nodes (Participants)**
Military Units: Each military unit or entity participating in the blockchain network acts as a node. These nodes could represent different branches of the military, such as army, navy, air force, etc.
- **Consensus Mechanism:**
Proof of Authority (PoA) or Practical Byzantine Fault Tolerance (PBFT): Given the sensitive nature of military operations, a consensus mechanism like PoA or PBFT can be more suitable than energy-intensive alternatives like Proof of Work (PoW). PoA relies on trusted nodes, while PBFT is known for its Byzantine fault tolerance.
- **Blockchain Protocol Layer:**
Smart Contracts: Smart contracts can be implemented for automating and enforcing specific military protocols, such as rules of engagement, logistics coordination, and supply chain management.
- **Communication Layer:**
Secure Communication Protocols: Military-grade encryption and secure communication protocols are crucial to protect the integrity and confidentiality of data transmitted between nodes. This layer ensures that sensitive information is not compromised during transmission.
- **Data Storage Layer:**
Distributed Ledger: The core of the blockchain, a distributed ledger ensures that data is replicated across multiple nodes, providing redundancy and resilience against attacks. It also maintains an immutable record of transactions and activities.
- **Identity and Access Management:**
Public Key Infrastructure (PKI): To ensure secure and authenticated access, PKI can be employed to manage digital identities and cryptographic keys for military personnel and devices.
- **Security and Authentication:**
Firewalls, Intrusion Detection Systems (IDS), and Anti-Malware Tools: These components help protect the network against cyber threats and unauthorized access. Multi-factor authentication and biometric measures can be integrated for additional security.
- **Integration with Legacy Systems:**
Middleware and APIs: To facilitate interoperability with existing military systems and databases, middleware and APIs can be employed. This ensures smooth integration with legacy technologies.
- **Supply Chain Integration:**
Blockchain-based Supply Chain Management: For military logistics, a blockchain-based supply chain management system can enhance transparency and traceability of critical resources, including ammunition, equipment, and fuel.
- **Monitoring and Analytics:**
Data Analytics Tools: Integrating analytics tools allows for real-time monitoring of the blockchain network, enabling the identification of

potential issues or anomalies. This layer can also support decision-making processes by providing insights into military operations.

- **User Interface (UI):**

Command and Control Interface: A user-friendly interface designed for military commanders to interact with the blockchain network, view analytics, and execute smart contract-based commands.

1.8 FUTURE RESEARCH DIRECTIONS

Future blockchain research for military applications will most likely focus on addressing specific challenges and improving defense and security system capabilities. Here are some potential research areas that could influence the future development of blockchain technology in the military context:

- **Scalability and Performance:**

Creating and optimizing scalable blockchain architectures that can handle the high transaction volumes and data throughput required by military systems while maintaining performance.

- **Interoperability with Legacy Systems:**

Investigating methods to ensure seamless interoperability between blockchain networks and traditional military systems, allowing for a smooth transition to blockchain-powered solutions.

- **Privacy-Preserving Technologies:**

Developing privacy-preserving techniques, such as zero-knowledge proofs and homomorphic encryption, to improve the confidentiality of military operations on the blockchain.

- **Energy-Efficient Consensus Mechanisms:**

To reduce the environmental impact of blockchain technology in military applications, researchers are investigating and developing energy-efficient consensus mechanisms (such as Proof-of-Stake and Practical Byzantine Fault Tolerance).

- **Resilience to Cyber Threats:**

To protect military blockchain networks from cyber threats, we are investigating and developing robust cybersecurity measures such as advanced encryption, anomaly detection, and consensus algorithm enhancements.

- **Smart Contracts for Automated Operations:**

Enhancing smart contract capabilities to securely and efficiently automate and execute complex military operations, logistics, and decision-making processes.

- **Dynamic Access Control:**

Investigating novel access control models that can adapt to changing military scenarios while ensuring authorized access and preventing unauthorized use of sensitive data.

- **Blockchain for Supply Chain Security:**
Developing blockchain-based supply chain solutions that provide increased transparency, traceability, and real-time monitoring to ensure military logistics integrity and security.

1.9 CONCLUSION

The use of blockchain technology in military applications represents a new paradigm for improving security, efficiency, and transparency in defense and national security operations. Blockchain's distinguishing features, such as decentralization, immutability, and cryptographic security, provide several advantages that can address key challenges faced by military organizations. The immutability of the blockchain ledger ensures the accuracy of military data, ranging from logistics and supply chain information to operational intelligence. Transparency is increased by keeping a shared and auditable record of transactions, which aids decision-making. Smart contracts allow for the automation of a variety of military processes, including supply chain management and complex operational workflows. This automation lowers the risk of errors, improves efficiency, and ensures that predefined rules are followed without the use of intermediaries.

In navigating the future of blockchain in military applications, ongoing research and collaboration among government agencies, defense contractors, and academic institutions will be critical. As the technology advances, addressing scalability, interoperability, and privacy concerns will be critical to blockchain's widespread adoption in defense and national security. Overall, blockchain has the potential to transform military operations by increasing their security, transparency, and efficiency in the face of evolving threats and challenges.

LIST OF ACRONYMS

IoT	Internet of Things
IoD	Internet of Drones
BoT	Blockchain of Things
UAV	Unmanned Aerial Vehicles
EW	Electronic Warfare
GPS	Global Positioning System
C&C	Command and Control
CSP	Cloud Service Provider
PRK	Private Key
MSK	Master Secret Key
HF	Hash Function
SHK	Shared Secret Key

RFID	Radio Frequency Identification
zk-SNARKs	Zero-Knowledge Succinct Non-Interactive Argument of Knowledge
PoA	Proof of Authority
PBFT	Practical Byzantine Fault Tolerance
PoW	Proof of Work
PKI	Public Key Infrastructure
IDS	Intrusion Detection System
API	Application Program Interface
UI	User Interface
PoS	Proof of Stake

REFERENCES

- Abualsaud, E. H. (2022). A hybrid blockchain method in internet of things for privacy and security in unmanned aerial vehicles network. *Computers and Electrical Engineering*, 99, 107847.
- Ahmad, R. W., Hasan, H., Yaqoob, I., Salah, K., Jayaraman, R., & Omar, M. (2021). Blockchain for aerospace and defense: Opportunities and open research challenges. *Computers & Industrial Engineering*, 151, 106982.
- Alladi, T., Chamola, V., Sahu, N., & Guizani, M. (2020). Applications of blockchain in unmanned aerial vehicles: A review. *Vehicular Communications*, 23, 100249.
- Al-madani, A. M., & Gaikwad, A. T. (2020, February). IoT data security via blockchain technology and service-centric networking. In *2020 International Conference on Inventive Computation Technologies (ICICT)* (pp. 17–21). IEEE.
- Angin, P. (2020). Blockchain-based data security in military autonomous systems. *Avrupa Bilim ve Teknoloji Dergisi*, 2, 362–368.
- Demertzis, K., Kikiras, P., & Iliadis, L. (2022, June). A Blockchained Secure and Integrity-Preserved Architecture for Military Logistics Operations. In Demertzis, K., Iliadis, L., Kikiras, P., & Tziritas, N. (eds.), *International Conference on Engineering Applications of Neural Networks* (pp. 271–283). Cham: Springer International Publishing.
- Golam, M., Lee, J. M., & Kim, D. S. (2020, October). A UAV-assisted blockchain based secure device-to-device communication in internet of military things. In *2020 International Conference on Information and Communication Technology Convergence (ICTC)* (pp. 1896–1898). IEEE.
- Harbi, Y., Medani, K., Gherbi, C., Senouci, O., Aliouat, Z., & Harous, S. (2023). A systematic literature review of blockchain technology for internet of drones security. *Arabian Journal for Science and Engineering*, 48(2), 1053–1074.
- Janardhana Swamy, G.B., Janardhana, D.R., Vijay, C.P., & Ravi, V. (2022). Blockchain-enabled IoT integrated autonomous sewage management system. In: Gururaj, H. L., Ravi Kumar, V., Goundar, S., Elngar, A. A., & Swathi, B.H. (eds.) *Convergence of internet of things and blockchain technologies*. Springer Nature
- Joshi, A. P., Han, M., & Wang, Y. (2018). A survey on security and privacy issues of blockchain technology. *Mathematical Foundations of Computing*, 1(2), 121–147.
- Krichen, M., Ammi, M., Mihoub, A., & Almutiq, M. (2022). Blockchain for modern applications: A survey. *Sensors*, 22(14), 5274.

- Krishna, B., Rajkumar, P., & Velde, V. (2021). Integration of blockchain technology for security and privacy in internet of things. *Materials Today: Proceedings*, 1–5.
- Lee, S., & Kim, S. (2021). Blockchain as a cyber defense: Opportunities, applications, and challenges. *IEEE Access*, 10, 2602–2618.
- Lv, Z., Qiao, L., Hossain, M. S., & Choi, B. J. (2021). Analysis of using blockchain to protect the privacy of drone big data. *IEEE Network*, 35(1), 44–49.
- Mohril, R. S., Solanki, B. S., Lad, B. K., & Kulkarni, M. S. (2021). Blockchain enabled maintenance management framework for military equipment. *IEEE Transactions on Engineering Management*, 69(6), 3938–3951.
- Rangappa, J.D., Manu, A.P., Kariyappa, S., Chinnababu, S.K., Lokesh, G.H., & Flammini, F. (2023). A lightweight blockchain to secure data communication in IoT network on healthcare system. *International Journal of Safety and Security Engineering*, 13(6), 1015–1024. <https://doi.org/10.18280/ijssse.130604>
- Reisman, R. J. (2019, January). Air traffic management blockchain infrastructure for security, authentication, and privacy. In *AIAA Scitech Forum* (No. ARC-E-DAA-TN63825).
- Sharma, G., Sharma, D. K., & Kumar, A. (2023). Role of cybersecurity and Blockchain in battlefield of things. *Internet Technology Letters*, 3, e406.
- Suma, V. (2019). Security and privacy mechanism using blockchain. *Journal of Ubiquitous Computing and Communication Technologies (UCCT)*, 1(01), 4554.
- Tariq, U., Ibrahim, A., Ahmad, T., Bouteraa, Y., & Elmogy, A. (2019). Blockchain in internet-of-things: A necessity framework for security, reliability, transparency, immutability and liability. *IET Communications*, 13(19), 3187–3192.
- Wrona, K., & Jarosz, M. (2019, April). Use of blockchains for secure binding of metadata in military applications of IoT. In *2019 IEEE 5th World Forum on Internet of Things (WF-IoT)* (pp. 213–218). IEEE.
- Wu, Y., Dai, H. N., Wang, H., & Choo, K. K. R. (2021). Blockchain-based privacy preservation for 5G-enabled drone communications. *IEEE Network*, 35(1), 50–56.
- Yahuza, M., Idris, M. Y. I., Ahmedy, I. B., Wahab, A. W. A., Nandy, T., Noor, N. M., & Bala, A. (2021). Internet of drones' security and privacy issues: Taxonomy and open challenges. *IEEE Access*, 9, 57243–57270.
- Yazdinejad, A., Parizi, R. M., Dehghantanha, A., Karimipour, H., Srivastava, G., & Aledhari, M. (2020). Enabling drones in the internet of things with decentralized blockchain-based security. *IEEE Internet of Things Journal*, 8(8), 6406–6415.
- Zhang, R., Xue, R., & Liu, L. (2019). Security and privacy on blockchain. *ACM Computing Surveys (CSUR)*, 52(3), 1–34.
- Zhong, S., & Huang, X. (2020). Special focus on security and privacy in blockchain-based applications. *Science China Information Sciences*, 63, 1–2.
- Zhu, Y., Zhang, X., Ju, Z. Y., & Wang, C. C. (2020, April). A study of blockchain technology development and military application prospects. *Journal of Physics: Conference Series*, 1507(5), 052018.

Document management across departments in defense using blockchain technology

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2.1 INTRODUCTION TO DOCUMENT MANAGEMENT SYSTEM

Document management plays a crucial role in organizations of all sizes and across various industries. However, traditional document management systems often suffer from inefficiencies, security vulnerabilities, and a lack of transparency. The use of a blockchain based Document Management System (DMS) has drawn a lot of attention as a solution to these problems. This background study aims to explore the benefits and considerations associated with adopting a blockchain solution for document management.

Numerous studies focused on traditional document management systems, examining their challenges, features, and benefits. These works often discuss the importance of efficient storage, retrieval, version control, collaboration, and security in document management. Research has explored numerous technologies and techniques used in document management, such as cloud-based systems and artificial intelligence for automated document classification and indexing.

1. Blockchain for Document Management: The application of blockchain technology in document management has gained attention in recent years. Research has indicated that blockchain technology may offer advantages such as enhanced security, immutable records, decentralization, and transparency. Research has explored various aspects of blockchain in document management, including blockchain-based access control, blockchain-enabled digital signatures, and blockchain-based document verification and provenance.
2. Blockchain-Based Document Management Systems: Several studies have proposed and implemented blockchain-based document management systems. These systems aim to leverage benefits of blockchain while addressing limitations of traditional document management.
3. IPFS: A promising new technique for distributed and decentralized file storage is IPFS. Some studies have explored the integration of IPFS with blockchain for document management purposes.

2.2 CHALLENGES IN TRADITIONAL DOCUMENT MANAGEMENT SYSTEMS

Traditional document management systems typically rely on centralized databases or file servers, which present several limitations. These consist of one single point of failure, exposure to data breaches, transparency in document modifications, and challenges in confirming the legitimacy of documents. These challenges contribute to increased operational costs, compliance risks, and reduced efficiency in the DM process.

2.3 BLOCKCHAIN TECHNOLOGY OVERVIEW

Blockchain, underlying technology behind cryptocurrencies like Bitcoin, offers decentralized, secure and transparent framework for various applications beyond finance. In the context of document management (DM), blockchain provides the following key advantages:

1. **Decentralization:** Blockchain operates as a distributed ledger, eliminating the need for central authority. Documents are stored across a network of interconnected nodes, ensuring redundancy and fault tolerance.
2. **Immutability and Data Integrity:** Once a document is recorded on the blockchain, it becomes virtually immutable, preventing unauthorized modifications or tampering. The use of cryptographic hashing ensures data integrity, as any change to a document will result in a different hash value.
3. **Transparency and Auditability:** Every transaction or modification within blockchain is recorded in a transparent and traceable manner. This audit trail enables participants to track document revisions, verify authenticity, and establish an unalterable record of actions.
4. **Enhanced Security:** Blockchain employs cryptographic algorithms and consensus protocols (Anupama and Sunitha 2022) [1] to secure document transactions and access control. Public-private key pairs ensure secure authentication and encryption, protecting sensitive documents from unauthorized access.

2.4 BENEFITS OF BLOCKCHAIN-BASED DOCUMENT MANAGEMENT

Implementing a blockchain-based document management system offers several significant benefits:

1. **Security and Trust:** Blockchain's inherent security features, including immutability, data encryption, and decentralized storage, enhance document security and instill trust among participants. This ensures

- the integrity and authenticity of documents, reducing risk of fraud or unauthorized modifications.
2. **Efficiency and Collaboration:** Blockchain streamlines document workflows by eliminating the need for intermediaries and manual processes. Smart contracts are self-executing agreements coded on blockchain, automate document-related tasks, enabling faster processing, reduced administrative burden, and improved collaboration between stakeholders.
 3. **Version Control and Tracking:** Blockchain's ability to maintain an immutable record of document versions facilitates easy tracking of changes over time. Participants can access a complete history of document revisions, enabling transparency, accountability, and simplified auditing processes.
 4. **Cost Savings:** By eliminating the reliance on physical storage, paper-based processes, and manual verification, a blockchain-based DMS can significantly reduce operational costs. Additionally, the decentralized nature of blockchain eliminates need for intermediaries, reducing associated fees.

Adoption of a blockchain-based DMS offers significant potential for improving security, transparency, efficiency, and collaboration in defense organizations. By leveraging blockchain's decentralized nature, immutability, and cryptographic security, defense organizations can mitigate traditional document management challenges and create a more secure and efficient document ecosystem. However, careful consideration of scalability, integration, compliance, and governance aspects is necessary for successful implementation. The benefits of blockchain-based document management systems make them a promising solution for defense organizations seeking enhanced document security, integrity, and operational efficiency.

2.5 LITERATURE SURVEY

Paper (Benet 2014) [2] presents IPFS, a revolutionary peer-to-peer DFS that aims to reshape the way files are stored, addressed, and shared on the internet. It outlines the core concepts of IPFS, such as content addressing and distributed hash tables, which allow files to be addressed and shared in a decentralized and resilient manner. Paper discusses design principles and mechanisms behind IPFS, including MerkleDAG, a directed acyclic graph structure, and the InterPlanetary Naming System (IPNS), which enables mutable file references. It explores the benefits of IPFS, such as improved data availability, reduced reliance on centralized servers, and resistance to censorship. The paper also discusses potential applications of IPFS in areas like distributed data storage, decentralized websites, and collaborative content sharing, with the goal of creating a more robust and scalable web infrastructure.

Paper (Battah et al. 2020) [3] provides fully decentralized blockchain based solution that uses multiple oracles to implement computationally expensive proxy re-encryption algorithms, allowing access to encrypted shared data stored on a public, decentralized storage platform like IPFS. Multi-party authorization is implemented using smart contracts in ethereum in this system.

Paper (Daniel and Tschorsch 2022) [4] gives technical synopsis of upcoming generation of data networks. Chosen data networks to highlight recent advancements and present broad ideas. A more thorough description of IPFS and a synopsis of Swarm, the Hypercore Protocol, SAFE, Storj, and Arweave were given.

Paper (Sun et al. 2020a) [5] provides an approach for the safe upkeep and storage of insurance records in insurance system's IPFS storage environment. To increase staff productivity, this strategy outsources insurance record encryption using the fog node. Moreover, to guarantee platform security and prevent a centralized mechanism's single point of failure, encrypted insurance records were kept on IPFS.

Paper (Bisht et al. 2023) [6] provides a new, effective, and dynamic PHR-sharing strategy. Integrated blockchain technology, searchable symmetric encryption and IPFS. Decentralized storage system ensures the privacy of PHRs, forward security, and validity of search results.

Blockchain based secure distributed data management (Shahjalal et al. 2022) [7] has been used to demonstrate a full end-to-end long-range wide area network. This system is used for variety of safe Internet of things applications. To maintain data security, integrity, and availability, data collected dynamically by many LoRa sensors is encrypted in LoRa server. Encrypted material is then automatically stored in IPFS.

The suggested approach consists of 3 layers: BC layer, storage layer, and message distribution layer. To guarantee that only authorized vehicles join the suggested network and communicate with one another, all vehicles in first layer are registered by a CA. Subsequently, in second layer, data transmitted by cars is kept at the IPFS (Javed et al. 2022) [8], an AI-powered system integrated with RSU. This guarantees a decrease in storage expenses and the accessibility of data. Also, by hiding the true identities of the cars, the privacy of the owners is protected. Additionally, the third layer of BC stores hashes of data that is stored in IPFS.

Paper (Zhou et al. 2022) [9] suggested a unique storage method that would reduce block sizes by compressing fixed-length fields of transactions and removing transaction data whose spent transaction output ratio was more than 67%. Then, to increase scalability of Bitcoin blockchain, freshly created block files were uploaded to IPFS private network.

Paper (Hasan et al. 2022) [10] suggested a blockchain-based solution that enables data chunks to be exchanged in decentralized, visible, traceable, dependable, safe, and trustworthy manner for IOT streaming devices with limited resources. protected the IoT data's privacy and security by using a

network of proxy re-encryption. Solved issue of large-scale data storage by using decentralized storage of IPFS to store and distribute IoT streaming data.

Paper (Dwivedi, Amin, and Vollala 2021) [11] suggested a revolutionary decentralized architecture that does not require a cloud server for the VANET. This paper also proposes a protocol that uses blockchain method to secure event data and authentication of vehicles. Registered user uses this protocol to safely retrieve event data from IPFS.

Paper (Alizadeh, Andersson, and Schelen 2020) [12] suggested storing data immutably and decentralized using IPFS distributed hash table (DHT) to reduce high cost of storage. Storage system that integrates blockchain technology with other storage systems ought to be built on unchangeable data and enable the DHT to be cleansed of data belonging to malevolent users.

Paper (Nizamuddin, Hasan, and Salah 2018a) [13] suggested a way to ensure the originality and legitimacy of digital content that is published and shared for free online, including music, movies, and novels. Approach makes use of a combination of recently developed technologies, the main ones being blockchain smart contracts and IPFS.

Paper (Das et al. 2022) [14] presented study which employs blockchain technology to enhance integrity of data in DM for construction applications. This is achieved through three methods: (1) smart contract technology for irreversible and irrevocable approval workflow logic; (2) blockchain ledger for irreversible recording of document changes; and (3) blockchain based data structure for document version history integrity.

The InterPlanetary Filesystem (Steichen et al. 2018a) [15] is enhanced in this work to enable access-controlled file sharing using smart contracts in Ethereum. Improved IPFS software enforces access control list, which is maintained by smart contract.

The suggested method makes use of blockchain (Sultana et al. 2023) [16] network, where every academic credential is kept on the network as digital asset. These digital certificates are timestamped, secured cryptographically and linked to distinct identities, like public keys or hashes, guaranteeing their integrity and unchangeability.

Goal of the technique being given is to introduce data aggregation based on IPFS (Subathra, Antonidoss, and Singh 2022) [17] and decentralized consensus blockchain for efficient classification of data and storage. Meta hyperparameter random forest (MHP-RF) classifier is used to identify attack. The transaction data is safely saved on the server using blockchain system based on smart contracts as soon as attack is identified.

Paper (Han et al. 2021) [18] suggested use of a decentralized document management system (DDMS) to strengthen digital document security. Initially, DDMS shares access rights across several users to increase document security. This is accomplished by DDMS using a symmetric key encryption and a Shamir's secret sharing mechanism to split the key.

Study developed attribute based encryption scheme for safe storage and effective exchange of electronic medical records (Sun et al. 2020b) [19] in an IPFS storage environment, based on ciphertext policy attribute based encryption system and blockchain technology.

Paper (Vimal and Srivatsa 2019) [20] provided a novel strategy to enhancing P2P file sharing system's effectiveness using blockchain technology and IPFS that takes proximity awareness and trustworthiness into account when transferring files. Discussed the problems that blockchain technology faces and how IPFS might be used to solve them.

Using blockchain (Sarang et al. 2023) [21] technology and the use cases involving digital document forms, secure online application has been developed for all document processing needs. The document is stored on an IPFS and web application securely stores the document's identification in the distributed Blockchain network (Polygon) upon request for predefined templates. Additionally, it offers the ability to confirm the document's legitimacy.

Smart contracts are used in construction projects to streamline workflows for document (Das, Tao, and Cheng 2021) [22] approval, supporting activities like design review and information requests. It suggested to use blockchain ledger data architecture for document version control and workflow tracking.

Paper (Choudaiah 2021) [23] recommended one solution to double-spending issue via a peer-to-peer network. Network timestamps transactions using hashing to create record that cannot be changed without reiterating proof-of-work.

In order to guarantee security and efficiency of the construction papers used in infrastructure project process, this study examines application of blockchain and develops a blockchain-based digital document management system (Zhao et al. 2022) [24]. The system completes the distributed file storage via the IPFS system, designs and writes smart contracts in response to queries from pertinent infrastructure units and leverages the blockchain's chain structure to enable file uploading throughout the concrete construction of the power grid infrastructure.

Paper (Shanmugaraja et al. 2022) [25] suggested blockchain system to store documents in electronic file format tautly on a blockchain network. The document will be permanently entered into the distributed databases and will be protected from duplication thanks to its immutable attribute.

Paper (Solarte-Rivera et al. 2018) [26] describes the creation of a test application that, in its place, permits the publication and distribution of embargo papers inside a document management system while maintaining the secrecy, accessibility, and dependability of all data stored in blockchain. It was discovered that very structure of blockchain made it easier to distribute and make the papers available as well as to keep track of and manage modifications to them.

Paper (Kumar, Marchang, and Tripathi 2020) [27] suggested a distributed off-chain medical data storage system utilizing blockchain technology and IPFS (Interplanetary File System). The suggested framework protects patient privacy and makes it simple for authorized parties, like medical providers, to access medical data (e.g., doctors and nurses).

Paper (Shapna Muralidharan 2019) [28] demonstrated, through experimental evaluation using Raspberry Pis, viability and benefits of an IoT framework in a Peer-to-Peer decentralized storage system utilizing IPFS.

Purpose of the blockchain-based solution that is presented in this article (Zaky and Nugraha 2019) [29] is to prevent system failures from causing activity time delays, which could reduce efficiency of business processes. Blockchain transforms official document system into a “Trusless Operation,” whereby a consensus mechanism ensures that every member of a separate organizational unit has a copy of transaction data.

With blockchain based system, process of managing digital documents (Islamay, Purwanto, and Ruriawan 2022) [30] can be done more openly by limiting its use to the read and write method, thereby preserving the legal worth of the documents that are saved. An end product of this research is a decentralized digital document processing application.

Literature review on Document management system in various papers are discussed in Table 2.1.

Table 2.1 Review on document management system

<i>References</i>	<i>Technology used</i>	<i>Functionality</i>
(Zou et al. 2019) [31]	IPFS, Blockchain	Suggested a blockchain based reporting system to be used in smart city administration.
(Huang et al. 2020) [32]	Ethereum, databases	Suggested Ethereum based system for protecting many domains participants' trustworthiness and access control.
(Nyaletay et al. 2019) [33]	IPFS and Hyperledger Fabric	Suggested a Block IPFS in order to track events of access and grant access of audit to IPFS stored files.
(Naz et al. 2019) [34]	Smart Contract, IPFS, RSA	Suggested a safe data sharing paradigm for digital asset delivery based on IPFS.
(Wang, Zhang, and Zhang 2018) [35]	Ethereum, IPFS and Attribute based encryption	Suggested an access control method for decentralized storage systems based on smart contracts.

(Continued)

Table 2.1 (Continued)

<i>References</i>	<i>Technology used</i>	<i>Functionality</i>
(Nizamuddin, Hasan, and Salah 2018b) [36]	Ethereum smart contract and IPFS	Proposed a remedy to ensure that original digital works published online are authentic.
(Ali, Dolui, and Antonelli 2017) [37]	IPFS, Sidechain, Consortium Blockchain	Suggested a modular consortium architecture to protect IoT data privacy.
(Steichen et al. 2018b) [38]	IPFS, Ethereum, Smart contract	Provided a file sharing access control list for the suggested IPFS system.
Proposed Chapter	IPFS, Ethereum, Smart Contract	Proposed a system to reduce the transaction cost of huge documents shared across the departments in defense

2.6 PROPOSED DOCUMENT MANAGEMENT SYSTEM

In today's digital landscape, defense organizations frequently encounter challenges with document authenticity and the timely transfer of ownership for digital documents. As departments across the organization transmit and modify common documents, ensuring integrity and accuracy of data becomes a promising concern. Additionally, average time taken for transactions poses a significant hurdle, leading to delays and inefficiencies in document ownership transfer.

To address these issues, there is need for an application that leverages blockchain technology, such as Ethereum, to reduce gas fees and minimize the average transaction time. This application should enable tamper-proof and secure document sharing, ensuring authenticity and integrity of data within. Moreover, it should streamline the transfer of ownership for digital documents, enabling swift and efficient processes across departments in defense. By developing such an application, defense organizations can enhance data security, improve operational efficiency, and expedite the ownership transfer of digital documents, thereby overcoming the challenges posed by document authenticity and transaction time.

Objectives Defined:

1. To provide Blockchain solutions for traditional Document Management System which ensures
 - The authenticity and integrity of shared documents.
 - Data security within the document management system.
2. Reduce gas fees on Ethereum platform associated with document transactions.
3. Minimize the average time taken for document transactions.
4. Enable secure and efficient transfer of ownership for digital documents.

2.7 HIGH LEVEL ARCHITECTURE

The system provides members in the military with the flexibility to upload documents in various file formats, such as Word, PDF and Excel. This adaptability ensures that the application can handle a diverse range of file types, making it convenient for military members to share a wide variety of materials with ease.

Moreover, the application goes beyond supporting individual file uploads. It allows military members to conveniently upload multiple files simultaneously. This time-saving feature enables users in defense to select and upload numerous files in one go, eliminating the need for separate uploads and simplifying the process of sharing extensive sets of materials, such as data sheets.

Military members benefit from the system's functionality to access and view documents uploaded by any member in defense. The application's user interface is thoughtfully designed to prioritize user-friendliness, ensuring that military members can effortlessly navigate and explore the documents at their disposal.

With a clean and intuitive interface, military members can easily search for specific documents, browse through different sections or categories, and access the relevant resources they need to support their journey.

In addition to document viewing, another valuable feature of the application is the capability for military members to grant ownership of documents to other users. This empowering functionality enables users in defense to delegate document management tasks or share collaborative responsibilities with trusted individuals. By assigning ownership, military members can authorize specific users, such as co-owner or assistants, to have control over documents, including editing permissions or managing updates. This fosters teamwork, streamlines document collaboration, and enhances the overall efficiency of resource management within the application.

Overall, the system's flexibility in accepting diverse file formats, the ability to upload multiple files at once, the user-friendly interface for military members, and the provision for granting document ownership contribute to a comprehensive and streamlined document sharing and management experience, catering to needs of military members.

2.8 SYSTEM ARCHITECTURE

Once a file is uploaded, it undergoes a multi-step process to ensure its security, integrity, and verifiability. Initially, the file is temporarily stored on a server. Subsequently, a new block is created on Ethereum blockchain network, and file is added to it as a transaction. This integration with blockchain technology ensures a decentralized and tamper-resistant record of the file's existence and history.

Following the creation of the block, it is transmitted to IPFS. IPFS functions as a DFS that leverages a P2P network for file storage and retrieval. As the block is sent to IPFS, it is broken into smaller fragments, which are then dispersed across multiple nodes in network. Each fragment is encrypted and stored redundantly on different nodes, ensuring both data integrity and availability.

Simultaneously, unique hash is generated for file stored on IPFS. Hash generated acts as a digital fingerprint of file, enabling easy verification of its authenticity and detecting any modifications. Additionally, the file undergoes the SHA256 algorithm, which produces a file hash. This hash serves as an additional layer of integrity verification, enhancing the overall security of the file.

The file hash and IPFS hash, along with relevant metadata, are then incorporated into a new block. This process simplifies future verification of the file's authenticity and enables users to validate the integrity of the file by comparing its hash values with the stored ones. If any alterations are made to the document, the file hash will change, signaling that modifications have occurred.

Moreover, the application maintains a comprehensive log of the users who have accessed and edited the files. This feature provides a robust audit trail, allowing users to track the activities associated with their uploaded files. Furthermore, authorized users are granted the ability to view and edit files uploaded by others, promoting collaborative workflows, and fostering seamless sharing of information within the application.

By combining blockchain technology, IPFS storage, file hashing, and activity tracking, the application ensures secure storage, integrity verification, and controlled access of uploaded files. These measures enhance data protection, facilitate collaborative work environments, and provide users with peace of mind regarding authenticity and integrity of their shared documents.

The architecture for document management across departments as shown in Figure 2.1 leverages blockchain to enhance security, transparency, and efficiency in handling defense documents:

1. **Blockchain Network:** The system utilizes a blockchain network, such as Ethereum, as a decentralized and immutable ledger to store and manage document-related transactions. The blockchain provides a transparent and tamper-resistant platform for recording document activities.
2. **Smart Contracts:** Smart contracts are employed to automate document management processes and enforce predefined rules and workflows. These contracts are written in languages like Solidity and define conditions and actions associated with document creation, access control, versioning, and verification.

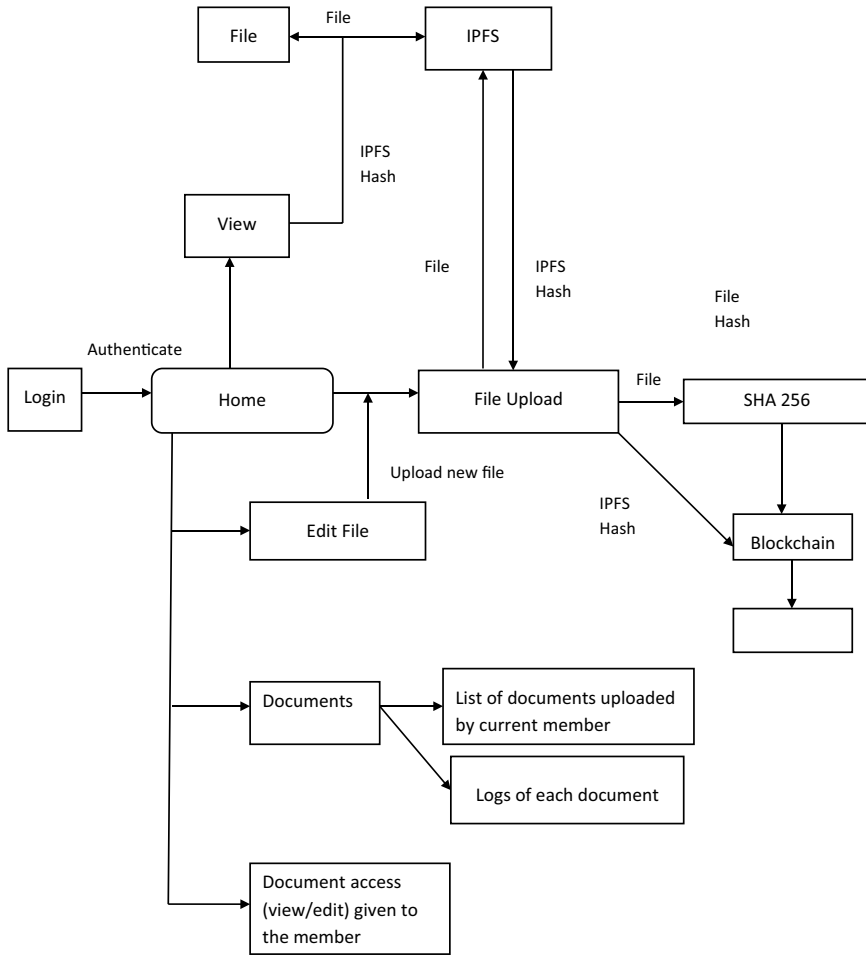


Figure 2.1 System architecture.

3. **Decentralized Storage:** Documents are stored in a decentralized manner using technologies like IPFS. IPFS breaks down documents into smaller pieces, encrypts them, and distributes them across multiple nodes in network. This redundancy ensures data integrity and availability.
4. **User Interface:** An intuitive user interface is developed to facilitate document management operations. The interface allows users to upload, view, search, and collaborate on documents, streamlining the document management workflow across departments.
5. **Blockchain Integration:** The system connects with the blockchain network using appropriate frameworks like Web3.js. This integration

enables seamless communication between the user interface and the blockchain, allowing users to interact with document-related functionalities while ensuring data consistency and integrity.

6. **Access Control and Permissions:** Blockchain-based access control mechanisms are implemented to regulate document access and permissions. Smart contracts enforce predefined rules, such as granting read or write permissions based on user's role and department. This ensures that only individuals who are authorized can access and modify documents.
7. **Document Verification and Auditing:** Blockchain's inherent immutability enables easy verification of document authenticity. Each document is assigned a unique hash, serving as a digital fingerprint. Any alterations to the document will result in a different hash, indicating tampering. The blockchain maintains an audit trail of document activities, providing a transparent history of document changes, views, and approvals.
8. **Testing and Enhancements:** Rigorous testing is conducted to ensure functionality and reliability of the document management system. Feedback from users is collected to identify areas of improvement, and iterative enhancements are made to optimize the system's performance and user experience.

By utilizing blockchain technology, smart contracts, decentralized storage, and secure access controls, the document management system enhances security, transparency, and collaboration across departments. It provides a robust and auditable framework for managing documents, streamlining workflows, and ensuring data integrity throughout the document lifecycle.

2.9 IMPLEMENTATION

A. Blockchain approach

The proposed blockchain approach for document management involves integration of IPFS and Ethereum blockchain. By leveraging IPFS, the system ensures decentralized and secure document storage and distribution. This eliminates the need for a centralized document management system and enables participants to access documents from anywhere. Use of blockchain technology, specifically Ethereum, adds an extra security layer, trust and immutability to process. All document-related activities and transactions are recorded on blockchain, creating an auditable and tamper proof record. This instills confidence in authenticity of documents and their revisions, reducing risk of unauthorized alterations or document tampering.

In addition to security, the proposed blockchain-based solution offers enhanced features for document management. System maintains different

versions of documents, allowing participants to track changes over time. This version control feature is particularly useful for documents that require frequent revisions, such as contracts or legal agreements. By providing a transparent history of all document versions, the system ensures transparency, accountability, and easy tracking of modifications. To protect documents from unauthorized access and manipulation, an access control mechanism is implemented. Editors have the authority to modify and update documents, while viewers can only access and view them. This access control mechanism ensures document confidentiality and integrity, safeguarding sensitive information from unauthorized individuals. Overall, the blockchain approach provides secure, reliable, and efficient solution for document management, reducing costs, promoting collaboration, and improving the overall workflow for participating entities.

How can gas fees be reduced?

Proposed blockchain approach for document management involving integration of IPFS and Ethereum blockchain can help reduce gas fees. Ethereum gas fees are transaction fees paid to miners for processing and validating transactions on Ethereum network. By utilizing IPFS for decentralized document storage and distribution, the system can reduce the frequency of on-chain transactions.

With IPFS, documents are stored and accessed directly from the distributed file system, reducing the need for frequent interactions with the Ethereum blockchain. Instead of storing the entire document on-chain, the IPFS integration allows for the storage of document hashes or references on the blockchain, pointing to the actual document stored in the IPFS network.

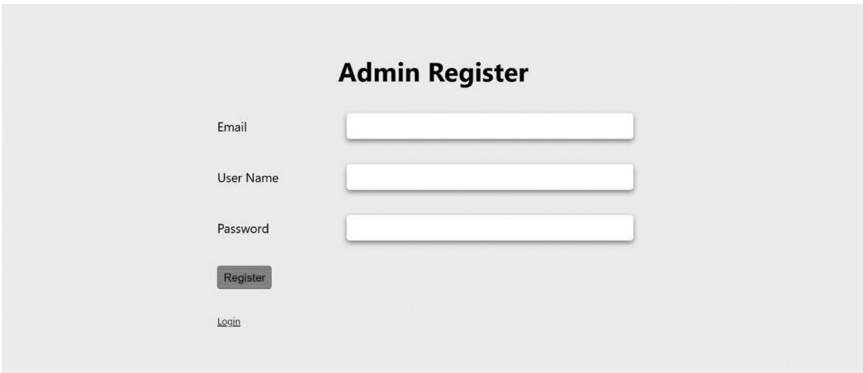
By minimizing the on-chain interactions, the frequency of transactions that incur gas fees is reduced, resulting in overall cost savings for document management processes. This approach allows defense organizations to optimize their use of Ethereum blockchain, minimizing impact of gas fees while still benefiting from the security and immutability provided by the blockchain technology.

B. Implementation Approach

The UI is designed by keeping end-users in mind, considering their needs, preferences, and skill levels. We Organized the UI elements in a logical and intuitive manner to enable users to navigate through different sections of the system effortlessly.

The Landing or Home Page, depicted in Figure 2.2, provides users with three primary options: Login, User Registration, and additional functionality related to document management.

Upon visiting the Landing or Home Page, users are presented with a visually appealing interface that serves as the gateway to the document management system. The page is designed to be intuitive and user-friendly, ensuring a seamless experience for individuals interacting with the system.



The image shows a web form titled "Admin Register". It contains three input fields: "Email", "User Name", and "Password". Below these fields is a "Register" button and a "Login" link.

Admin Register

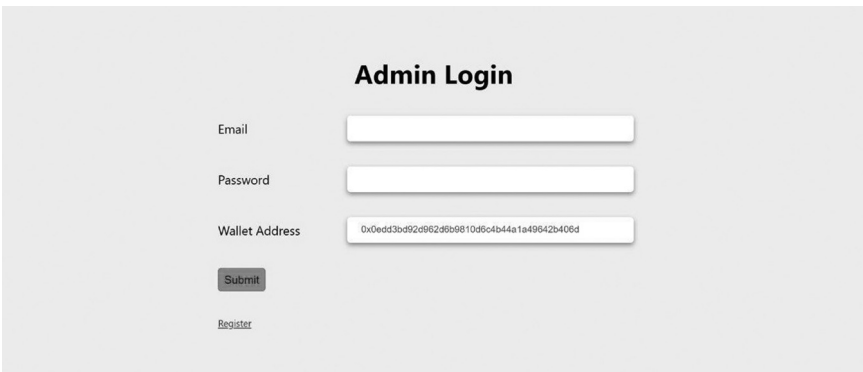
Email

User Name

Password

[Login](#)

Figure 2.2 Register page.



The image shows a web form titled "Admin Login". It contains three input fields: "Email", "Password", and "Wallet Address". Below these fields is a "Submit" button and a "Register" link.

Admin Login

Email

Password

Wallet Address

[Register](#)

Figure 2.3 Login page.

The Login option allows registered users to securely access their accounts by providing their unique credentials as shown in Figure 2.3. It ensures that only authorized individuals can access the document management functionalities and perform relevant actions based on their assigned roles and permissions.

Alternatively, users who are new to the system can opt for the User Registration option. This enables them to create a new account by providing necessary information and following a streamlined registration process. Once registered, they can proceed to utilize the document management system and benefit from its features.

After entering their Email ID and password, users can click on the Submit option located on the home page. This action will direct them to the initial user interface, which is designed for registered users. In this interface, users can log in to access their profiles and utilize various document management features, including document upload to the blockchain.

Once logged in, users gain visibility into their profiles, which allows them to manage their personal information and settings within the system. They can also take advantage of document management capabilities by uploading their own documents securely to the blockchain network.

Furthermore, registered users have the privilege to access documents for which they have been granted read or edit permissions. This enables seamless collaboration and ensures that users can conveniently view and modify the relevant documents associated with their accounts.

For individuals who are not yet registered, the initial user interface provides an option to register using their email address and create a password for authentication.

User interface (UI) where users can upload files and define access permissions for other users. The UI allows the user who uploads a file to designate other individuals as either editors or viewers of that file. This enables collaborative document management within the system.

Moreover, the UI supports the uploading of multiple files simultaneously, as illustrated in Figure 2.4. This functionality enables users to efficiently add multiple documents to the system in one go, enhancing productivity and streamlining the document upload process.

In addition to setting permissions, the UI provides two buttons for users: “View my Docs” and “View eligible docs.”

The “View my Docs” button allows users to conveniently access and review all files they have uploaded or have been assigned as editors or viewers. This feature provides users with a comprehensive overview of their document portfolio within the system.

On other hand, the “View eligible docs” button allows users to explore files uploaded by other users for which they have been granted permission to view. This feature facilitates seamless access to relevant documents and promotes collaboration among users with shared access privileges.

The screenshot displays a web interface for uploading multiple files. At the top right, there are two buttons: "View My Docs" and "View Eligible Docs". Below these, the main area contains a form with three identical rows for file uploads. Each row includes a "Choose file" button, a "No file chosen" status indicator, a "Views" dropdown menu with the text "Select the Views", an "Editors" dropdown menu with the text "Select the Editors", and an "Upload" button. At the bottom of the form, there is an "Add More" button.

Figure 2.4 Multiple files upload page.

Within the application, users can access logs that provide crucial information about the files or documents they have uploaded. These logs serve as a valuable resource for tracking and monitoring various activities and interactions related to files.

By referring to the logs, users can gain insights into important events associated with their uploaded files. These events may include actions such as file creation, modification, access requests, approvals, rejections, and other relevant activities. The logs offer a comprehensive record of these events, enabling users to maintain a detailed overview of the lifecycle of file.

Furthermore, the logs serve to monitor user interactions with the files. Users can review and analyze information such as who accessed the files, when they accessed them, and the actions performed. This level of visibility allows for better accountability and facilitates effective collaboration among users.

The logs also contribute to security and compliance measures. They provide an audit trail of file-related activities, ensuring transparency and traceability of actions taken within the application. In case of any disputes, discrepancies, or unauthorized access attempts, the logs can serve as evidence and aid in investigations.

To facilitate easy access to the logs, the application provides a dedicated interface or section. This interface offers a user-friendly display of the log entries, organized in a clear and structured manner. Users can navigate through the logs, search for specific events, apply filters, and generate customized reports if needed.

Overall, the availability of file logs within the application empowers users with valuable insights into file-related activities. It enhances transparency, accountability, and security, while also providing a means for compliance and investigations if required. Users can rely on these logs to track and monitor events associated with their uploaded files, ensuring efficient management and control of their document assets.

2.10 COMPARING COST EFFICIENCY

When storing a 1MB file in Ethereum, it would cost approximately 6380 rupees or 0.04132 ETH. However, when using IPFS (Inter Planetary File System), you do not store the entire file on blockchain but rather just hash of the file.

IPFS is decentralized file storage system that uses a distributed network of nodes to store files. When you add a file to IPFS, it generates unique hash for that file, which serves as its identifier. This hash can be stored on blockchain, providing a reference to file without storing its content on-chain.

Compared to storing the entire file on Ethereum, using IPFS significantly reduces the cost of gas fees. Storing just hash on blockchain requires less computational resources and storage space, resulting in lower transaction fees.

When comparing cost of storing a 1MB file on Ethereum testnet, it was found that it costed approximately 32000 wei, which is equivalent to 0.000000000000032 ether or ₹4.9e-8.

By leveraging IPFS, you can take advantage of its distributed file storage capabilities while still benefiting from the immutability and security of blockchain. IPFS allows you to store large files or even entire websites in a decentralized manner, with the content being distributed across IPFS network.

When using IPFS to store files, it is typically only necessary to store hash of file on blockchain. Storing just hash on testnet would result in significantly lower gas fees compared to storing the entire file on Ethereum. Cost of storing the hash would depend on the specific gas price and gas limit set for the transaction.

When retrieving a file stored on IPFS, you use the stored hash as the reference and fetch the file from IPFS network. This process is separate from blockchain and doesn't involve any transaction fees.

2.11 CONCLUSION

In conclusion, the challenges of document authenticity, timely ownership transfer, and transaction efficiency can be effectively addressed by leveraging blockchain. By integrating IPFS with Ethereum blockchain, defense organizations can ensure secure and decentralized document storage and distribution, eliminating the need for centralized systems.

This integration provides tamper-proof records, immutability, and transparent document history through blockchain's auditable ledger. Additionally, by optimizing gas fees and implementing gas optimization techniques, defense organizations can reduce transaction costs associated with document management on the Ethereum network. Overall, adopting blockchain solutions empowers defense organizations to enhance data security, streamline processes, and foster trust and efficiency in document ownership transfer across departments, thereby driving business success in the digital landscape. Transaction cost of huge documents shared across the departments in the proposed system is less when compared to existing systems. Storing a 1MB file in Ethereum would cost approximately 6380 rupees or 0.04132 ETH in the existing systems. Whereas in the proposed system it costs 0.000000000000032 ETH or 0.00000004 rupees.

FUTURE SCOPE

The proposed blockchain-based document management system opens exciting avenues for further advancements in the field. Some potential future directions:

1. **Integrating with Artificial Intelligence:** Integrating blockchain with artificial intelligence technologies can enhance document management system's capabilities. AI algorithms can be employed to automate tasks such as document classification, metadata extraction, and content analysis, thereby improving the overall efficiency and accuracy of the system.
2. **Enhanced Collaboration Features:** Future developments could focus on expanding the collaboration features of the document management system. Integration with collaborative tools, such as real-time editing and commenting capabilities, can facilitate seamless teamwork and improve productivity among users.
3. **Integrating with Emerging Technologies:** As new technologies emerge, there are opportunities to integrate them into the document management system. For example, integration of blockchain with emerging technologies like IoT devices can enable automated document capturing, indexing, and organization, further streamlining the document management process.
4. **Smart Contracts for Workflow Automation:** Smart contracts can be leveraged to automate document-related workflows. By defining pre-defined conditions and triggers, smart contracts can automatically initiate actions such as document approval, notification, and distribution, reducing manual intervention and improving process efficiency.
5. **Expansion to Other Industries:** While the current focus is on document management, the blockchain based approach can be extended to other industries and sectors that requires secure and transparent data management. This includes areas such as supply chain management, legal documentation, healthcare records, and more, offering a versatile and scalable solution.
6. **Interoperability and Integration:** Future developments can explore interoperability with other blockchain networks or existing document management systems, enabling seamless data exchange and collaboration between different platforms and organizations.

These future directions hold potential to further enhance security, efficiency, and functionality of the blockchain-based document management system, paving way for broader adoption and transforming way organizations manage their documents.

LIST OF ABBREVIATIONS

DFS	Distributed File System
IPFS	Interplanetary File System
DM	Document Management
DMS	Document Management System

DHT	Distributed Hash Table
ETH	Ethers
IPNS	Interplanetary Naming System
PHR	Patient Health Record
LoRaWAN	Long Range Wide Area Network
BC	Blockchain
CA	Certificate Authority
RSU	Road Side Units
AI	Artificial Intelligence
IOT	Internet of Things
VANET	Vehicle Adhoc Network
MHP-RF	Meta Hyper Parameter Random Forest
DDMS	Decentralized Document Management System

REFERENCES

- [1] B. S. Anupama and N. R. Sunitha, "Analysis of the Consensus Protocols Used in Blockchain Networks - An Overview," *IEEE Int. Conf. Data Sci. Inf. Syst. ICDSIS 2022*, 2022, doi: 10.1109/ICDSIS55133.2022.9915929
- [2] J. Benet, "IPFS - Content Addressed, Versioned, P2P File System," no. Draft 3, 2014, [Online]. Available: <http://arxiv.org/abs/1407.3561>
- [3] A. A. Battah, M. M. Madine, H. Alzaabi, I. Yaqoob, K. Salah, and R. Jayaraman, "Blockchain-Based Multi-Party Authorization for Accessing iPFS Encrypted Data," *IEEE Access*, vol. 8, pp. 196813–196825, 2020, doi: 10.1109/ACCESS.2020.3034260
- [4] E. Daniel and F. Tschorsch, "IPFS and Friends: A Qualitative Comparison of Next Generation Peer-to-Peer Data Networks," *IEEE Commun. Surv. Tutorials*, vol. 24, no. 1, pp. 31–52, 2022, doi: 10.1109/COMST.2022.3143147
- [5] J. Sun, X. Yao, S. Wang, and Y. Wu, "Non-Repudiation Storage and Access Control Scheme of Insurance Data Based on Blockchain in IPFS," *IEEE Access*, vol. 8, pp. 155145–155155, 2020, doi: 10.1109/ACCESS.2020.3018816
- [6] A. Bisht, A. K. Das, D. Niyato, and Y. Park, "Efficient Personal-Health-Records Sharing in Internet of Medical Things Using Searchable Symmetric Encryption, Blockchain and IPFS," *IEEE Open J. Commun. Soc.*, vol. 4, no. October, pp. 1–1, 2023, doi: 10.1109/ojcoms.2023.3316922
- [7] M. Shahjalal, M. M. Islam, M. M. Alam, and Y. M. Jang, "Implementation of a Secure LoRaWAN System for Industrial Internet of Things Integrated With IPFS and Blockchain," *IEEE Syst. J.*, vol. 16, no. 4, pp. 5455–5464, 2022, doi: 10.1109/JSYST.2022.3174157
- [8] M. U. Javed, A. Jamal, E. H. Alkhamash, M. Hadjouni, S. A. Bahaj, and N. Javaid, "Secure Message Handling in Vehicular Energy Networks Using Blockchain and Artificially Intelligent IPFS," *IEEE Access*, vol. 10, no. August, pp. 82063–82075, 2022, doi: 10.1109/ACCESS.2022.3194513
- [9] K. Zhou, C. Wang, X. Wang, S. Chen, and H. Cheng, "A Novel Scheme to Improve the Scalability of Bitcoin Combining IPFS With Block Compression," *IEEE Trans. Netw. Serv. Manag.*, vol. 19, no. 4, pp. 3694–3705, 2022, doi: 10.1109/TNSM.2022.3199346

- [10] H. R. Hasan, K. Salah, I. Yaqoob, R. Jayaraman, S. Pesic, and M. Omar, "Trustworthy IoT Data Streaming Using Blockchain and IPFS," *IEEE Access*, vol. 10, pp. 17707–17721, 2022, doi: 10.1109/ACCESS.2022.3149312
- [11] S. K. Dwivedi, R. Amin, and S. Vollala, "Blockchain-Based Secured IPFS-Enable Event Storage Technique with Authentication Protocol in VANET," *IEEE/CAA J. Autom. Sin.*, vol. 8, no. 12, pp. 1913–1922, 2021, doi: 10.1109/JAS.2021.1004225
- [12] M. Alizadeh, K. Andersson, and O. Schelen, "Efficient Decentralized Data Storage Based on Public Blockchain and IPFS," *2020 IEEE Asia-Pacific Conf. Comput. Sci. Data Eng. CSDE 2020*, no. 2, 2020, doi: 10.1109/CSDE50874.2020.9411599
- [13] N. Nizamuddin, H. R. Hasan, and K. Salah, "IPFS-Blockchain-Based Authenticity of Online Publications," *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, vol. 10974 LNCS, no. June, pp. 199–212, 2018, doi: 10.1007/978-3-319-94478-4_14
- [14] M. Das, X. Tao, Y. Liu, and J. C. P. Cheng, "A Blockchain-Based Integrated Document Management Framework for Construction Applications," *Autom. Constr.*, vol. 133, no. October, p. 104001, 2022, doi: 10.1016/j.autcon.2021.104001
- [15] M. Steichen, B. Fiz, R. Norvill, W. Shbair, and R. State, "Blockchain-Based, Decentralized Access Control for IPFS," *Proc. - IEEE 2018 Int. Congr. Cybermatics 2018 IEEE Conf. Internet Things, Green Comput. Commun. Cyber, Phys. Soc. Comput. Smart Data, Blockchain, Comput. Inf. Technol. iThings/Gree*, no. July, pp. 1499–1506, 2018, doi: 10.1109/Cybermatics_2018.2018.00253
- [16] S. A. Sultana, C. Rupa, R. P. Malleswari, and T. R. Gadekallu, "IPFS-Blockchain Smart Contracts Based Conceptual Framework to Reduce Certificate Frauds in the Academic Field," *Inf.*, vol. 14, no. 8, pp. 1–19, 2023, doi: 10.3390/info14080446
- [17] G. Subathra, A. Antonidoss, and B. K. Singh, "Decentralized Consensus Blockchain and IPFS-Based Data Aggregation for Efficient Data Storage Scheme," *Secur. Commun. Networks*, vol. 2022, 2022, doi: 10.1155/2022/3167958
- [18] J. Han, H. Kim, H. Eom, and Y. Son, "A Decentralized Document Management System Using Blockchain and Secret Sharing," *Proc. ACM Symp. Appl. Comput.*, pp. 305–308, 2021, doi: 10.1145/3412841.3442077
- [19] J. Sun, X. Yao, S. Wang, and Y. Wu, "Blockchain-Based Secure Storage and Access Scheme for Electronic Medical Records in IPFS," *IEEE Access*, vol. 8, pp. 59389–59401, 2020, doi: 10.1109/ACCESS.2020.2982964
- [20] S. Vimal and S. K. Srivatsa, "A New Cluster P2P File Sharing System Based on IPFS and BLOCKCHAIN Technology," *J. Ambient Intell. Humaniz. Comput.*, no. 0123456789, 2019, doi: 10.1007/s12652-019-01453-5
- [21] S. Sarang, D. Rana, D. Rana, and S. Patel, "Document Management System Empowered by Effective Amalgam of Blockchain and IPFS of Blockchain IPFS," 2023.
- [22] M. Das, X. Tao, and J. C. P. Cheng, "A Secure and Distributed Construction Document Management System Using Blockchain," *Lect. Notes Civ. Eng.*, vol. 98, no. March, pp. 850–862, 2021, doi: 10.1007/978-3-030-51295-8_59
- [23] M. S. Choudaiah, "Block Chain Based Document Management System," vol. 12, no. 08, pp. 101–106, 2021, [Online]. Available: www.jespublication.com

- [24] X. Zhao et al., "Blockchain Technology Based Digital Document Management System Design," *Proc. - 2022 7th Asia Conf. Power Electr. Eng. ACPEE 2022*, pp. 440–446, 2022, doi: 10.1109/ACPEE53904.2022.9783819
- [25] P. Shanmugaraja, K. S. Susmitha, S. Swadha, R. Vijay, and G. Naveen, "Document Management System Using Blockchain," *Proc. - Int. Conf. Augment. Intell. Sustain. Syst. ICAISS 2022*, pp. 888–891, 2022, doi: 10.1109/ICAISS55157.2022.10011076
- [26] J. Solarte-Rivera, A. Vidal-Zemanate, C. Cobos, J. A. Chamorro-Lopez, and T. Velasco, "Document Management System Based on a Private Blockchain for the Support of The Judicial Embargoes Process in Colombia," *Lect. Notes Bus. Inf. Process.*, vol. 316, no. 1122, pp. 126–137, 2018, doi: 10.1007/978-3-319-92898-2_10
- [27] R. Kumar, N. Marchang, and R. Tripathi, "Distributed Off-Chain Storage of Patient Diagnostic Reports in Healthcare System Using IPFS and Blockchain," *2020 Int. Conf. Commun. Syst. NETworks, COMSNETS 2020*, pp. 1–5, 2020, doi: 10.1109/COMSNETS48256.2020.9027313
- [28] H. K. Shapna Muralidharan, "An InterPlanetary File System (IPFS) Based IoT Framework," *2019 IEEE Int. Conf. Consum. Electron*, pp. 1–2, 2019, [Online]. Available: <http://arxiv.org/abs/1407.3561>
- [29] A. Zaky and I. G. B. B. Nugraha, "Increase Activity Time Efficiency in Official Documents Management using Blockchain-Based Distributed Data Storage," *Proc. Int. Conf. Electr. Eng. Informatics*, vol. 2019-July, no. July, pp. 81–86, 2019, doi: 10.1109/ICEEI47359.2019.8988882
- [30] B. F. Islamay, Y. Purwanto, and M. F. Ruriawan, "Implementation of Blockchain and Peer-to-Peer Network for Digital Document Management," *ICACNIS 2022 - 2022 Int. Conf. Adv. Creat. Networks Intell. Syst. Blockchain Technol. Intell. Syst. Appl. Hum. Life, Proceeding*, pp. 1–6, 2022, doi: 10.1109/ICACNIS57039.2022.10055538
- [31] S. Zou, J. Xi, S. Wang, Y. Lu, and G. Xu, "Reportcoin: A Novel Blockchain-Based Incentive Anonymous Reporting System," *IEEE Access*, vol. 7, pp. 65544–65559, 2019, doi: 10.1109/ACCESS.2019.2915956
- [32] H. Huang, S. Zhou, J. Lin, K. Zhang, and S. Guo, "Bridge the Trustworthiness Gap amongst Multiple Domains: A Practical Blockchain-Based Approach," *IEEE Int. Conf. Commun.*, vol. 2020-June, no. June, 2020, doi: 10.1109/ICC40277.2020.9148828
- [33] E. Nyalety, R. M. Parizi, Q. Zhang, and K. K. R. Choo, "BlockIPFS - Blockchain-Enabled Interplanetary File System for Forensic and Trusted Data Traceability," *Proc. - 2019 2nd IEEE Int. Conf. Blockchain, Blockchain 2019*, no. September, pp. 18–25, 2019, doi: 10.1109/Blockchain.2019.00012
- [34] M. Naz et al., "A Secure Data Sharing Platform Using Blockchain and Interplanetary File System," *Sustain.*, vol. 11, no. 24, pp. 1–24, 2019, doi: 10.3390/su11247054
- [35] S. Wang, Y. Zhang, and Y. Zhang, "A Blockchain-Based Framework for Data Sharing with Fine-Grained Access Control in Decentralized Storage Systems," *IEEE Access*, vol. 6, no. c, pp. 38437–38450, 2018, doi: 10.1109/ACCESS.2018.2851611
- [36] N. Nizamuddin, H. R. Hasan, and K. Salah, "IPFS-Blockchain-Based Authenticity of Online Publications," *Lect. Notes Comput. Sci. (including Subser. Lect. Notes Artif. Intell. Lect. Notes Bioinformatics)*, vol. 10974 LNCS, no. September, pp. 199–212, 2018, doi: 10.1007/978-3-319-94478-4_14

- [37] M. S. Ali, K. Dolui, and F. Antonelli, "IoT Data Privacy via Blockchains and IPFS," *ACM Int. Conf. Proceeding Ser*, no. October, 2017, doi: 10.1145/3131542.3131563
- [38] M. Steichen, B. Fiz, R. Norvill, W. Shbair, and R. State, "Blockchain-Based, Decentralized Access Control for IPFS," *Proc. - IEEE 2018 Int. Congr. Cybermatics 2018 IEEE Conf. Internet Things, Green Comput. Commun. Cyber, Phys. Soc. Comput. Smart Data, Blockchain, Comput. Inf. Technol. iThings/Gree*, no. September, pp. 1499–1506, 2018, doi: 10.1109/Cybermatics_2018.2018.00253

Constrained networking middleware for military operations

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3.1 INTRODUCTION

Communication in military settings is essential for almost every aspect of tactical decision making. A commander needs to communicate about evolving situations in the field, and get updates from their soldiers. This entails using text, telemetry, and real-time voice and video calls for fast information interchange. The networks used for this communication can be unreliable and potentially low bandwidth. This creates a need for a way to communicate rapidly, in a way that efficiently utilizes the network resources while minimizing loss of information. It was with this understanding that the customer wanted to collaborate on this work in creating a middleware service that sits between the application and the underlying transport layer protocols. This would serve as an API to be used by developers of networking applications to communicate over sockets in a configurable and predictable way.

The main reason for wanting to incorporate configurability is due to limitations of the commonly used Transmission Control Protocol (TCP).

TCP is designed for reliable communication between two hosts on a packet-switched network. Its goal is to make sure data is recovered from being damaged, lost, duplicated or delivered out of order. For this it has many tools, however this work was especially concerned with how TCP used acknowledgements and retransmissions, so called congestion control (Transmission Control Protocol 1981). Acknowledgements (ACKs) are small packets that are sent to the other host to confirm that data has been received and verified to be in good shape. These are positive ACKs, there are also negative ACKs which can be used, but shall not be further discussed here. To ensure correct ordering of the delivered data, TCP uses a sequence number. The ACK confirms received good packets up to a certain sequence number. If the ACKs sent from the receiver are delayed by enough time, the transmitter will assume lost data, and retransmit (Transmission Control Protocol 1981). It was from this interaction between ACKs, wait timers and retransmissions that the main problem tackled in this work originated from. In civil computer networks one may always assume a route between two endpoints exists (at all) or with a certain connection speed or latency.

In military networking environments, this may not be true. Sub-networks may be disconnected altogether for a certain amount of time, as troops move in and out of communication coverage or radio shadows [1].

Transmission equipment may be slow and/or unreliable, and total network bandwidth may be limited as in the case of narrow-band radio communication. When these network statistics fluctuate wildly or change in a short time span, one cannot guarantee that TCP will behave properly. TCP may expect a certain latency for its ACKs, but if the network situation has changed, this latency may be much larger, leading to retransmissions, when in reality it was just a delayed ACK; the data made it through just fine. The test on the network with unlimited connection ended up having no retransmissions or duplicate ACKs. It completed in under a tenth of a second. The test with the limited network was more interesting however. After about 30% of the data transfer being finished, the connection was cut due to a reset packet being sent from the client from a time-out. Out of the 22 packets sent, 5 were retransmitted, a retransmit rate of 22%. Compared to the network packet loss configured for 1%, this was high, leading to unnecessary network traffic. From the 7 packets sent from the host to the client, 3 were normal, and 4 were retransmissions for one of the ACKs. The team believed this to have been caused by the client trying to push 100 kB of data through the network, congesting it completely and choking the host's ability to respond in time. The client waited for too long, tried retransmitting and gave up, sending a reset packet to the host. This was a problem, as the host never went offline, it was waiting for the data through the slow link [2].

The research work is organized into several distinct sections for clarity and coherence. Section 2 furnishes an overview of related work within the field, offering valuable context for the study. In Section 3, the materials utilized in the research are elaborated upon, providing insight into the experimental setup. Section 4 meticulously outlines the proposed method and delineates the dataset employed for experimentation. Following this, Section 5 explain the conclusions drawn from the study are encapsulated in the concluding section, offering insights and implications derived from the research findings.

3.2 RELATED WORKS

A survey of the literature on SDN research issues, difficulties, and solutions was given by Govindarajan et al. [3]. They classified the difficulties in SDN research, listed the related current works, and compared them. Despite not being structured as a methodical literature review, this study offers a thorough overview and discussion of the technologies, approaches, and unresolved concerns, including security, scalability, and load balancing. Additionally, Xia et al. [4] offered a survey on SDN in 2015 and covered the most recent advancements in this field of study at that time. Instead of

delving deeply into any one subject, the paper offers a broad and thorough review of SDN theory together with relevant research initiatives on a range of subjects, including processor optimization, policy, and rule validation, network virtualization, and maintenance. Alsmadi et al. [5] categorized research papers published on SDN, investigated current opportunities and problems, emphasized the development of the SDN paradigm, and identified key figures in the field by doing a thorough study of the literature.

A survey of SDN was presented by Jammal et al. [6], with an emphasis on the examination of the state-of-the-art and research problems at the moment. The open-flow architecture, network function virtualization, SDN applications, difficulties, and current solutions were among the topics covered by the writers. Goel et al. [7] on network virtualization approaches and Lopes et al. [8] on trends and challenges in programming for SDN environments are two other important surveys in the subject of SDN. For SDN taxonomy, research directions, difficulties, and recommendations for future study, see Jarraya et al. [9]; for programmable networks with a focus on SDN, see Mendonca et al. [10]. Additionally, Hu et al. [11] concentrated particularly on SDN/OpenFlow designs and architectures, Anderson et al. [12] examined resilience supporting mechanisms for SDN, and Bennesby et al. [13] presented a study on current methods for lowering BGP (Border Gateway Protocol) appropriate inter-domain routing convergence delay on the Internet. Using SDN for emergency response and military networks, Mihailescu et al. [14] proposed a prototype framework for (i) network control, (ii) traffic priority management, and (iii) radio/protocol integration. A two-tier hierarchical SDN controller configuration that exposes a modular interface for wireless network management is the article's first contribution. The authors also suggested a framework that permits the fusion and compatibility of various wireless network technologies.

Furthermore, in order to enable the best possible network reactivity to wireless link fluctuations, Phemius et al. [15] presented an architectural framework for traffic management in tactical networks that combines components of Mobile Edge Cloud (MED) and SDN. Four layers comprise the proposed framework: the (i) "waveform layer," which is made up of several deployed programmable radios; the (ii) "SDN layer," which has an SDN controller and a virtual switch; the (iii) "application management layer," which is mapped to the MED where the applications are executed; and the (iv) "mobile edge controller," a module that connects the other three layers. L. Campioni et al. [16] presented Aceso prototype smart city middleware that supports all HADR (hazard assessment and disaster response) processes and provides location- and context-aware services. Aceso is a package of features that may be quickly implemented to expedite HADR operations during unanticipated and unfavorable situations. These features range from resource identification to secure data transmission. Shams Al Ajrawi et al. [17] introduced a platform to evaluate the potential benefits of mobile ad hoc networks in tactical military scenarios. The study suggests employing

reactive (on-demand) routing protocols for route discovery, where nodes only need to transmit data, eliminating the need for each node to store and manage routing tables. Fendy Santoso et al. [18] explained the algorithm which is tailored to mitigate the cybersecurity risks associated with the Robot Operating System (ROS), a widely utilized middleware platform in civilian and military sectors. To validate its effectiveness, we perform penetration testing involving real-time man-in-the-middle cyberattacks on the GVR-BOT ground vehicle. This vehicle serves as a replica of a military ground robot, developed by the United States Army Combat Capabilities Development Command (CCDC), Ground Vehicle Systems Center. This studies [19, 20] explores the topic of cyberattacks directed toward smart grids and suggests countermeasures to avoid Distributed Denial of Service (DDoS) assaults. Scientists are presently traversing a paradigm change, which calls for a careful analysis and reduction of the variables impacting it. The discussion covers the transition from traditional to smart grids, emphasizing the technologies and possible ways to reduce the hazards involved.

The existing work on Constrained Networking Middleware for Military Operations exhibits several disadvantages that warrant attention and improvement. Firstly, there may be a lack of adaptability within the middleware, particularly in dynamically changing network environments, leading to suboptimal performance under varying conditions. Additionally, manual configuration requirements pose challenges, as the middleware may necessitate time-consuming and error-prone adjustments, hindering its efficiency. Scalability issues could also arise, impeding the middleware's ability to accommodate larger networks or increased traffic demands effectively. Furthermore, potential security vulnerabilities may exist within the middleware, leaving it susceptible to cyberattacks or unauthorized access, posing significant risks to military operations' integrity and confidentiality. Moreover, the complexity of the middleware architecture or implementation may hinder its understandability, maintenance, and extension, further complicating its deployment and management. Compatibility limitations with existing military systems or protocols may also be present, necessitating additional integration efforts to ensure seamless operation within military networks. Performance overhead and resource consumption concerns could further detract from the middleware's effectiveness, impacting the overall operational efficiency of constrained devices or systems. Addressing these disadvantages is essential for enhancing the reliability, security, and adaptability of Constrained Networking Middleware for Military Operations.

3.3 PROPOSED METHODOLOGY

In accordance with the requirements and recommendations from the customer, the initial iteration of the system architecture (illustrated in Figure 3.1)

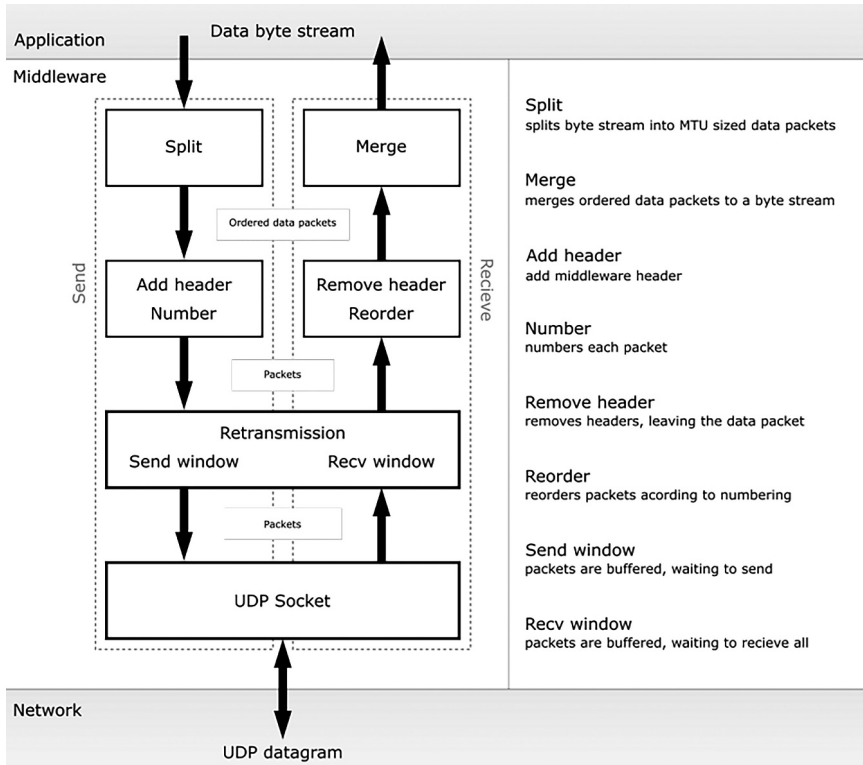


Figure 3.1 Proposed system architecture.

was structured around an underlying UDP (User Datagram Protocol) socket. The primary aim of the proposed work is to detail the process by which data originating from the application would be partitioned into packets. Moreover, it aims to elucidate how these packets would subsequently be dispatched as UDP datagrams. To mimic how applications interact with UDP, the middleware would expose an API which enables sending and receiving a byte stream (shown at the top of Figure 3.1). This byte stream would then be divided into an ordered list of “data packets” (fragments) to fit within the Maximum Transmission Unit (MTU) of the network (shown as “Split” and “Merge” in Figure 3.1). IP fragmentation would normally take care of this, however, military networks do not always support fragmentation, and the customer therefore required that the middleware should handle this.

To reassemble the fragmented packets at the receiving end, a header is prepended with a unique identifier when sending, and packets are reordered using this information when receiving (shown as “Add header, Number” and “Remove header, Reorder”). The final part of the diagram, before packets are sent to the UDP socket, shows how packet buffering was planned for both sending and receiving. This was to enable more

control over when packets are sent, and how long they are kept before reassembly. This was also the “module” that handled retransmission for when the middleware was configured for reliable communication. With the exception of “Retransmission”, the above explanation and architecture diagram (Figure 3.1) holds true for the middleware when configured for unreliable communication.

Due to reliable communication not needing manual fragmentation, the system architecture was reworked. The new system architecture is shown in Figure 3.2 and is centered around the idea of a “Middleware API” that supports both unreliable and reliable communication. The diagram can be split into three parts: unreliable communication (left), configuration (middle) and reliable communication (right). Unreliable communication is realized by allowing the user to instantiate instances of Middleware Unreliable sockets. These sockets utilize an underlying UDP socket to communicate over the network and implements fragmentation and defragmentation (Figure 3.2).

The middle part of the diagram depicts configuration, which is responsible for allowing the user to adapt the middleware to different network environments. This is done by editing a configuration file, which is then loaded by the middleware at startup and used by both unreliable and reliable communication. The final part of the diagram is reliable communication, which is realized by configuring an underlying TCP socket. The specifics of this configuration are described which deals with the implementation of unreliable and reliable fragmentation.

With this new system architecture, the most interesting part is unreliable communication. An updated architecture diagram for unreliable communication is shown in Figure 3.3. The diagram is somewhat similar to the draft presented in Figure 3.1, with some notable changes. Firstly, the diagram

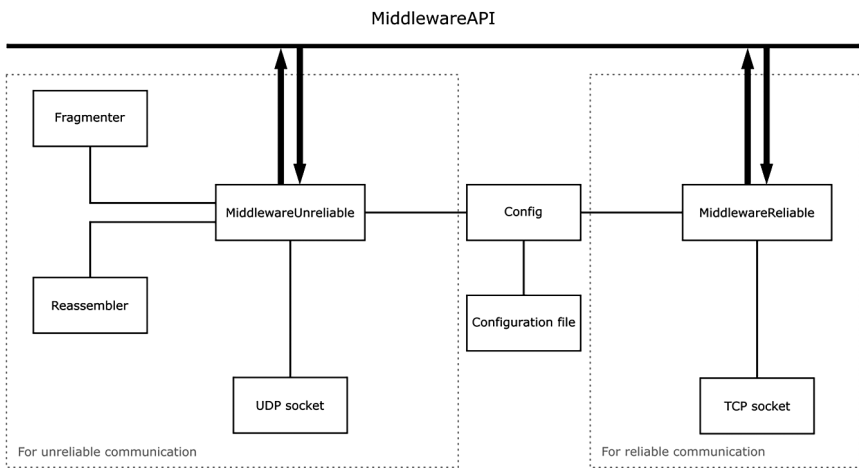


Figure 3.2 Extended system architecture.

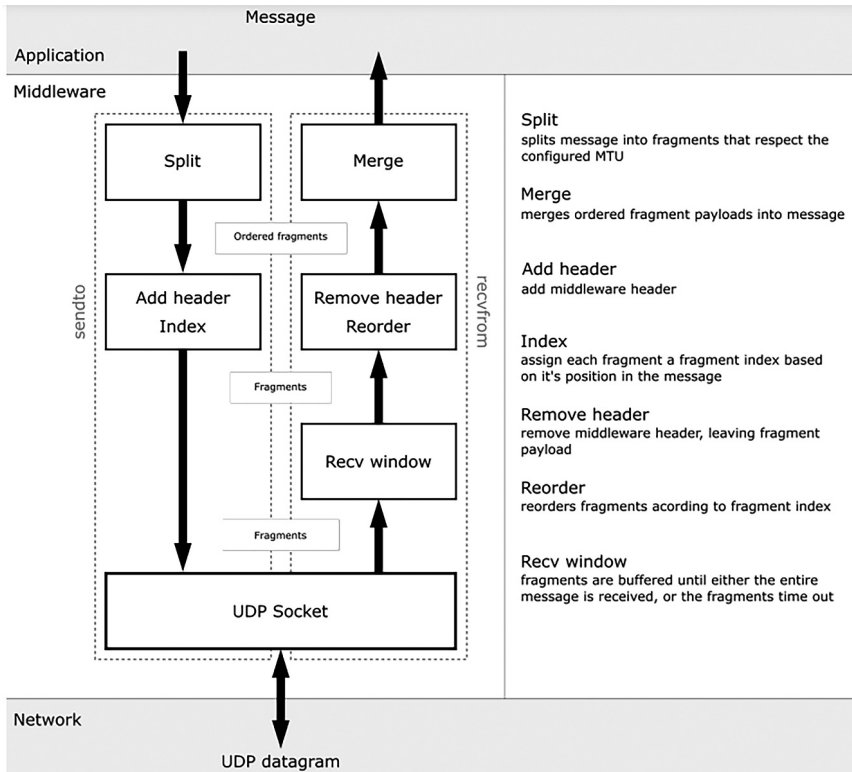


Figure 3.3 Unreliable communication architecture.

depicts a message oriented communication API, as opposed to the byte stream oriented API presented in the draft. This is an important change from the draft, as it allows unreliable communication to more closely reflect how UDP works, as well as unburdening reliable communication from having to emulate a message-oriented API on top of TCP. Secondly, the send window is removed, as UDP has no concept of congestion control, and therefore does not use a send windows, as opposed to TCP.

3.3.1 Initial design

The initial design is shown in Figure 3.4. This was a naive approach that was functional but quite inefficient. The team implemented a 4 byte identification field. If the identification has a non-zero value, it indicates that the packet is a fragment in a sequence and needs to be reassembled using other received fragments sharing the indicated identification. This is done using the fragment id and final fragment id. The reassembly of the fragments occurs by removing the header, then appending them in order of fragment id. On the other hand if identification has a value of 0, it indicates that the

(a) First iteration

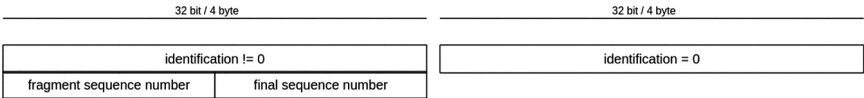


Figure 3.4 The first iteration of the header format, shown both for fragmented (left) and unfragmented packets (right).

packet is not fragmented and can be used as is. In this case, the header can be stripped, and passed on to the application immediately.

This approach was quick to put together and allowed the team to quickly demonstrate a working prototype of fragmentation to the customer. However, a number of concerns quickly became evident: The inclusion of the entire final sequence number in the header was inefficient, as it had to be included for every fragment. Considering that the aim was to support MTU values down to 100 bytes, any reduction in header size would save significant overhead. Furthermore, the identification field would have to be unique on the receiving end for fragments to be reassembled correctly. This could not be guaranteed without extensive added complexity if a host received data from multiple sources simultaneously. While this would not be a problem for TCP, this is a use case that is supported by UDP.

3.3.2 Limiting IP fragmentation

It was found that the problem could be solved by imitating IP fragmentation (Internet Protocol 1981), by indicating the final package with a single bit flag instead of including the sequence number of the final packet. With this change the new header looked as shown in Figure 3.5. During the work on this iteration it had been decided that traffic would be received on separate ports based on configuration and destination service. The team there thus no longer had to have identification be globally unique amongst all received fragments; instead it would only have to be unique among all received fragments from a given source IP address and port combination – which can be guaranteed by the sending instance of the middleware, thus sidestepping some complexity. As a consequence the identification field size was reduced the to three bytes.

(b) Second iteration

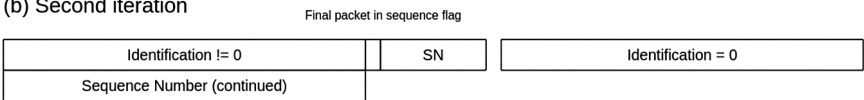


Figure 3.5 The second iteration of the header format.

3.3.3 Introducing a length field

The first two iterations had only been tested within unit tests, and had worked well under those conditions. As fragmentation was integrated into the service API, another issue was quickly encountered, however. The Python socket library only provides a byte stream of received data, that is, it does not return the packets. This means that the middleware would have to extract the individual packets from the byte stream itself, something the unit tests so far had just assumed to happen.

Multiple options were considered to make packets separable, but it was concluded that the easiest solution would be to include a length field in the header from which the receiver could then determine at what position in the byte stream the next packet header would be found. The identification field size was also further reduced to two bytes, matching that of IP fragmentation. The resulting design is shown in Figure 3.6, and resulted in 7 bytes of header overhead for fragments and 4 bytes of overhead for unfragmented packets.

A major problem encountered in iteration 3 was how packets were received as a byte stream, and this made reassembly more difficult. TCP, by design, does not delineate between data received from separate segments. For messages to be separable over a TCP connection, the start and end has to be encoded in the message data, which requires either specifying the message length before the data, or using a terminator sequence (e.g. the null terminator) to mark the end of messages.

Additionally, a related problem with the current approach was discovered: data sent with TCP is buffered. This means that even though the application might send the data of each fragment separately, TCP might buffer these operations and send them in a batch. This is normally not a problem, but if IP fragmentation is to be avoided, using a fixed MTU, buffering by TCP with no restrictions on the buffer size will not respect the set MTU.

Furthermore, fragmentation in unreliable transmissions can also be simplified, while simultaneously making it more robust. This can be shown by first analyzing what the system has to handle, and then extrapolating the necessary information that has to be exchanged for correct reassembly on the receiving end. Firstly, on the sending side messages to be sent over the network have to be divided into fragments, such that the resulting IP packets respect the configured MTU. Secondly, on the receiving end, the fragments have to be reassembled to the original message. Since IP packets can be dropped, for reassembly, the receiver must know both how many

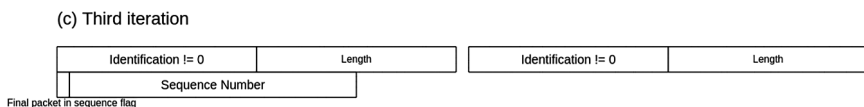


Figure 3.6 The third iteration of the header format.

fragments constitute a message, and which message they belong to. Information identifying which message a fragment belongs to has to be included in every fragment from said message, while information about the total number of fragments can be transferred in a multitude of ways. Additionally, since IP packets can arrive out of order, each fragment must include information relating them in a total order.

An approach that partially solves this is to add timing information in the fragment header. This allows the receiver to know when a fragment was sent, instead of only when it was received. In combination with a fragment id, the receiver can forcibly time out an older message when a newer message with duplicate id arrives. This solves the problem of reassembling corrupt messages in some cases, but not all. There is no easy way of separating two messages with identical id using only timing information, since the receiver does not know how much time has elapsed between messages. One way of solving this problem in some cases is to use the timing information from successive ids to separate messages with duplicate ids by time. However this adds a lot of latency and does not solve the problem in all cases (e.g. all successive messages might be dropped).

3.4 EXPERIMENTAL RESULTS

To evaluate the performance of the proposed fragmentation scheme, multiple tests were run with equal amount of data transfer, but differing fragmentation. A baseline with regular UDP and IP fragmentation was established for 1 000 000 datagrams of size 146 bytes, 100 000 datagrams of size 1 460 bytes and 10 000 datagrams of size 14 600 bytes. The same tests were run with the middleware configured for unreliable communication with the MTU set to 500 bytes, 1 000 bytes and 1 500 bytes. Result shows the time taken to send 146 MB under the different configurations between a Linux virtual machine running on a windows host and a Linux sever connected to the windows host via a router. Results shows how much slower the middleware is compared to IP fragmentation when measuring IP packets transferred per second. This shows that the middleware is slower than IP fragmentation by 7% to 8% in most configurations. This slowdown does not seem to be caused by fragment time out and checking for completed messages, as testing has shown these speed up transfer by only 0.3%. The major culprit seems to be memory allocation, as changing the receive buffer size for UDP datagrams to a larger number seems to significantly impact performance. In the test case of sending 1 460 bytes with a configured MTU of 1 500 bytes, changing the receive buffer size from the configured MTU to 2 048 bytes resulted in an additional 5% slowdown. An additional 548 bytes of memory allocation per UDP datagram therefore almost doubled

the slowdown compared to regular IP fragmentation. An obvious solution to this would be to reuse the memory instead of allocating for every receive call. This can be done by using an alternative receive function for Python sockets that overwrites an input array, up to a specified size, instead of allocating its own byte array and returning the result. The problem with this is that the received UDP datagrams need to be stored in order to be reassembled, and reusing the receive buffer memory will therefore only make the copy explicit, instead of implicit. This is however not the only cause for slowdown, as can be seen in Figure 3.7. The graph shows raw throughput over the network, and as can be seen, there are multiple dips in throughput which grow larger over the runtime of the test. The dips seem to happen at a regular interval, roughly one second apart. This interval is much greater than the interval between each successive message reassembly (which is on the order of milliseconds). The cause of these dips could again be garbage collection, as it is a process which is in many languages running on a timed interrupt on the order of seconds. However, this would likely not explain why the dips grow greater as time progresses. This is due to the dips increasing significantly 15 seconds into the test (circa 9 seconds into transmission) which is after roughly 75% of the data has been sent. If the dips were purely caused by garbage collection overhead, they would likely start growing earlier. This is however pure speculation and would need a full memory trace to prove one way or another. Future iterations of fragmentation reassembly

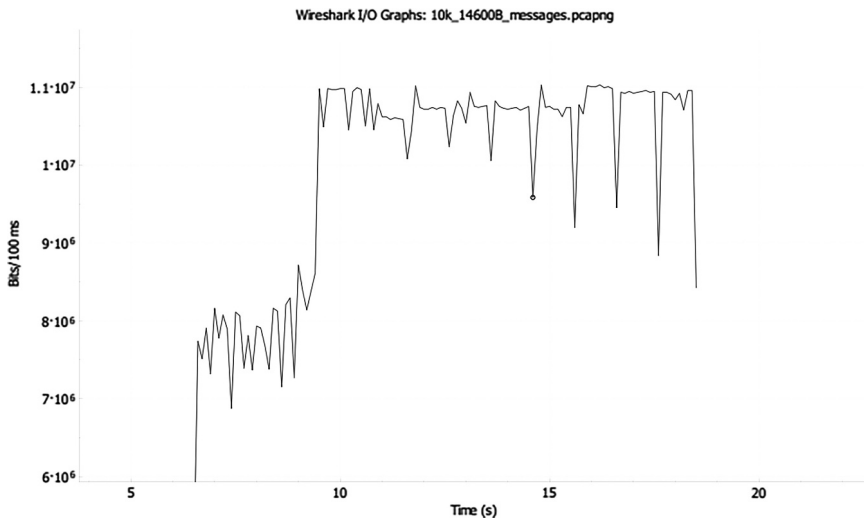


Figure 3.7 Number of bits sent over the network under one of the test cases, where the message length was 14 600 byte with and MTU of 1 500 bytes (which equals 10 fragments per message).

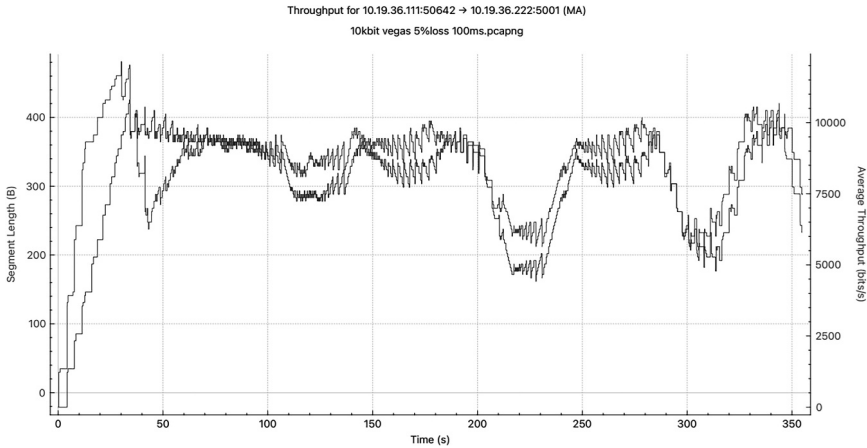


Figure 3.8 TCP Vegas throughput over a CNR-like network. Brown represents throughput – total amount of data sent. Green represents ‘goodput’ – total amount of ACKed data.

would therefore likely benefit from focusing more on memory usage, and e.g. use an object pool to not put as much pressure on the garbage collector.

Figure 3.8 illustrates the outcome of testing conducted on the congestion algorithm TCP Vegas. One notable observation is the initial spike in the transmit rate, which is believed to stem from the congestion algorithm’s overenthusiastic behavior. Consequently, the algorithm adjusts its behavior dynamically to align with the constraints of the limited network capacity. The subsequent dip in transmission rate represents this corrective action, as evidenced by the relatively constant goodput at the network bandwidth limit thereafter. Additionally, the dips occurring at approximately 200s and 300s are attributed to packet drops and/or duplicate ACK packets, prompting TCP to retransmit data accordingly to mitigate the effects of these network events.

In Figure 3.9, one can see the time taken for communication to happen; Round Trip Time (RTT). Measured in milliseconds, this is the time taken from when the data is sent by one client, to a response is received. For most of the communication stream, this was under 5×10^3 ms. Notably, spikes can be observed of up to 1.5×10^4 ms, a three-fold increase in the RTT. These spikes coincide with the timings of the drops in throughput as discussed above. This is another representation of a loss event, where packets were lost, and the algorithms need to retransmit old data to catch up. This leaves the sender waiting on their newest data, as old data is still being responded to, which in turn increases the RTT as shown.

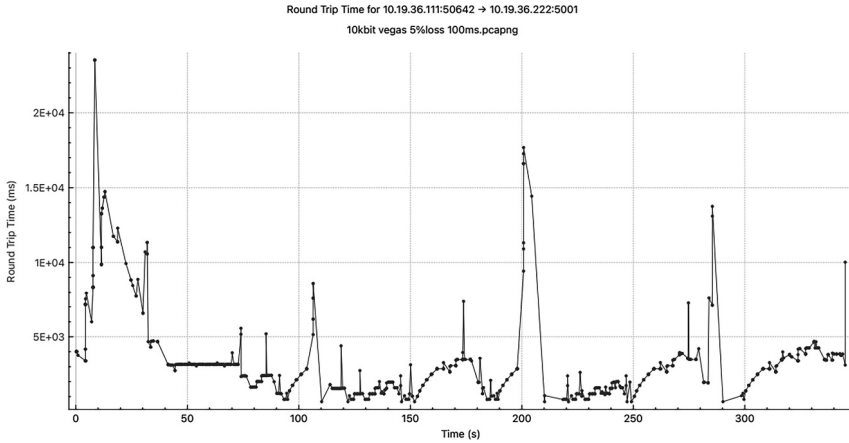


Figure 3.9 TCP Vegas Round Trip Time (RTT) over a CNR-like network. RTT is the time from one client sends a packet to the time it receives a response.

3.5 CONCLUSION

The final product managed to deliver on every requirement except one, which was deemed low priority. The paper can be attributed to its appropriately scoped objectives, which struck a balance between ambition and feasibility, considering the time constraints. This pragmatic approach ensured that work goals were achievable within the allocated timeframe. Similarly, the testing phase was efficiently conducted, albeit limited in scope to specific environments. Looking ahead, there is potential for enhancing the middleware and its methodology. One area for improvement involves automating the adaptation process to accommodate evolving network conditions. Introducing intelligent mechanisms to dynamically adjust parameters such as congestion algorithms based on real-time network metrics like throughput and latency could significantly enhance the middleware's effectiveness and adaptability. By fostering greater interoperability, the middleware can facilitate information sharing and coordination across multi-domain operations, enhancing overall mission effectiveness and operational agility. Furthermore, ongoing research into intelligent routing algorithms and data management strategies holds promise for optimizing bandwidth utilization and prioritizing critical data transmission in bandwidth-constrained environments. Finally, efforts to improve human-machine interaction interfaces and decision support tools can empower military personnel with actionable insights and situational awareness, enabling more informed decision-making and mission execution. Overall, the future scope for the Constrained Networking Middleware for Military Operations is vast and multifaceted,

offering numerous opportunities for innovation and advancement to support the evolving needs of modern military operations.

LIST OF ACRONYMS

API	Application Programming Interface
ACK	Acknowledgment
CNR	Carrier-to-Noise Ratio
CCDC	Capabilities Development Command
DDoS	Distributed Denial of Service
GVR-BOT	Ground Vehicle Robot - Battery Operated Tactical
HADR	Hazard Assessment and Disaster Response
IP	Internet Protocol
MED	Mobile Edge Cloud
MTU	Maximum Transmission Unit
ROS	Robot Operating System
RTT	Round Trip Time
SDN	Software-Defined Networking.
TCP	Transmission Control Protocol
UDP	User Datagram Protocol

REFERENCES

1. Suri, Niranjan, Maggie Breedy, Roberto Fronteddu, Alessandro Morelli, Eelco Cramer, Jan Nilsson, Anders Hansson, Kelvin Marcus, and Andreas Martens. "Evaluating the scalability of group communication protocols over synchronized cooperative broadcast." In *2021 International Conference on Military Communication and Information Systems (ICMCIS)*, pp. 1–9. IEEE, 2021.
2. Kniberg, Henrik. "Lean from the trenches: Managing large-scale works with Kanban." *Lean from the Trenches* 2011, 1–178.
3. Govindarajan, K.; Meng, K.C.; Ong, H. A literature review on Software-Defined Networking (SDN) research topics, challenges and solutions. In *Proceedings of the 2013 Fifth International Conference on Advanced Computing (ICoAC)*, Chennai, India, 18–20 December 2013; pp. 293–299.
4. Xia, W.; Wen, Y.; Foh, C.H.; Niyato, D.; Xie, H. A Survey on Software-Defined Networking. *IEEE Commun. Surv. Tutor.* 2015, 17, 27–51.
5. Alsmadi, I.M.; AlAzzam, I.; Akour, M. A Systematic Literature Review on Software-Defined Networking. In *Information Fusion for Cyber-Security Analytics*; Springer International Publishing: Cham, Switzerland, 2017; pp. 333–369.
6. Kreutz, D.; Ramos, F.M.V.; Verassimo, P.E.; Rothenberg, C.E.; Azodolmolky, S.; Uhlig, S. Software-Defined Networking: A Comprehensive Survey. *Proc. IEEE* 2015, 103, 14–76.
7. Jammal, M.; Singh, T.; Shami, A.; Asal, R.; Li, Y. Software defined networking: State of the art and research challenges. *Comput. Netw.* 2014, 72, 74–98.

8. Lopes, F.A.; Santos, M.; Fidalgo, R.; Fernandes, S. A Software Engineering Perspective on SDN Programmability. *IEEE Commun. Surv. Tutor.* 2016, 18, 1255–1272.
9. Goel, N.; Gupta, A.; Singh, S.N. A study report on virtualization technique. In *Proceedings of the 2016 International Conference on Computing, Communication and Automation (ICCCA)*, Noida, India, 29–30 April 2016; pp. 1250–1255.
10. Jarraya, Y.; Madi, T.; Debbabi, M. A Survey and a Layered Taxonomy of Software-Defined Networking. *IEEE Commun. Surv. Tutor.* 2014, 16, 1955–1980.
11. Nunes, B.A.A.; Mendonca, M.; Nguyen, X.N.; Obraczka, K.; Turletti, T. A Survey of Software-Defined Networking: Past, Present, and Future of Programmable Networks. *IEEE Commun. Surv. Tutor.* 2014, 16, 1617–1634.
12. Da Silva, R.B.; Mota, E.S. A Survey on Approaches to Reduce BGP Interdomain Routing Convergence Delay on the Internet. *IEEE Commun. Surv. Tutor.* 2017, 19, 2949–2984.
13. Hu, F.; Hao, Q.; Bao, K. A Survey on Software-Defined Network and Open Flow: From Concept to Implementation. *IEEE Commun. Surv. Tutor.* 2014, 16, 2181–2206.
14. Mihailescu, M.; Nguyen, H.; Webb, M.R. Enhancing wireless communications with software defined networking. In *Proceedings of the 2015 IEEE Military Communications and Information Systems Conference (MilCIS)*, Canberra, ACT, Australia, 10–12 November 2015; pp. 1–6.
15. Phemius, K.; Seddar, J.; Bouet, M.; Khalifé, H.; Conan, V. Bringing SDN to the edge of tactical networks. In *Proceedings of the 2016 IEEE Military Communications Conference (MILCOM)*, Baltimore, MD, 1–3 November 2016; pp. 1047–1052.
16. Campioni, Lorenzo, Filippo Poltronieri, Cesare Stefanelli, Niranjan Suri, Mauro Tortonesi, and Konrad Wrona. “Enabling civil–military collaboration for disaster relief operations in smart city environments.” *Future Generation Computer Systems* 2023, 139, 181–195.
17. Al Ajrawi, S., Tran, B. Mobile wireless ad-hoc network routing protocols comparison for real-time military application. *Spat. Inf. Res.* 2024, 32, 119–129.
18. Santoso, Fendy, and Anthony Finn. “Trusted Operations of a Military Ground Robot in the Face of Man-in-the-Middle Cyber-Attacks Using Deep Learning Convolutional Neural Networks: Real-Time Experimental Outcomes.” *IEEE Transactions on Dependable and Secure Computing* 2023.
19. Chandan, Radha Raman, Awatef Balobaid, Naga Lakshmi Sowjanya Cherukupalli, Gururaj HL, Francesco Flammini, and Rajesh Natarajan. “Secure Modern Wireless Communication Network Based on Blockchain Technology.” *Electronics* 2023, 12, no. 5, 1095.
20. Gururaj, H. L., B. H. Swathi, R. Trupti, Urs R. Darshan, A. B. Rajendra, and K. Paramesha. “Analysis of Preventive Measures Against DDoS Attacks in Smart Grid.” *Journal of the Institution of Engineers (India): Series B* 2023, 104, no. 1, 297–303.

Artificial intelligence for military applications with blockchain

Arsh Krishanater, Gururaj H L, Gowtham M, and Hong Lin

4.1 INTRODUCTION

4.1.1 Introduction to blockchain

Blockchain, is a digital transformation introduced in the year 1991. It functions as a distributer and decentralized ledger. This technology securely logs transactions using a network of computers, eliminating centralized systems and preventing any potential points of failure. By using network of computers blockchain securely logs the transactions, preventing any potential points of failure by eliminating the centralized systems. By using mechanisms like Proof of Stake or Proof of Work the ledger is kept consistent and cannot be altered across all the nodes, it uses the underlined mechanisms to incorporate and verify new transactions. Five basic principles underlying the technology:- 1. Distributed Database 2. Peer-to-Peer transmission 3. Transparency with pseudonymity 4. Irreversibility of records 5. Computational Logic [1].

4.1.2 Introduction to artificial intelligence for military applications

Artificial Intelligence (AI) is revolutionizing military applications, introducing new capabilities. AI is not meant to function as a standalone application. Rather, it is a valuable technology that enhances and supports existing functional applications. It relies on intricate algorithms that are tailored to solving specific problems by collecting, organizing, processing, analyzing, and transmitting large sets of data. Additionally, it is designed to respond to the human intellect's complex cognitive abilities and operations, approaching them with precision and accuracy [2]. To enhance efficiency in defense operations it encircles machine learning, computational algorithms autonomous systems. Key aspects includes systems like drone, data analysis for intelligence, predictive analytics for threat anticipation, and AI-driven cybersecurity measures.

In military training, AI facilitates realistic simulations, helping to improve preparedness in real-world scenarios. Decision support systems and optimized logistic processes leverage AI for informed choices and efficient

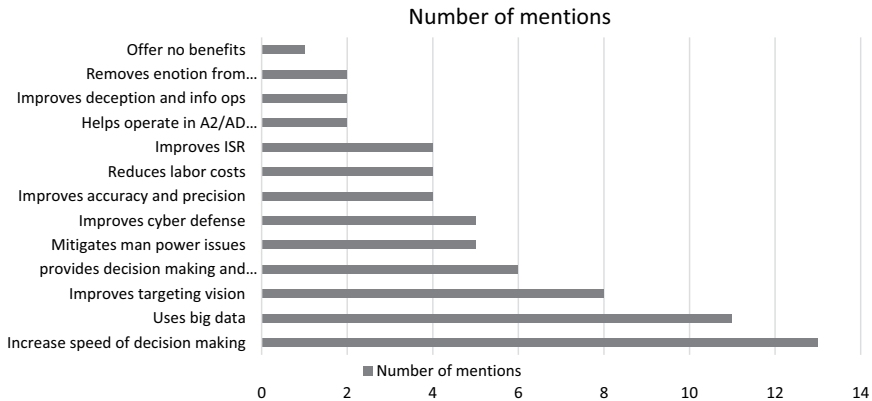


Figure 4.1 AI in defense – A graphical overview.

resource distribution. Human augmentation through AI technologies enhances soldiers' cognitive and physical abilities on the battlefield.

While advancements offer numerous benefits, ethical concerns emerge, particularly regarding the use of AI in warfare and the deployment of autonomous weapons. As AI continues to advance, its influence on military capabilities grows shaping the future of defense operations and global security. Responsible development of AI in the military is crucial for ensuring positive contributions to global security as shown in Figure 4.1.

4.1.2.1 AI Applications in Military Operations

Artificial Intelligence (AI) has emerged as a cornerstone in revolutionizing military operations, offering a spectrum of applications that augment strategic capabilities and decision making processes.

1. **Autonomous Systems:** The development and deployment of autonomous systems are one of the primary applications of AI in military. Without direct human control systems like drone, unmanned vehicles and robotic systems equipped with AI algorithms can execute a variety of tasks, from reconnaissance to surveillance. This autonomy not only enhances operational reach but also mitigates human exposure in high-risk environments.
2. **Data Analysis for Military Intelligence:** AI's capability to rapidly analyze vast datasets has significantly transformed Military intelligence. Complex information, decision-making, threat detection, and providing actionable insights for strategic planning can be processed and interpreted by Advanced algorithms. The speed and accuracy of AI driven data analysis contribute to a more effective response to dynamic and evolving threats.
3. **Predictive Analytics:** Military operations benefit from AI's predictive analytics capabilities, which allow for the anticipation of potential

threats and outcomes. By leveraging historical data and real-time information, AI algorithms can forecast scenario, aiding military planners in preparing for various situations and adapting strategies accordingly.

4. **Cybersecurity:** In the digital age, cybersecurity is paramount for military operations. AI contributes by detecting and responding to cyber threats in real-time. Machine learning algorithms analyze network traffic patterns, identify anomalies, and autonomously neutralize potential cyberattacks, fortifying the military's cyber resilience.
5. **Simulation and Training:** AI facilitates realistic simulations for military training, providing immersive environments for soldiers to hone their skills. Virtual scenarios help improve decision-making, strategic thinking, and teamwork, enhancing overall preparedness for real-world situations.
6. **Decision Support Systems:** AI-powered decision support systems assist military leaders in analyzing complex situations and making informed choices. These systems integrate real-time data, intelligence, and predictive analytics, offering commanders a comprehensive view for strategic planning and mission execution.
7. **Logistics and Supply Chain Optimization:** AI optimizes logistics and supply chain management by streamlining resource distribution, maintenance, and transportation. Predictive analytics forecast demand, ensuring timely and efficient allocation of resources, reducing costs, and improving overall operational efficiency.
8. **Human Augmentation:** AI technologies contribute to human augmentation, enhancing the cognitive and physical abilities of military personnel. Wearable devices and exoskeletons powered by AI provide soldiers with improved situational awareness, resilience, and overall performance on the battlefield.

4.2 BLOCKCHAIN TECHNOLOGY OVERVIEW

Blockchain technology emerged as a fundamental aspect of the revolutionary digital currency, Bitcoin. Interestingly, Bitcoin's founder did not refer to it as "blockchain" in his initial whitepaper. Upon reading the paper, it becomes apparent that the creator was not presenting a completely new technology, but rather utilizing various existing technologies to develop a strictly decentralized form of digital money [3].

1. **Decentralization:** At the core of blockchain lies a critical principle: decentralization. Unlike conventional centralized systems where a single entity holds the reins of the database, blockchain delegates control to a web of participants, known as nodes. This distribution of power guarantees the absence of any weak links, bolstering the security and dependability of the system.

2. **Distributed Ledger:** Data in a blockchain is securely stored in inter-connected blocks, creating an unchangeable and timestamped chain. Distributed nodes work together to maintain this ledger, guaranteeing that all users have an up-to-date and consistent version of the information. Every block also includes a cryptographic code from the previous block, ensuring a strong and unbroken sequence of data.
3. **Consensus Mechanism:** In order to successfully validate and append fresh transactions to the blockchain, it is imperative for participants to come to a consensus on a particular algorithm. Popular methods for reaching this agreement include Proof of Work (PoW) and Proof of Stake (PoS). These mechanisms work toward guaranteeing that every node in the network is in alignment before introducing a new block, ultimately safeguarding against deceitful activities and preserving the authenticity of the ledger.
4. **Security through Cryptography:** In today's world, security, and information privacy are crucial components of our existence. Information plays a significant role in our daily lives, making personal authentication a major concern that demands attention [4]. In the world of blockchain, cutting-edge cryptographic techniques are utilized to safeguard transactions. Every individual involved in the network is assigned a unique public and private key. These keys play a critical role in safeguarding the confidentiality, integrity, and authenticity of transactions. Furthermore, the implementation of cryptographic hashes makes it nearly impossible to manipulate the information within a block, ensuring an unparalleled level of security.
5. **Transparency and Immutability**
The exciting world of blockchain employs advanced cryptographic methods to protect transactions. Each member in the network has their own exclusive public and private key, playing a vital role in securing the confidentiality, reliability, and legitimacy of transactions. With the added layer of cryptographic hashing, the data within a block is virtually impenetrable, ensuring an unparalleled level of protection.
6. **Smart Contracts:** The revolutionary technology of blockchain allows for the seamless development and implementation of smart contracts, which are essentially automated contracts with predetermined rules and requirements. These smart contracts operate on the blockchain platform and are able to enforce the terms of the agreement without the need for third-party intermediaries. This advanced feature promotes efficient automation and decreases reliance on external parties.

4.3 BLOCKCHAIN IN MILITARY OPERATIONS

Blockchain is a cutting-edge technology that offers a plethora of benefits, including streamlined, reliable, and tamper-proof data records. What's more, it is an economically efficient solution as it eliminates the need for

intermediaries to oversee transactions [5]. The world of Supply Chain Management (SCM) is vast and intricate, spanning across numerous parties and tasks. From production to distribution, SCM is responsible for overseeing the flow of materials from initial suppliers to the final customer. This complex process, known as a multi-tier supply chain, consists of various stages before the end product is achieved. To illustrate, Diagram 2 depicts the eight essential processes that must be carried out within and between firms in the supply chain [6].

Blockchain technology is a game-changing innovation, revolutionizing military operations with its powerful combination of heightened security and transparency. This segment delves into the diverse range of ways that blockchain is being utilized in the armed forces, showcasing its ability to transform communication, data control, and supply chain logistics.

1. **Secure communication:** One of the primary applications of blockchain in the military is secure communication. Blockchain's decentralized and cryptographic nature ensures that messages and sensitive information remain confidential and tamper-proof. This feature is critical for safeguarding military communications against cyber threats and espionage.
2. **Tamper-Proof Record-Keeping:** Blockchain's distributed ledger technology provides an immutable and transparent record-keeping system. In military operations, this feature proves invaluable for maintaining accurate and unalterable records of mission-critical data, such as troop movements, intelligence reports, and logistics information. The tamper-proof nature of the blockchain enhances data integrity and mitigates the risk of unauthorized alterations.
3. **Supply Chain Management:** The military's complex and extensive supply chain can benefit significantly from blockchain integration. Blockchain facilitates real-time tracking and traceability of military assets, equipment, and resources throughout the supply chain. This transparency ensures that the origin, movement, and status of critical supplies are easily verifiable, reducing the risk of fraud, counterfeiting, and logistical errors.
4. **Integration Challenges:** While the potential benefits of blockchain in military applications are vast, integration is not without challenges. The decentralized nature of blockchain can pose interoperability issues with existing centralized systems. Additionally, concerns related to scalability, regulatory compliance, and adapting legacy systems need careful consideration.
5. **Future Prospects:** Looking ahead, the integration of blockchain in military applications is poised for continued growth and refinement. Advancements in blockchain technology, such as improved scalability and interoperability, may address current challenges. The decentralized and secure nature of blockchain aligns with the military's need for trustworthy and resilient systems, making it a promising technology for future defense strategies.

4.4 INTEGRATION OF AI AND BLOCKCHAIN

By leveraging the capabilities of blockchain technology, we can propel decentralized marketplaces and coordination platforms for the different facets of AI, such as data management, algorithm development, and computing resources. Such an approach will undoubtedly drive the growth and implementation of AI to new heights. Furthermore, integrating blockchain with AI will enhance the transparency and credibility of AI-powered decisions. Notably, the public nature of blockchain data will ensure user confidentiality and privacy, making AI a crucial safeguard for data protection [7].

By integrating the capabilities of AI and blockchain, a multitude of shortcomings faced by each innovative system can be effectively addressed. AI relies on the utilization of data in order to learn, gather insights, and make informed decisions. The effectiveness of AI algorithms is greatly enhanced when the data is derived from a reliable, trusted, secure, and credible source. This is where blockchain technology comes into play, serving as a distributed ledger where data can be securely stored and executed using cryptographic signatures that have been validated and agreed upon by all mining nodes. Due to the nature of blockchain, data is stored with utmost integrity and resilience, making it impossible to alter. By utilizing smart contracts, AI algorithms can be empowered to make decisions and conduct analyses, resulting in outcomes of unparalleled precision and reliability. The perfect amalgamation of AI and blockchain technology creates a robust, immutable, and decentralized framework for safeguarding highly sensitive data collected, stored, and utilized by AI-driven systems. This innovative concept presents significant advancements in securing data across various domains, including healthcare, personal information, finance, banking, trade, and legal records. The utilization of multiple blockchain platforms empowers AI with the ability to execute algorithms and track data stored on decentralized P2P storage systems. This data is primarily generated by intelligent connected devices from a variety of sources. Additionally, the cloud's capabilities and services can be effectively utilized for off-chain AI processing. Shown in Figure 4.2.

Challenges and Future Outlook

1. Challenges in Integration: Despite the promising prospects, integrating AI and blockchain in military applications is not without challenges. Interoperability issues, scalability concerns, and the need to adhere to stringent military regulations pose significant hurdles. Additionally, the resource-intensive nature of AI algorithms may impact the efficiency of certain blockchain networks.
2. Future Outlook: The integration of AI and blockchain in military applications is a rapidly evolving field with a promising future. Ongoing research and development are addressing current challenges, and as technology advances, the combined power of AI and blockchain is likely to redefine military operations. The collaboration between these

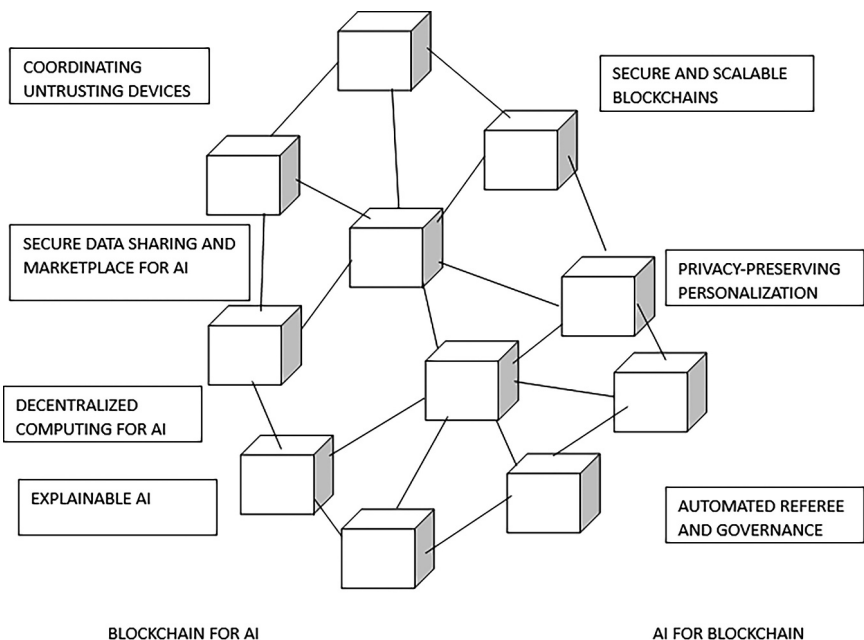


Figure 4.2 Integration of AI and blockchain.

two transformative technologies is expected to bring about increased efficiency, transparency, and security in defense strategies, making them more resilient and adaptive in the face of evolving threats.

4.5 SUPPLY CHAIN MANAGEMENT

Numerous industry associations are actively exploring the integration of blockchain solutions into supply chain operations. The issue of security in supply chain management has become a top priority, particularly as the use of commercial off-the-shelf (COTS) components for embedded software systems continues to rise. This presents a potential concern, as these components could potentially house intentional vulnerabilities that could be exploited by adversaries at their discretion. The novel Ghost Fleet has brought attention to the potential severity of a threat where China purposely causes the entire fleet of F-35 aircraft to fail by using flawed commodity circuit boards. However, there is a solution in blockchain technology which enables tracking of each circuit board, processor, and software component from its production to usage. By utilizing blockchains, card design companies can record the evolution of circuit designs, manufacturers can track all models and serial numbers of their products, and distributors can monitor

sales to system integrators who then distribute the circuits to specific aircraft assemblies. Essentially, blockchains act as a permanent record of the transfer of assets between owners, ultimately creating a derivative.” Weapon systems are typically built to last for over 30 years, but the computer technology utilized in these systems is only meant to endure for approximately a decade. As a consequence, it becomes increasingly challenging to replace outdated components as time passes. In certain countries, there are laws in place that prohibit the use of components with unknown origins. This means that even if the parts are in good working order and highly sought after, they are rendered useless. This creates a financial incentive for resellers to track their commercial components and preserve their origin via a block, ultimately increasing their value. The Hungarian Defense Forces currently do not have a distinct focus on decentralized technologies, but there is ongoing international research and development in this area. In fact, NATO’s C4ISR and the US Department of Defense (DARPA - DoD) have already initiated their own blockchain initiatives. Specifically, they have started developing a secure and decentralized messaging application for military use, known as SBIR 2016.2 [8] as shown in Figure 4.3.

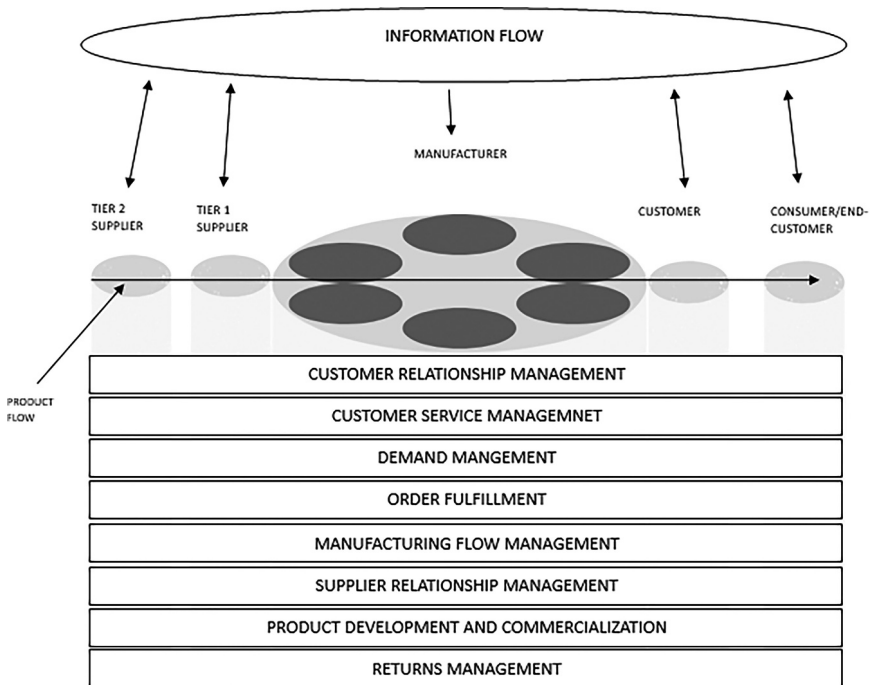


Figure 4.3 Supply chain management processes.

4.6 TRUST MANAGEMENT

At present, CPS (Computer-Physical System) face a heightened level of susceptibility to a multitude of threats. These include highly evolved adaptive attacks that target physical layer links, resulting in catastrophic consequences. Furthermore, CPS are at risk of both physical and digital assaults, with the former being the more prevalent and the latter being the most frequent. Malicious software, such as malware and ransomware, is the weapon of choice for cyberattacks. These attacks infiltrate network systems in order to cause harm. On the other hand, physical attacks target the manipulation and exploitation of tangible components. While active attacks can alter data through methods such as inference or data poisoning, passive attacks involve hackers gathering information by intercepting publicly available data from the CPS. Criminals often seek to alter normal data to carry out data poisoning attacks. A prevalent type of this attack is false data injection which can be directed at CPS networks. As a result, safeguarding the integrity and protection of data is an essential aspect of CPS. To address this issue, a three-layer privacy-preserving architecture has been proposed, consisting of the application, network, and device layers. Specifically, the device layer is responsible for gathering data from IIoT sensing devices. Then, in order to ensure data security, the data's authenticity [9] is simultaneously verified.

Security and Privacy Concerns:

Block chain and artificial intelligence (AI) are quickly becoming the focal points of the tech world, earning notable spots on every chief information officer's list of revolutionary technologies that are primed to transform entire industries [10]. The integration of Artificial Intelligence (AI) and blockchain introduces a transformative paradigm in various domains, including security and privacy. This section critically examines the implications of this convergence and explores strategies to address potential risks while ensuring compliance with privacy regulations (Figure 4.4).

Security Implications

1. **Smart Contract Vulnerabilities:** Smart contracts, a core feature of blockchain, may be susceptible to vulnerabilities if not designed and audited meticulously. Flaws in the code could be exploited, leading to security breaches. Employing rigorous code audits and secure coding practices is imperative to mitigate such risks.
2. **Decentralized Network Security:** Blockchain's decentralized nature is a strength, but it also introduces challenges. As the network relies on consensus mechanisms, any compromise in the consensus process could undermine the security of the entire system. Implementing robust consensus mechanisms, such as Proof of Stake or Proof of Work, and regularly updating security protocols is vital.
3. **Data Privacy Concerns:** While blockchain ensures data integrity, the transparent nature of the ledger can pose privacy challenges. Sensitive

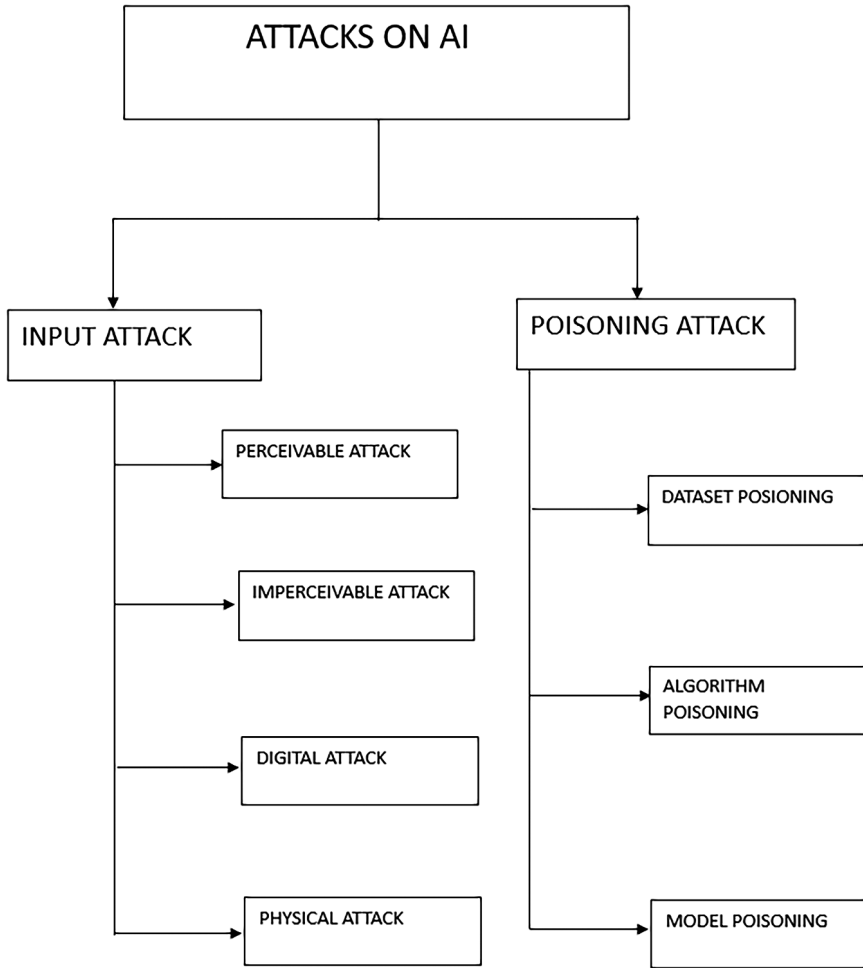


Figure 4.4 Attacks on AI.

information stored on the blockchain may be visible to all participants. Implementing privacy-focused solutions, such as zero-knowledge proofs or privacy-focused blockchain platforms, can mitigate these concerns.

Privacy Implications

1. **Data Ownership and Control:** The decentralized nature of blockchain raises questions about data ownership and control. Individuals may lose control over their personal data once it is stored on a blockchain. Implementing clear data ownership frameworks and enabling user-centric control mechanisms is crucial for preserving privacy.
2. **Identity Protection:** The use of AI in conjunction with blockchain may involve the creation of decentralized identities. While enhancing security, it also requires careful handling to prevent the misuse or exposure

of personal information. Employing encryption techniques and decentralized identity management systems can safeguard individual identities (Figure 4.4).

Strategies for Mitigation and Compliance

1. **Privacy by Design:** Incorporating privacy measures into the design phase of AI and blockchain systems is essential. To safeguard sensitive information, it is crucial to employ various security measures such as encryption, access controls, and data anonymization techniques.
2. **Regulatory Compliance:** Staying abreast of evolving privacy regulations and ensuring compliance with frameworks such as GDPR or HIPAA is crucial. Adapting the design and operation of AI and blockchain systems to align with these regulations helps mitigate legal and reputational risks.
3. **Ethical AI Development:** Incorporating ethical AI principles into our practices is crucial. These principles, including fairness, transparency, and accountability, are paramount to ensure the ethical use of AI. To build trust and address concerns regarding data misuse, it is imperative to establish precise guidelines for the ethical application of AI within blockchain systems. The development of AI exhibits four characteristics that indicate a principled approach may not have a significant impact on design and governance. When compared to medicine, the field of AI lacks a shared set of goals and ethical responsibilities, a well-established professional background and conventions, effective strategies for implementing principles into practice, and strong legal and professional accountability measures.
4. **Continuous Monitoring and Auditing:** By consistently monitoring and regularly auditing for security, potential vulnerabilities can be swiftly identified and resolved. This proactive strategy greatly fortifies the security of the AI and blockchain system.

Challenges in Integration

1. **Fog Computing Paradigm:** This revolutionary property boasts the ability to perform localized computing for data generated by both IoT devices and customers. These innovative fog nodes serve to enhance the speed and efficiency of cloud computing [11]. To fully harness the power of Blockchain and AI, these nodes must be equipped with Blockchain interfaces, as well as the cutting-edge capabilities of machine learning and artificial intelligence. Furthermore, the fog nodes also play a crucial role in controlling and accessing the data.
2. **Smart Contract:** Ensuring the solidity and security of a smart contract can be a daunting task, with the potential for attacks and bugs to threaten its implementation. The information and code on the network must be safeguarded against vulnerabilities. This vulnerability stems from a lack of thorough programming studies in the language

used for writing smart contracts. This issue has given rise to a critical need for vulnerability testing tools specific to smart contracts [11], as well as measures to secure the code itself. Furthermore, the execution of smart contracts cannot rely on probability, which poses a significant challenge for decentralized AI initiatives where decision-making algorithms based on ML and AI must be executed. In such cases, the outcome of smart contract execution is often deterministic.

3. **Data Protection Laws:** Military operations involve vast amounts of data, and the integration of AI and blockchain requires compliance with data protection laws. Navigating these regulations necessitates transparent data governance, encryption, and mechanisms for secure data handling to protect individual privacy rights.

Potential Drawbacks

1. **Scalability and Performance:** The resource-intensive nature of AI algorithms and the computational demands of blockchain networks may pose challenges in terms of scalability and performance. Striking a balance between computational requirements and operational efficiency is crucial to avoid potential bottlenecks.
2. **Interoperability:** Integrating AI and blockchain systems from different vendors or across military branches may encounter interoperability challenges. Developing standardized protocols and interfaces can facilitate seamless integration and communication between diverse systems.
3. **Cost Implications:** The adoption of advanced AI and blockchain technologies requires substantial financial investments. Balancing the potential benefits with budget constraints and ensuring cost-effectiveness in the long run is essential for sustainable deployment.

4.7 CASE STUDY

In-Depth Analysis of Successful Deployments: AI and Blockchain in Military Scenarios

The successful integration of Artificial Intelligence (AI) and blockchain technology in military applications has yielded valuable insights and practical lessons. Through in-depth analysis of prominent cases, this examination uncovers key understandings and best practices for utilizing these cutting-edge technologies in defense.

Case 1 Secure Communication – U.S. Department of Defense (DoD)

Overview:

The implementation of blockchain technology by the U.S. Department of Defense has significantly bolstered the security of communication within its

network. Using blockchain, sensitive information is safeguarded through its integrity and confidentiality, creating a secure avenue for messaging and data transmission.

Lessons Learned and Best Practices:

- **Immutability:** The immutable nature of blockchain technology guarantees that communication data remains unaltered, ensuring a superior level of data integrity.
- **Decentralized Key Management:** Blockchain implementation enables the distribution of key management, mitigating the vulnerability of cryptographic systems to single points of failure.

Case 2 NATO

Overview:

NATO has tapped into the power of AI and blockchain technology to enhance its supply chain operations. This clever integration allows for seamless real-time monitoring, complete transparency, and thorough traceability of military resources, ultimately resulting in more efficient logistics.

Lessons Learned and Best Practices:

- **Enhanced Traceability:** The transparent ledger of Blockchain allows for real-time tracing of supplies, mitigating the chance of errors and amplifying overall accountability.
- **AI Predictive Analytics:** Through the power of AI algorithms, supply chain demands are accurately foreseen, leading to efficient distribution of resources and reduced bottlenecks. This undeniable partnership between AI and blockchain highlights their remarkable potential in streamlining processes.

4.8 FUTURE TREND AND DEVELOPMENT

As technology moves forward at an accelerated pace, its influence on military operations remains a constant force. The fusion of Artificial Intelligence (AI) and blockchain technology is leading the way in shaping the future of military applications. As we shift our gaze to what lies ahead, a multitude of fresh developments and advancements are on the horizon, poised to revolutionize the capabilities and tactics of defense systems.

As the fields of AI and blockchain continue to progress, their integration in military operations is poised to greatly impact the evolution of defense systems. With various trends and advancements already in motion, it is apparent that a future is being forged where these technologies not only

bolster operational effectiveness, but also account for ethical and diplomatic considerations. This promises a responsible and secure application of AI and blockchain in military scenarios (Figure 4.5).

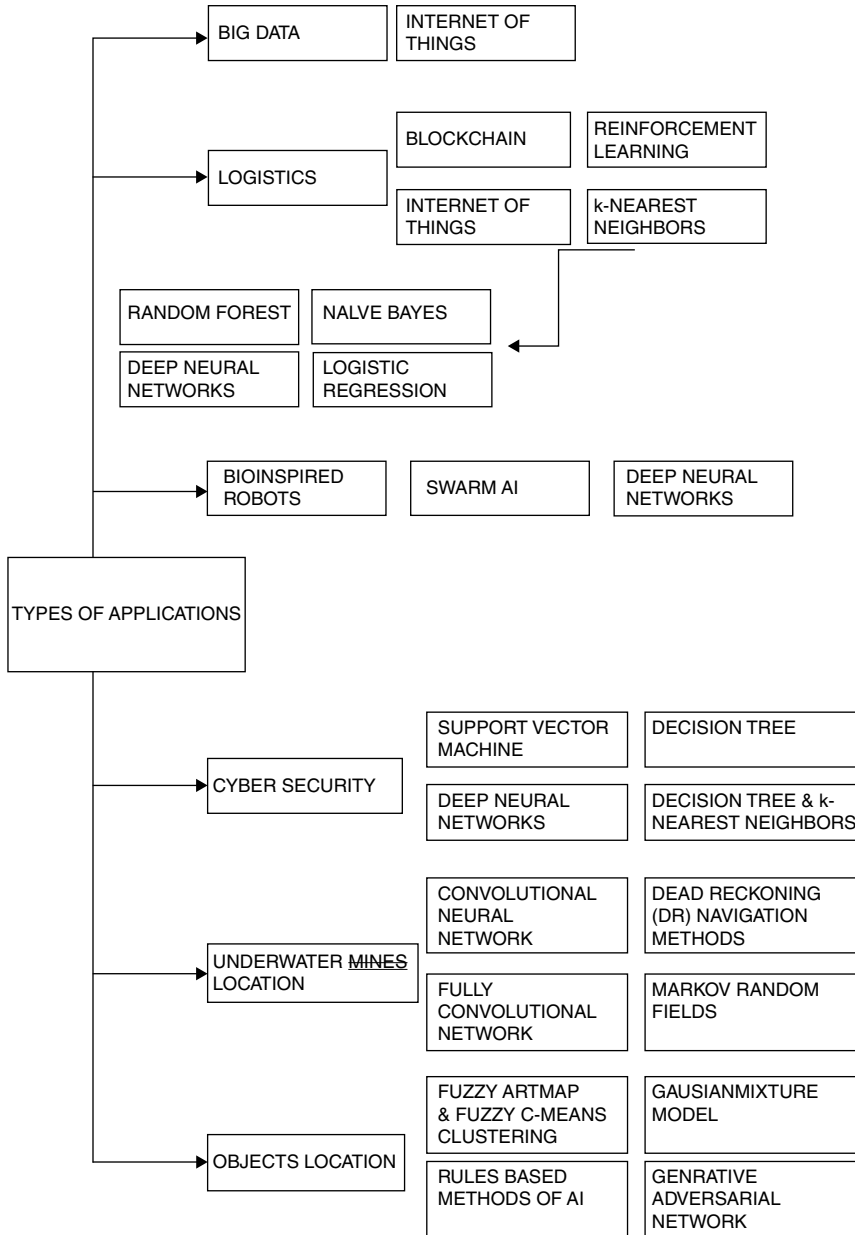


Figure 4.5 Types of applications.

4.9 CONCLUSION

To summarize, the incorporation of AI and blockchain in military contexts offers enormous potential for improving security, promoting transparency, and optimizing operations. By harnessing the power of AI in autonomous functionality, data interpretation, and forecasting, and merging it with the highly secure and decentralized features of blockchain technology, military endeavors can attain unparalleled levels of efficiency and success.

Our integrated approach tackles cybersecurity, logistics, and decision-making hurdles, providing a holistic answer to the demands of today's military. At the same time, the ethical implications and regulatory obstacles highlight the crucial need for responsible implementation.

The rapid advancement of technology has paved the way for exciting possibilities in military practices. The intersection of AI and blockchain has been a prominent focus, offering opportunities for improved defense operations. However, as we delve into this uncharted territory, it is crucial to find a balance between innovation, security, and ethical considerations. Only then can we fully utilize the potential of these groundbreaking technologies to safeguard global security.

REFERENCES

1. Iansiti, Marco, and Karim R. Lakhani. "The truth about blockchain." *Harvard Business Review* 95.1 (2017): 118–127.
2. Szabadföldi, István. "Artificial intelligence in military application – Opportunities and challenges." *Land Forces Academy Review*, vol. 26.2 (2021): 157–165. <https://doi.org/10.2478/raft-2021-0022>
3. Ammous, Saifedean. "Blockchain technology: What is it good for?." *Available at SSRN* 2832751 (2016).
4. Mohan, Jinu, and R. Rajesh. "Enhancing home security through visual cryptography." *Microprocessors and Microsystems* 80 (2021): 103355.
5. Ahmad, Raja Wasim, et al. "Blockchain for aerospace and defense: Opportunities and open research challenges." *Computers & Industrial Engineering* 151 (2021): 106982.
6. Rahayu, Syarifah Bahiyah, Nur Diyana Kamarudin, and Afiqah Mohammad Azahari. "Military blockchain for supply chain management." *Journal of Education and Social Sciences* 13.1 (2019): 353–361.
7. Dinh, Thang N., and My T. Thai. "AI and blockchain: A disruptive integration." *Computer* 51.9 (2018): 48–53.
8. Selvarajan, Shitharth, et al. "An artificial intelligence lightweight blockchain security model for security and privacy in IIoT systems." *Journal of Cloud Computing* 12.1 (2023): 38.
9. Mittelstadt, Brent. "Principles alone cannot guarantee ethical AI." *Nature machine intelligence* 1.11 (2019): 501–507.

10. Gulati, Prerna, et al. "Approaches of blockchain with ai: Challenges & future direction." *Proceedings of the international conference on innovative computing & communications (ICICC)*. 2020.
11. Tyagi, Amit Kumar, S. U. Aswathy, and Ajith Abraham. "Integrating blockchain technology and artificial intelligence: Synergies perspectives challenges and research directions." *Journal of Information Assurance and Security* 15.5 (2020): 1554.

Blockchain for military applications

Opportunities, challenges, and its applications

Sudeep J, Shivashankara S, Ananth G S, Girish S C, and Hong Lin

5.1 INTRODUCTION

Technology plays a pivotal role in country development by driving innovation, increasing productivity, and enhancing infrastructure. It empowers nations to leapfrog traditional limitations, fostering economic growth and competitiveness on a global scale. It also promotes inclusive development and improves living standards of citizens through education and connectivity. At the same time, the role of technology has a significant impact on defense and military systems. Technology on the battlefield enhances the military capabilities through advanced weaponry, machinery, surveillance system and communication tools, leading to increased security and strategic management. However, the economic impact is significant, as the development and deployment of these technologies requires substantial investments which can strain country resources [1, 2].

In the modern era, balancing military innovation with economic development is a crucial challenge for nations. In a rapidly evolving digital landscape, the military sector is exploring innovative technologies to enhance security and efficiency. Blockchain is one such technology that is renowned for its decentralization and immutable nature and has gained significant attention in the military field. Blockchain technology has transpired as a game changer in the military sector offering numerous advantages [3]. In military operations, the accuracy and security of information can be a matter of life and death. Blockchain cryptographic and consensus mechanism make it nearly a tamper proof securing sensitive military data. The decentralization nature of blockchain means there is no single point of failure, enhancing operational resilience. In the era of cyberwarfare, this is critical for maintaining operational capabilities. Supply chain management is a cornerstone of military applications, and blockchain can revolutionize it. Blockchain transparency and traceability can ensure that critical supplies, from ammunition to rations, are genuine and secure. In the military, secure communication is a paramount and blockchain can ensure that message remain confidential and unaltered. This can be critical in safeguarding sensitive information, plan and strategies. Blockchain can enhance the

management and security of military personnel records. Sensitive information, such as service histories and health records, can be securely stored and accessed. Blockchain can protect against cyber threats through its decentralized and encrypted networks. It can also be used to trace and mitigate cyberattacks in real time. The logistics and assets management within the military can be vastly improved using blockchain technology. The real time tracking of assets ensures they are in the right place at the right time. While the current military applications developed on blockchain may not be fully matured for deployment, it is anticipated that the imminent implementation of military blockchain technology will primarily focus on defense logistics and data security [4, 5]. Finally, the implementation of blockchain in the military sector can encompass the following areas such as

- Intrusion detection
- Infrastructure monitoring
- Battles management
- UAV management
- Supply chain management
- Encrypted communications
- Security and privacy
- Risk and Time management

Figure 5.1 shows how blockchain implementation areas that can be applied in military sector. The purpose of the chapter is to explore the potential opportunities, challenges, and real world applications of blockchain technology within military sectors. The chapter aims to provide an in-depth understanding of how blockchain can enhance data security, supply chain

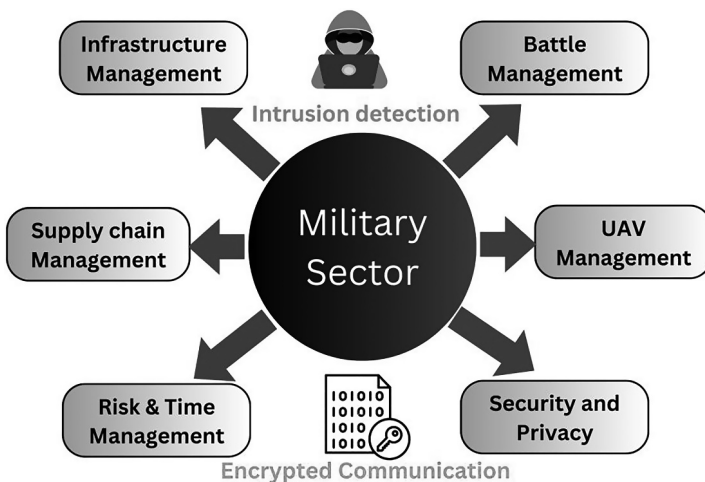


Figure 5.1 Blockchain Implementation areas in military sector.

management, logistics, communication and overall operational efficiency in the military context. It also aims to highlight the significant role the blockchain can play in modernizing and strengthening military capabilities while addressing the complexities and nuances involved in development.

The main contributions of our chapter are as follows

- We provide a comprehensive overview of blockchain technology and its basic principles, highlighting key components and factors that ensures security and trust in military applications.
- We provide a concise view of various application scenarios of blockchain in military sectors to secure and provide efficient solutions for storing information. We outline the opportunities, challenges and risks involved in blockchain for data storage in military applications.
- We highlight the security consideration of blockchain by protecting the sensitive military data. We discuss future trends and developments along with some case study.

The rest of the chapter is organized as follows. Section 1 provides a brief introduction about the importance of blockchain in military applications. Section 2 provides an in-depth understanding of blockchain technologies by highlighting significant factors of blockchain along with its basic principles. Section 3 throw light on military applications of blockchain. Section 4 provides insights on various opportunities for military applications using blockchain. Section 5 discuss on challenges and risks in military application. Section 6 highlights about future trends and developments of blockchain in military sector. Section 7 throw light on security consideration of using blockchain technology in military applications. Section 8 shed light on case study that explain how blockchain technology can be applied to military sectors.

5.2 UNDERSTAND BLOCKCHAIN TECHNOLOGIES

5.2.1 Basic principles of blockchain

Over the past few decades, we have maintained a swift pace of digital transformation. Blockchain is one such emerging technology camp; highly debated technological advancements, much like AI and quantum technology, have gained a significant attention in recent times. Its importance surged exponentially as it found extensive application within the financial sector, with bitcoin and Ethereum emerging as well-known names among virtual currency enthusiasts. Nonetheless, the military has also found applications for this technology in various areas. In essence, this technology revolves around securely and trustworthily organizing and storing information

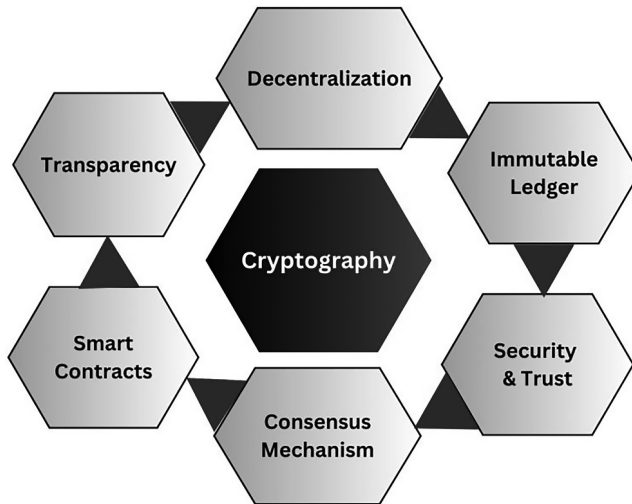


Figure 5.2 Basic principle of blockchain.

through intricate mathematical principles. Let's understand blockchain technology in detail. Figure 5.2 shows a schematic view of basic principles of blockchain.

Blockchain is a distributed ledger technology that affords a secure and transparent approach for recording and verifying transactions

At its core, it operates on several basic principles. These fundamental principles underpin the technology and make blockchain a powerful tool to widespread choice of applications, from the crypto-currencies like from Bitcoin to military and supply chain management and ahead. They are as follows:

- **Decentralization:** Blockchain eliminates the need for the central authority, such as bank or government, to validate transactions. Instead it relies on nodes (computers) that reach consensus through consensus algorithms.
- **Immutable Ledger:** On blockchain, once the data is recorded, then it will not be modified or removed. This immutability confirms a permanent, and also tamper resistant transactions records.
- **Cryptography:** Complex cryptographic algorithms are used to secure the data on the blockchain. Public and private keys ensure the authenticity of participants, while hashing ensures integrity of data.
- **Transparency:** The entire blockchain is publicly accessible, and everybody is possible to view all the transactions. This translucence builds belief in the system because it reduces the potential for fraud or corruptions.

- **Smart Contracts:** Smart contracts are the self-execution treaties by means of the contract terms, written to code directly. This certainly executes, when specified circumstances met, removing the need for intercessors.
- **Consensus Mechanism:** Various consent method, such as Proof of Work (POW) and Proof of Stake (POS), are used for validating and adding new businesses to blockchain. These mechanism ensure agreements among participants and protect against double spending
- **Security and Trust:** The combination of cryptographic security, transparency, and decentralization makes blockchain a highly secure and trustable systems for recording and verifying transactions [6].

5.2.2 Significant factors of a blockchain

The significant factors of blockchain are necessary for its functionality and security. Such factors together perform for creating a decentralized and also tamper-less ledger to verify and to record the transactions. The precise execution and features may vary depending on the blockchain platform [7, 8]. The Key components of blockchain system include the following:

- **Nodes:** These are individual computers or devices connected to the blockchain networks. The nodes play a crucial role in validating and storing transactions, ensuring the network security.
- **Blocks:** Transactions are congregated as masses; they are the ampules for data. Every mass naturally consists a reference to the previous block (forming the chain) and a unique identifier, known as cryptographic hash.
- **Transactions:** These are fundamental units of data on a blockchain. They represent actions or exchange of values, such as cryptographic transfers or smart contract executions.
- **Consensus mechanism:** This is the protocol or algorithm used for achieving the agreement amongst the network participants above on the transactions validity and also the directive in which they have been added into the blockchain. Conventional consent mechanism consists POW and POS.
- **Cryptographic Hashes:** These are mathematical functions that create fixed sized unique representation of data. They have been using for securing and linking blocks together, creating it awfully problematic to modify the past data.
- **Smart Contracts:** Self execution contracts with the specified instructions and conditions. They have been stored on the blockchain and mechanically accomplish when conditions are met.
- **Public and Private keys:** These cryptographic keys are used for authentication and digital signatures. Public keys serve as addresses to receive

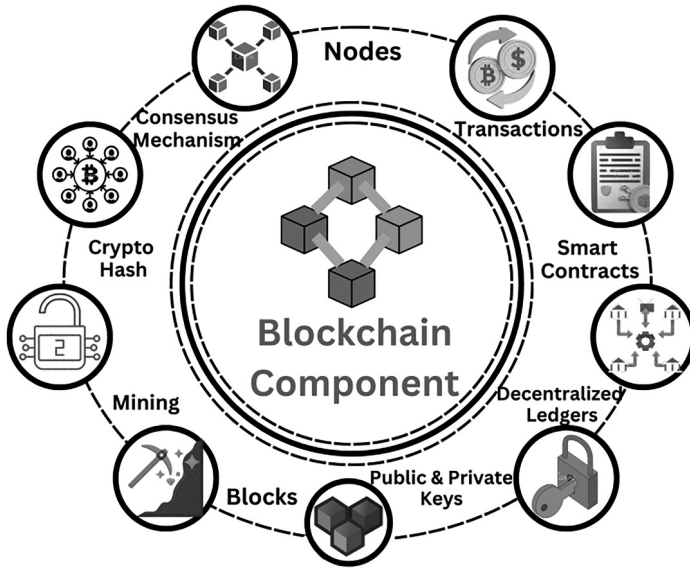


Figure 5.3 Key components of blockchain systems.

funds, while private keys are used for signing transactions and proving ownerships.

- **Decentralized Ledger:** The ledger is distributed across all network nodes, and it contains immutable history of all transactions. It ensures transparency and trust in the system
- **Mining (in POW):** In POW based Block chains like Bitcoins, miners strive for solving intricate mathematical puzzles for adding fresh blocks for the chain and are rewarded with cryptocurrency for their efforts. Figure 5.3 depicts the key components of blockchain systems.

5.2.3 How blockchain ensures security and trust

Blockchain technology is renowned for its ability to ensure security and trust in a wide range of applications. At its fundamental level, blockchain serves as a decentralized and distributed ledger, meticulously documenting the transactions in an unalterable and openly accessible manner. This fundamental architecture plays a pivotal role in fortifying the system security and reliability [9].

One key element of blockchain security is immutability. Once the transaction has been added to blockchain, it is cryptographically wrapped and also linked for the earlier transactions in a chain of blocks. This brands it extremely tedious for altering or deleting past entries, as any changes may need the consent of the most of the network contestants, creating fraudulent

activities or data manipulation nearly impossible. This immutability provides a top level of trust in an accuracy and also integrity of the recorded information [10].

Blockchain ensures security through its decentralization nature. Conventional integrated systems are susceptible to sole point of ruin, making them susceptible to cyber-attacks and data breaches. In contrast, blockchain functions on a network of computers, which independently authenticate and stores all the transactions. This decentralization decreases the chance of sole view of ruin and increases the security since the attacker may require for compromising a most of the network for manipulating the data. Furthermore, transparency of the blockchain ensure that any suspicious activity is easily detectable by network participants, further enhancing trust.

Additionally, blockchain uses cryptographic approaches for securing data and controlling the access. Every partaker in the network has his private and public key, allowing to secure authentication and encryption of transactions. This cryptographic security ensures that only authorized parties can access and update the blockchain, adding another layer of trust to the system.

Overall, blockchain technology security and trust mechanism are underpinned by its immutability, decentralization and cryptographic techniques. These features make it a reliable resolution to the widespread choice of application, from the financial transaction to supply chain management, where trust and security are of utmost importance [9, 10].

5.3 MILITARY APPLICATIONS OF BLOCKCHAIN

In today's digital era, safeguarding communication and data sharing have become increasingly significant. Blockchain technology has emerged as an innovative and reliable solution to address these concerns. The military is one such sector that is undergoing a revolution with the adoption of blockchain technology, which offers secure and efficient solutions for sharing and storing data [11].

5.3.1 Secure communications and data sharing

Information security has remained a critical aspect of information technology and communication since the inception of internet. In old warfare system, military communication involved commanders transmitting the orders to their subordinates, often resulting in significant delays, due to the isolated locations of troops. Communication progressed over time, and the introduction of the telegraph stood out as one of the most significant communication methods employed in military applications.

On its fundamental, Blockchain is a Distributed Ledger Technology (DLT), which enables people for creating a secure, transparent and tamperproof

records. This technology works by grouping transactions into blocks which are linked together chronologically and encrypted using advanced algorithms. Since every node in the network maintains a copy of entire history of transactions, it becomes nearly difficult to any specific or entity for manipulating the data.

One of the primary reasons why blockchain holds potential for transforming secure communication and data sharing lies in its decentralization nature. Unlike traditional systems that stores sensitive information on centralized servers, blockchain distributes the data around the computer network. This decentralization not only increases transparency within the network but also eliminates single point of failure, thereby making it less susceptible to hacking attacks. Moreover, blockchain technology ensures data integrity through its consensus mechanism. In order to validate transactions or make modifications to existing records, a majority consensus among the nodes in the network must be achieved. This system gives confidence that once their information is stored on the blockchain, it cannot be altered without mutual agreement [12].

One real time example of secure communication and data sharing using blockchain technology in military sector can be found in the context of secure logistics and supply chain management. The use of blockchain technology can enhance the security, transparency and efficacy of military supply chains.

Scenario: In a hypothetical case study, let's consider a military organization "Alpha Defense Forces (ADF)" that is responsible for managing and maintaining a secure supply chain for crucial military equipment, including ammunition, weapons, spare parts, and other critical resources. ADF has to implement blockchain technology for ensuring the integrity and security of supply chain operations. ADF has to adopt a private, permissioned blockchain for its supply chain management, where only authorized participants (military units, manufacturers and logistic suppliers) can access and update the distributed ledgers [13, 14]. Figure 5.4 illustrates the military operations of blockchain through secure communication and data sharing. Here is how the system works

- **Data Recording:** Each time a new shipment of military equipment is prepared, a record is created on the blockchain. This records includes details such as the item description, quantity, origin, destination, and expected delivery time.
- **Smart Contracts:** Smart contracts are used to systematize distinct viewpoints of supply chain. For instance, when a consignment achieves a precise checkpoint, the smart contract can trigger automatic notifications or initiate the release of payment to the logistics providers
- **Transparency:** The blockchain ledger is transparent to all the authorized participants, ensuring that everyone involved can monitor the status of shipments and verify their authenticity



Figure 5.4 Military operations of blockchain.

- **Immutable records:** Once the data is appended to the blockchain, then it turns into immutable, making it awfully hard for altering or tampering with any transaction history. This enhances data integrity and prevents fraudulent activities [13, 14].

5.3.2 Identity management and authentication

Identity management and authentication are critical components of military operations, where security and trust are paramount. Blockchain technology holds significant promise for enhancing these aspects in military applications. Blockchain with its decentralized and tamper-resistant ledger, offers a robust solution for managing the identities of military (army) personnel, vehicles, and equipment while ensuring a secure authentication process.

Authentication in the military domain is equally vital, as it ensures that only authorized individuals or entities gain access to classified information, systems, and resources. Soldiers and equipment can use cryptographic keys stored on the blockchain to prove their identity, thereby granting them specific access to these systems or areas. This reduces the risks of unauthorized access, espionage, and cyber-attacks, making military operations more secure and efficient.

In the military context, verifying the identity of individuals is of paramount significance. Blockchain technology can serve as a secure and unalterable database for storing digital identities, guaranteeing that every soldier,

officer, or civilian staff member possesses an tamperproof record. This enables the military to maintain a precise database of its personnel, their qualification, security clearances, and access permissions. The decentralization nature of blockchain minimizes risks of data breaches and unauthorized access, safeguarding sensitive information [15, 16].

Scenario: A real world example of identity management and authentication using blockchain technology in military applications can be found in the management access control for secure military facilities and the authentication of military personnel. In this scenarios, a blockchain system is employed to ensure that only authorized individuals with necessary security clearances can access classified areas or resources.

Let's consider a military base where soldiers, officer and civilian personnel need access to various sections, each with different security levels. In the traditional approach, access might be managed through physical key cards or authentication codes, which can be vulnerable to theft, duplication, or unauthorized use.

With Blockchain technology, each military personnel identity and access permissions are securely stored on a decentralized ledger. When an individual needs to access a specific area, they use their unique cryptographic key, which is tied to their identity on the blockchain. The blockchain then verifies the person's identity and their authorization to access the requested area by checking the smart contracts governing access control. Figure 5.5 illustrates the identity management and authentication process using blockchain

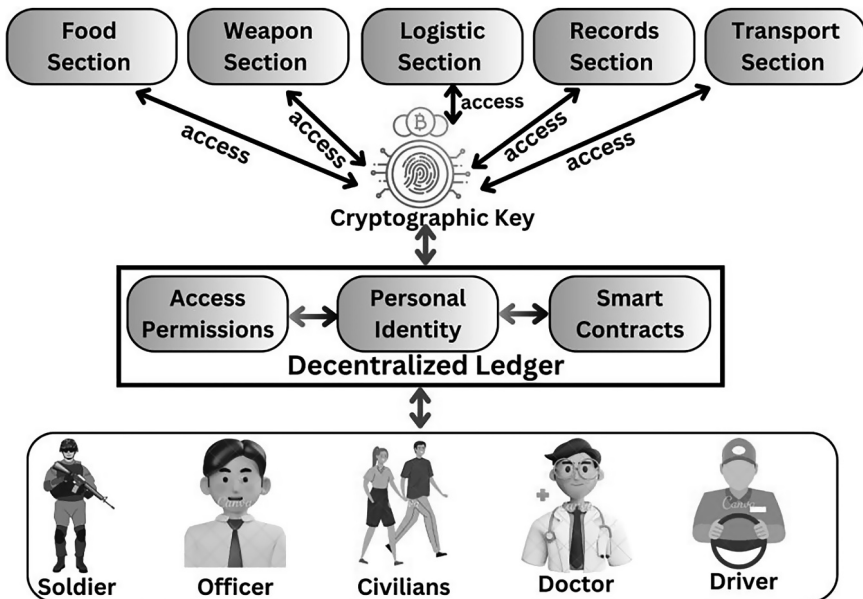


Figure 5.5 Identity management and authentication process using blockchain.

This real world example demonstrates how blockchain technology can significantly improve the security and efficiency, helping to safeguard classified information and resources while enabling rapid and secure access for authorized personnel.

5.3.3 Smart contracts in military operations

Smart contracts, a core components of blockchain technology, offer significant potential for enhancing military operations. These self-executing code based agreements can automate and streamline various processes in the military domain, resulting in improved efficiency, security and accountability. One of the key areas where smart contracts make a substantial impact is in the procurement and supply chain management of military equipment and resources. These contracts could be automated to activate routinely when specified conditions are met, such as ordering spare parts when inventory falls below certain level or scheduling maintenance for military vehicles after certain number of hours in operation. This automation reduces the administrative burden, minimizes human errors, and ensures that critical equipment is well maintained and readily available when needed.

Furthermore, smart contracts can revolutionize the payment and financial transactions within the military ecosystem. They enable the creation of trust less, transparent, and highly secured payment systems

Smart contracts have the potential to revolutionize military operations by automating process, enhancing financial transparency, securing data sharing, and improving accountability. Their adoption in the military domain can lead to more efficient and secure operations, ultimately contributing to the readiness and effectiveness of armed forces. As blockchain technology continuous for evolving, it is expected that smart contracts becoming an integral part of modern military practices [17].

Scenario: A real-world example of the use of smart contracts in military operations can be found in the management of logistics and supply chain for military base. Consider a military base that needs to ensure a continuous supply of critical resources such as fuel, ammunition, food supplies and spare parts. Smart contracts can be used to automate and secure the procurement and delivery processes, making the logistics more efficient and transparent. In this scenarios, military base enters contracts with various suppliers, specifying the terms and conditions for the procurement of essential resources. Smart contracts are created on a blockchain, and these contracts are automated for executing specified activities when predefined circumstances met. For instance, when an inventory of ammunition falls below a certain level, a smart contract can automatically trigger an order for replenishment. The contract can specify the quantity, quality, and delivery timeline ensuring that the base remains adequately supplied without manual interventions. Once the order is placed, the blockchain records the

transactions, and automatically initiates the payment to the suppliers upon successful delivery and verification.

The smart contract can also facilitate the tracking and monitoring of the entire supply chain process. For example, they can use Internet of Things (IoT) devices and sensors to record the location, condition and handling of sensitive military equipment during transportation. Any deviations from the agreed upon conditions, such as unauthorized access to a shipment or damage to supplies, can trigger automatic alerts and actions, including notifications to relevant personnel or even rerouting the shipments.

The use of smart contracts in military logistics not only streamline the procurement process, but also enhance the transparency and security. It reduces the risk of human errors, ensures timely resupply, and provides an immutable record of all transactions and actions taken throughout the supply chain [18–20]. Additionally, it can improve the cost efficiency and accountability in the management of military resources, allowing military personnel to focus on more strategic and critical tasks. Figure 5.6 illustrates the smart contract in military operations using blockchain technology.

This real world example demonstrates how smart contracts in military operations, powered by blockchain technology, can optimize and secure the logistics and supply chain processes, ultimately contributing to more efficient and effective military readiness

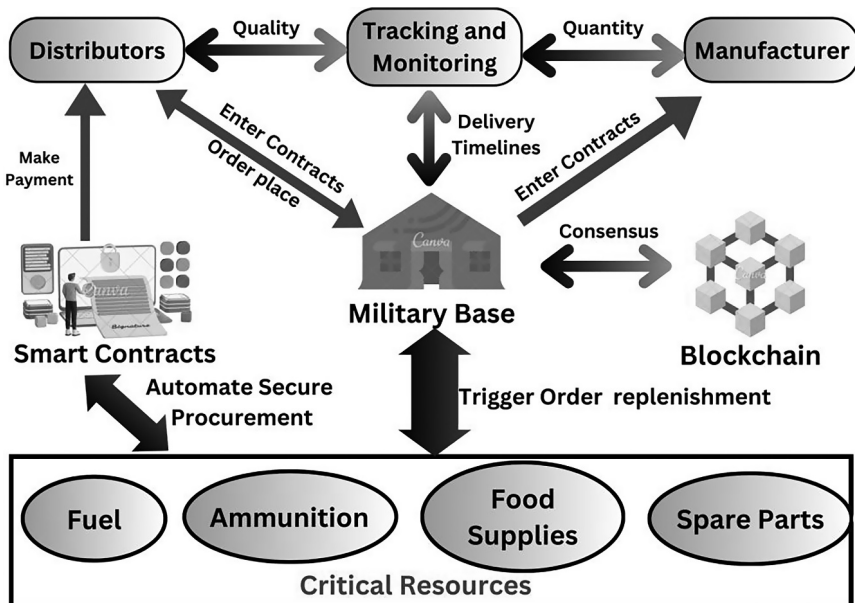


Figure 5.6 Smart contracts in military operations using blockchain technology.

5.4 OPPORTUNITIES FOR THE MILITARY APPLICATIONS

Blockchain technology is yet in an initial stage of implementation, but blockchain technology has the probable for revolutionizing like military operates. It is emerged as a ground-breaking innovation with transformative potential, and the military section is no exception. The decentralized and secured nature of blockchain presents unique opportunities for enhancing military applications ranging from logistics to cyber security [21, 22]. Some of these are listed as follows

- **Secure Communication:** Blockchain can be employed to create a secure and tamper resistant communication channel for military personnel. This ensures that sensitive information, orders and intelligence remain confidential and unaltered.
- **Supply Chain Management:** Utilizing blockchain in military supply chains enhances transparency and accountability. It enables real-time tracking of flow of resources, ammunition, and equipment reducing the risk of theft, fraud, and unauthorized alterations.
- **Smart Contracts for Logistics:** Smart contracts, self-execution contracts by means of the agreement terms written to code the directly, can streamline logistic operations. These contracts can mechanically activate and enforce agreements, viz deliveries or payments when pre-defined conditions are met.
- **Identity Management:** Blockchain can improve the identity management system within the military. Soldier personal data, certifications and access permissions can be securely stored on a blockchain, decreases the hazard of identity stealing or illegal access.
- **Cybersecurity Enhancements:** Implementing blockchain in military networks can enhance cybersecurity by providing a decentralized and immutable ledger. This makes it more challenging for malicious actors to compromise the integrity of the system, protecting critical infrastructure from cyber threats. For example, blockchain based encryption algorithms can be used to protect military data from cyberattacks
- **Swarm Intelligence:** Blockchain can be used to develop new and innovative ways to control and coordinate swarms of autonomous vehicles and weapons.
- **Battle field operation management:** Blockchain can be used to develop new and innovative ways to manage battlefield operations. For example, blockchain based command and control systems could be used to coordinate the actions of different military units in real time
- **Decentralized Command and Control:** Blockchain can support decentralized command and control structures, ensuring that
- **Immutable Records for Accountability:** The immutability of blockchain records can be leveraged for accountability purposes. Military operations and actions can be recorded on a blockchain, creating

an indisputable and transparent record of events for a review or an investigation.

- **Cross Border Operations and Transactions:** For multinational military operations involving different currencies and jurisdictions, blockchain can facilitate seamless and transparent financial transactions, removing the necessity for intermediaries and decreasing the risks of errors or fraud.
- **Intellectual Property (IP) Protection:** Blockchain is used for protecting the IP associated with military technologies. This ensures that sensitive information and innovations are securely recorded and controlled, preventing unauthorized use or replication.
- **Interoperability among Allies:** Blockchain can enhance the interoperability among allied forces by providing a standardized and secure platform for information exchange. This fosters collaborations and coordination in joint military operations [23].

5.5 CHALLENGES AND RISKS

While the integration of blockchain technologies into military application offers numerous advantages, it is not without its challenges and risks. Understanding and mitigating these issues are crucial for the successful implementation of blockchain in defense operations. Some of challenges and risks are included below

5.5.1 Technical challenges

Implementing blockchain in defense operations presents several technical challenges that needs to be addressed to ensure its successful integration into highly sensitive and complex military environments. Figure 5.7 illustrates the essential technical challenges of integrating blockchain technology into military applications. Some key challenges include

- **Security Concerns:** Blockchain relies on cryptographic algorithms for securing transactions, but the defense sector demands even higher level of security. Ensuring robust encryption methods and protection against quantum computing threats becomes imperative to safeguard sensitive military information
- **Scalability:** Defense operations involve massive amount of data and transactions. Scaling blockchain to handle large volumes of transactions, while maintaining performance and efficiency, is a significant challenge. The technology must be able to accommodate the scale and complexity of military operations.
- **Interoperability:** Military systems and technologies are diverse and may not be inherently compatible. Achieving interoperability between

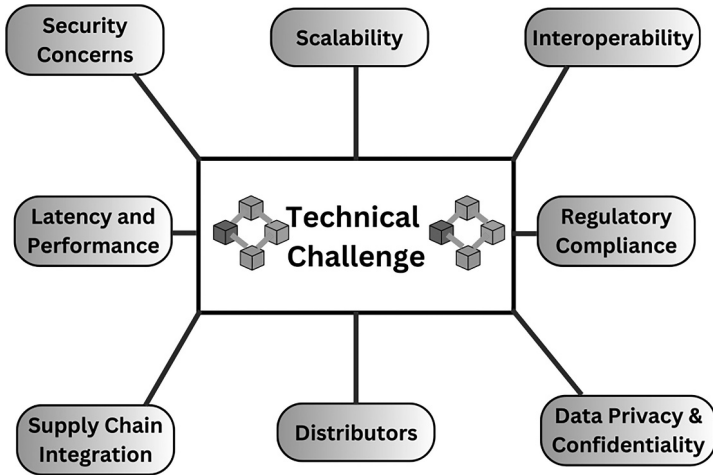


Figure 5.7 Technical challenges of integrating blockchain technology into military application.

existing defense systems and blockchain technology is crucial for seamless integration. Standardization efforts must be undertaken to ensure different components can communicate effectively.

- **Latency and Performance:** Defense operation require real-time decision making, and any latency in blockchain transactions can be detrimental. Improving the performance of blockchain networks to meet the stringent time requirements of military actions is a challenge that needs to be addressed particularly in scenarios where split-second decisions are critical.
- **Regulatory Compliance:** The defense sector is subject to strict regulations and compliance standards. Adapting blockchain to meet these regulatory requirements, including the data protection and adherence to international laws, is a complex task that requires careful consideration.
- **Supply chain Integration:** Defense operations involve intricate and globally distributed supply chains. Integrating blockchain into these supply chains requires overcoming challenges related to tracking and verifying the provenance of goods, ensuring transparency and preventing counterfeit or tampered products from entering systems.
- **Data privacy and Confidentiality:** Blockchain transparent and decentralized nature can be at odds with the need for classified and confidential information in defense. Striking a balance between the transparency and maintaining the privacy and confidentiality of sensitive military data poses a significant technical challenge

- **Adoption and Training:** Introducing blockchain technology requires a shift in mindset and expertise within the defense workforce. Providing adequate training and fostering a culture of understanding and trust in blockchain technology is essential for successful adoption [24, 25].

Addressing these technical challenges is crucial for harnessing the full potential of blockchain in defense operations, offering increased efficiency, transparency, and security in military processes.

5.5.2 Organizational challenges

The integration of blockchain into defense operations faces not only technical obstacles but also organizational challenges that requires careful navigation. Figure 5.8 depicts the organization challenges faced in defense operation. These challenges include

- **Cultural Resistance:** The defense sector often operates within a traditional and hierarchical organization culture. The introduction of blockchain may face resistance from personnel unfamiliar with decentralized technologies. Overcoming these resistance requires effective communications and a comprehensive change management strategy.
- **Coordination among stakeholders:** Defense operation involve multiple stakeholders, both within and outside the organization. Achieving consensus and coordination among these diverse entities to adopt



Figure 5.8 Organizational challenges faced in defense operation.

and implement blockchain solutions can be a formidable organizational challenge. Cooperation is essential for the success of blockchain activities

- **Legacy System Integration:** Defense organizations typically have extensive legacy systems in place. Integrating blockchain with these existing systems pose a challenge, as it requires a careful transition strategy to ensure minimal disruption to ongoing operations while maximizing the benefits of blockchain technology
- **Leadership support and Understanding:** Successful blockchain implementation in defense operation requires strong leadership support and a thorough understanding of the technology. Leaders will be willing to invest in the necessary resources and drive the cultural shift needed for blockchain adoption across the organization.
- **Governance and Regulation:** Establishing governance structure and navigating regulatory frameworks for blockchain in defense sector can be complex. Clear policies and guidelines need to be developed to govern the use of blockchain, ensuring compliance with existing regulation and addressing political legal challenges.
- **Skill Gaps and Training:** Implementing blockchain technology necessitates workforce with the relevant skills. The defense sector may face challenges in recruiting or upskilling personal with expertise in blockchain development, maintenance, and security. Comprehensive training programs are essential for ensuring that the workforce is adequately prepared.
- **Data Standardization:** The diverse nature of data in defense operations, often structured differently across various departments and units, poses a challenge for the blockchain implementation. Standardizing data formats and structures is crucial to ensure seamless integration and effective information exchange
- **Cost Implications:** While blockchain offers long term benefits, the initial investment and operational costs associated with its implementation can be significant. Convincing decision makers about the cost effectiveness and long term advantages of blockchain solutions requires careful financial planning and communication [26, 27].

Addressing these organizational challenges is vital for successful adoption of blockchain in defense operations. A strategic and collaborative approach, coupled with effective leadership and commitment to change, is essential to overcome these hurdles and unlock the potential benefits of blockchain technology in the defense sector.

5.5.3 Operational challenges

The implementation of blockchain in defense operation is accompanied by a set of operational challenges that must be carefully managed to ensure a

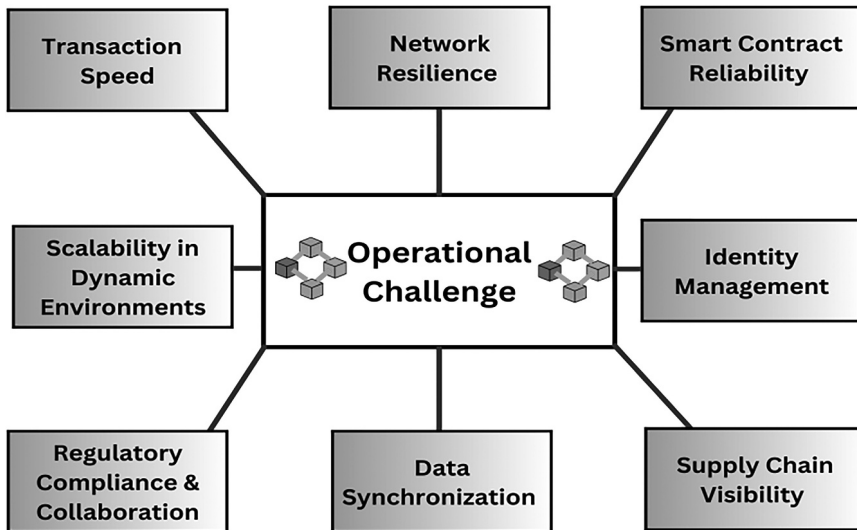


Figure 5.9 Operational challenges of integrating blockchain in defense operations.

smooth integration. Figure 5.9 depicts the operational challenges of integrating blockchain in defense operations. These challenges include

- **Transaction Speed:** The inherent design of blockchain, particularly in public networks, can result in slower transaction speeds compared to traditional centralized systems. In defense operations where real-time decision making is critical, addressing latency challenges is essential to maintain operational effectiveness.
- **Network Resilience:** Military operations often occurs in diverse and challenging environments, including areas with limited network connectivity. Ensuring the resilience of blockchain networks in such conditions is crucial to prevent disruptions and maintain continuous access to critical information
- **Smart Contract Reliability:** Smart Contracts, self-execution agreements coded on blockchain, plays a major role in making automated process. However, ensuring reliability and smart contract security is a challenge, as vulnerabilities, or coding errors could have a severe operational consequences.
- **Integration with Existing Systems:** Defense operation rely on a myriad of existing systems, each with unique specification and requirements. Integrating blockchain seamlessly with these systems poses a challenges, necessitating careful planning and execution to avoid disruptions to the ongoing operations.
- **Identity Management:** Establishing and managing digital identities securely is crucial in defense operations. Blockchain can provide a

solution, by ensuring the integrity and security of identity information, especially in dynamic military environments, requires robust operational protocols.

- **Supply Chain Visibility:** Achieving end to end visibility in the military supply chain is a complex operational challenge. Blockchain can enhance transparency, but ensuring that all stakeholders, including suppliers and logistic partners, consistently and accurately record transactions poses a logistical challenge.
- **Data Synchronization:** Maintaining consistent and synchronized data across decentralized blockchain nodes is vital for accurate and reliable information. Operational challenges arise in ensuring that all nodes are updated simultaneously, especially in scenarios where rapid, synchronized decision making is critical.
- **Regulatory Compliance and International Collaboration:** Defense operations often involve collaboration with international partners, each subject to their own set of regulations. Ensuring that blockchain implementations comply with diverse regulatory frameworks while prompting interoperability is an operational challenge that requires careful negotiations.
- **Scalability in Dynamic Environments:** Military operations are dynamic, with requirements changing rapidly. Ensuring that blockchain networks can scale up or down efficiently to accommodate varying operational demands is a challenge that requires a flexible and adaptive approach [28].

Addressing these operational challenges requires a combination of technological innovation, operational planning, and collaboration among stakeholders. Successful blockchain integration in defense operations hinges on the ability to adapt the technology to the unique demands of military environments while maintaining operational efficiency and security.

5.5.4 Additional risks

Beyond technical, organizational, and operational challenges the implementation of blockchain in defense operations also brings forth additional risks that demand careful considerations

- **Cybersecurity Threats:** Despite the fact, blockchain is renowned for its security characteristics, it is not invulnerable to cyber threats. The quantum computing emergence, sophisticated hacking techniques, or vulnerabilities in specific blockchain implementations could pose significant risks to the confidentiality and integrity of defense-related data.
- **Insider Threats and Malicious Actors:** Blockchain transparency can be a double-edged sword. Insider threats, where individuals with the

organization exploit their access to manipulate or compromise blockchain data, as well as external malicious actors attempting to disrupt or exploit the system, pose heightened risks in defense operations

- **Regulatory and Compliance Risks:** The defense sector is subjected to strict regulatory frameworks, and the blockchain implementation may inadvertently violate certain regulations. Failure to navigate these legal landscapes accurately can result in regulatory penalties, legal challenges, and damage to the reputation of the defense organization
- **Lack of Standardization:** The absence of industry wide standards for blockchain in defense operations may lead to interoperability issues, making it challenging to collaborate seamlessly with allies or integrate blockchain systems across different defense entities. This standardization lack could hamper the extensive implementation and effectiveness of the technology.
- **Dependency on Third Party Providers:** Many blockchain solutions rely on third party service providers for infrastructure, maintenance, or specific functionalities. Depending on external entities introduces dependencies and vulnerabilities, and disruptions or security breaches at third party level could impact the reliability and security of the defense blockchain network
- **Data Privacy Concerns:** Blockchain decentralization nature raises concerns about the exposure of sensitive defense data to unintended parties. Striking the right balance between transparency and data privacy, especially in military operations involving classified information, is crucial to prevent unauthorized access or leaks
- **Adaptation Challenges:** The defense sector's traditional structure and processes may face resistance in adapting to the decentralized and transparent nature of blockchain. Overcoming resistance and fostering a culture of innovation are critical to ensure that personnel effectively embrace and use blockchain technology.
- **Supply Chain Vulnerabilities:** While blockchain can enhance the transparency and traceability of the military supply chain, it also introduces new vulnerabilities. A single point of failure or compromise at any stage of blockchain enabled supply chain could have a cascading effects on the operational capabilities
- **Environmental Impact:** Certain blockchain implementations, especially those relying on energy intensive consensus mechanism like proof of work, can have environmental implications. The energy consumption required for mining or validating the transactions may conflict with sustainable goals, and the defense organizations need to weigh the environmental impact of their blockchain choices [29].

Mitigating these additional risks requires a comprehensive risk management strategy that involves ongoing monitoring, regular assessments, and adaptability to emerging threats. As the defense sector explores blockchain

solutions, a proactive approach to risk mitigation is essential to ensure the technology successful and secure integration into military operations

5.6 FUTURE TRENDS AND DEVELOPMENTS

Future trends and developments across various domains are shaped by ongoing technological advancements, societal changes and global challenges. Some of these are listed below

5.6.1 Emerging technologies and their impact on military blockchain

Emerging technologies are playing a transformative role in shaping the landscape of military operations, and one significant aspect of this evolution is in the integration of blockchain technology. Blockchain, with its decentralized and secure nature, is finding applications that enhance various aspects of military operations. Several emerging technologies are synergizing with blockchain to create a more robust and efficient military ecosystem.

- **Artificial Intelligence (AI):** The integration of AI with military blockchain is amplifying the capabilities of autonomous systems and decision making process. AI algorithms can analyze vast amount of data stored on the blockchain, enabling quicker and more informed responses to dynamic situations. Additionally, AI can improve blockchain network security by detecting and qualifying potential threats in real-time, contributing to a more resilient military infrastructure.
- **Internet of Things (IOT):** The combination of blockchain and IOT in military applications is revolutionizing asset management, logistic and supply chain operations. IOT devices can generate real-time data that is securely documented on the blockchain, confirming the integrity and traceability of critical military assets. This incorporation enhances situational awareness, reduces the risk of tampering, and optimizes the overall efficiency of military logistics.
- **Quantum Computing:** As quantum computing capabilities advance, there is a growing need to address the potential vulnerabilities it poses to traditional cryptographic methods. In the context of the military blockchain, the integration of quantum resistant cryptographic algorithms becomes crucial for maintaining the integrity and security of sensitive data. Blockchain platforms that incorporate quantum resistant cryptography are poised to withstand the evolving threat landscape posed by quantum computing.
- **Edge Computing:** Edge computing is decentralized data processing and storage bringing computation closer to the data source. In military

blockchain applications, edge computing can enhance the speed of transaction processing and reduce latency. This is particularly advantageous in scenarios where real time data analysis is imperative, such as tactical decision making on the battle field.

- **5G Technology:** The deployment of 5G networks is enhancing communication capabilities in military operations. In conjunction with blockchain, 5G can provide a necessary bandwidth and low latency connectivity for secure and rapid data transfer. This synergy facilitates the implementation of blockchain based solution in scenarios while timely and reliable communication is paramount such as command and control systems [30].

While the integration of emerging technology with military blockchain presents opportunities for increased efficiency and security. It also introduces new challenges. The military must navigate the complexities of interoperability, cybersecurity, and ethical considerations to harness the full potential of these synergies. As the technological landscape continues to evolve, a holistic and adaptive approach to integrate emerging technologies with military blockchain will be essential to maintain a strategic edge and ensure the success of military operation in the future.

5.6.2 Potential advancements in blockchain for military use

In the recent years there has been a growing interest in exploring the use of blockchain in military applications. The decentralized and secure nature of blockchain offers several potential advancements that could enhance the efficiency, security, and transparency of military operations. As the research and development in this field progress, it is essential for military organizations to carefully assess the risks and benefits of implementing blockchain solutions and adapt these technologies to meet the unique demands of defense and national security [31].

- Secure communication and data sharing
- Supply chain management and logistics
- Identity Management and Access Control
- Cyber security and threat Intelligence
- Smart Contracts for automated processes
- Decentralized Command and Control
- Interoperability and coalition Operations

A thoughtful and phased approach to the adoption of blockchain technology can ensure that the military reaps the full benefits while mitigating risks

5.6.3 International collaboration and standardization efforts

International collaboration and standardization efforts play a crucial role in fostering cooperation, interoperability, and the harmonious development of technologies across borders. In an increasingly interconnected world, where innovation knows no geographical boundaries, these initiatives are essential for addressing global challenges, ensuring compatibility between systems and promoting the responsible use of emerging technologies.

- **Technology Interoperability:** In the realm of technology, international collaboration is vital for achieving interoperability between different systems and devices. Standardization efforts work to create a common technical standards, protocols, and specifications that enable seamless communication and interaction. This is particularly important in areas such as telecommunications, information technology and emerging technologies like blockchain and IOT where devices from various manufacturer need to work together efficiently.
- **Trade and Commerce:** Standardization contributes to the facilitation of international trade by establishing common norms and quality standards. Harmonized standards ensure that products and services meet specific criteria, enhancing consumer safety, and promoting fair competition. Organization such as International Organization for Standardization (ISO) provide platform for nations to collaborate on developing and adopting standards that benefit industries and consumers globally
- **Climate Change and Sustainable Development:** Addressing global challenges such as climate change requires coordinated international efforts. Standardization in areas like environmental management systems, renewable energy technologies, and sustainable practices enable nations to work together toward common goals. Frameworks like Paris agreements on climate change exemplify how countries can collaborate to set international standards for environmental protection.
- **Healthcare and Pharmaceuticals:** The covid-19 pandemic highlighted the importance of international collaboration in healthcare. Standardization efforts in the development and distribution of vaccines, diagnostic test and treatment protocols are crucial for a coordinated global response to health crisis. Organization like World Health Organization (WHO) play a pivotal role in facilitating such collaboration and setting international health standards
- **Emerging Technologies:** As new and disruptive technologies emerge, international collaboration become even more critical. Standards for technologies like Artificial Intelligence (AI), Blockchain and Quantum computing are essential to ensure ethical practices, data privacy and the responsible development and deployment of these innovations. Collaborative efforts can prevent fragmentation in global

technology landscape and promote a shared understanding of ethical considerations.

- **Space Explorations:** In the realm of space explorations, international collaborations is evident through initiatives like the Indian Space Station (ISS). Standardization in space missions, satellite communication protocols, and space debris management fosters cooperation among nations with varying levels of space capabilities, contributing to the peaceful and responsible use of outer space [32].

5.7 SECURITY CONSIDERATIONS

Military applications using blockchain technology introduce unique security considerations due to the sensitive nature of military operations and the critical importance of securing data, communication and logistics. Some of the key security considerations for military applications are listed below:

- Data Integrity and Immutability
- Secure communications
- Supply chain Security
- Automation through smart contracts
- Resilience against cyberattacks
- Zero knowledge proofs for privacy
- Selection of Consensus Mechanism
- Redundancy and backups
- Incident Response and Auditing
- Restricted Access Controls
- Physical Security

In military application, the security of blockchain is paramount and careful consideration of these factors can contribute to the successful implementation of blockchain solutions, while safeguarding sensitive military operation and information. The collaboration with cybersecurity experts, adherence to military standards, and ongoing security assessments are critical elements of maintaining a robust security posture in military blockchain applications [33].

5.8 CASE STUDY

The evolution of blockchain as a disruptive technology has not only opened new avenues for exploration by scholars and industry professionals but also presented opportunities to delve into intricacies of this technology, offering solutions that contribute to the society. This section will shed light on case study that explain how blockchain technology can be applied to military sectors

5.8.1 A framework for managing maintenance of military equipment, utilizing the capabilities of blockchain technology

The increasing potential of data driven decision making are fostering a greater understanding and significance of data management. Over the past few decades, various efforts have been undertaken to address the challenges in data management, particularly focusing on security. Numerous data security technologies, including data encryption, data masking, and tokenization were suggested and put into practice. The defense sector stands out as one such domain, with the capacity to undergo transformation through the expanding utilization of data driven techniques in pivotal decision making process. Hence, similar to other industry sector, the defense industry must be proficiently handle critical security data. Nevertheless, given the magnitude of data produced by the defense organizations, the complexities in data management, particularly concerning data security, are manifold compared to other industrialized domains. Blockchain is one such technology that offers a significant solution to address various issues related to the data management in an innovative way.

- **Data management maintenance from military perspectives:** War readiness and sustainability are fundamental pillars of a nation's military capabilities. To enhance these metrics, military authorities consistently seek information on the current state of equipment, assess mission reliability, formulate efficient maintenance strategy, and establish effective war reserve policies. Methodically managed maintenance data is crucial for all data driven decision building processes. Effectiveness of statistical data is dependent on both its quality and quantity. Any loss of the components within this maintenance data diminishes its effectiveness and may result in inaccurate estimations. Therefore, data intended for decision making within the military organization must exhibit high integrity, ensuring the utmost accuracy and consistency. Likewise, in the formulation of maintenance strategies, it is essential to monitor the real age of each equipment and its components. A majority of equipment operates in various locations within diverse operating environments, determining effective age requires estimation based on the operational data from different landscapes and upkeep records. The maintenance of defense equipment during a war like emergencies poses a formidable challenge. Some of them are as follow
 - a. The utilization of hand written upkeep logs is fairly common. However, it is crucial to electronically record maintenance data in proper format, ensuring accuracy and comprehensive details.
 - b. In conventional maintenance management system, the military critical data must be stored in a highly secure environment such that data is upheld with desired accuracy and granularity.

- c. The third party service providers engaged in the inventory management of defense equipment necessitate access to usage data. This entails establishment of controlled, monitored and genuine channel for engagement of external organizations with limited access granted to ensure security.

Considering above scenarios of maintaining functionality in military organization, a blockchain enabled framework is required to manage the military equipment. Figure 5.10 depicts the scenario of military equipment maintenance using blockchain technology

In context of military organizations, considering the implementation of blockchain for managing maintenance data, the federated blockchain is preferred. Unlike private blockchain, it is not central and only authorized nodes can take part, making it a permissioned decentralized blockchain. All nodes within this blockchain are recognized as clients. Additionally, regardless of the network size, all transactions are recorded in milliseconds. To attain agreement among all the nodes on transaction, consensus is essential. Achieving consensus in a scattered environment poses a challenge prior to the implementation of blockchain. To address this challenge, blockchain employs various consensus algorithms. For example, the bitcoin crypto currency utilized Proof of Work (POW) as a consensus algorithm. Numerous other consensus algorithm exists, including proof of burn, proof of stake, proof of activity, proof of capacity etc. The consensus algorithm used in

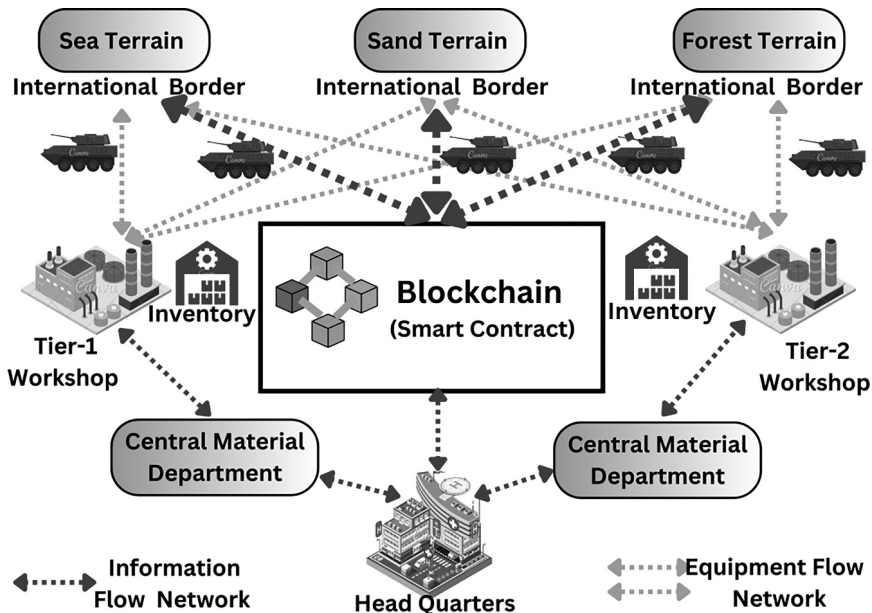


Figure 5.10 Military equipment maintenance scenarios using blockchain.

permissioned blockchain is proof of elapsed time (POET). These consensus algorithm enable all the blockchain nodes to come to agreement on the current state of the ledger. When any node within the blockchain seeks to append information, it is crucial for all other nodes to reach consensus on this data before it is eternally added to the blockchain. The consensus on including each transaction in the blockchain enhances transparency and consequently fosters trust in the network. The selection of consensus algorithm should align with the type of blockchain and the expectations of organizations regarding the speed, efficiency and confidence in blockchain operations

- **Policy of Access control:** To control the access to the envisioned blockchain based framework from a security stand point, an access controller policy is devised. This policy governs the permissions for engaging with the distributed ledger (DL) and validating transactions by respective nodes. Furthermore, the policy assigns an authentication level to each node, determining the node classifications as partial node or full node. A full node has access to the entire ledger, while a partial node possesses only a specific subset of distributed ledger (DL) [34].

5.9 CONCLUSION

The exploration of blockchain in military applications has unveiled a realm of opportunities and challenges that demand careful consideration. The rapid integration of blockchain technology into military applications signifies a transformative era in defense operations. As we navigate through the opportunities and challenges presented by blockchain, its potential to revolutionize secure communication, data sharing, supply chain management, identity authentication and the implementation of smart contracts in military contexts becomes increasingly evident. This chapter has provided a foundational understanding of blockchain principles, emphasizing its core attributes that ensures security and trust in data management.

By exploring real world examples and case study, we have witnessed how blockchain addresses critical needs within military, showcasing its efficacy in enhancing operational efficiency and security. While the chapter underscores the promising opportunities that blockchain brings to military applications. It also acknowledges the challenges such as scalability issues and integration complexities that need to be addressed for successful adoption.

As we look ahead, the chapter assess emerging trends and developments, recognizing the evolving landscape of cutting edge technologies and their impact on the future of military blockchain applications. The journey through this multifaceted exploration serves as a guide for military stakeholders and technologist alike, fostering a deep understanding of the potential risks, and rewards that blockchain technology brings to the forefront of modern defense operations.

REFERENCES

- [1] Reyneke, Michael A., Barry E. Mullins, and Mark G. Reith. 2023. Review of LoRaWAN & the Helium Blockchain: A Study on Military IoT Deployment. In *18th International Conference on Cyber Warfare and Security, 2023*, edited by Richard L. Wilson and Major Brendan Curran, 18:327–337. Academic Conferences International Limited Curtis Farm, Kidmore End, Nr Reading, RG4 9AY, United Kingdom. <https://papers.academic-conferences.org/index.php/iccws/article/view/944>
- [2] Sharma, Gaurav, Deepak Kumar Sharma, and Adarsh Kumar. 2023. “Role of Cybersecurity and Blockchain in Battlefield of Things.” *Internet Technology Letters*, January. <https://doi.org/10.1002/itl2.406>
- [3] Mekawy, Ibrahim. 2023. “Blockchain-Powered Wireless Sensor Networks: Enhancing Security and Privacy in the IoT Era.” *Big Data and Computing Visions* 3 (2): 70–75. <https://doi.org/10.22105/bdcv.2022.371381.1084>
- [4] Zhu, Y., X. Zhang, Zh Y. Ju, and Ch Ch Wang. 2020. “A Study of Blockchain Technology Development and Military Application Prospects.” *Journal of Physics: Conference Series* 1507 (April): 052018. <https://doi.org/10.1088/1742-6596/1507/5/052018>
- [5] Calzada, Paul E., Md Sami, Kimia Zamiri Azar, Fahim Rahman, Farimah Farahmandi, and Mark Tehranipoor. 2023. “Heterogeneous Integration Supply Chain Integrity through Blockchain and CHSM.” *ACM Transactions on Design Automation of Electronic Systems* 29 (1): 1–25. <https://doi.org/10.1145/3625823>
- [6] “Overview of the Basic Principles of Blockchain | IEEE Conference Publication | IEEE Xplore.” n.d. ieeexplore.ieee.org. Accessed November 18, 2023. <https://ieeexplore.ieee.org/document/9653563>
- [7] Shoemaker, Phillip. 2023. “Key Components of a Blockchain Network.” *Identity*. April 24, 2023. <https://www.identity.com/key-components-of-a-blockchain-network/>
- [8] Sudeep, J., S. C. Girish, Ganapathi Karthik, K. Raghavendra, Hong Lin, and P. Subramanya. 2021. “Blockchain: A New Era of Technology.” *Convergence of Internet of Things and Blockchain Technologies*, September, 3–22. https://doi.org/10.1007/978-3-030-76216-2_1
- [9] Zhang, P., and M. Zhou. 2020. “Security and Trust in Blockchains: Architecture, Key Technologies, and Open Issues.” *IEEE Transactions on Computational Social Systems* 7 (3): 790–801. <https://doi.org/10.1109/TCSS.2020.2990103>
- [10] Zhang, Rui, Rui Xue, and Ling Liu. 2019. “Security and Privacy on Blockchain.” *ACM Computing Surveys* 52 (3): 1–34. <https://doi.org/10.1145/3316481>
- [11] Subbiah, Kannan, Benno Ferrarini, Julie Maupin, Marthe Hinojales, Rahul Guhathakurta, Sanatan Kulshrestha, and Danika Wright. 2018. “The Age of Blockchain: A Collection of Articles.” Google Books. IndraStra Global. https://books.google.co.in/books?hl=en&lr=&id=6_NRDwAAQBAJ&oi=fnd&pg=PA21&dq=military+applications+of+blockchain&ots=ZG3wRDFXph&sig=or1eKkyGMkvdjC7ATjyMCfx3T2Q#v=onepage&q=military%20applications%20of%20blockchain&f=false
- [12] Manogaran, Gunasekaran, Mamoun Alazab, P. Mohamed Shakeel, and Ching-Hsien Hsu. 2021. “Blockchain Assisted Secure Data Sharing Model for Internet of Things Based Smart Industries.” *IEEE Transactions on Reliability*, 1–11. <https://doi.org/10.1109/tr.2020.3047833>

- [13] Mohril, Ram S., Bhupendra S. Solanki, Bhupesh Kumar Lad, and Makarand S. Kulkarni. 2021. "Blockchain Enabled Maintenance Management Framework for Military Equipment." *IEEE Transactions on Engineering Management*, 1–14. <https://doi.org/10.1109/tem.2021.3099437>
- [14] Kumar, Ajay, Kumar Abhishek, Muhammad Rukunuddin Ghalib, Pranav Nerurkar, Sunil Bhirud, Waleed Alnumay, S. Ananda Kumar, Pushpita Chatterjee, and Uttam Ghosh. 2020. "Securing Logistics System and Supply Chain Using Blockchain." *Applied Stochastic Models in Business and Industry* 37 (3): 413–428. <https://doi.org/10.1002/asmb.2592>
- [15] Nusantara, Hardjanto, Ruli Supriati, Nur Azizah, Nuke Puji Lestari Santoso, and Sabda Maulana. 2021. "Blockchain Based Authentication for Identity Management." *IEEE Xplore*. September 1, 2021. <https://doi.org/10.1109/CITSM52892.2021.9589001>
- [16] Fan, Pingzhi, Yazhen Liu, Jiyang Zhu, Xiongfei Fan, and Liping Wen. 2019. "Identity Management Security Authentication Based on Blockchain Technologies." *Int. J. Netw. Secur.* 21 (January): 912–917.
- [17] cassienstxl. 2023. "Blockchain for Military Logistics: What You Should Know." *NSTXL*. April 4, 2023. <https://nstxl.org/blockchain-for-military-logistics/>
- [18] Mcabee, Ashley, Murali Tummala, and John Mceachen. n.d. "Military Intelligence Applications for Blockchain Technology." Accessed November 18, 2023. <https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/b55707a4-8cd5-47b5-81d1-37b24fa142f6/content>
- [19] Bahiyah, Syarifah, Rahayu Norizam, Jusoh Rmn, Nur Diyana, Kamarudin Afiqah, and Mohammad Azahari. n.d. "Military blockchain for supply chain management." *Journal of Education and Social Sciences* 13 (1). https://www.jesoc.com/wp-content/uploads/2019/08/KC13_015.pdf
- [20] Lora, Francisco. 2020. "What Is Military Supply Chain Management? | SCM Globe." *www.scmglobe.com*. August 20, 2020. <https://www.scmglobe.com/what-is-military-supply-chain-management/>
- [21] Lilly, Bilyana, and Sale Lilly. 2021. "Weaponising Blockchain." *The RUSI Journal*, March, 1–11. <https://doi.org/10.1080/03071847.2021.1886871>
- [22] Wasim Ahmad, Raja, Haya Hasan, Ibrar Yaqoob, Khaled Salah, Raja Jayaraman, and Mohammed Omar. 2021. "Blockchain for Aerospace and Defense: Opportunities and Open Research Challenges." *Computers & Industrial Engineering* 151 (January): 106982. <https://doi.org/10.1016/j.cie.2020.106982>
- [23] Jadey, Sudeep, S. C. Girish, K. Raghavendra, G. Prasanna Kumar, H. R. Srinidhi, and K. M. Anilkumar. 2022. "Introduction to Cyber Security." *Methods, Implementation, and Application of Cyber Security Intelligence and Analytics*, June, 1–24. <https://doi.org/10.4018/978-1-6684-3991-3.ch001>
- [24] Habib, Gousia, Sparsh Sharma, Sara Ibrahim, Imtiaz Ahmad, Shaima Qureshi, and Malik Ishfaq. 2022. "Blockchain Technology: Benefits, Challenges, Applications, and Integration of Blockchain Technology with Cloud Computing." *Future Internet* 14 (11): 341. <https://doi.org/10.3390/fi14110341>
- [25] Eswaran, Sumathy. n.d. "Challenges in Blockchain Technology | Blockchain Technology Fundamentals." *Witspry Witscad*. <https://witscad.com/course/blockchain-fundamentals/chapter/challenges-in-bct>
- [26] Alam, Tanweer. 2022. "Blockchain-Based Internet of Things: Review, Current Trends, Applications, and Future Challenges." *Computers* 12 (1): 6. <https://doi.org/10.3390/computers12010006>

-
- [27] Uddin, Muen, Shitharth Selvarajan, Muath Obaidat, Shams Ul Arfeen, Alaa O. Khadidos, Adil O. Khadidos, and Maha Abdelhaq. 2023. "From Hype to Reality: Unveiling the Promises, Challenges and Opportunities of Blockchain in Supply Chain Systems." *Sustainability* 15 (16): 12193. <https://doi.org/10.3390/su151612193>
 - [28] Sharma, Kapil, and Deepakshi Jain. 2019. "Consensus Algorithms in Blockchain Technology: A Survey." *2019 10th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, July. <https://doi.org/10.1109/icccnt45670.2019.8944509>
 - [29] "Managing Risk in Blockchain Deployments." 2022. *Trail of Bits Blog*. June 24, 2022. <https://blog.trailofbits.com/2022/06/24/managing-risk-in-blockchain-deployments/>
 - [30] StartUs Insights. 2021. "Top 10 Military Technology Trends for 2022." *StartUs Insights*. November 5, 2021. <https://www.startus-insights.com/innovators-guide/top-10-military-technology-trends-2022/>
 - [31] Sudhan, Amool, and Manisha J. Nene. 2017. "Employability of Blockchain Technology in Defence Applications." *IEEE Xplore*. December 1, 2017. <https://doi.org/10.1109/ISS1.2017.8389247>
 - [32] "Digital Regulation Platform." n.d. *Digitalregulation.org*. <https://digitalregulation.org/cross-border-collaboration-in-the-digital-environment-2/>
 - [33] Cassetto, Orion. 2022. "21 Top Cyber Security Threats and How Threat Intelligence Can Help." *Exabeam*. June 1, 2022. <https://www.exabeam.com/information-security/cyber-security-threat/>
 - [34] Mohril, Ram S., Bhupendra S. Solanki, Bhupesh Kumar Lad, and Makarand S. Kulkarni. 2021. "Blockchain Enabled Maintenance Management Framework for Military Equipment." *IEEE Transactions on Engineering Management*, 1–14. <https://doi.org/10.1109/tem.2021.3099437>

Role of ubiquitous AI and blockchain security for military & healthcare

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6.1 INTRODUCTION

Military and civilian medical planners must prepare for prolonged field care in austere conditions, lacking facilities and infrastructure for current medical planning and evacuation support. In future conflicts and prolonged civilian disasters, limited air access, extended geographic supply lines, and high casualties may overwhelm available resources. Medical data is expanding rapidly, and storing sensitive data is challenging due to limited space and unauthorized access. Decentralized storage and blockchain technology are proposed as solutions to these challenges [1].

Blockchain's decentralized nature provides an innovative solution to the challenges of managing medical records and maintaining confidentiality of health-related information. By distributing healthcare data across a network of nodes, blockchain ensures data remains secure, tamper-resistant, and easily auditable. This introduction explores the pivotal role of blockchain security in military healthcare, enhancing data integrity, mitigating cyber threats, and improving the quality of healthcare services provided to military personnel [2].

The combination of Blockchain and Artificial Intelligence (AI) has the potential to create new and innovative medical systems in military healthcare. Blockchain technology offers a data format with built-in security features that guarantee transaction trust, such as decentralization, consensus, and cryptography. AI, with its capacity for data analysis, pattern recognition, and predictive modeling, holds immense promise in optimizing healthcare delivery and decision-making processes. In the military context, where rapid and accurate medical interventions are critical, the synergistic integration of AI and blockchain can lead to safer and more efficient medical systems (Figure 6.1) [3].

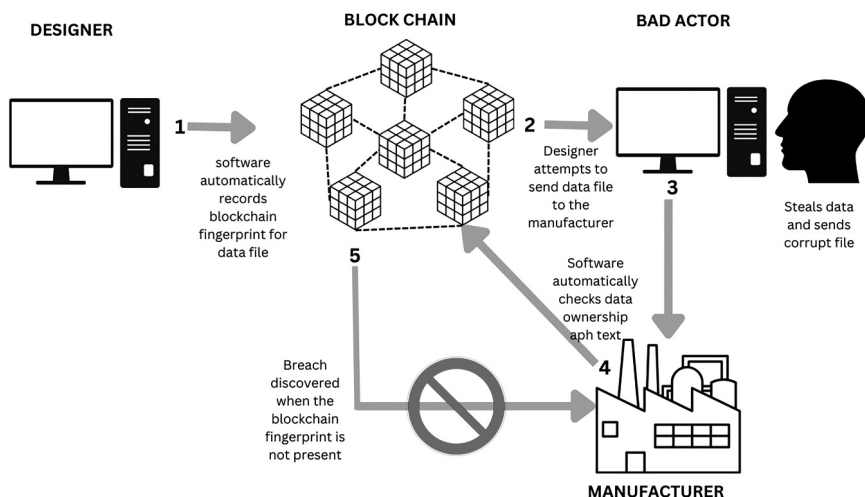


Figure 6.1 Basics of blockchain security.

6.2 BASIC BLOCKCHAIN SECURITY

Blockchain is a popular distributed, decentralized ledger consisting of blocks with unique hashes, each containing transaction information.

Blockchain technology creates a secure data structure based on cryptography, decentralization, and consensus principles. It is structured into blocks, each containing a transaction or bundle. Each block connects to previous blocks in a cryptographic chain, making them nearly impossible to tamper with.

Blockchain security is a crucial risk management technique aimed at protecting vital information from unauthorized access. It involves cybersecurity, authorized services, and ethical users to secure the transactions and the entire network in blockchain, making it a prime concern in the growing dependency on blockchain networks [3].

6.2.1 What makes security necessary for blockchain?

Blockchain is a decentralized, unchangeable ledger that employs cryptography to conceal information, making it harder for hackers to manipulate individual blocks. However, because blockchain networks are vulnerable to fraud and cyberattacks, loopholes enable malevolent users to carry out malicious activities including both outside and network-based attacks.

A public blockchain network offers numerous ways for attackers to disrupt consensus, including sybil attacks, where bogus participants are added to the network, gaining control over most participants [4].

6.2.2 Public blockchain security

Public blockchains, like Bitcoin, are decentralized, secure and accessible for anyone, making them secure than private or acquiescence blockchains. They can be slow and expensive, but they are arduous for malicious actors to attain a 51% attack on than private blockchains.

Public blockchains, such as Ethereum and Bitcoin, are open, permission-less networks that allow anybody to validate a transaction. Their codebase is open source, continuously reviewed by a community of developers for bugs and vulnerabilities [5].

6.2.3 Private blockchain security

Private Blockchains are distributed databases with permissions, allowing only authorized users to access data and complete transactions, unlike public Blockchain like Bitcoin, which allow anyone to join. Exclusive networks with restricted access, or private blockchains, are more centralized and may be more resilient to outside attacks. Because of their single point of failure, institutions must put robust security measures in place. They might not gain as much from decentralized security as public blockchains, but because they require less computing power, they are typically faster and more effective. However, they also pose a theoretical risk of network shutdown or manipulation (Figure 6.2) [5].

6.2.4 How is blockchain used for security?

An authorized participant inputs a transaction, which is authenticated by technology. This creates a block representing the transaction, which is sent to all network nodes. Authorized nodes validate and add the block to the blockchain, finalizing the transaction.

Blockchain is a computer network-managed decentralized database, providing increased transparency and security. It functions like a digital ledger,

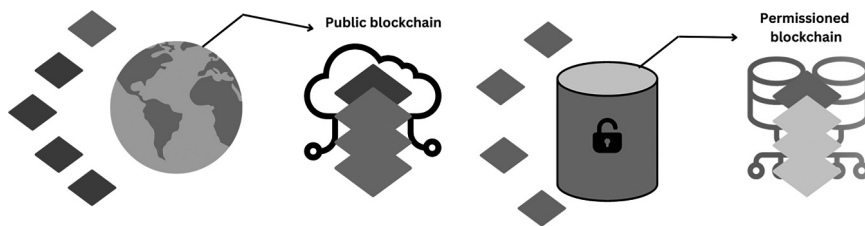


Figure 6.2 Public & private security in blockchain.

with no central authority managing transactions. Each new transaction is recorded on the Blockchain, which is verified by intricate mathematical algorithms that are shared by all parties. A permanent record of every transaction is created, as once confirmed, it cannot be altered or removed. This makes blockchains ideal for applications requiring enhanced security and transparency in areas like supply chain management and finance. As a promising new technology, blockchains are attracting a high number of applicants for security jobs and projects, with the potential to revolutionize industries and change our interactions with technology [4].

It is being increasingly used by enterprises, but security concerns must be addressed to protect data. Key management, data privacy, secure communication between nodes, smart contract auditing, and transaction endorsement are essential security measures. IAM ensures authorized users have access, key management protects sensitive information, secure communication prevents eavesdrop attacks, auditing smart contracts prevents vulnerabilities, and transaction endorsement increases security by needing several people to approve each transaction [4].

6.2.5 Blockchain security and intelligence solutions

Sybil attacks are a common threat in public blockchains, as attackers can disrupt consensus through various ways. As digital currencies gain traction, businesses, law enforcement agencies, financial institutions, and regulatory bodies need tools to keep an eye on transactions and fight cybercrime. Chainalysis offers security solutions for forensics, looking through blockchain data to find money, track down transactions, and make sure regulations are followed. Additionally, an incident response retainer helps in collaborating with law enforcement in case of hacks [6].

6.2.6 How secure are blockchains?

A distributed digital ledger system made up of nodes that verify and log transactions underpins blockchain technology. Every cryptocurrency transaction, such as sending or receiving, is recorded on a block and needs to be confirmed by consensus through the use of two well-liked techniques: Proof-of-Work and Proof-of-Stake. In Proof-of-Work, participants lock up tokens to run nodes, while in Proof-of-Stake, miners solve computationally demanding algorithms to validate transactions. Rewards are given to miners and stakers to encourage them to protect the network. A block is sealed when it is filled and connected to the one before it by a cryptographic code to create a chain. This technology, which powers well-known cryptocurrencies like ether and bitcoin, has enormous promise for the development of digital transactions in the future [6].

6.3 UBIQUITOUS COMPUTING

Ubiquitous computing is a concept in software engineering, hardware engineering, and computer science where computing can be present anytime and anywhere, using any device, location, and format. It supports various technologies such as the Internet, sophisticated middleware, operating systems, sensors, microprocessors, new I/O and user interfaces, computer networks, mobile protocols, positioning and location, and novel materials [6].

The pervasive computing integrates connectivity functionalities into all objects, automating routine tasks and requiring minimal human effort to complete and follow machine instructions [6].

6.3.1 AI in ubiquitous computing

Artificial intelligence (AI) is transforming various sectors like healthcare, finance, transportation, and entertainment. Ubiquitous computing, integrating computational capabilities into everyday objects, is a significant advancement. AI aids in seamless interaction between devices and users, enabling seamless learning and adaptation.

Ubiquitous computing, or pervasive computing, is the integration of computing capabilities into everyday objects, making them context-aware and enabling communication. Introduced by Mark Weiser in the late 1980s, this concept has been developed by researchers and engineers, with AI playing a crucial role in achieving this vision (Figure 6.3) [6].

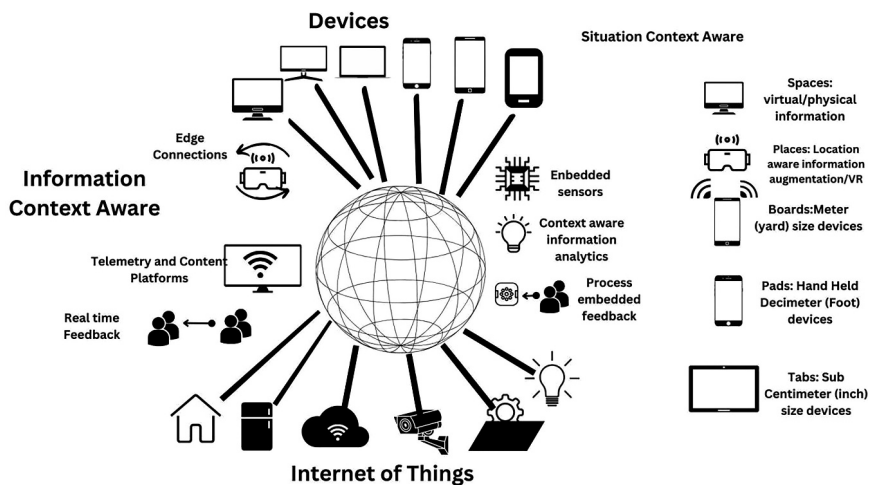


Figure 6.3 Ubiquitous computing architecture.

6.3.2 Computing challenges and concerns surrounding AI use in the context of defense

The Defense Innovation Board (2019) provides various definitions of AI in defense, including:

1. An artificial system is one that can learn from experience or carry out tasks without human supervision.
2. An artificial intelligence (AI) system built into computer hardware or software that can accomplish tasks involving human-like perception, thought, planning, learning, communication, or physical action.
3. AI systems, such as neural networks and cognitive architectures, that are created to behave or think like humans.
4. A collection of methods intended to approximate cognitive tasks, such as machine learning.
5. A rational AI system, such as an embodied robot or intelligent software agent.

AI classifications vary within civilian and defense contexts, with minimal difference in application and types. Gillath et al. (2021) provide a summative definition for general purposes [7].

6.4 AI'S APPLICATION IN WARFARE

The creation of computer systems that can carry out operations that call for human intelligence, such as speech recognition, visual perception, decision-making, and language translation, is known as artificial intelligence (AI). Artificial intelligence systems are becoming more and more useful in many aspects of life. Recent advances in AI include natural language processing (NLP), computer vision, decision making, and autonomous systems. NLP allows humans to communicate with machines using typical grammar and syntax, achieving new levels of accuracy and fluency. These developments also present opportunities for additional military applications of AI.

Strategic research and simulation are utilized in the transportation sector to enhance understanding of drone swarms, thereby enhancing cybersecurity and enhancing safety measures [7].

6.4.1 Blockchain technology in military healthcare

Along with emphasis on population health and preventive care, information sharing has been crucial in helping to enhance the quality of care as the health IT sector continues to develop. Increasing and increasing volumes of healthcare data are being produced by developments in health information technology, which can be used by tools for data sharing, processing, and analysis [8].

The health IT business is seeing significant growth, and with it comes a large number of providers and solutions. As a result, interoperability, standardization, data storage, access controls, and governance are issues that require ongoing attention. Interoperability and the safe exchange of healthcare data have been major industry buzzwords for blockchain technology, which is used to secure and share financial transactions in cryptocurrencies [8].

Blockchain technology may offer a standards-based method that eliminates vendor interdependencies and makes it possible for patients, medical professionals, and researchers to safely exchange electronic health data. Nationwide use cases are the main focus of blockchain research. Nonetheless, there hasn't been any public study on how blockchain can affect the Military Health System (MHS) [9].

Blockchain technology is transforming the military sector by offering safe, effective methods for exchanging and storing data. It has the ability to resolve significant issues with data security, personnel record management, supply tracking, and other aspects of military logistics.

When it comes to testing, monitoring, and verifying the manufacturing procedures of potential medications during their production and research and development (R&D), blockchain technology might be the most appropriate choice. In this sense, a long-term solution to preventing illicit drugs could be an automated drug management system (DDCS). Three major pharmaceutical companies, Sanofi, Pfizer, and Amgen, started working together on a pilot project to use a blockchain-based DDCS for the inspection and evaluation of experimental medications [9].

Military personnel may obtain current information fast and securely using blockchain technology, all without having to worry about accuracy or privacy concerns. This would facilitate effective resource management while safeguarding private data from unauthorized access [8].

Blockchain technology, as a secure ledger system, has the potential to collect and verify the accuracy of vital patient data as it travels from the field hospital to the base hospital, the rehabilitation facility, and the neighborhood pharmacy. More significantly, it might facilitate and enhance health IT interoperability between the several hospital systems (military and civilian) that need prompt access to patient data [4].

The commercial and civilian sectors, including organizations like the Centers for Disease Control and the Department of Health and Human Services (HHS), are already utilizing blockchain technology [8].

There are numerous possible uses for blockchain technology in the field of **military healthcare**. Enhancing security, transparency, and data management efficiency is one of its main benefits. Here are some applications for blockchain technology:

- **Unchangeable and secure medical records** for military personnel can be achieved with blockchain technology. This guarantees that private health information is shielded from manipulation or illegal access.

- **Interoperability:** A number of providers frequently provide healthcare to military personnel. Blockchain technology can help healthcare systems communicate with one another more easily and securely, enabling the secure exchange of medical data between military medical facilities [10].
- **Supply Chain Management:** Pharmaceutical and medical equipment supply chains can be tracked and managed with blockchain technology. By doing this, the supply chain's integrity is maintained, preventing the sale of fake medications and guaranteeing that the military has access to real, secure medical supplies [3].
- **Research and Clinical Trials:** Blockchain can make managing and conducting clinical trials more efficient. It offers an open and impenetrable record of trial data, facilitating results verification and guaranteeing the accuracy of study conclusions [1].
- **Identity Management:** By guaranteeing that only individuals with the proper authorization can access particular medical records, blockchain technology can improve identity management for military personnel. This can be especially important when handling sensitive health information or in emergency situations.
- **Healthcare Services and Smart Contracts:** Smart contracts, which are self-executing agreements with terms encoded directly into code, have the potential to automate and optimize a number of military healthcare procedures. They can be used, for example, to automate insurance claims, billing, and other administrative work.
- **Decentralized Healthcare Applications:** Blockchain makes it possible to create decentralized healthcare apps that let military personnel and healthcare professionals communicate directly and securely. When immediate medical attention is needed in remote or combat situations, this can be extremely helpful.
- **Unchangeability:** By connecting, accessing, and timestamped a ledger with earlier documents, the blockchain data structure is constructed. Because the blockchain is decentralized, it can store any kind of record in any setting where authenticity, auditability, and banking are crucial, including supply chains, healthcare, and banking [3].

6.5 THE ROLE OF AI IN MILITARY HEALTHCARE

One of the biggest obstacles military medical staff members encounter is the requirement to offer prompt, precise diagnosis and treatment plans in high-stress, time-sensitive scenarios. This procedure could be completely changed by AI, which can swiftly analyze enormous volumes of data and provide accurate recommendations based on that analysis. AI systems, for instance, can be used to evaluate medical pictures, such CT or X-rays, in order to recognize ailments and suggest the best course of action. This can save up a

lot of time and money, enabling medical staff to concentrate on giving their patients the best treatment possible [11].

AI has the potential to be extremely important in the field of military medicine when it comes to the creation of sophisticated exoskeletons and prosthetics. With the use of these gadgets, wounded soldiers may be able to restore their freedom and mobility, which will enhance their quality of life and, in certain situations, enable them to resume active service [12].

AI can be used to optimize these gadgets' operation and design, making them as efficient and user-friendly as feasible. To establish the best force distribution and movement patterns, for instance, AI algorithms can evaluate data from sensors included into the prosthetic or exoskeleton, enabling the device to adjust to the user's specific requirements and preferences. AI can also be utilized to increase the military's medical supply chains' efficacy and efficiency. Artificial intelligence (AI) systems can assist in forecasting future demand and maximizing resource allocation by evaluating data on the distribution and use of medical goods. This can lessen the possibility of shortages and treatment delays by ensuring that medical staff have access to the materials they require when they need them [13].

AI has the potential to improve military medical personnel's education and training in addition to these direct uses. Medical professionals can practice their abilities in a safe and regulated setting by using realistic, immersive training simulations created by AI-powered virtual reality (VR) and augmented reality (AR) technologies. These simulations may be customized to meet each person's unique needs and learning goals, guaranteeing that they receive the most pertinent and efficient instruction available [12].

Lastly, AI can help soldiers receive mental health care both on and off the battlefield. Chatbots driven by artificial intelligence and virtual therapists can offer assistance and direction to military personnel who are facing mental health challenges including depression or post-traumatic stress disorder (PTSD). When support and assistance are most required, these AI tools can help close the gap between soldiers and mental health specialists [12].

Drones and artificial intelligence can work together to share threat intelligence, detect dangers, and keep an eye on border regions so that response teams can be informed. AI improves military personnel's security during times of conflict as a result. Artificial Intelligence (AI) has led to significant advancements in military healthcare. The following examples show how AI is used in military healthcare:

- **Health Diagnosis:** AI is used in medicine to diagnose patients more quickly and accurately. To find anomalies and aid in diagnosis, machine learning algorithms can examine medical images from CT, MRI, and X-ray scans.
- **Analytics that predicts:** By evaluating a variety of data sources, such as medical records, environmental variables, and individual health metrics, artificial intelligence (AI) assists in anticipating possible health

problems among military personnel. This makes proactive management of healthcare possible.

- **Trauma Management and Emergencies:** By examining data from combat injuries, artificial intelligence (AI) facilitates prompt decision-making in emergency scenarios. This entails estimating the degree of injuries, suggesting course of treatment, and supporting medical staff in delivering efficient and timely care.
- **Security of Computer Systems in Healthcare:** Military healthcare is susceptible to cyberattacks, just like any other digital system. Artificial Intelligence is utilized to improve cybersecurity protocols, identify and address possible security breaches, and protect private health data [13].

6.6 USE-CASES OF BLOCKCHAIN IN CYBER SECURITY

Let's now examine a few of the key use cases of blockchain technology in cyber security.

1. Decentralized Storage medium

A safer environment for all the personal information we exchange on a medium (computer or internet) is imperative as more and more people move faster and more online. The majority of businesses still store their data on centralized platforms, which makes it simpler for hackers to access and steal data. Given that decentralized storage platforms based on blockchain technology can stop hacking and theft of company data. Hackers no longer have a single point of entry to access all data when using blockchain as a medium for data storage. The primary reason why businesses view blockchain as essential to data security is because of this characteristic.

2. DNS Security

Since most DNS is centralized, hackers can disrupt the link that identifies a domain name with an IP address and even take down a website. Recently, hackers have attempted to combine DDoS and DNS attacks to make websites unavailable for a predetermined amount of time. Because your data is stored immutably on a decentralized, distributed ledger and immutable smart contracts are there to power connections, using blockchain technology can make it more difficult for hackers to access your data.

3. Secured Private Messaging Platform

Since the world is designed to be commercial, social media data exchanges by customers reveal a great deal about them. However, because many fall behind the roof, several social media platforms offer end-to-end data encryption. A significant number of social media platforms have begun utilizing blockchain technology to protect user account information and private conversations. Blockchain technology

is a fantastic tool for securing data transfers and allowing secure communication between various messaging apps.

4. Veracity Data

Regardless of the industry, detecting bad data and tracing frauds come at a significant cost. The majority of the time, algorithms used to combat fraud are quite good at spotting discrepancies, but the cost is higher. Due to numerous verification processes, data that is filtered and stored using decentralized blockchain technology is typically more reliable, secure, and 100% accurate. Therefore, when employing blockchain technology, the bar for data veracity is higher [14].

Case Study 1: How Blockchain Technology Supports the Healthcare Domain?

Health care: a vast sea of data in medical domain

I'm assuming you are already aware of how vast the health care network is—we could even refer to it as an ocean of data. Every day, there are more and more patients, medications, doctors, hospitals, and other things. Furthermore, I have no doubt that this growth will never stop. Due to the drastic changes in people's lifestyles and the scarcity of healthful foods, the number of illnesses and patients is rising. This turns into a pain point, which hospitals began to leverage for their expansion [15].

Data Breach in Healthcare industry

A very recent report by Verizon's 2018 Data breach Investigation Reports stated that the health care industry "is the worst in pretending the insider data breaches." Approximately 750 cyber incidents occurred in the previous year, or in 2017–18. 37% of cyber incidents involve the misuse of personal information, and 63% involve the abuse of privilege.

About 56% of the actors in this cyber incident were internal, and the remaining 44% were external.

I hope this brief overview of the health care industry's history and present situation has helped.

Let's now talk about how blockchain technology will be the best solution to all of the problems the sector is currently facing.

Blockchain Started to Hear the HeartBeat

Blockchain technology was mainly developed for cryptocurrencies like bitcoins. It was thought that this technology was public ledger & the data gets encrypted and stored in a secure block in cloud, making it accessible to only a select group of authorized users. A significant revolution in bitcoin cryptocurrency exchange transactions was brought about by this technology and its applications. After witnessing the enormous technology's security, integrity, and transparency, numerous industries began snatching it up for their

improved techniques for retrieving & storing data. It was initially used by the banking industry, then by e-commerce, SCM and now the health industry. The food, fashion, telecommunication, power and energy supply, and marketing sectors will all remain a part of this chain [15].

How Blockchain Facilitates the Healthcare Domain?

Let's prior examine the issues facing the healthcare sector and how blockchain technology might be able to address them.

While there are many difficulties, maintaining the data is the biggest problem facing hospitals and the healthcare industry. Thus, following six facts were determined to be factors risk that hospital management must deal with in order to maintain the data challenge [6, 15].

1. Meeting the HIPAA Compliance
2. Digital Data Computing
3. Breach of patient data
4. Stopping the Insider Data Vulnerability
5. The incompatibility of clinical data and administration
6. Absence of expertise in tracking and operational analytics

We may infer that "maintaining data in the healthcare ecosystem requires a back painning effort" from the previously discussed points.

Blockchain Modernizes Healthcare Sector by Eliminating Risks.

According to Experts, Health information exchanges (HIE) should improve in terms of security, effectiveness, manageability, and interoperability by leveraging this enormous technology in the healthcare system. Private blockchain implementation, or permissioned blockchain, is the technology used to accomplish this goal; only individuals with permission can access data. The costs and difficulties faced by middlemen in the exchange of medical data will be decreased by this type of Blockchain-based system. This is a list of objectives that blockchain technology can help us accomplish [15].

1. Data Integrity and Provenance
2. Data Safety
3. Transparency of Data
4. Medication Traceability
5. Simple to uphold Health Electronic Records (HER)
6. Management of Patient Data
7. Simple SCM maintenance for medical accessories
8. Stops Data Breach (particularly when sharing patient information with unaffiliated parties)
9. Distillation of the medical insurance claims process is planned.
10. Results of clinical research are easily maintained.
11. Can Acquire Immutable Health Data
12. Better Security Measures [15].

Case Study 2: How blockchain technology can modernize government sectors?

Blockchain is a widely used technology that is essential to the development of the digital economy of the future. The development of the most transparent government-citizen relations could be fueled by blockchain technology.

Let's now examine how blockchain technology is being used in government sectors and how government sectors are being reconstructed through the use of blockchain.

The following ideas about blockchain in government will be discussed in this section.

1. Challenges In Government Sectors

- i. Lack of Transparency
- ii. Slow Step-by-step Approach
- iii. Lack of Privacy
- iv. Uncontrollable Corruption
- v. Expensive Process

2. Blockchain in Government

A growing number of government sectors are moving toward implementing blockchain technology in order to meet public demands for accountability, openness, better functioning, and lower costs across the board [16].

Blockchain technology offers many opportunities to bring more security and transparency to the current government sectors. An improved approach, blockchain-based government provides the following benefits:

- A safe place to store data for both citizens and businesses.
- A lower likelihood of theft and corruption.
- Less labor-intensive procedures without going over budget
- A rise in public confidence in online civil systems and government.

Benefits:

The following are advantages of a digital government built on blockchain technology.

- ❖ Data protection
- ❖ Offers efficient procedures
- ❖ A decrease in waste, fraud, and abuse
- ❖ Consequently, boosting public trust [16].

3. Blockchain Technology used the following government:

1. The Bank of England sees 'tens of billions' saved in settlement costs by utilizing blockchain technology.

2. By implementing blockchain technology, Dubai is able to save 5.5 billion dirhams a year in document processing costs alone.
3. Zug ID: Examining the First Blockchain Identity to Be Publicly Verified
4. Future of Government with Blockchain

Blockchain technology has the potential to improve government sectors in many ways, including speed, immutability, privacy, security, and transparency. Therefore, the greatest way to do away with the centralized nature of government sectors will be through blockchain-based government.

Blockchain technology can therefore be used to provide quick fixes for healthcare issues, a reliable tax system, and ever-faster processing of government benefits and allowances for all citizens. The greatest way to create a transparent and contemporary government will undoubtedly be through blockchain-based government. Let's hope that the adoption of blockchain technology by several government agencies in India will significantly alter the country's government [16].

6.6.1 AI Tokens used in blockchain

The creation of AI tokens marks a paradigm shift in the way blockchain technology and artificial intelligence are integrated. These cryptographic tokens are essential to decentralized AI ecosystems because they facilitate smooth transactions and encourage cooperative innovation. The potential for innovation and cooperation is enormous as AI tokens continue to gain popularity. AI tokens' decentralized structure promotes an inclusive AI environment where businesses and individuals can both contribute to and profit from the field's advancements. The incorporation of AI tokens is set to revolutionize how we approach and utilize AI as we set out on this revolutionary journey, opening up new avenues and influencing a future that is more open and accessible [17].

6.6.2 Future outlook on blockchain & AI

Blockchain-based cryptographic tokens created especially for use in artificial intelligence applications are known as AI tokens. Within AI ecosystems, these tokens act as a medium of exchange, facilitating smooth interactions and transactions between different AI entities. AI tokens function on decentralized networks, which promote efficiency, security, and transparency in contrast to traditional currencies.

Many businesses began offering AI-powered token development services as early adopters of these AI tokens. Among them, top token development company Bideal offers state-of-the-art token solutions, such as creative AI token development solutions [18].

6.7 CONCLUSION

In the military, is blockchain a revolution or an evolution? Whereas a revolution would fundamentally alter the weapons themselves, evolution would only involve changing how the military's current arsenal is employed. Practically speaking, the data currently points to an evolution rather than a significant shift on the scope of the "Military Affairs Revolution." Blockchain improves military logistics and communication security. From now on, blockchain will support and enhance the military's efficacy. If blockchain technology is implemented correctly and many more military applications are found, along with cost-effective and prudent use, it will eventually be used in the military as a revolutionary tool. But instead of witnessing a genuine revolution in the way the national DoD and armed forces operate, we are witnessing a significant advancement at the moment. We can all agree that, if the biggest military players start using blockchain extensively, it will change the security and efficacy of the military's current capabilities in a gradual but unquestionable way. As we've already observed, there is active competition between state actors in this domain. Blockchain will need to be adopted by the majority of the defense industry, if not all of them, in order to see a revolution. To achieve and understand the range of blockchain technologies for tactical sustainment challenges, the military should closely examine the potential of blockchain solutions to the issues related to in-transit visibility, data integrity, additive manufacturing, reporting, operational contracting, and logistic estimation, among others.

REFERENCES

1. Rahayu, Syarifah Bahiyah, Nur Diyana Kamarudin, and Afiqah Mohammad Azahari. "Military blockchain for supply chain management." *Journal of Education and Social Sciences* 13.1 (2019): 353–361.
2. Hadlington, Lee, et al. "The use of artificial intelligence in a military context: Development of the attitudes toward AI in defense (AAID) scale." *Frontiers in Psychology* 14 (2023): 1164810.
3. Tagde, Priti, et al. "Blockchain and artificial intelligence technology in e-health." *Environmental Science and Pollution Research* 28 (2021): 52810–52831.
4. Wenhua, Zhang, et al. "Blockchain technology: security issues, healthcare applications, challenges and future trends." *Electronics* 12.3 (2023): 546.
5. Marsalek, Alexander, and Thomas Zefferer. "A correctable public blockchain." *2019 18th IEEE International Conference on Trust, Security and Privacy in Computing and Communications/13th IEEE International Conference on Big Data Science And Engineering (TrustCom/BigDataSE)*. IEEE, 2019.
6. Shuaib, Khaled, et al. "Blockchains for secure digitized medicine." *Journal of Personalized Medicine* 9.3 (2019): 35.
7. Kreinbrink, Justin L. *Analysis of artificial intelligence (AI) enhanced technologies in support of cyber defense: Advantages, challenges, and considerations for future deployment*. Diss. Utica College, 2019.

8. Wrona, Konrad, and Michal Jarosz. "Use of blockchains for secure binding of metadata in military applications of IoT." *2019 IEEE 5th World Forum on Internet of Things (WF-IoT)*. IEEE, 2019.
9. Randall, David, Pradeep Goel, and Ramzi Abujamra. "Blockchain applications and use cases in health information technology." *Journal of Health & Medical Informatics* 8.3 (2017): 8–11.
10. Friedewald, Michael, and Oliver Raabe. "Ubiquitous computing: An overview of technology impacts." *Telematics and Informatics* 28.2 (2011): 55–65.
11. Oniani, David, et al. "Adopting and expanding ethical principles for generative artificial intelligence from military to healthcare." *NPJ Digital Medicine* 6.1 (2023): 225.
12. Crump, Cindy, and Loretta M. Schlachta-Fairchild. "Achieving a trusted, reliable, AI-ready infrastructure for military medicine and civilian care." *Artificial intelligence and machine learning for multi-domain operations applications II*. Vol. 11413. SPIE, 2020.
13. Miletić, Tomislav. "Military medical enhancement and autonomous AI systems: Requirements, implications, concerns." *Ethics of medical innovation, experimentation, and enhancement in military and humanitarian contexts* (2020): 175–194.
14. Rawat, Danda B., Vijay Chaudhary, and Ronald Doku. "Blockchain: Emerging applications and use cases." *arXiv preprint arXiv:1904.12247* (2019).
15. Zhang, Peng, et al. "Blockchain technology use cases in healthcare." *Advances in computers*. Vol. 111. Elsevier, 2018. 1–41.
16. Alketbi, Ahmed, Qassim Nasir, and Manar Abu Talib. "Blockchain for government services—Use cases, security benefits and challenges." *2018 15th Learning and Technology Conference (L&T)*. IEEE, 2018.
17. Lee, Jei Young. "A decentralized token economy: How blockchain and cryptocurrency can revolutionize business." *Business Horizons* 62.6 (2019): 773–784.
18. Nguyen, Quoc Khanh, and Quang Vang Dang. "Blockchain technology for the advancement of the future." *2018 4th International Conference on Green Technology and Sustainable Development (GTSD)*. IEEE, 2018.
19. Gonçalves, Daniel Jorge Viegas. "Ubiquitous computing and AI towards an inclusive society." *Proceedings of the 2001 EC/NSF Workshop on Universal Accessibility of Ubiquitous Computing: Providing for the Elderly*, 2001.

Current applications and future direction of artificial intelligence using blockchain for military application

Suhas K C, Mahesh N, Asifullakhan, Kotresh Naik D, and Francesco Flamminie

7.1 INTRODUCTION

The military capacity of a state plays a crucial role in determining its position in the global arena. According to the United States Bureau of Defense, military potential [1] refers to the ability to achieve a particular military objective, such as winning a conflict, prevailing in a conflict, or destroying a particular target. This capability is influenced by factors such as the state's organizational structure, level of modernization, readiness, and ability to sustain military operations. The degree of modernization is primarily influenced by the technical sophistication of the state's military equipment, weapon systems, and overall military technology.

A typical Second World War is moving into digital space and gradually disappearing from memory. According to research, cyber threats on government and private organizations have increased frequently [2]. Researchers [3, 4] predict that in the future, creative automatic systems and Artificial Intelligence (AI) will be an integral part of armed conflicts.

A significant amount of data is needed for the majority of contemporary AI algorithms [5, 6], such as those that process natural language. They can function more effectively, quicker, and more efficiently, but they need access to big databases to function properly. The development of this field of study has been made possible by the accessibility of extensive resources for data and the growing computing power of machines. The advancement of ITS (Intelligent Transport Systems) the capacity to identify informative tweets, forecast and assess atmospheric phenomena (comprising pertinent facts) from non-informational ones (comprising rumors or incomplete data), and the expectation of thriving real estate and FX markets are just a few of the topics on which Neural network applications are continuously gaining popularity today.

Artificial intelligence has many applications, contributing to its profound and diverse effects on society. The paper discusses how AI has contributed to the development of new norms for human conduct, interpersonal interactions, diplomacy, and state operations.

Blockchain technology is finding its way into more and more important facets of international life, such as aviation and maritime. It is currently utilized by military combatants and defense forces worldwide.

Blockchain is a technological marvel that is fast gaining popularity as the internet of the future—a more potent version of the internet than we currently know and use. In the little more than ten years that blockchain has existed, all of this reputation has grown. Not too bad, huh? Naturally, skeptics mocked the blockchain and the virtual currencies it hosted when they first appeared in the financial market. But these days, it is harder and harder to ignore blockchain's power—from its application in voting systems to the military.

The blockchain is an incredibly powerful technology. It has already been utilized for everything from combating internet piracy to certifying artwork to providing unparalleled security for digital file storage. But what about the military's role in providing worldwide defense. Many nations throughout the world, from the US and other NATO members to major military foes China and Russia, are showing interest in integrating blockchain technology into their armed forces to make them more effective, safe, and—most importantly—stronger than the competition.

7.2 MILITARY USE OF BLOCKCHAIN WILL BE ESPECIALLY USEFUL IN DEFENSE

First, the scattered organization of blockchain ensures its ease of use. It also makes these tools less costly. Second, its resilience, and security, are mainly helpful for storing information and are strong assets for many forces applications. This is why an aerospace and defense diligence executive survey by Accenture in 2017, blockchain was mentioned as one of the hottest new technologies. They want to encourage driving business expansion and efficiency. Additionally, defense sections around the planet are more and more drawn in by the control of blockchain., As per UK Defense Matter (14th edition, 2017), Over the next 8 years, the defense research community expects to find submissions concerning the forces relied on the methodology of blockchain, with strong applicant in sections like cyber defense. Networking of Communications, Resilient Communications, Logistics Support, and Defense Internet of Things [7]. A methodology that uses blockchain possesses a lot concerning defense segment, since it is possible used in both operational and hold roles. This section mostly addresses three types of employs highlighted via Barnas (2016): virtual defense and information honesty, supply chain coordination, and adaptable dialogue. It appears from a technical standpoint that private blockchains will be most helpful in the defense industry. Having an open blockchain, we have unrestricted access to the chain, which can be perilous for protecting delicate data. Because confidential blockchains are characterized by obstacles to entrance, in which an management defines the rules for accepting participants and chains (per-mission read and write), it is best suited for defense use. Right to use and guidelines of system can be

forbidden by one each one: the Chief of Staff of the Army. In terms of inter-service governance, hybrid blockchains will also be promising [8].

7.3 CYBER SECURITY AND DATA INTEGRITY

The numerous advantages of Blockchain formulate it a dominant tool to protect adjacent to fake attack and clarify the machinery be useful within various areas related to cyber defense. In 2017, North Korea stole 235 GB of confidential in rank belonging to South Korean and US intelligence agencies. That day, the European mission reported to “additional than 4,000 attack are conceded elsewhere each date, and 80% of European companies experience at least one cyber defense event [9]. The financial impact of cybercrime has increased fivefold in the last four existence lifetime only. The US central administration have targeted additional than 60,000 imitation attacks on the energy division, which is vulnerable owing to its confidence and reliance on compute tools (Mire, 2018). Store big amount of sensitive data in one position is mainly dangerous. It preserve direct towards worn to portray the mass theft of confidential in turn by strange actor. In this regard, the toughness accessible by blockchain, with its scattered character and capacity to perceive and chunk any intrusion attempts, can help significantly. Soldiers on the arena must ensure that the guidelines and in order they receive are correct and precise. Centralized organizations responsible for digital communications are additional susceptible to attack that can disrupt or alter communications [10]. Furthermore, if division of the system is affect, the integrity of the arrangement is not guaranteed and the complete organization may collapse. Yet another challenge is the solution. By sharing information horizontally, it democratizes the battlefield and creates a safe and sound atmosphere where the malfunction of one end does not denote the breakdown of others. As for dangerous attach system, conventional weapons are increasingly integrated with military digitalization. For example [11]. The US armada is functioning to upgrade its aging auspices battle club using blockchain to provide a more efficient centralized command and control system that connects sensors to the weapons in the system.

Countless defense industry actor want to rely on blockchain to provide a particular, mutual, and unchallengeable foundation of fact. As one more Accenture account points out, “erroneous, manipulate or prejudiced information can contain sweeping consequences such as shady production insight and decisions”. According to the account, 86% of the defense of communication and supply chains will be strengthened [12]. A crowd-sourced and decentralized executive move towards protect these operation as of cyberattacks. Blockchains are completely protected in addition to immutable due to superior cryptographic technique. Strictly, this dissimilar field preserve benefit from the data reliability that blockchain allows. Block cannot be changed lacking the approval of all participants or chain administration given and must come from a genuine performing artist (Figure 7.1).

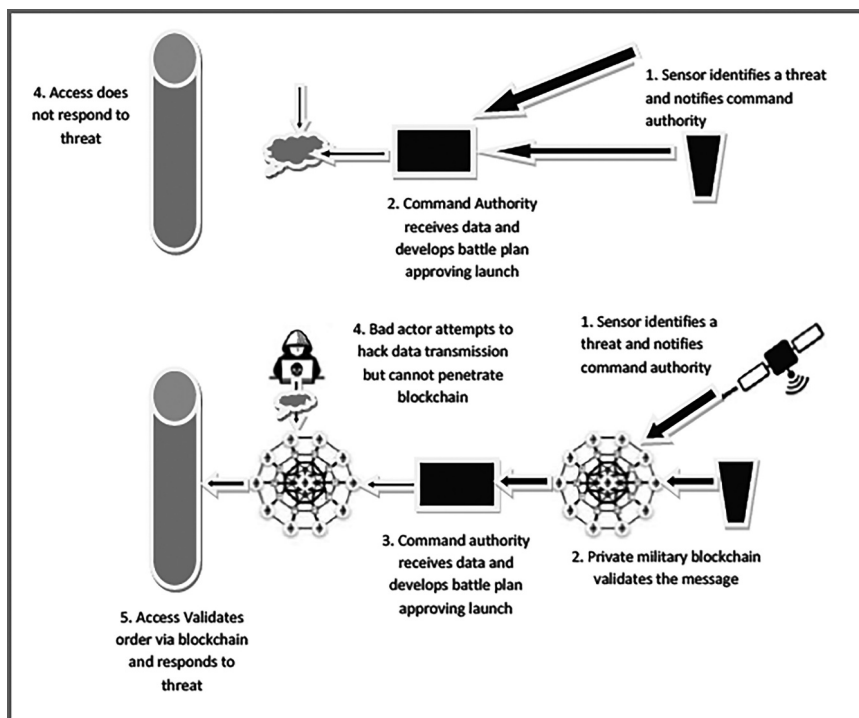


Figure 7.1 Current vs blockchain-based control of a basic weapons framework.

Working principle of the framework steps:-

- Step 1:-** The Blockchain based sensor identifies the threat like (Phishing, Sybil attacks etc) and notifies the commanding authority about the threat it has been identified.
- Step 2:-** The Commanding authority receives data from the sensors and creates a battle plan and access does not respond to threat.
- Step 3:-** Repeat the same steps 1 and 2 along with a new agent involving into the process i.e private military blockchain this will validate the message regarding the sensor data which notifies that it encountered a threat.
- Step 4:-** The bad actor (a person who is a reason for harmful work) attempts to hack data transmission but cannot penetrate into the blockchain.
- Step 5:-** The agent access and validates order via blockchain and responds to the threat.

Besides, the defense ensured by block chain isn't based on mystery or belief, as well as close by are no passwords or cryptographic key to ensure it. Ultimately, Blocking employment the whole control of the whole framework to proactively deny endeavors by noxious performing artists. Supply chain management [3].

Nowadays, military coordination's and contribute chains fetch jointly hundreds of distinctive armed and private on-screen characters [13]. This environment is getting to be more complex due to collaboration with designers, new companies, and industry players. In this manner, the focus of grinding with numerous members cannot be maintained at a strategic distance. This leads to pointless costs, mistakes, or focus of disappointment. In any case, blocking can unravel this issue. When it comes to military coordination, it offers different points of interest: tall swiftness, traceability, protection, in addition to fetched diminishment.

Individual of the foremost critical employments of blockchain in provide to create extra versatile. Straightforward, solid, and proficient. To begin with, when weapons and basic frameworks are at hazard, personality confirmation is a basic component within the defense industry. There are too follows of parts and materials utilized within the defense segment. exceptionally vital. Blockchain innovation can offer assistance follow the root and anticipate the forging of weapons from production to dispersion. Typically accomplished by consolidating sensors, announcing forms, and Web of effects innovation at each footstep to screen and account everywhere the source resources are, what time and anywhere they are shaped, how they are transported and put away, the same as fine when repairs or support. As a result, Resistance Company can join blockchain innovation with others such as false insights, compute, and expanded authenticity to extend their capabilities [14].

Information sharing and information investigation. Coordination of blockchain into each step of the fabricating handle, counting plan, prototyping, testing, and generation, will offer a secure supply chain. Things are followed at each step, giving DoD coordination groups a diagram of item courses. Nowadays, such a need for permeability leads to disarray and miscommunication among supply chain members, as well as a need for belief among them (McPherson, Eskaravage Boozallen). Instep, a private key utilized by a supply chain accomplice would permit that accomplice to be distinguished. Delicate data such as the sort, root, and goal of the thing is scrambled and as it were unmistakable to authorized parties. A few companies have chosen to square this sort of utilization. For case, at 2018 [15], The Researcher made a framework base on this innovation to secure in addition to rationalize contribute chains within the aviation in addition to defense businesses. In case a portion is failing, it can be found quicker outstanding on the way to the input circuit and blocking way. Moreover, as defense frameworks progressively utilize non-commercial components for computer program frameworks, there's developing anxiety that this machinery may contain deliberateness vulnerabilities that can be misused by enemies (Barnas, 2016). Blockchain can be the arrangement, recognizing the occurrence of each processor, circuit board, and computer program from "Dark red to the cockpit". In the war zone, the following and the permeability of materials along the item line are basic. The Armed force must moreover be certain in its pre-deployed capabilities and operational capabilities

to succeed in its mission. Blockchain innovation not as it were empowers more noteworthy perceivability in fabric supply chains but can moreover make strides in nourishment security and on the front line. Following and following the nourishment provide chain can anticipate connected episodes and offer assistance to fathom the basic issue of following basic merchandise such as pharmaceuticals. The military workforce must know what assets are accessible, in travel, or on ask at each level of arranging. Observation of materials, merchandise, and kit. Monetary misfortunes and squandering be issues within the defense division and amid operations. For illustration, the misfortune in Afghanistan is assessed at 17 billion dollars for the Joined together States. (Disruptor Day by day,) It is very important to recognize exactly anywhere the reserves are departing. Within the case of, misfortunes of \$2.7 billion were ascribed to airborne observation inflatables, and \$150 million went through on private estates for military staff. Blockchain-enabled resources following will offer assistance to avoid [5].

7.3.1 Tough connection and other employment

Tough communication is valuable within the military, particularly amid operations. This requires certain get to the organization amid this handle. Furthermore, blocking can give steady communication in high-level struggle settings. In this strife, defense Units must be prepared for foe assaults within the electromagnetic range particularly basic communication frameworks such as satellites, submarine cables, or strategic information joins (Barnas, *ibid*). Adversaries will too attempt to control the information to break the murder chain. To meet this risk, the military needs the capacity to safely make, ensure, and share information, made conceivable by blockchain as depicted over [16].

Also, blocking can be valuable to secure the informing framework. Blockchain cryptographic encryption strategies will empower the usage of computerized measures that can diminish costs and move forward connect-and inner communication. In like manner, in 2016, the US Defense Progressed Investigate Ventures Office propelled a call for tenders for the “Secure Informing Stage” to send messages through a decentralized protocol (Mire, 2018). It moreover appears a readiness to assist NATO individuals execute blockchain arrangements. Sensible due to the topographical area of individuals on both side of the Atlantic, dependable communication is a critical case both at yearly gatherings and at a separate.

Blockchain’s numerous focal points make it valuable for other defense applications as well. To begin with, the protection of individual data can be guaranteed with the assistance of this innovation. Advanced personalities put away on the blockchain cannot be manufactured. Even though it has been utilized by the Joined Together Countries to assist Syrian outcasts purchase nourishment, it might be more valuable in a defense setting to make combat. These will be disseminated among units, permitting the warriors to be coordinated. Check their correct area on the front line. Moment, blockchain offers more noteworthy arranged exactness through savvy contracts.

Utilizing it, partners can boost high-level defense ventures and consequently designate stores to concurred activities and a common foundation. At long last, as 3D printers are progressively utilized to construct buildings, as glowing as to create rambles, explosive launchers, body armor, and other gadgets, blockchain can be able to gotten up to live an additional safe information exchange device due to its disseminated nature [7].

7.4 PROSPECTS FOR MILITARY APPLICATIONS OF BLOCK CHAIN INNOVATION

Military command: The command data framework is a critical strategy and hardware for realizing exact and programmed command on the front line in real-time amid the war and makes a difference in the command at all levels to carry out logical and successful control of dependent units and weapons in standard times. As a natural “human-machine” framework that takes commanders as like center along with computers, and employments computers along other data innovation gear as a necessity and fabric ensure, the framework naturally incorporate different command along with control strategies and commanders facing empower high-level robotization in gathering, channeling, handling and utilize of military command information. It in this way guarantees information judgment, accessibility, and security of the command data framework and, in the guideline, ensures an increment in combat viability and the victory of taking the activity in war [17].

Blockchain command data framework: Military necessities: To begin with, the command data framework has centralized systems and a database. As a vital target in times of war and indeed in times of peace, it is helpless to adversaries or programmers. They can utilize distinctive data implies to carry out arranged and electrical assaults that cause the loss of motion of the whole data framework, burglary, misrepresentation of character data, and control of critical information [18]. Subsequently, war-fighters confront extraordinary dangers in information genuineness and may indeed make off-base choices with pernicious information. Moment, when commanders depend on the command information framework for decision-making, there’s a chance that they will issue inaccurate orders or untrue orders caused by foe information control.

Specialized rule: The command data framework derived from blockchain innovation can be divided into information, organize, agreement, motivation, contract and application layers. Including them, the information layer ensures the information or insights unwavering quality, reliability, and safe future of the military data framework; the organizing layers ensures self-organizing along with Spring Universal Conference in use with Defense Innovation 2020 decentralized components of military data frameworks; the agreement layer typifies different sorts of agreement calculations to realize independent and trusted decision-making; the motivation layer employs a programmable motivation instrument to maintain a strategic

distance from all sorts of awful behavior and accomplish positive behavioral motivating forces; the contract layer makes a difference robotize and shrewdly military data frameworks and diminishes the vulnerability, differing qualities, as well problem that human and other variables bring to combatant command and control [8].

Arrangement: Utilizing the points of interest of decentralized and disseminated capacities, agreement component, independent characteristics, and credit instrument of deviated encryption algorithm of blockchain innovation, ready to make a programmed and safe mechanism of command and control. In the future, it may be paired with military IoT and false insights to begin shifting the strategic level of battle management from a centralized to a decentralized one.

It is also possible to accomplish trusted fight command, which involves recording instructions from commanders at all levels while making the decrypted information verifiable and traceable. As it were in case the foe or programmers alter more than 51% of the hub data at the identical instance, the command data framework information can be changed. Much obliged to the time arrangement information, collective upkeep, and programmable and solid highlights of blockchain innovation, energetic and changeless recording in systems or databases can be realized and viably avoid wrong data and programmers. This removes the risk of malicious assaults which are inherent in command data frameworks by enabling us to instantly check if a database has been updated or whether a military architecture has been observed.[19].

Case in point: The DARPA is attempting to generate a trustworthy and safe data stage built on advancements in blockchain. This would live a secure data benefit framework competent of viably ensuring touchy information and anticipating programmers.

Blockchain + Cluster operations of UAVs Military essentials: UAV cluster frameworks have five normal characteristics, specifically decentralization, independent control, cluster recuperation, work enhancement, and zero misfortune. In any case, current UAVs in clusters need a general discernment of the outside environment and need successful data sharing and coordination of activities between person UAVs and UAV arrangements.

Arrangement: Utilizing the agreement mechanism of blockchain innovation, we will change from individual and regulation beliefs to machine beliefs. On long-standing time battlefields, to link the operations, combat command information will be transmitted in a distributed way with the UAV group control and command structure. In contrast with unmanned operations, the ability to preserve combat capability irrespective of any misfortunes is of immense significance for the establishment of a modern “human-machine/machine-machine” control and command administration [19].

Security is the primary issue. Every UAVs in a cluster that may operate as an intended hub have been included in the “developed signature” of blockchain through “open and secret key encryption and unscrambling”. Every hub transmits and maintains an identical record, ensuring the authenticity of communication information and verifying the identity of the others in the cluster.

The moment includes decentralized decision-making. The conveyed making decisions calculation is essential to the efficient functioning of UAV cluster frameworks, and the agreement component guarantees that all hubs within the disseminated framework concur on the decision-making objective [20]. In genuine warfare, every single thing within the cluster need to concur regarding operational assignments, goals like gathering, designing, way of arranging, and judicial shirking.

Control over development in blockchain sidechain innovation empowers more block chains to exist interconnected progressively. On the one hand, UAV's completely different chains can work according to the protocol preset on the chain where they are found essentially participation between chains encourages exchanging between diverse cluster arrangements. Within the preparation of facilitated look, surveillance, and protest, the UAV cluster can alter arrangements to secure itself and murder the foe.

The next one is almost a decentralized independent cluster. Later on, each substance within the UAV cluster knows how to be well-thought-out and independent specialists with the employment of recognition, thinking, and making judgements. Through clever contracts, these experts will establish various decentralized independent clusters to independently form optimal decisions.

7.5 MILITARY MANAGEMENT

The conventional structure of military administration might live a top-down “command and control” framework that offers with issues for example congested education and high administration costs, it needs the combat framework to be more responsive and dexterous, but education with various levels of administration works wastefully. A huge sum of information such as plan plans, and tests concerning, the situation of combat gear must be recorded all through the complete life cycle of weapons, from venture show, improvement, generation, conveyance, to benefit, to retirement. Data is effectively lost or controlled in this preparation.

Arrangement: According to blockchain innovation, we will construct a weapon as well as gear life cycle and an administration framework together worked and commonly controlled by designers, producers, and clients. Using the framework, as we are going to track and handle the requirements of the hardware plan, test data, the condition of the battle gear, and support documents. Because no substance can be deleted or controlled, the security of information, comfort, and reliability are increased. All administration activities will depend on shrewd contracts for open and straightforward choices, lessening administration progression, making a level administration mode, and at long last moving forward productivity. At the same time, each piece of weapons and gear can be followed reverse toward its beginning, which makes a difference in resolving the debate over open obtainment contracts and constructs a total unbreakable observing, administration, and control framework, making strides the safety, comfort, and validity of administration.

Blockchain + touchy information administration [21]. Utilizing the “untamable” highlight of blockchain innovation, ready to provide an arrangement to the issue of “hard-to-maintain proof” within the administration of delicate information in the military forces review and observation, human assets, and restorative and healthcare. An “honest record” among all data can be accomplished by “witness of the entire arrangement”, abstain from fraud of reports, lost records, and control of data.

7.6 MILITARY COORDINATION

Military Prerequisites: In military coordination’s, arrange, information capacity, framework upkeep, traceability, and quality control issues amid bundling, stacking and emptying, transportation, and destroying are troublesome to fathom.

Arrangement: Utilizing blockchain innovation, we will make an isolated, secure, shared, and changeless auditable record, track and review the exchanges of diverse supply chains and between all operational accomplices, and perform successful lifecycle administration of the defense supply chain and acquirement and coordination’s framework.

By incorporating blockchain innovation into the military coordination organization, we will construct a decentralized independent arrangement of personnel and materials in coordination frameworks. The data related to the generation, buying, transportation, and conveyance of materials within the frameworks can be put away completely different pieces in a uniform manner, which can incredibly progress the security of military coordination data and empower military coordination hardware and vitality to meet the necessities of diverse administrations and offices, hence continuously within the best condition [22].

Modern military coordination’s incorporate clever capacity, bundling, transportation, and conveyance. Different processes are shaping into a little military Web of Things based on an energetic independent organization of individuals and objects. Utilizing hubs, ready to communicate specifically or through transfers and oversee critical information within the military coordination chain, such as client needs, putting away merchandise, stacking, transportation and dispersion, and travel. The upkeep of the blockchain will be managed by all hubs within the whole organization and the unlawful operation of a few person hubs will be rejected and stood up to by most hubs. In this way, the security and comfort of transactions will be expanded and the time that went through clever military coordination will be decreased. It can also offer assistance resolve arrange, communications, information capacity, and framework support issues amid pressing, stacking, emptying, shipping, and destroying in military coordination’s, subsequently guaranteeing legitimate and proficient framework operation. This can be considered the foremost promising military application of blockchain [23].

Case in point: In April 2016, the US Division of Defense and its NATO allies started looking at the potential defense applications of blockchain

innovation, counting the programmed execution of savvy contracts, the secure capacity of delicate records, and the diminishment of mistakes and interruptions amid the execution of a defense contract. In expansion, blockchain innovation can also be utilized for crisis reaction when calamities happen and progress the straightforwardness of acquiring crude materials within the supply chain and cargo transportation within the coordination's prepared. In June 2017, the US Naval Force utilized blockchain innovation to make strides in the security of added substance fabricating frameworks. They recorded the complete handle of the portion plan, prototyping, testing, fabricating, and last handling, so clients seemed to see any particular information and caution the portion when it was harmed or after its life cycle [24].

7.7 MILITARY SECURITY

Quantum blockchain. Military prerequisites: The security of information transmission in the war zone must be guaranteed and the issues of flag interferences, decoding, and surveillance must be illuminated.

Arrangement: By utilizing quantum key dispersion as a substitute for the initial private key structure, and utilizing quantum cryptography's anti-eavesdropping and listening stealthily features within the blockchain organize, we will enormously make strides in the defense capabilities of the blockchain organize and take advantage of the troublesome impacts on the issue of flag capture attempts, decoding, and detection.

Case in point: The Russian Quantum Centre and the Russian Institute of Sciences have effectively tried the primary quantum blockchain system. Blockchain + versatile communication. Military necessities: The security of information transmission in the war zone must be ensured.

Arrangement: Based on the distributed characteristics of blockchain innovation, a wide-coverage, disaster-resistant, and exceedingly secure communication framework can be built to realize secure and versatile communication within the complex war zone environment [25].

Case in point: In May 2017, a DARPA-funded Indiana-based Innovation and Fabricating Company utilized blockchain innovation to create an "inaccessible informing and exchanging platform" for the military. This stage isolates the creation and transmission of data to guarantee that information sent and gotten cannot be breached and empowers secure communications between central command and ground strengths, as well as between the Division of Defense and insights authorities. In July 2019, as a portion of DOD's computerized modernization procedure, DARPA started utilizing blockchain innovation to construct a more effective, capable, and secure communications stage to encourage the secure exchange of data for any command, control, and communications frameworks, workers track exchanges through a conveyed record channel, and they ensure the security of communication between central command and the nation [26].

Analysis of various Machine Learning Algorithms and Blockchain Technology in Military Applications shown in the Tables 7.1 and 7.2.

Table 7.1 Comparison of various artificial intelligence algorithms for military applications [27, 28]

<i>Sln</i>	<i>Algorithm</i>	<i>Description</i>	<i>Military Application</i>	<i>Strength</i>	<i>Weakness</i>
1	Decision Trees	Method for learning rules that categorizes data by asking a number of answers that are yes or no.	Recognizing targets, evaluating threats, and allocating resources.	Interpretable and computationally effective	Susceptible to excessive fitting and noise sensitivity.
2	Support Vector Machines	Determines which hyperplane is most effective in dividing points of information into different categories.	Cyber security, fault prediction, anomaly detection.	Impressive generalization, suitable for data with multiple dimensions.	Expensive to compute for huge datasets and challenging to tune
3	Artificial Neural Networks	Emulating the networked layers of neurons found in the human brain.	Self-driving automobiles, training, modelling, and image recognition	Flexible, versatile, and able to pick up intricate relationships	Enormous datasets needed for training, high computing expenses, and black box nature
4	Genetic Algorithms	Iteratively evolves solutions to an issue, drawing inspiration from natural selection.	False detection, supply optimization, and arranging routes	Robust, capable of locating worldwide ideal, and ideal for challenging issues	Slow integration, needs parameter adjustment, and problem-specific programming.
5	Reinforcement Learning	In a setting with incentives and consequences, agents learn by making mistakes	Autonomous weaponry, tactical choices, and automation control	Able to deal with ambiguity and adapt to changing circumstances	Significant computation expenses, prone to catastrophic failures, and careful incentive design are required

Table 7.2 Comparison of various blockchain technologies for military applications [29, 30]

<i>Slno</i>	<i>Feature</i>	<i>Permissioned vs. Permissionless</i>	<i>Consensus Mechanism</i>	<i>Scalability</i>	<i>Privacy</i>
1	Security	Extreme - Limited participant authority and access	Byzantine Fault Tolerance (BFT): Even when there are defective nodes present, the majority of nodes must agree	Limited: The scale of a network is determined by its size and the consensus method	Excellent Control over Data Usage and Participant Authentication
2	Public Blockchain	Poor: Anyone is welcome to sign up and take part	Proof-of-Work (PoW): Excessive consumption of power and sluggish operation speeds	Excessive: It can scale indefinitely in theory, although there may be resource limitations	Poor: each exchange is available to everyone.
3	Consortium Blockchain	Combination: Authorized entry through a group of reliable groups	Compared to BFT, Practical Byzantine Fault Tolerance (PBFT) is faster but necessitates greater node confidence	Mild: As the number of nodes increases, scalability improves but may be constrained by consortium size	Variable: Privacy is dependent upon consortium regulations and particular implementation
4	Private Blockchain	Confidential: Access permissions limited to a particular group or enterprise.	Proof-of-Authority (PoA) is dependent on reputable validators and is quicker and more economical than proof-of-work (PoW).	Strong - Sustainability relies on the system's architecture and consensus process selected	Excessive: Within the community, every transaction is confidential

7.8 CONCLUSION

This chapter demonstrates the use of Blockchain technology and Artificial Intelligence (AI) in military applications is an important advance that is going to have a significant effect on defense systems. The applications that are currently in use show improved security, efficiency, and decision-making skills, enabling military forces to adjust to the changing face of contemporary conflict. The combination of AI and Blockchain solves important issues in military operations by offering a strong framework for data integrity, transparency, and decentralized control.

The route that this union will go in the future seems very promising. AI algorithms will keep developing, getting more complex and adaptable, allowing military systems to learn on their own and react to changing threats. Assuring the integrity and traceability of data across military networks, blockchain's importance will grow. This chapter also provides a comprehensive study of various Artificial Intelligence and Blockchain methodologies that can be used for military applications.

ACRONYMS AND ABBREVIATIONS

AI	Artificial Intelligence
ITS	Intelligent Transport Systems
NATO	North Atlantic Treaty Organization
PoW	Proof of Work
DARPA	Defense Advanced Research Projects Agency
UAV	Unmanned aerial vehicle
SVM	Support Vector Machine
BFT	Byzantine Fault Tolerance
PBFT	Practical Byzantine Fault Tolerance
PoA	Proof-of-Authority
DT	Decision Trees
DLT	Distributed Ledger Technology
DoD	Division of Defence
US	Unites States
IoT	Internet of Things
BTC	Bitcoin

REFERENCES

1. Vitu, J.C. (2020). Supply Chain et blockchain: un cyber-mariage parfait. *Economie Matin*, published on February 23, 2020, accessible at: <http://www.economiematin.fr/news-blockchain-supply-chain-technologie-developpement-securite-vitu>; <https://finabel.org/wp-content/uploads/2020/09/FFT-Blockchain.pdf>

2. Wood, M. (2019). South Korea reveals blockchain plans for defense and arms procurement. Ledger Insights, published on August 2019, accessible at: <https://www.ledgerinsights.com/south-korea-blockchain-defense-arms-procurement/>
3. Young, L. and Desai, J. (2020). Blockchain's Promise for Defense Agency Supply chains, Chains. *Booz Allen*. Available at: <https://www.boozallen.com/s/insight/blog/blockchain-promise-for-defense-agency-supply-chains.html> [last accessed on 13/03/2020]
4. <https://www.nationaldefensemagazine.org/articles/2019/12/11/blockchain-tech-has-numerous-applications-for-defense>
5. <https://www.analyticsinsight.net/how-blockchain-is-being-used-by-global-militaries/>
6. McAbee, A. S. M., Tummala, M., and McEachen, J. C. Military Intelligence Applications for blockchain Technology. *Proceedings of the 52nd Hawaii International Conference on System Sciences* (2019), pp. 6031–6040. <https://www.c4isrnet.com/opinion/2021/04/05/america-and-its-military-need-a-blockchain-strategy/>
7. <https://www.pwc.com/gx/en/aerospace-defence/pdf/blockchain-defence.pdf>
8. <https://www.linkedin.com/pulse/how-blockchain-transforming-defence-prashant/>
9. <https://www.sciencedirect.com/science/article/pii/S0360835220306525>
10. <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9654201>
11. Artificial Intelligence the Frontline of a New Age in Defense. (n.d.). Breaking Defence. Available at [https://cdn2.hubspot.net/hubfs/2097098/MCM120_BreakingDefense_AI_ebookR1%20\(1\).pdf](https://cdn2.hubspot.net/hubfs/2097098/MCM120_BreakingDefense_AI_ebookR1%20(1).pdf), accessed on 29 March 2021.
12. European Commission. (2020). European Defence Industrial Development Programme (EDIDP). Available at: https://ec.europa.eu/research/participants/data/ref/other_eu_prog/edidp/wp-call/edidp_call-texts-2020_en.pdf, accessed on 09 April 2021.
13. Asharaf, S. and Adarsh, S. (2017). *Decentralized Computing using Blockchain Technologies and Smart Contracts: Emerging Research and Opportunities*, IGI Global, Hershey, Pennsylvania.
14. Lian, L., and Zhu, Q.C. (2019) *Decoding military applications of blockchain technology*. PLA Dail.
15. Huang, L.J., Wu, S.J., and Cao, F., et al. (2018) *Blockchain Security Technology Guide*. China Machine Press.
16. Lu, Y., Huang, X., Dai, Y., Maharjan, S., and Zhang, Y. (2019). Blockchain and federated learning for privacy-preserved data sharing in industrial IoT. *IEEE Transactions on Industrial Informatics*. <https://doi.org/10.1109/TII.2019.2942190>
17. Sharma, P. K., Kumar, N., and Park, J. H. (2020). Blockchain technology toward green IoT: Opportunities and challenges. *IEEE Network*. <https://doi.org/10.1109/MNET.001.1900526>
18. Yang, W., Aghasian, E., Garg, S., Herbert, D., Disiuta, L., and Kang, B. (2019). A survey on blockchain-based internet service architecture: requirements, challenges, trends, and future. *IEEE Access* 7, 75845–75872. doi:10.1109/ACCESS.2019.2917562
19. Johnson, J. (2019). Artificial intelligence & future warfare: implications for international security. *Def. Secur. Anal.* 35, 147–169.

20. Pietrow, D. and Matuszewski, J. (12–14 September 2017). Objects detection and recognition system using artificial neural networks and drones. In *Proceedings of the 2017 Signal Processing Symposium (SPSymposium)*, Jachranka, Poland.
21. Sanchez, S.L. Artificial intelligence (AI) enabled cyber defence. Available online: [https://www.eda.europa.eu/webzine/issue14/cover-story/artificial-intelligence-\(ai\)-enabled-cyber-defence](https://www.eda.europa.eu/webzine/issue14/cover-story/artificial-intelligence-(ai)-enabled-cyber-defence) (accessed on 25 January 2021).
22. Mahesh, N. and Prasanna, Keshava. (2023). A brief survey on lung cancer detection using conventional neural network. *European Chemical Bulletin* 12 (Special Issue-8), 14.
23. Anand Babu, J. and Suhas, K. C. (2023). A survey on diabetes prediction models using machine learning over big data. *European Chemical Bulletin* 12 (Special Issue-8): 14.
24. Fan, M. and Zhang, X. (2019). Consortium blockchain-based data aggregation and regulation mechanism for smart grid. *IEEE Access* 7, 35929–35940.
25. Feng, J., Richard Yu, F., Pei, Q., Chu, X., Du, J. and Zhu, L. (2020). Cooperative computation offloading and resource allocation for blockchain-enabled mobile-edge computing: a deep reinforcement learning approach. *IEEE Internet of Things Journal* 7(7), 6214–6228.
26. Tanwar, Sudeep, Bhatia, Qasim, Patel, Pruthvi, Kumari, Aparna, Singh, Pradeep Kumar, and Hong, Wei-Chiang. (2019). Machine learning adoption in blockchain-based smart applications: the challenges, and a way forward. Dec. 2019/ACCESS.2019.2961372
27. Bhardwaj, Rohan, Nambiar, Ankita R., and Dutta, Debojyoti. (2017). A Study of Machine Learning in Healthcare. *2017 IEEE 41st Annual Computer Software and Applications Conference*.
28. Suzuki, K. (2017). Overview of deep learning in medical imaging. *RadiolPhysTechnol* 10(3), 257–273. doi: 10.1007/s12194-017-0406-5
29. Nalepa, J. and Kawulok, M. (2019). Selecting training sets for support vector machines: A review. *ArtifIntell Rev* 52, 857–900. doi: 10.1007/s10462-017-9611-1
30. Zheng, Z., Xie, S., Dai, H.N., and Wang, H. Blockchain challenges and opportunities: A survey. Available online: <http://inpluslab.sysu.edu.cn/files/blockchain/blockchain.pdf> (accessed on 10 February 2019)

Integration of AI principles with blockchain technology for effective applications

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8.1 INTRODUCTION

Technology is evolving exponentially and in the midst of this, two technologies stand out as disruptive forces, the Artificial Intelligence (AI) and Blockchain.

AI works to imbibe computers with intelligence and has contributed in all of the sectors starting from retail industry to healthcare to military. With the incept of massive language models and generative AI [1], it has redefined the manner we approach and tackle problems with smart decisions.

On the alternative hand, blockchain technology is now creating IT revolution and is being considered as the fifth pillar [2]. Blockchain is all about storing data in decentralized manner with distributed networks. It has revolutionized as such that the blockchain is owned and monitored by everyone but controlled by none. It has found its applications ranging from bitcoins to accountancy [2] to marketing [3, 4], supply chain and ultra-modern applications in innovative industries [5, 6] too. The question of do we need blockchain, is very well addressed in depth using three use cases - supply chain management, payments within interbank, and Decentralized Autonomous Organizations in [7].

However, the synergy of AI and blockchain brings out genuine capability of innovation. The powerful amalgamation of these 2 disruptive technologies has given birth to a disruptive synergy known as Intelligent Blockchain or Smart Blockchains. It will reshape the way we work, interact and live [8]. The amalgamation of AI and blockchain is trusted to make numerous potentialities. Such strategies offer scientists and authorities with a greater accuracy [9]. The amalgamation of these two technologies is not only a collaboration of equipment and instruments, however a fruitful synthesis that holds the promise of reshaping the society and industries.

The importance of this convergence extends beyond the limitations and challenges faced by these individual technologies. It bridges the gap and makes a strategic move that unlocks unparalleled opportunities for innovation. The convergence of AI with blockchain technology has significantly improved the traditional blockchain models with powerful added features

[6]. The fundamental principles of AI can further strengthen the competences of blockchain and has found applications in healthcare and agriculture [10], smart towns [11], business innovation [12] and many others.

Overall, the applications of AI can be a boon to blockchain technology to be an effective tool to address real life privacy, transparency, back tracking and tampering issues.

As we move into this chapter, in section II we see the impact of synergizing AI and blockchain, in section III, we will take a look at its implications through numerous lenses by taking 4 use cases such as healthcare, agriculture, financial sector and finally military sector. For each of the use case the structure followed is first to explain the blockchain aspects, then AI followed by the impact of synergy with example applications. This section is followed by a conclusion and reference sections.

8.2 THE SYNERGY OF AI AND BLOCKCHAIN TECHNOLOGIES

The synergy of AI and blockchain has given birth to the intelligent blockchain. The inherent blockchain technology itself has multiple advantages and has revolutionized many industries with its core principles of security, scalability and transparency through its immutable, decentralized and distributed network. In spite of these advantages, the blockchain technology do face certain challenges with regards to costs, security and scalability. It is found that when the blockchain is governed by AI or if AI capabilities are utilized to build blockchain, it is bound to deliver the world with an automated and optimized way of content and economy sharing platforms. AI has impacted and enhanced the core concepts of blockchain as follows:

- **Increased Security:**

Blockchain Benefits: Blockchain provides inherent security because of its decentralized feature and immutable ledger.

AI enhancement: AI can further improve security by using Intelligent anomaly detection algorithms by monitoring the network in real time for any unusual activity or attacks.

- **Increased scalability:**

Blockchain Benefits: Blockchain are highly scalable, however it has been a challenge for some blockchain networks.

AI enhancement: AI can help streamline resource allocation and network management, making blockchain systems more scalable.

- **Effective data analysis:**

Blockchain Advantages: Blockchain can effectively can store and maintain massive data in the form of records.

AI enhancement: AI can do predictive analysis on the huge data involved in blockchain, offering useful insights and decisions.

- **Creating Smart Contracts:**

Benefits of Blockchain: Blockchain supports in implementation of smart contracts for automated and trustless transactions.

AI enhancement: AI can improve the performance and flexibility of smart contract execution by generating even better intelligent decisions based on real-time data and peripheral events.

- **Enhanced Tokenomics:**

Tokens are a fundamental issue in blockchain ecosystems, which gives blockchain advantages.

AI Enhancement: AI-powered models can design tokenomics and crypto-economics that correspond with project goals and incentivize ideal behaviors such as governance participation.

- **Interoperability:**

Blockchain Advantage: Blockchain networks can be remoted from one another, limiting interoperability.

AI enhancement: AI can facilitate interoperability via translating records and transactions between distinctive blockchain networks, enabling seamless go-chain communicate.

The combination of blockchain with AI provides synergies that can result in more stable and reasonable decentralized packages and structures. However, in order to fully reap these benefits, it is critical to carefully analyze the precise use case, implementation, and ethical considerations while integrating those technologies

8.3 USE CASES

The Combination of AI and Blockchain leads to many applications in various sectors, each bringing unique advantages. Foreach of the use cases, we first detail on blockchain aspects, then AI aspects and then finally how the synergies can impact and in what ways. Here are some of the key use cases:

8.3.1 Healthcare sector

Blockchain and AI together have the required potential to renovate the healthcare sector by improving data integrity, efficiency, data security, record-sharing, and patient enrolment.

A. Blockchain in healthcare

Blockchain provides a safe and distributed platform for storing and managing healthcare data. For example, “Connecting Care”, is a blockchain-based record-sharing platform used to safeguard hospital with their medical records data. It includes an access control list that guarantees approved users to have access to the data. Clinical studies and medications can also be tracked using blockchain. It can provide

consistent, secure, and scalable storage and retrieval of medical lessons, which can aid in the identification and treatment of illnesses. In general, blockchain is utilized in five key extents if health sector [13] as follows.

- Blockchain technology has the potential to modernize interoperability and facilitate seamless data sharing within the electronic health records domain. By creating a framework to interconnect diverse personal records, blockchain provides a solution to the challenge of connecting heterogeneous data. Moreover, it can enhance data sharing by providing direct connections between data owners and consumers, thereby enhancing efficiency and transparency in the industry.
- Blockchain in the drug supply chain ensures transparency by creating a traceable record of each product's origin and journey. This effectively builds trust among stakeholders as they can track the movement of drugs accurately, mitigating the risk of counterfeit products infiltrating the market.
- For health insurance, blockchain can simplify the procedure of managing the insurance claims and assist stakeholders to estimate medical reimbursements.
- For genomics, blockchain provides a secure and auditable way of sharing genomic data by connecting data owners and buyers and directly.
- For consent management, blockchain's immutable ledger offers all network members a transparent and timestamped record of consent, fostering trust and accountability in the process. This ensures that each consent instance is securely documented and cannot be altered, boosting transparency and integrity.

B. *AI in healthcare*

AI, can be used to learn from data, understand it, and draw conclusions. AI is utilized in drug discovery and disease research. AI is mostly used to detect variances in scan reports, which AI can do at least as well as people and ensure a higher degree of personalized therapy [14].

C. *AI and blockchain synergy in healthcare*

The integration of Blockchain and AI can upshot in a secure, immutable, decentralized method for storing delicate data. Blockchain technology acts as a repository of data with high integrity and robustness that cannot be tampered with. Machine learning algorithms perform better when this data is obtained from a safe and trusted platform, such as blockchain Combining smart contracts and machine learning in healthcare can enable professionals to access patient records securely. This fusion of technologies enhances service efficiency, lowers costs, and democratizes healthcare by leveraging advanced algorithms and vast data sets for decision-making. Blockchain AI framework is proposed for healthcare records management [15].

In addition to significantly transforming the healthcare segment, the merging of AI and blockchain addresses various challenges toward enhancing the privacy, efficiency, and interoperability of health data. Some of such challenges on which the combined values of AI and blockchain can impact healthcare are:

Data security and privacy:

Blockchain: Offers a tamper-resistant ledger that can certify the security, privacy and integrity of health data. Patient data and records would be stored on a blockchain, and access can be controlled through cryptographic keys.

AI: Utilizes advanced encryption techniques and secure channels for data processing. AI algorithms could be applied to analyze health data while preserving patient privacy.

Interoperability:

Blockchain: Facilitates a unified and standardized system for storage and sharing health data across various healthcare providers and systems.

AI: Helps in extracting meaningful insights from diverse and interoperable datasets, leading to more comprehensive and accurate analyses.

Data integrity and traceability:

Blockchain: Makes all record of transactions immutable, ensuring the traceability and provenance of health data. This is crucial for maintaining the reliability and accuracy of patient records.

AI: Can help identify patterns and anomalies in data, ensuring that the data of records stored on the blocks of blockchain is accurate and up-to-date.

Clinical trials and research:

Blockchain: Simplifies clinical trial data management while assuring transparency and traceability. Clinical trial agreements can be automated and enforced using blockchain smart contracts.

AI: Speeds up drug research and development by processing massive amounts of biomedical data. AI systems can more efficiently identify possible clinical trial candidates.

Pharmaceutical supply chain:

Blockchain improves the traceability and transparency by minimizing the problem of counterfeit medications and ensuring the supply chain's integrity.

AI: Increases supply chain efficiency by forecasting demand, improving logistics, and addressing supply chain issues.

Claims and billing processes:

Blockchain: Streamlines and automates the claims and billing processes by creating a transparent and auditable record of transactions.

AI: Could be used to analyze claims data for fraud detection, improving the accuracy and efficiency of the billing system.

Patient-generated data:

Blockchain: Allows patients to own their own health data and securely share it with healthcare professionals. Smart contracts enable patients to authorize certain pieces of their data access.

AI: Analyzes data provided by patients, such as wearables and IoT devices, in order to deliver individualized insights and better preventive treatment.

Tokenization of healthcare assets:

Blockchain: Allows for secure and transparent transactions by tokenizing healthcare assets such as patient data or research findings.

AI can be utilized evaluate patterns and trends in tokenized healthcare data, giving significant insights for research and tailored medicine.

Combined strengths of blockchain and artificial intelligence could successfully handle these encounters, and can overcome data silos, enhance security, and foster innovation, ultimately improving patient care. However, there are yet to be addressed challenges: In terms of scalability [16], the medical records and data are growing exponentially and the size of the blockchain grows equally. This must be handled in the future needs. In terms of security [17] sharing of healthcare data and substantiation of medical records is still a matter of challenge. Blockchain requires high bandwidth for its smooth functioning and sometimes IoT devices are required which are computationally limited. These challenges with regards to environment and architectures also need to be addressed in the future

8.3.2 Agriculture sector

The agriculture sector has seen substantial progressions with the amalgamation of technology, leading to the advent of AgriTech (agricultural technology). AgriTech leverages innovations in areas such as AI and blockchain.

8.3.2.1 Blockchain in agriculture

Blockchain is utilized at several phases of the agriculture lifespan for better progression. In the agriculture ecosystem, the crucial data such as farm status, inventories, contracts, and management are stockpiled in the blockchain. This way of storing data empowers farmers to achieve a decentralization and automation [18]

8.3.2.2 AI in agriculture

Utilization of AI in agriculture [19] is helping in evolving agricultural methods capable of increasing crop yield and reducing the traditional challenges.

The application of AI is found in digital agriculture, precision agriculture and using machine learning for detecting diseases, weeds etc.

8.3.2.3 AI and blockchain synergy in agriculture

The use of blockchain and AI in agriculture can provide various advantages. Smart farming integrated with food industry have been implemented with blockchain and AI [20].

8.3.2.3.1 Precision agriculture

Artificial intelligence (AI) can be used to evaluate data from numerous sources (such as satellite imaging, weather and soil data) to give agriculturalists with insights about optimal planting times, watering schedules, and fertilizer usage. This can result in higher agricultural yields and less waste.

8.3.2.3.2 Smart contracts

Blockchain's smart contracts can automate various processes in agriculture, such as payments to farmers, distribution of crops, and insurance claims.

AI:AI can help in creating more complex and intelligent smart contracts based on various conditions.

8.3.2.3.3 Farm management

AI can examine data received from IoT devices installed in farms to monitor crop health, soil conditions, and weather patterns. This can aid agriculturalists in making knowledgeable decisions and can also help in predicting any problems that might affect the crops.

8.3.2.3.4 Food safety and quality control

Blockchain can be utilized to trace foodstuff from farm to consumer, guaranteeing its safety and quality.

AI can aid in the analysis of blockchain data to spot abnormalities and ensure food safety requirements.

While blockchain and artificial intelligence have the potential to change agriculture, there are significant hurdles and restrictions to their use:

- **Adoption Roadblocks:** One of the greatest important problems is a lack of understanding and mindfulness of blockchain and AI and their applicability in agriculture. This lack of understanding can stymie their uptake. If policymakers, farmers, and other stakeholders do not fully understand the paybacks of these technologies.
- **Data Security:** Because blockchain technology is heavily reliant on the Internet of Things (IoT) in agriculture, it is subject to data security

risks. Data security in a blockchain system is critical, and any leak might damage trust in the system.

- **Technical Difficulties:** Implementing blockchain and AI in agriculture can be technically difficult. This involves data integration difficulties, system compatibility, and the requirement for advanced algorithms to evaluate and comprehend the huge amounts of data created by these technologies.
- **Regulatory and Compliance Issues:** The adoption of blockchain and AI in the agriculture sector can also face regulatory and compliance issues. These technologies may not be fully understood or regulated by existing laws and regulations, which can create uncertainty and potential legal risks.
- **Cost:** Implementing blockchain and AI technologies can be expensive. This includes the cost of hardware, software, and the skills required to implement and manage these systems. This cost can be a barrier to adoption, particularly for small and medium-sized enterprises.

To summarize, while blockchain and artificial intelligence have the potential to provide various benefits to the agricultural sector, their implementation is not without hurdles.

8.3.3 Financial sector

Artificial intelligence (AI) and blockchain technologies are both contributing significantly to finance, and their combination has the potential to create more advanced and efficient financial systems.

8.3.3.1 Blockchain in financial sector

Blockchain technology has the potential to enhance financial sector offering various advantages in cryptocurrencies, smart contracts, trade finance, identity management, etc. Built on blockchain technology, decentralized finance (DeFi) systems provide a variety of financial services such as lending, borrowing, and asset management without the need for traditional intermediaries such as banks.

8.3.3.2 AI in financial sector

AI has had a significant impact on financial sector, transforming various aspects of the industry by increasing productivity, improving decision-making and enabling innovation.

8.3.3.3 AI & blockchain synergy in financial sector

Artificial Intelligence together with Blockchain technology are transforming the capabilities and services of finance companies by improving the trust

factors, eliminating the resistance from multiparty collective transactions, and reducing the time taken for processing the transactions through AI-based automation. This includes all the processes from opening the account, debit, credit, all types savings, insurances etc. to closing the account.

Throughout various processes, applicants authorize access to their personal data stored on blockchain, fostering trust in data and automated evaluation processes that expedite application approvals and enhance customer satisfaction. Collective trust in a transparent and secure view of truth on blockchain holds promise in eradicating crimes, scams, and fraud cases entirely.

Leading financial institutions are spearheading blockchain adoption, collaborating to eliminate longstanding barriers, innovate new solutions, and deliver tangible business benefits. The convergence of blockchain with artificial intelligence (AI) unlocks a realm of possibilities, particularly in finance, by providing a secure, decentralized platform for managing vast datasets and automating decision-making processes.

AI enriches blockchain by embedding decision-making capabilities into autonomous processes, enabling rapid analysis of customer data for unbiased decisions and predictions. Conversely, blockchain fortifies AI by bolstering data privacy and security, reducing risks of data manipulation and enhancing trust in AI systems.

Together, these technologies bolster fraud detection, risk assessment, and customer identification while optimizing inventory through predictive analytics and generating new revenue streams. CFOs stand to benefit from improved operational efficiency, accuracy, data security, and privacy facilitated by the fusion of AI and blockchain, addressing a myriad of challenges they encounter. In the financial services industry, blockchain and AI are transforming various aspects and are as follows

8.3.3.3.1 Encryption

Blockchain Advantage: Blockchain affords transparency, however the same transparency can offer a threat for the reason that all data is visible to members.

AI Improvement: AI is used to achieve calculations on encrypted data (homomorphic encryption) or to implement zero-expertise proofs, warranting privacy of data while allowing AI-driven review and processing.

8.3.3.3.2 Trade finance

Blockchain technology can reduce extensive paperwork by digitalizing trade documents, reducing slow and cumbersome processes by modernizing processes, and provide a tamper-proof setting for making secure transactions.

8.3.3.3.3 *Asset management*

Smart contracts are programmable automated treaties that execute actions when specific conditions are met, streamlining tasks like trade clearance and settlement in asset management. Tokenization involves transforming real assets into digital tokens on a blockchain, which boosts liquidity, enables fractional ownership, and broadens access to investment options.

8.3.3.3.4 *Insurance*

Storing policy and claims information on a blockchain platform provides insurers a transformative opportunity to streamline their operations. Through the implementation of smart contracts, insurers can automate various aspects of the claims process, from verification to payout, resulting in expedited claim resolutions and minimized administrative overheads. This automation not only accelerates the pace at which policyholders receive their payouts but also significantly reduces the costs associated with manual claims processing.

Moreover, leveraging blockchain technology allows insurers to access a wealth of verifiable data, enabling them to conduct more accurate risk assessments and pricing strategies. By tapping into this extensive pool of data stored securely on the blockchain, insurers can gain deeper insights into customer behavior, claims history, and market trends. This enhanced understanding of risk factors empowers insurers to make more informed underwriting decisions, thereby improving their overall profitability and sustainability.

In summary, the integration of blockchain in insurance operations offers tangible benefits such as faster claim processing, cost savings, and improved risk assessment capabilities. By embracing this technology, insurers can enhance customer satisfaction, optimize their processes, and stay competitive in a rapidly evolving market landscape.

8.3.3.3.5 *Forensic investigations of financial crimes*

Blockchain technology can also help in forensic investigations of financial crimes. Blockchain can provide a credible trail of evidence for fraudulent activity such as embezzlement or money laundering by maintaining an immutable record of all transactions. This can be very helpful in the investigation and prosecution of financial crimes further AI can make appropriate predictions.

In summary, the amalgamation of blockchain and AI ML supports a tool for enhancing fraud exposure and risk valuation in the finance companies. By providing a secure, transparent, and tamper-resistant platform for transactions and by using AI to monitor transaction history and identify

anomalies, financial institutions can significantly improve their fraud detection and risk assessment capabilities.

8.3.4 Military sector

In these years, the rapid advancements in AI and blockchain have unlocked new potentials for a diverse range of industries. Realm of defense and military operations is one such major area in which these technologies together are capable of making a significant impact. As nations around the world seek to maintain their security and protect their interests, the integration of blockchain and AI into defense strategies could revolutionize the manner military operations are executed and help ensure a more secure future.

8.3.4.1 Blockchain in military sector

Blockchain has found itself as viable solution in many of the military aspects, ranging from qualifying cyber security susceptibilities to managing battlefield logistics and supply chain [21]. Blockchain technology has also been used as a viable solution for ensuring integrity and attribution of data to support military maneuvers using networks and maintaining sustainability of these networks [22]. A study on understanding the military application potential of blockchain technology in order to improve combat effectiveness of the army and promote its transformation and development has been conducted effectively and proved [21].

8.3.4.2 AI in military sector

AI has already begun to transform the way militaries operate, with its ability to process massive volumes of information and make choices. It has makes decisions with great speed and accuracy that far surpasses the human capabilities. This has led toward building the advanced independent systems, such as drones and robots, which could be used in the field to perform the tasks which would otherwise be too dangerous or time-consuming for human soldiers. These advanced systems could be efficiently programmed to perform a broad range of functions, from reconnaissance and surveillance to combat and logistics support, allowing military forces to operate more effectively and efficiently.

8.3.4.3 AI and Blockchain synergy in military sector

Witnessing the prowess of individual technology, one can imagine how effective it shall be when both technologies are utilized in synergy. Blockchain and AI have been used in tandem to support in defense management and police organizations [23]. AI has been widely used in intelligent automation, however with vulnerability in security and trust [24]. Blockchain helps in such case to alleviate such risks. Further, AI can enhance blockchain technology in military applications in several ways as follows

8.3.4.3.1 *Enhancing security and cybersecurity*

AI can be used to enhance the security of blockchain systems in military applications. By analyzing data from various sources, AI algorithms can identify potential security threats and take appropriate mitigation measures. This can help prevent cyberattacks and espionage, which are important concerns for military operations.

- AI for predictive defense: AI can assess probable security risks in blockchain systems, allowing positive measures to be taken before the attack to include enforcement predict potential cyberattacks, identify weaknesses in the system and determine appropriate security measures to include
- AI for Threat Identification: AI can be used to investigate data from various sources to classify possible security threats in real time. This may include detection of unusual activity, identification of potential hackers.
- AI for anomaly detection: AI can be used to classify anomalies in blockchain systems that can indicate security threats. This data may include identifying unusual patterns, identifying potentially fraudulent transactions, and issuing alerts to appropriate personnel.
- Build a decentralized cybersecurity intelligence sharing platform using blockchain and AI.
- Integrating AI and blockchain for continuous monitoring for tamper resistant logs to ensure the veracity of military system components.

8.3.4.3.2 *Enhancing supply chain management*

AI is utilized to enhance supply chain management in military operations [25] by using blockchain to trail and verify the movement of items. This ensures the integrity of supply chains and prevent unauthorized access to sensitive information. Overall, the AI-driven blockchain solutions can optimize military logistics, enhance supply chain security, and ensure the efficient distribution of resources in defense operations.

- AI for Demand Forecasting: AI can be used to forecast future demand for sure sources, optimizing the usage of assets, and reducing waste. Thereby, bearing in mind extra green supply chain management. This can encompass predicting future call for positive demands, optimizing the use of resources, and lowering waste.
- AI for Inventory Management: AI is utilized to manipulate inventory greater efficaciously, lowering the hazard of stockouts or overstocking. It can balance the production and distribution of army equipment.
- AI for Supply Chain Optimization: AI can be used to enhance the deliver chain routes, enhancing efficacy and lowering expenses. This

can include optimizing using assets, lowering waste, and enhancing the overall effectiveness of the supply chain.

- Implementing smart contracts for computerized verification of the authenticity of military components inside the logistics chain.
- Integrating blockchain for stable and transparent economic transactions within army logistics structures.

8.3.4.3.3 Improving efficiency and decision-making in battle management

AI is utilized to improve the competence and decision-making capabilities of blockchain systems in military operations. By processing large amounts of data and making decisions with speed and accuracy far beyond human capabilities, AI can help streamline operations and make more informed decisions in real-time.

- AI for data analysis: AI can be used to analyze large amounts of data from blockchain systems, enabling more informed decisions. This includes analyzing interaction data, identifying patterns, and providing understandings that can assist decision-making.
- AI for predictive analytics: AI is utilized to predict future trends based on data from blockchain systems. This can include predicting future demand for certain products, identifying potential disruptions in the supply chain, insights that can guide decision making and exploring the use of AI for predictive analytics to monitor change a robust approach in combat situations, including blockchain to share secure data between commanders
- Implementing Smart Contracts for Automated Transparent Decision Processes in the Military Command System.
- Integrating Blockchain and AI for Adaptive Decision Support

8.3.4.3.4 Blockchain and AI in military training simulations

The use of AI and blockchain technology in military training simulations, ensures a safe and authentic virtual environment for training exercises.

- Using AI-powered virtual agents in military training simulations, with blockchain for safe and transparent operational control.
- Use of blockchain to maintain decentralized and non-transformative records of individual and collective performance in military training exercises.
- Implementing smart contracts for scoring and feedback systems used in AI-enhanced military training.

- Use AI for dynamic scenario generation in military training simulations, and blockchain for secure version control of simulation environments.
- Blockchain integration to create a decentralized platform for collaborative AI-enhanced military training scenarios.
- Adaptive systems to dynamically changing battlefield conditions.

8.3.4.3.5 Enhancing communication networks

AI is utilized to enhance communication networks in military applications Blockchain is utilized to create encrypted messaging systems that are nearly impossible to intercept or tamper with This can help ensure the secrecy of information need to be confidential and secure.

- AI for encryption: AI can be used to improve the encryption of networks, making them more secure. This may include implementing advanced encryption algorithms, identifying potential security threats and taking appropriate actions to mitigate.
- AI for Data Encoding: AI is utilized to encode data in a secure and efficient manner, so that it becomes difficult for unauthorized persons to intercept or alter data.
- AI for network management: AI can be used to better manage communication networks, improving efficiency and reducing costs. This may include resource efficiency, waste reduction, and improvements throughout the network.
- Using Smart Contracts to Implement Secure Key Management Processes in Military Communications Systems.
- Implement a decentralized authentication system using AI for secure military communication channels.
- Using Blockchain to Create an Immutable Record of Transaction Transactions for Analysis and Accountability.

8.3.4.3.6 Enhancing autonomous systems

AI is utilized to improve the efficiency of autonomous systems in military operations. The utilization of AI to control these systems will enable armies to operate more efficiently and effectively, accomplish tasks that would otherwise be too dangerous or time-consuming for human forces, and respond more quickly and precisely to emerging threats.

- AI for autonomous control: AI can be used to control autonomous systems, making them more efficient and effective. This may include control of autonomous vehicles, control of autonomous weapon systems, and to guide the actions of automated devices.

- AI for autonomous decision-making: AI is utilized for autonomous decision-making, enabling the autonomous system to operate more efficiently and effectively. This may include deciding on the best strategy, determining how best to use resources, and guiding autonomous systems actions,
- AI for autonomous learning: AI can be used to enable autonomous systems to learn and adapt, improving their performance over time. This can include learning from past experiences, adapting to new circumstances and improving their performance in the long run.

8.3.4.3.7 Blockchain-AI paradigm for military intelligence data handling

1. The combination of AI and blockchain [26, 27] is changing how military intelligence data is used, solving privacy challenges, and ensuring the secrecy of sensitive information.
2. Using blockchain for secure and auditable storage of military intelligence data with AI-powered access controls.
3. Use of AI algorithms for real-time analysis of scientific data, including blockchain for transparent sharing between authorized entities.
4. Implementing smart contracts to automate data-sharing agreements and access to military intelligence networks.
5. Using AI to detect anomalies in intelligence data to identify potential security threats, including blockchain for data management with security.
6. Blockchain integration for decentralized and robust military intelligence data storage and analysis.

8.3.4.3.8 Federated learning and blockchain in military surveillance systems

The use of federated learning integrated with blockchain to enhance privacy in military surveillance systems, allows collaborative model training without compromising sensitive data [28]. Federated learning can be used as follows

- Implementing federated learning for collaborative AI model training among military surveillance devices, with blockchain for data aggregation.
- Applying blockchain to guarantee the veracity and authenticity of surveillance data collected by distributed AI-powered sensors.
- Utilizing federated learning to enhance AI-based object recognition in military surveillance, with blockchain for privacy-preserving data sharing.
- Implementing smart contracts for transparent and automated data-sharing agreements among federated learning participants in military surveillance networks.

8.3.4.3.9 Unmanned aerial vehicles (UAV) management

Military personnel may be in danger of losing their lives in the complex and challenging modern warfare. The gathering of mission-critical information to guide tactical choices is an essential component of military operations that are successful. UAV's are vital for obtaining reconnaissance information from enemy positions and giving friendly forces vital situational awareness. But it is crucial to protect this information from possible hackers who might use it against you. Furthermore, traditional communication networks are hampered by the enormous amount of data that UAVs gather, including high-definition photos and surveillance videos. Therefore, the synergy of AI and blockchain can bring in a lot of benefits to the highly needed UAV's.

The role of AI and blockchain in securing UAVs, addressing privacy concerns, and fortifying the overall cybersecurity framework for unmanned military vehicles can be as follows.

- Use Case 1: Implementing blockchain for secure and transparent storage of UAV flight data with AI-driven anomaly detection.
- Use Case 2: Using AI algorithms for real-time threat detection during UAV missions, with blockchain for secure communication.
- Use Case 3: Applying blockchain for control and coordination of multiple UAVs in military operations.
- Use Case 4: Utilizing AI for adaptive encryption of UAV communication channels with blockchain as a secure communication infrastructure.
- Use Case 5: Integrating blockchain for immutable records of UAV maintenance and software updates to ensure operational integrity.
- **Block-USB:** Block0USB [29], is a blockchain and machine learning (ML) based secure and intelligent UAV communication system built to run in a 6G network [30], as a novel solution to these problems. Block-USB addresses the data scalability issues while guaranteeing the integrity and confidentiality of extremely sensitive military operations data collected by UAVs. It uses machine learning and blockchain to produce an intelligent communication framework for unmanned aerial vehicle operations. The essential elements of Block-USB are as follows:
 - **Blockchain Protection:** Block-USB's blockchain component makes sure that UAV data is stored securely and irreversibly. It protects the obtained data from unwanted access and manipulation by utilizing decentralized and tamper-resistant ledger technology. Data integrity is maintained by the ease with which attempts to modify the data can be identified.
 - **Machine Learning:** To enhance the intelligence of UAV operations, ML algorithms are fused into the system [31]. ML models can help with threat detection, decision-making, and real-time data analysis, which increases the system's adaptability and responsiveness to changing operational environments.

- **6G Network:** Block-USB is made to function in 6G networks, which provide noticeably faster data rates, reduced latency, and improved connectivity. This is essential to guaranteeing data transfer and real-time communication for UAVs operating in the field.
- **Ensuring the security of UAV-Based Military Operations:** A case study that inspects multiple situations involving malevolent controllers or UAVs is used to assess the suggested system. Measures of classification accuracy, latency in communications, and scalability of data is utilized to assess Block-USB's performance. Block-USB continuously exhibits superior performance with regards to data security, system intelligence, and operational efficiency in comparison to non-IPFS-based solutions and traditional baseline 4G/5G systems in the evaluation.

In summary, Block-USB is an important development for protecting and enhancing UAV- based military operations. This system protects the confidentiality, integrity, and scalability of mission-critical data by fusing blockchain technology, machine learning, and 6G network capabilities. Solutions such as Block-USB are essential for protecting critical military data and improving the efficiency of unmanned aerial reconnaissance operations in a rapidly changing modern warfare scenario.

8.4 CONCLUSION

The convergence of AI with blockchain technology has significantly improved the traditional blockchain models with powerful added features. The synergies offer an, efficient, and intelligent decentralized applications and systems. However, it's important to carefully consider the specific use case, implementation, and ethical considerations when integrating these technologies to realize these advantages fully. Overall, applications of AI can be a boon to blockchain technology to be an effective tool to address real life privacy, transparency, back tracking and tampering issues. This chapter serves as a manual through the complicated interaction of AI and blockchain, unraveling the capability advantages, demanding situations, and ethical concerns associated with this transformative convergence. We are trying to pave the manner of how AI can improve the competences of blockchain by means of taking various use instances of agriculture, healthcare. Finance and military applications.

The synergies offer an, efficient, and intelligent decentralized applications and systems. However, it's important to carefully consider the specific use case, implementation, and ethical considerations when integrating these technologies to realize these advantages fully.

REFERENCES

1. Chui, Michael, Lareina Yee, Bryce Hall, and Alex Singla. "The state of AI in 2023: Generative AI's breakout year." (2023).
2. Han, Hongdan, Radha K. Shiwakoti, Robin Jarvis, Chima Mordi, and David Botchie. "Accounting and auditing with blockchain technology and artificial Intelligence: A literature review." *International Journal of Accounting Information Systems* 48 (2023): 100598.
3. Marthews, Alex, and Catherine Tucker. "What blockchain can and can't do: Applications to marketing and privacy." *International Journal of Research in Marketing* 40, no. 1 (2023): 49–53.
4. Joo, Mingyu, Seung Hyun Kim, Anindya Ghose, and Kenneth C. Wilbur. "Designing distributed ledger technologies, like Blockchain, for advertising markets." *International Journal of Research in Marketing* 40, no. 1 (2023): 12–21.
5. Malik, Nikhil, Gil Appel, and Lan Luo. "Blockchain technology for creative industries: Current state and research opportunities." *International Journal of Research in Marketing* 40, no. 1 (2023): 38–48.
6. Harris, Justin D., and Bo Waggoner. "Decentralized and collaborative AI on blockchain." In *2019 IEEE international conference on blockchain (Blockchain)*, pp. 368–375. IEEE, 2019.
7. Lilly, Bilyana, and Sale Lilly. "Weaponising blockchain: Military applications of blockchain technology in the US, China and Russia." *The RUSI Journal* 166, no. 3 (2021): 46–56.
8. Hewa, Tharaka, Gürkan Gür, Anshuman Kalla, Mika Ylianttila, An Bracken, and Madhusanka Liyanage. "The role of blockchain in 6G: Challenges, opportunities and research directions." *2020 2nd 6G Wireless Summit (6G SUMMIT)*, pp. 1–5, 2020.
9. Hussain, Adedoyin A., and Fadi Al-Turjman. "Artificial intelligence and blockchain: A review." *Transactions on Emerging Telecommunications Technologies* 32, no. 9 (2021): e4268.
10. Vyas, Sonali, Mohammad Shabaz, Prajjawal Pandit, L. Rama Parvathy, and Isaac Ofori. "Integration of artificial intelligence and blockchain technology in healthcare and agriculture." *Journal of Food Quality* 2022 (2022).
11. Singh, Jagendra, Mohammad Sajid, Suneet Kumar Gupta, and Raza Abbas Haidri. "Artificial intelligence and blockchain technologies for smart city." *Intelligent Green Technologies for Sustainable Smart Cities* (2022): 317–330.
12. Wang, Zeyu, Mingyu Li, Jia Lu, and Xin Cheng. "Business Innovation based on artificial intelligence and Blockchain technology." *Information Processing & Management* 59, no. 1 (2022): 102759.
13. Velmovitsky, Pedro Elkind, Frederico Moreira Bublitz, Laura Xavier Fadrique, and Plinio Pelegrini Morita. "Blockchain applications in health care and public health: increased transparency." *JMIR Medical Informatics* 9, no. 6 (2021): e20713.
14. Jiang, Fei, Yong Jiang, Hui Zhi, Yi Dong, Hao Li, Sufeng Ma, Yilong Wang, Qiang Dong, Haipeng Shen, and Yongjun Wang. "Artificial intelligence in healthcare: past, present and future." *Stroke and Vascular Neurology* 2, no. 4 (2017).

15. Wehbe, Youssef, Mohamed Al Zaabi, and Davor Svetinovic. "Blockchain AI framework for healthcare records management: constrained goal model." In *2018 26th Telecommunications forum (TELFOR)*, pp. 420–425. IEEE, 2018.
16. Mazlan, Ahmad Akmaluddin, Salwani Mohd Daud, Suriani Mohd Sam, Hafiza Abas, Siti Zaleha Abdul Rasid, and Muhammad Fathi Yusof. "Scalability challenges in healthcare blockchain system—a systematic review." *IEEE Access* 8 (2020): 23663–23673.
17. McGhin, Thomas, Kim-Kwang Raymond Choo, Charles Zhechao Liu, and Debiao He. "Blockchain in healthcare applications: Research challenges and opportunities." *Journal of Network and Computer Applications* 135 (2019): 62–75.
18. Krithika, L.B. "Survey on the applications of blockchain in agriculture." *Agriculture* 12, no. 9 (2022): 1333.
19. Singh, Parvinder, and Amandeep Kaur. "A systematic review of artificial intelligence in agriculture." *Deep Learning for Sustainable Agriculture* (2022): 57–80.
20. Ahamed, N. Nasurudeen, and R. Vignesh. "Smart agriculture and food industry with blockchain and artificial intelligence." *J. Comput. Sci* 18, no. 1 (2022): 1–17.
21. Zhu, Y., X. Zhang, Zh Y. Ju, and Ch Ch Wang. "A study of blockchain technology development and military application prospects." *Journal of Physics: Conference Series*, 1507, no. 5 (2020): 052018.
22. Sudhan, Amool, and Manisha J. Nene. "Employability of blockchain technology in defence applications." In *2017 International Conference on Intelligent Sustainable Systems (ICISS)*, pp. 630–637. IEEE, 2017.
23. Huszár, Viktor. "Application possibilities of decentralization and blockchain technology using computer vision and artificial intelligence in defense management, military and police organizations." *Honvédségi Szemle: A Magyar Honvédség Központi Folyóirata* 148, no. 1.-SI (2020): 4–14.
24. Lockey, Steven, Nicole Gillespie, Daniel Holm, and Ida Asadi Someh. "A review of trust in artificial intelligence: Challenges, vulnerabilities and future directions." (2021).
25. Rahayu, Syarifah Bahiyah, and Afiqah Azahari. "Military supply chain management and blockchain development." *Journal of Education and Social Sciences* (2019).
26. Wüst, Karl, and Arthur Gervais. "Do you need a blockchain?." In *2018 crypto valley conference on blockchain technology (CVCBT)*, pp. 45–54. IEEE, 2018.
27. Dinh, Thang N., and My T. Thai. "AI and blockchain: A disruptive integration." *Computer* 51, no. 9 (2018): 48–53.
28. Brik, Bouziane, Adlen Ksentini, and Maha Bouaziz. "Federated learning for UAVs-enabled wireless networks: Use cases, challenges, and open problems." *IEEE Access* 8 (2020): 53841–53849.
29. Jadav, Nilesh Kumar, Tejal Rathod, Rajesh Gupta, Sudeep Tanwar, Neeraj Kumar, Rahat Iqbal, Shadi Atalla, Hijji Mohammad, and Saba Al-Rubaye. "Blockchain-based secure and intelligent data dissemination framework for uavs in battlefield applications." *IEEE Communications Standards Magazine* 7, no. 3 (2023): 16–23.

30. Dahmane, Sofiane, Mohamed Bachir Yagoubi, Chaker Abdelaziz Kerrache, Pascal Lorenz, Nasreddine Lagraa, and Abderrahmane Lakas. "Toward a secure edge-enabled and artificially intelligent internet of flying things using blockchain." *IEEE Internet of Things Magazine* 5, no. 2 (2022): 90–95.
31. Shinde, Rucha, Shruti Patil, Ketan Kotecha, and Kirti Ruikar. "Blockchain for securing ai applications and open innovations." *Journal of Open Innovation: Technology, Market, and Complexity* 7, no. 3 (2021): 189.

AI in military applications

Case studies

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9.1 INTRODUCTION

Artificial Intelligence (AI) has become a transformative force across various sectors and the military domain is no exception. With its ability to process enormous amounts of data, analyze complex patterns, and make instant decisions, AI has revolutionized military technology. However, the defense sector deploys AI in so many ways that it is often difficult to understand on how revolutionary it is. Whether it is facilitating the development of autonomous vehicles or playing a critical cybersecurity role, AI underpins many of the significant defense sector developments of recent years.

9.1.1 Autonomous vehicle and weapons system

This section elaborates various weapons and vehicle system used for military applications in India and US.

9.1.1.1 Indian vehicle systems

Sapper Scout: This is an AI-based mine detector, which comes in the UGV (Unmanned Ground Vehicle) category, that can detect land mine and marks the land mines using an illuminated spray. Some prominent features are mine detection, wide-range surveillance, metal detection, and minefield marking. The Indian Military can use this UGV for: *Route Clearance:* Sapper Scout can be deployed ahead of the route to get rid of potential mine threats, ensuring safe passage through the route without casualties. *Urban Warfare:* In a war situation there will be the risk of explosive traps, Sapper Scouts can be used for the detection of traps and reconnaissance to enhance the safety of military personnel. *Patrols in High-Risk Areas:* Deploy Sapper Scout on autonomous or semi-autonomous patrols in areas where the risk of encountering mines or explosive devices is high.

Silent Sentry: It is an AI-based surveillance system that is installed on fences and AIOS (Anti-Infiltration Obstacle System). Some prominent features are AI-based Object Detection, Wireless Control and Telemetry, and Autonomous and full-time Patrolling. It is currently deployed on the India-Pakistan Border in Jammu and Kashmir, and it can be used for *Border Surveillance*: Deploy Silent Sentry along railway lines in border regions for continuous surveillance. The AI is capable of human detection, facial recognition, and identifying potential threats, providing real-time information to a military command center for rapid responses. *Anti-Terrorism Operations*: Employ Silent Sentry in anti-terrorism operations, particularly in areas with high-risk railway infrastructure. The AI technology can assist military personnel in identifying suspicious activities and detecting potential security breaches contributing to defensive measures and ensuring safety passage. *Critical Infrastructure Protection*: Silent Sentry can be utilized to patrol and monitor the critical infrastructure along the railway network, such as bridges, tunnels, and railway stations. The AI is capable of detecting security breaches and threats or damages in the route.

Autonomous Fast Intercept Boat (AFIB): This is an AI-based vehicle used in the marine segment and it is first of its kind used in the Indian Forces. The vehicle can perform its autonomous features even in dense maritime traffic and in shallow water areas. The Indian Forces can use AFIB in: *Maritime Patrol and Surveillance*: Deploy AFIBs for maritime patrol and surveillance in coastal regions and strategic waterways. The autonomous capability allows these boats to patrol large areas efficiently, monitoring any suspicious activities, and providing real-time data to enhance awareness for the commandos. *Border Security*: Deploying AFIBs to patrol and monitor maritime borders will enhance border security. The AI capabilities make them effective in preventing unauthorized vessels from crossing borders and engaging in illegal activities. *Rapid Response to Emerging Threats*: With the support of data given by AFIBs, there will be rapid responses to emerging maritime threats. The autonomous nature of these boats allows them to quickly adapt to changing situations, making them effective to respond for threats or unexpected incidents as mentioned in Table 9.1.

Table 9.1 Specifications of AFIB

Parameters		Units
Speed	35 ± 1 Knots	
Range	150 Nm with 25% reserve	
Main Engine	2 × Cummins QSC8.3 500 BHP @2600 RPM Diesel Engine	
Propulsion	2 × Waterjet Kamewa FF310	

9.1.1.2 US vehicle systems

Husky Vehicle: It is an AI-based mine detection system employed on UGVs that is specialized for route clearance and mine detection. *Route Clearance:* The Husky Vehicle System's main goal is to remove landmines and other explosive risks from the route. It is frequently used in locations where military troops or vehicles must travel. *Safety for Personnel:* Using the Husky Vehicle improves military personnel safety because it can withstand the impact of explosive devices without risking the lives of military people. *Operator Control:* The vehicle is typically operated by a human who can control it remotely from a safe distance. The operator receives feedback from the sensors on the vehicle and makes decisions regarding route adjustments or potential threats.

RQ-4 Global Hawk: It is an Unmanned Aerial Vehicle designed for high-altitude, reconnaissance, and surveillance missions. It is operated by the USAF. *Persistent Surveillance:* This vehicle can fly at high altitudes for extended time periods, allowing it to provide determined surveillance over a large area. It is used for monitoring activities on the ground and tracking the threats if found. *Reconnaissance and Intelligence Gathering:* This vehicle is equipped with a variety of sensors, including radar, infrared cameras, and signals intelligence equipment. It can collect a wide range of information that will be useful for military personnel like images, signals, etc. *Strategic and Tactical ISR Support:* This vehicle will provide both strategic and tactical ISR support (Walter Pincus 2023). It is used for gathering information in support of ongoing military operations, humanitarian activities, and other missions where situational awareness is critical.

Sea Hunter: It is an Autonomous Unmanned Surface Vehicle (USV) designed to operate for long periods of time without crew members on board. It is at an experimental level and designed by the Défense Advanced Research Projects Agency (DARPA). Its capabilities are: *Anti-Submarine Warfare (ASW):* The Sea Hunter is designed to operate autonomously. It can be used in warfare situations to detect and track submarines. It allows for extended patrols in areas of interest. *Escort and Protection:* The Sea Hunter can be used to escort and protect larger ships, submarines, and other naval assets. It can operate alongside the manned vehicles, providing extra protection. *ISR* (Walter Pincus 2023): The Sea Hunter can be equipped with sensors and communication systems to perform surveillance and reconnaissance missions. It will collect data on maritime activities, monitor sea lanes, and provide information for the command center.

9.1.1.3 Indian weapon systems

Smart-Counter Measure Dispensing System: CMDS protects airplanes/fighter jets/helicopters from incoming infrared/radar guided missile threats by delivering payloads (flares/chaffs) based on data received from the various sensors installed on the aircraft. When the aircraft flies over a hostile area that is vulnerable to attack, SMART-CDMS automatically releases

appropriate payloads in an appropriate sequence against active threats and prevents the threat.

Advantages of CDMS for the Indian Military

- i. **Enhanced Survivability:** CMDS significantly improves the survivability of military platforms by diverting incoming threats, and ensuring the safety of aircraft, ground vehicles, and military personnel inside the aircraft.
- ii. **Rapid Response:** CMDS systems deploy countermeasures quickly and offer a real-time defense mechanism against incoming threats.
- iii. **Adaptability:** CMDS can be customized to counter different types of threats, providing an adaptable defense against radar-guided and infrared-homing missiles.
- iv. **Reduced Risk to Personnel:** The automated deployment of countermeasures minimizes the need for manual intervention, decreasing the risk to personnel onboard.
- v. **Operational Flexibility:** CMDS systems can be seamlessly integrated into a range of military vehicles, including aircraft, helicopters, ships, and ground vehicles, offering operational flexibility in diverse deployment scenarios (Shaurya Karanbir Gurung 2019).

Disadvantages of CDMS for the Indian Military

- i. **Weight and Space Constraints:** The hardware required for CMDS systems can add weight and take up valuable space on military platforms, potentially impacting their performance and payload capacity.
- ii. **Maintenance and Reliability:** CMDS systems need regular maintenance to ensure they are functioning correctly. The reliability of these systems is critical, as any failure could lead to disastrous events.
- iii. **Limited Effectiveness:** While CMDS can greatly enhance survivability, they are not perfect. Advanced threats and counter-countermeasure technology can reduce their effectiveness.
- iv. **Signature Reduction:** Some countermeasures used in CMDS, like flares, can increase the thermal and visual signature of the system, potentially making it more vulnerable to some threats.
- v. **Operational Training:** The effective use of CMDS systems requires well-trained personnel who can respond quickly and appropriately to threats. Training can be time-consuming and costly (Shaurya Karanbir Gurung 2019).

BRAHMOS Weapon System: The BRAHMOS weapon system is a supersonic cruise missile system that provides a precision strike weapon for the army, navy and air force. The name of this supersonic missile BrahMos combines the names of the Brahmaputra River in India and the Moskva River in Russia. These missiles can be deployed in ships, mobile launchers, submarines and aircraft. BrahMos is characterized by accuracy, reliability, ease of

Table 9.2 Specification of BrahMos

Parameters	Units
Range	Up to 290 kms
Speed	2.8 Mach
Flight Altitude	5m to 15 km
Warhead	200-300kg Conventional
Length	9000 mm
Diameter	700 mm
Weight	3000 kg

use and faster reaction time. The specifications of BrahMos Weapon System are mentioned in Table 9.2.

Advantages of BrahMos: Speed and Flexibility: BrahMos has a speed of 2.8 Mach, and it can gain up to 3.5 Mach which is three times the speed of sound. This supersonic speed makes it difficult for enemy air defenses to intercept. The missile has a high level of flexibility, making it capable of avoiding enemy defenses. Range and Precision: BrahMos has a range of up to 290 kms initially but increased to 650 kms which is a long operational range, allowing the Indian Military to fix targets at a considerable distance. It is also known for its precision and accuracy, enabling it to strike targets with high lethality. Dual-Role Capability: The BrahMos missile is a ship-launched as well as land-based missile system, this dual-role capability provides flexibility for the Indian Military and will allow it to counter a range of threats and operational requirements. Attack Capability: The BrahMos has the attacking capability to strike surface targets as low as 5 meters in altitude and strike targets as high as 15 kilometers in altitude.

Disadvantages of BrahMos: Limited Range for Air-Launched Variant: the Brahmos missile from the Air-Launched variant has a range of 500 kms which is less than the land-based missile system. Strategic imbalance: The deployment of this highly advanced missile system might lead to concerns among neighboring countries, potentially increasing tensions. Generally, there were several accidents that took place where BrahMos malfunctioned and affected our neighboring countries. Dependency on Foreign Suppliers: While BrahMos is a joint venture between India and Russia, there might be concerns about dependency on foreign suppliers for critical components. Changes in geopolitical relations with the partner country could impact the supply chain.

9.1.1.4 United States weapons systems

Predator and Reaper UAVs: There are two MQ-1 Predator and MQ-9 Reaper which are UAVs that are capable of attacking, carrying, and employing precision-guided. These aircraft UAVs are used for surveillance and striking missions as mentioned in Tables 9.3 and 9.4.

Table 9.3 Specifications of predator

<i>Parameters</i>	<i>Units</i>
Range	675 nautical miles
Power Plant	Rotax 914F four-cylinder engine
Height	2.1 meters
Weight	512 kgs
Payload	204 kgs

Table 9.4 Specifications of reaper

<i>Parameters</i>	<i>Units</i>
Range	1000 nautical miles
Power Plant	Honeywell TPE331-10GD turboprop engine
Height	3.8 meters
Weight	2223 kgs
Payload	1701 kgs

Advantages of Predator and Reaper for US Military

Real-time Surveillance: The UAV provides real-time video feeds and intelligence to ground operators, enabling rapid decision-making. **Armed Capability:** These drones are armed with precision-guided munitions, providing the capability to engage and strike targets. **Greater Payload Capacity:** These drones can carry a larger payload of sensors and munitions, enhancing their versatility.

Disadvantages of Predator and Reaper for the US Military

Limited Payload Capacity: The payload capacity for sensors and munitions is smaller compared to larger UAVs like the MQ-9 Reaper. **Limited Stealth Features:** While the Reaper is equipped with certain features to reduce its radar cross-section, it is not a stealth aircraft and can be detected by advanced air defense systems. **Ethical and Legal Concerns:** The use of armed UAVs raises ethical and legal concerns, particularly regarding targeted strikes and collateral damage.

SWORDS: SWORDS that stands for Special Weapons Observation Reconnaissance Detection System, is a remotely operated robotic system that is used for both armed reconnaissance and combat missions.

Advantages of SWORDS for the US Military

Remote Operation: SWORDS is remotely operated, reducing the risk to human operators in potentially dangerous or hostile environments. **Manned-Unmanned Teaming:** SWORDS can operate in coordination with manned military units, providing additional support and capabilities on the

battlefield. EOD (Explosive Ordnance Disposal): One of the intended applications of SWORDS is in EOD missions, where it can be used to remotely handle and dispose of explosive devices.

Disadvantages of SWORDS for the US Military

Limited Deployment: The deployment of SWORDS in active combat situations has been limited, and the system faced challenges in achieving universal use. **Technological Limitations:** The technology available at the time may have had limitations in terms of mobility, agility, and the ability to adapt to dynamic battlefield conditions. **Ethical and Legal Concerns:** The deployment of armed robotic systems raises ethical and legal concerns, particularly regarding the potential for autonomous decision-making and adherence to rules of engagement.

The chapter is structured as follows **section 2** elaborates intelligent command and control systems used in India and US military applications, **section-3** explains predictive maintenance using AI in military applications, **section-4** deals with various means to improve security in military applications, **section-5** depicts usage of AI in surveillance operations and finally the chapter is concluded in **section-6**.

9.2 INTELLIGENT COMMAND AND CONTROL SYSTEMS

9.2.1 Indian command and control system

The Indian Military is using C4ISR (Command, Control, Communication, Computers, Intelligence, Surveillance, and Reconnaissance) systems will play an important role in improving the country's military capabilities. These systems are designed to make sure to have effective communication, decision-making, and information gathering on the battlefield. India has been investing in developing various components of C4ISR to modernize its armed forces (Col. Everett Bud Lacroix 2023). The country is also developing and advancing satellite communication, electronic warfare, and surveillance capabilities. The IAF and Indian Navy have dedicated satellites for their C4ISR needs (Brigadier Sanjeev Chopra 2022). In C4ISR systems there are some capabilities: *AI-based Intercept Management System (IMS)*: It is a software developed to analyze adversaries. It uses visualization techniques and AI to interpret critical data. *AI-based Motion Detection and Target Identification System*: Proactive Real-time Intelligence and Surveillance Monitoring System (PRISM) is an AI-based system capable of generating real-time audio and visual alerts from multiple surveillance data when suspicious hostile movements are detected. *Deep Learning Toolkit for Aerospace and Defence*: Computer Vision is one of the most powerful types of AI. This deep learning toolkit masters computer vision, Natural Language Processing (NLP), reinforcement learning, tracking and fusion. *Adversary Network*

Analysis Tool (ANANT): Law enforcement agencies need to understand and analyze data in two dimensions, namely spatial and temporal data. ANANT provides spatio-temporal analysis to analyze adversarial networks with multiple nodes and multiple relational networks. *Detection and classification of enemy aircraft activity:* In an air defense scenario, there is a high probability of threats appearing in single or multiple groups and formations from different parts of the enemy area. The AI can help identify enemy aircraft and even their action plan. This facilitates evasive maneuvers or an effective counterattack. Currently, this solution has been developed for the nation's air defense system and is to be implemented in IACCS (Integrated Air Command and Control System).

9.2.2 United States command and control system

The US Military uses the C5ISR^[8] (Command, Control, Communications, Computers, Cyber, Intelligence, Surveillance, and Reconnaissance) system is a comprehensive framework that integrates various elements to enhance the military's operational capabilities. The US military uses C5ISR in a wide range of applications to achieve greater situational awareness, operational effectiveness, and decision-making (Col. Everett Bud Lacroix 2023). There are some military applications where C5ISR^[8] capabilities are used: Command and Control (C2): Some tools of C5ISR^[8] systems are used by military commanders to command and control their forces, these tools provide real-time monitoring of troop movement, logistical information, and the ability to issue orders and directions. Communications: The system enables secure and reliable communication between military units on all land, sea, and air. This includes voice communication, data exchange, and video conferencing, ensuring that information can be shared quickly and properly. Intelligence Gathering and Analysis: C5ISR facilitates the collection, analysis, and distribution of intelligence data. This includes information from various sources such as satellites, reconnaissance aircraft, surveillance systems, and human intelligence.

Surveillance and Reconnaissance: The system includes a variety of sensors, including radar, cameras, and other reconnaissance tools, to provide continuous surveillance of the battlefield. This information is crucial for identifying and tracking potential threats and targets.

Cyber Operations: C5ISR systems include components to monitor and defend against cyber threats. This is important as modern military operations are heavily dependent on networked and computerized systems (Col. Everett Bud Lacroix 2023).

In the USA, there is a center that develops new technologies and integrates existing mechanisms. The C5ISR Centre of the Combat Capabilities Development Command (CCDC) is the United States Army's Centre for Information Technologies and Integrated Systems. The C5ISR Center develops new technologies and adapts technologies from other Army R&D centers and laboratories, Department of Defence partners, state and national laboratories, academia and industry (Col. Everett Bud Lacroix 2023).

9.3 PREDICTIVE MAINTENANCE USING AI

Predictive maintenance can be defined as a computerized technique used to predict the future failure and seizure of a component of a weapon, machine, or delivery system so that the Department of Defence and the military services can plan to replace components before they fail. Predictive maintenance begins with the collection of data on the condition of components, the algorithms of which feed into an AI system that predicts maintenance needs and is also dependent on climate events (Flysight 2023).

China takes great importance to AI-supported autonomous systems. It has already begun developing AI-supported cruise missile systems with high accuracy. This is believed to be a response to the ‘semi-autonomous’ long-range anti-ship missiles to be deployed by the US in the Pacific region. The high level of support for research and development in the field of robotics and unmanned systems has led to the establishment of numerous institutes in China’s defence industry (Brigadier Sanjeev Chopra 2022).

India is at a comparatively developing stage in terms of its R&D efforts in AI, especially in the defence sector—the lag in this area is strong when viewed with respect to the size of India’s economy and defence needs. The Indian armed forces need to focus more on a ‘bottom-up’ approach to automation, followed by support services such as medical, inventory management, logistics, repair and maintenance, professional military training, autonomous surveillance and combat systems (Brigadier Sanjeev Chopra 2022).

The Indian Army is developing an automated program that informs it in real time about the service life and current condition of its tanks, vehicles, guns and aircraft, as well as any upcoming problems. A program called Project Beehive has been initiated, that is a centralized and automated system-work based in Delhi under the Army’s EME corps. The EME will run around 2000 workshops under its wing (Raunak Kunde 2023). The EME is entrusted with carrying out repair, refurbishing and service the Army’s equipment, ranging from tanks, infantry combat vehicles, artillery guns including the new M777 Ultra Light Howitzers, small arms, radars and aviation assets. The workshops would consist of two-year old automated program named wasp, which would compile all data of equipment such as repairing and their remaining life. The wasps will be integrated with the Beehive and thereby provide every data to it.

“The concept is same like a wasp that collect nectar for their beehive. And then through the program we can draw data out of this nectar,” as explained by Lieutenant General Anil Kapoor, Director General EME, who is heading the program’s development. The project was launched on September 1, 2019, and is expected to be completed by October 15 2020 after it undergoes an army cyber test, Kapoor added (Raunak Kunde 2023). The army is in works with exceptional coders from the Electronics and Information Technology ministry to develop this program, which has eight modules

(Yin et al. 2020). The first one is completed and the second one is being developed. “Every module will contain the power of automation,” the officer said. So basically, the program is that the army will have access to real time availability of data of all the equipment at the workshops. “As DG of EME, I will have information about critical equipment at the click of a button. This information will include how much mileage a particular type of vehicle has or how many gun or tactical equipment are remaining for inspection. We can also know how much life is there in a particular equipment. This is the advantage of automation, that at any point of time we can know what’s happening,” Kapoor explained (Shaurya Karanbir Gurung 2019). The program has been designed for accepting specific queries for specific answers. For example, between a particular time period how many radiators have failed and in which terrain. “These answers will help us in managing our tasks far better and improve decision making,” he said, adding that at different levels functionaries can see the equipment readiness state (Shaurya Karanbir Gurung 2019).

In the year 2023, Hindustan Aeronautics Limited has taken initiative to develop AI-based Predictive Maintenance Tools for Fighter Jets for the Indian Army. As called, the Health Access and Monitoring System, this revolutionary system harnesses the power of AI and drastically reduces the time it takes to analyze data. With this, aircraft maintenance teams experience faster data processing and analysis, which results in faster flight maintenance time. Along with this, the system also provides maintenance teams with better insights and forecasts for preventive maintenance, allowing them to predict potential problems and ensure optimal aircraft performance. The benefits of this AI-based predictive maintenance have been proven very efficient. By analyzing huge volumes of data, AI algorithms can detect similarities and problems that may go unnoticed by the human eye. This allows maintenance teams to detect potential failures or damage to aircraft components, allowing them to take preventive measures to prevent failures and improve safety (Raunak Kunde 2023).

9.3.1 AI in logistics

In fact, military logistics contains more functions than many people realize. In modern warfare, this also means that huge amounts of data will have to be swamped to be able to decide on the delivery of supplies, the possibility of their transportation, the availability of communications, and so on. The use of AI and ML in one or more areas of logistics may accelerate this procedure and offer more flexibility to this. Even though AI and ML has a low option in military logistics, the military concentrates on AI and ML options. The military, meanwhile, has been slow to change and adapt often accepting yellow status for its logistics in active war zones (Sarkar and Gul 2023). However, there are problems with AI integration into military logistics despite their disutility in comparison with AI usage for surveillance,

reconnaissance or weapons. AI and ML technologies suffer from their unpredictability and vulnerability to hacking. Data scientists and engineers demonstrate significant progress in work with AI and ML, but they are by no means perfect. Previous doubt regarding the possibility of failure significantly levelled out the number of AI and ML case uses; however, the lack of uncertainty determined overwhelming AI infrastructure. With the help of data analytics techniques, Pro-HM+ was developed, an AI-based aviation health monitoring software application that can examine trends, patterns and correlations that allow it to predict future events such as equipment failures and aircraft behavior. Pro-HM+ transforms the collected data into business and operational intelligence. The descriptive, predictive and prescriptive data analytics layers keep an eye on operational trends and forecast maintenance needs with a high degree of probability. This can drastically reduce the total cost of ownership of the plant and extend its lifespan (Millicent Abadicio 2019).

i. Maximizing Supply Chain Management

The AI sustainable impact on military supply chain management is indeed the game-changer. As per a study, the ai in supply chain management can boost the efficiency by 20% and sometimes even more. Its capability to crunch Big-data and forecast future trends and resource allocation needs proves to be beneficial for the military logistics. The military can leverage ai algorithms to accurately predict the needs of soldiers and ensure that crucial supplies are delivered to the right location at the right time. Further, the predictive analysis can benefit the military operations by appropriately assigning personnel and equipment. The predictive analytics in military logistics are capable of foreseeing and predicting the need to replace any parts of a vehicle, which makes Predictive Maintenance a reality, thus understanding any potential failure before it happens (Heba Soffar 2016). This method reduces the chance of significant expenses related to unexpected repairs and improves operational safety by minimizing the chance of an accident. As an outcome, centralized information and resource access and use will result in superior performance, shorter delivery times and better supply chain availability (Millicent Abadicio 2019; Col. Everett Bud Lacroix 2023).

ii. Counterargument

While the integration of AI into military logistics offers many benefits, there are also legal concerns and potential disadvantages to consider. Some may argue that relying on artificial intelligence can lead to an overemphasis on technology at the expense of human experience and intuition, which are crucial in complex and unpredictable situations. There is a risk that AI may also provide a false sense of security, leading to overconfidence and strategic errors. On top of that, the significant

costs associated with implementing AI technology, such as infrastructure upgrades, software development and ongoing maintenance, can outweigh the potential benefits. Budget constraints and competing priorities can make it difficult to allocate sufficient resources to the integration of AI, which can limit its effectiveness. Another issue is the vulnerability of AI systems to cyber-attacks and manipulation by competitors. As AI-based logistics systems become increasingly important to military operations, they also become valuable targets for adversaries seeking to disrupt or damage military capabilities and operations. Developing strong cybersecurity measures is critical, but there is no guarantee that these defenses will always be effective against rapidly evolving threats. Consider also the ethical issues of artificial intelligence in military logistics. The use of artificial intelligence can lead to biased decisions, lack of transparency, or unintended consequences, (Indian Defence Industry- A Global Reach 2023). Responsibility for the operation of AI systems must be clearly defined to ensure accountability in the event of errors or failures. Nevertheless, the integration of ai into military logistics may have unintended consequences for the military logistics profession. Even though the automation of certain tasks can increase efficiency, it can also lead to loss of job and the need for extensive workforce retraining. It is important to ensure that the military can adapt to these changes and retain a skilled workforce, but this will require continued effort and also investment (Col. Everett Bud Lacroix 2023).

9.4 TIGHTENING CYBERSECURITY

In the experience of calculations, we apply intelligent plans that ideas accompanying each one. These systems help us in attaining troublesome tasks, but they too reveal us to potential issues. If someone accompanying distressing aims accesses our methods, it could enhance an important question. But you can trust, we have AI. It's like a hero that uses creative calculating programs to recognize and resolve computer questions. These programs resolve our netting traffic, find feeble points, and block them (Gp Capt AK Sachdev 2023). This considerably provides to assuring our calculating networks from warnings and security breaches. Even therefore AI doesn't stop skilled. It learns from past happenings. If entity bad occurred before, AI remembers it and enhances even more persuasive at halting similar issues from persisting. Like bearing a hero friend that's continuously knowledge and levelling up. So, presenting machine intelligence to reinforce computer protection in India's military isn't almost fast agreeing issues. This ensures that our calculations stay shielded from all the smart and complicated calculating problems out there.

9.4.1 Benefits

- i. Improved Detection of Threats: AI-driven systems excel at rapidly analyzing datasets, facilitating early identification of potential digital threats that human analysts could overlook. This capability improves the speed of response and enables a proactive defense posture.
- ii. Quick Response: AI systems demonstrate real-time responsiveness, automatically implementing measures to counteract or mitigate digital threats. This capability assists in the prevention or minimization of damage resulting from cyberattacks (Kim Jones 2007; Radovanović et al. 2023).
- iii. Continuous Learning and Evolution: AI systems possess the capability to learn from past incidents, adjust to emerging threats, and enhance their skills over time. This continuous learning experience enhances their effectiveness in identifying and responding to digital threats.
- iv. Mitigation of Human Error: AI helps to minimize the impact of human errors in cybersecurity. By automating processes, it reduces the mistakes and ensures the consistent execution of security measures (Douglas Scott 2019).
- v. Versatility: AI exhibits outstanding flexibility, efficiently managing big volumes of data. This flexibility is crucial for safeguarding the complex and interconnected systems within the military.
- vi. Cost-Efficiency: Over the long term, AI proves to be a cost-effective alternative to maintaining extensive teams of human cybersecurity experts. AI systems operate seamlessly without the breaks, contributing to cost savings.
- vii. Increased Awareness: AI enhances the overall awareness of military leaders, providing a broader understanding of the cybersecurity landscape. This awareness helps officers in making well-informed decisions.

9.4.2 Drawbacks

- i. Challenging Implementation: The integration of artificial intelligence into military systems poses challenges due to its complex nature and costs, requiring significant resources and specialized expertise.
- ii. Lack of Human Insight: While artificial intelligence excels at threat detection, it falls short on judgment and analytical understanding that human experts bring. This can result in misleading positives of emerging threats.
- iii. Vulnerability to Attacks: AI systems themselves are vulnerable to adversarial attacks, wherein attackers manipulate input data to mislead the system into making incorrect decisions.
- iv. Data Security Concerns: Artificial intelligence heavily dependent on extensive data, leading to concerns about data privacy and the potential for misuse of the important information.

Table 9.5 Advancement in Indian military technology initiatives

Technology	Description
Project I-Hari	The Indian Armed Forces have launched Project I-Hari (Information Centre for all Deployed Data) to enhance awareness of situations and capabilities. This initiative integrates AI, big data analysis, and machine learning for real-time threat assessment and navigation.
Cyber Operations Centre (COC)	The Indian Naval force has established a Digital Tasks Community to monitor and secure critical organizations and systems. Leveraging advanced technologies like AI and machine learning, this center efficiently identifies and responds to cyber threats.
AI-based Border Surveillance	India's border security forces have deployed AI-powered surveillance systems along the country's borders. These systems analyze sensor data, including thermal imaging and motion sensors, using AI algorithms to detect and respond to potential threats and intrusions.

- v. Skills Gap: Deploying AI in the military requires a skilled workforce to survive, operate, and oversee the systems, posing a challenge in developing this expertise.
- vi. Continuous Learning and Evolution: AI systems can learn from past experiences, adjust to emerging threats, and refine their skills over time.
- vii. Technology Reliance: Excessive dependence on artificial intelligence can become a weakness if the technology malfunctions or is compromised. Finding the right harmony among human and artificial intelligence capacities is important.
- viii. Global Standards and Regulations: The use of AI in military settings should comply with international standards and regulations to avoid diplomatic and legal repercussions associated with non-compliance (Table 9.5).

9.4.3 Case Studies

Project I-Hari: The Indian Armed Forces have launched Project I-Hari (Information Centre for All Deployed Data) to enhance its awareness of the situation and communication capabilities. This initiative integrates AI, big data analysis, and machine learning to process and analyze extensive information from various sources efficiently. This enables real-time threat assessment and navigation.

Table 9.6 AI used by United States and America in their military

Initiative/Program	Description
Project Maven	An Initiative by The U.S. Department Of Defense that utilizes Artificial Intelligence And Machine Learning For analyzing Extensive Imagery And Data.
Digital Order	The U.S. Digital Order integrates AI into Its Digital Tasks, Emphasizing Activities Like Threat Detection, Incident Response, And Securing Critical Military Organization And Systems
Ai-Enhanced Robotics	The U.S. Military Incorporates Ai Into Its Robotic Technology, Enabling Autonomous Flight and Providing Advanced Capabilities

Cyber Operations Centre (COC): The Indian Naval force has laid out a Digital Tasks Community to screen and get its basic organizations and frameworks. Leveraging advanced technologies such as AI and machine learning, this center effectively identifies and responds to cyber threats.

National Cyber Coordination Centre (NCCC): The NCCC, an essential drive by the Indian government, intends to improve the country's online protection capacities. Tasked with continuous threat analysis and information sharing between government agencies, the NCCC relies on AI and machine learning for rapid threat detection and response.

Secure Communication Systems: India is actively developing secure communication systems that leverage artificial intelligence to protect military communications from cyber threats (Neha Grover (2023)). These systems use encryption and AI-driven threat detection to ensure the safeguarding of sensitive data.

AI-Based Border Surveillance: India's border security forces have deployed AI-powered surveillance systems along the country's borders so that they can detect and respond to intrusions. These systems use AI algorithms to analyze sensor data, these systems also include thermal imaging and motion sensors, to identify potential threats (Table 9.6).

9.4.4 AI used by United States of America in their military

- i. Project Maven: Project Maven, an initiative by the U.S. Department of Defense, which uses artificial intelligence and machine learning to analyze extensive imagery and data, this was primarily for the identification and tracking of targets. Its essential goal is to work on the effectiveness of military activities in U.S.
- ii. Digital Order: The U.S. Digital Order integrates man-made consciousness and AI into its digital tasks, which has a specific accentuation on

- exercises like distinguishing dangers, answering episodes, and getting basic military organizations and frameworks.
- iii. AI-Enhanced Robotics: The U.S. military integrates AI into its robotic technology, enabling autonomous flight and providing advanced capabilities
 - iv. Public safety Organization (NSA): The NSA^[7] use computer-based intelligence to screen worldwide web traffic for network protection purposes, which is helping with the recognizable proof of expected dangers and weaknesses.
 - v. DIUx: The Safeguard Development Unit Exploratory (DIUx) inside the U.S. Division of Safeguard teams up with innovation organizations to foster computer-based intelligence arrangements custom-made for military applications, with a specific accentuation on network safety.

9.4.5 AI used by China in their military

Public simulated intelligence Vital Outline: China has concocted an exhaustive public strategy for the improvement of man-made brainpower i.e computer-based intelligence, incorporating the tactical space. Individuals' Freedom Armed force has been proactively coordinating simulated intelligence into its protection frameworks. Simulated intelligence Empowered Automated Elevated Vehicles (UAVs): China's military utilizes man-made intelligence in its automated aeronautical vehicles, improving their capacities for observation, surveillance, and battle missions. These artificial intelligence driven UAVs can work independently, displaying progressed functionalities. Quantum Correspondence Speculations: China has dispensed assets to propel quantum correspondence innovations, reinforcing military online protection by establishing a safer correspondence climate that is moving for foes to block or think twice about.

Committed Digital Units: Inside the PLA, China has laid out particular digital units that influence artificial intelligence for danger discovery, digital protection, and the advancement of hostile digital abilities (Millicent Abadicio 2019). *Man-made consciousness in Satellite Frameworks:* China uses artificial intelligence in its satellite frameworks to expand space-based capacities, zeroing in on secure correspondence and knowledge gathering. *Associations with Tech Undertakings:* China's military participates in cooperative endeavors with homegrown innovation organizations to foster artificial intelligence arrangements custom fitted for safeguard applications, including network protection measures and independent military frameworks. Both the US and China effectively exploit simulated intelligence and other state of the art advancements to upgrade their tactical ability and sustain online protection. Be that as it may, nitty gritty data and results may not necessarily be freely unveiled because of public safety contemplations as depicted in Figure 9.1.

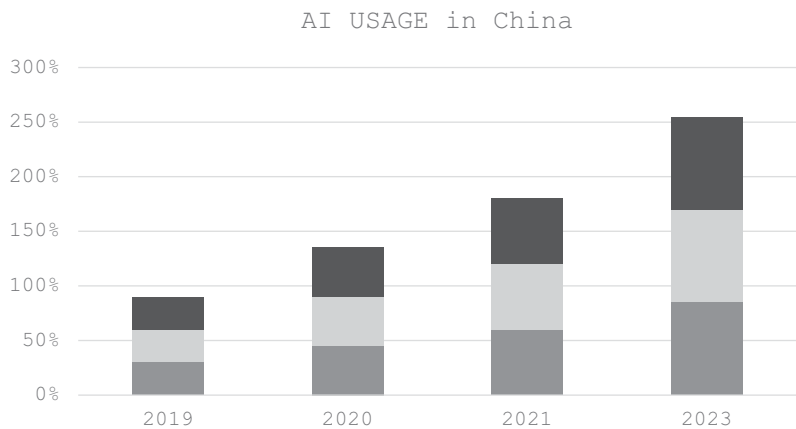


Figure 9.1 Chart showing the data of AI usage in previous years and the current year for China.

9.5 INTELLIGENCE AND SURVEILLANCE

In this age, artificial intelligence is changing the way that we assemble data and watch out for things. Utilizing PC stunts, we can now take loads of pictures and recordings, which makes it simpler to find and follow targets. Man-made intelligence calculations assist with going through a lot of information from better places, similar to virtual entertainment, to track down significant data and figure out what’s going on. This is a major step in the right direction by the way we comprehend and utilize data. Throughout recent decades, we saw that our PCs make a lot of information. In any case, in those days, we didn’t have great ways of grasping it. Presently, with cool artificial intelligence innovation, we can take a gander at this information and learn significant things to keep an eye out for issues and make our knowledge frameworks shockingly better. It resembles we’ve arrived at an unheard-of degree of being savvy about what’s happening (DDPMOD 2022). In India, artificial intelligence is changing the way that we get data and watch out for things. In view of shrewd PC stunts, we can now take a gander at bunches of pictures and recordings, which help us find and track targets all the more without any problem. Man-made intelligence calculations resemble partners that go through a lot of information from better places, similar to web-based entertainment, to track down significant data and comprehend what’s happening.

This is a major step in the right direction by the way we comprehend and utilize data. Throughout recent decades, we saw that our PCs make a great deal of information. In any case, in the days of yore, we didn’t know great

ways of grasping it. Presently, with extravagant artificial intelligence tech, we can take a gander at this information and learn significant things to look out for issues and make our knowledge frameworks stunningly better. It resembles we've gotten way better at grasping what's going on, even in India.

9.5.1 Benefits of artificial intelligence in Indian knowledge and observation

- **Better at Spotting Risks:** Artificial intelligence helps rapidly and precisely track down potential dangers, so we can move quickly to guard our country.
- **Know What's Happening:** AI checks loads of data from different places, such as web-based entertainment, assisting us in understanding what's happening in real-time and staying alert to potential dangers.
- **Find Targets Quicker:** AI-powered systems are adept at finding and tracking targets in images and videos, making it easier to gather intelligence and keep watch over things.
- **Immediately See Enormous Information:** AI programs can sift through and comprehend vast amounts of data super quickly, assisting us in making decisions promptly based on valuable information.
- **Always Learning:** AI systems continue learning from past occurrences, improving over time to handle new threats and challenges.

9.5.2 Hindrances and difficulties

- **Doing the Right Thing:** Involving artificial intelligence in intelligence and surveillance raises questions about doing things the right way, especially concerning security and how collected data might be misused.
- **Guarding Data:** Relying more on AI for handling crucial information increases the chances of data breaches and cyber-attacks, which could jeopardize our nation's security.
- **Costs and Needs:** Setting up AI systems for intelligence and surveillance requires substantial investment and skilled personnel, which may be challenging for managing resources.
- **Not Depending A lot on Tech:** Relying too much on AI could be risky. If the technology fails or someone manipulates it, our intelligence work may be impacted, and our national security could be at risk.
- **Sorting Out the Principles:** Establishing the right regulations and rules for involving artificial intelligence in intelligence is a challenging task, and ensuring everyone adheres to good ethical standards is an ongoing challenge as shown in Figure 9.2.

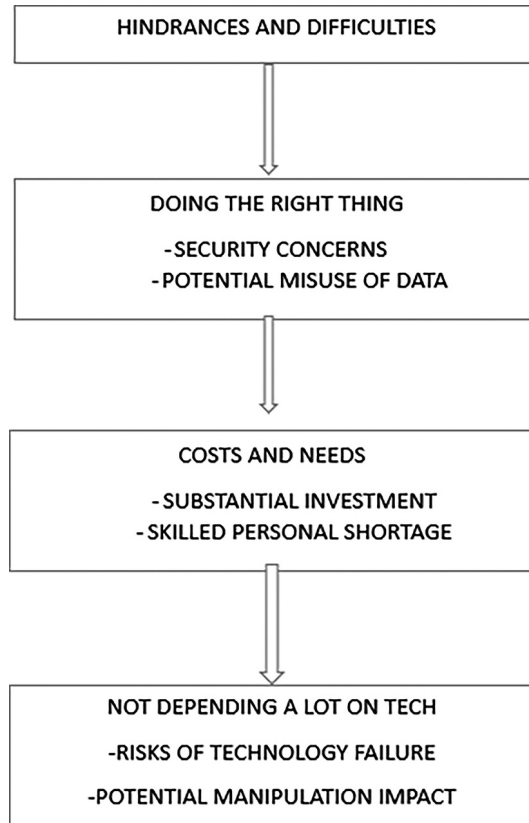


Figure 9.2 Challenges in integrating AI into intelligence and surveillance.

9.6 CONCLUSION

The applications of artificial intelligence in military technology are diverse and are evolving constantly. We have yet to fully understand the technology's full potential, and the defense sector is continuously developing with new applications and functions for combat as well as defence operations. From optimizing logistics and transportation planning to providing realistic AR training to combatants, AI is the basic technology used on which we will develop the next generation of defense and combat sector's vehicle systems, weapons systems, logistics and security.

LIST OF ABBREVIATION

USAF	United States Air Force
ISR	Intelligence, Support, and Reconnaissance

CMD5	Smart-Counter Measure Dispensing System
IAF	Indian Air Force
EME	Electronics and Mechanical Engineers
PLA	People's Liberation Army
NSA	National Security Agency
C5ISR	Command, Control, Communications, Computers, Cyber, Intelligence, Surveillance, and Reconnaissance

REFERENCES

- Flysiht. (2023). *Artificial Intelligence in Military Technology – 8 Applications & Examples*. Flysiht Publisher.
- Walter Pincus. (December 2023). *Is AI the right fit for predictive military maintenance?*. The Cipher Brief Publisher
- Shaurya Karanbir Gurung. (2019). *For better operations, army developing real-time equipment Database*. The Economic Times Publisher.
- Brigadier Sanjeev Chopra. (2022). *Artificial Intelligence in Military Operations*. Claws Publisher.
- Raunak Kunde. (2023). *HAL Developing AI-based Predictive Maintenance Tools for Fighter Jets*. Hindustan Aeronautics Ltd Publisher.
- Millicent Abadicio. (2019). *Artificial Intelligence for Military Logistics–Current Applications*. Emerj Publisher.
- DDPMOD. (2022). *Artificial Intelligence in Defence–The New Age of Defence*. DDPMOD Publisher
- Col. Everett Bud Lacroix. (2023). *Future of Army LogisticsExploiting AI, Overcoming Challenges, and Charting the Course Ahead*. US Army Publisher.
- Neha Grover. (2023). *National Cyber Coordination Centre–Science & Technology Notes*. Prepp Publisher.
- Yin, Rui, Li, Wei, Wang, Zhi-qiang, Xu, Xin-xin. (2020). *The Application of Artificial Intelligence Technology in UAV*. 238–241. 10.1109/ISCTT51595.2020.00048
- Sarkar, Nurul I., & Gul, Sonia. (2023). *Artificial Intelligence-Based Autonomous UAV Networks: A Survey*. MDPI Publisher. Advances of Unmanned Aerial Vehicle Communication.
- Heba Soffar. (2016). *Unmanned ground vehicle (UGV) advantages and disadvantages*. Online-Sciences Publisher.
- Indian Defence Industry- A Global Reach. (2023). Defence Ministry Publisher.
- Gp Capt AK Sachdev. (2023). *C4ISR and Autonomous Capabilities. Lt Gen Sunil Srivastava (Retd) Director CENJOWS Publisher*
- Kim Jones. (2007). *Special Weapons Observation Remote recon Direct Action System (SWORDS)*. NATO International Publisher
- Radovanović, Marko, Petrovski, Aleksandar, Žnidaršič, Vinko, & Aner, Behlić. (2023). The C5ISR System Integrated with Unmanned Aircraft in the Large-Scale Combat Operations. *Vojenské rozhledy* 32. 98–118. 10.3849/2336-2995.32.2023.02.098-118
- Douglas Scott. (May 2019). *Army introduces next-generation capability for radio interoperability with allies*. US Army Publisher.

Analysis of various cyber security challenges, issues, and counter measures in military application

Sashi Pritam Manandi Anand, Vibha Karen Dsouza, Gururaj H L, Gowtham M, and Dalibor Dobrilovic

10.1 CYBERWARFARE: WHERE CODE BECOMES THE BATTLEFIELD

10.1.1 Introduction

Imagine a missile launch code intercepted, classified intel leaked online, or troops paralyzed by a communications blackout. These aren't fictional scenarios, but stark realities of the modern cyberwarfare landscape. Information technology (IT) empowers modern warfare, but it also creates a digital battleground where vulnerabilities lurk. **Cyberattacks aren't mere inconveniences; they are threats with real-world consequences.**

From state-sponsored hackers to lone wolves, malicious actors exploit vulnerabilities in military IT, manipulating data, disrupting operations, and wielding digital weapons with devastating consequences [1]. While IT has revolutionized warfare, it has also opened a new front - one fought in the silent realm of code and encrypted networks (Figure 10.1).

Enter blockchain technology, a potential game-changer in this evolving threat landscape. Its unique features offer a powerful arsenal to combat the very issues plaguing military cybersecurity:

- **Data Integrity:** Immutability and transparency ensure data remains tamper-proof and traceable, safeguarding sensitive information.
- **Secure Communication:** Encrypted channels and permissioned networks shield communication from eavesdropping and unauthorized access.
- **Resilience:** Decentralized architecture makes it immune to single points of failure, enhancing network stability and resistance to attacks [2].

Imagine an immutable military network: a blockchain ledger where sensitive information is secured by cryptography and distributed across a network of nodes, making it virtually untouchable. Imagine secure communication

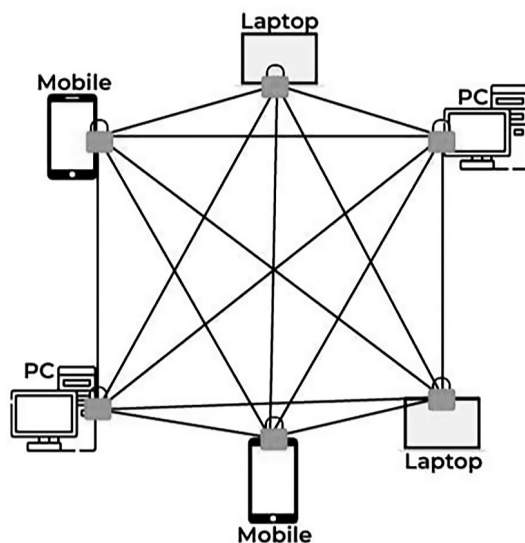


Figure 10.1 Encrypted Network.

channels shielding military communication from prying eyes, and permissioned networks granting access only to authorized personnel. This is the transformative potential of blockchain, offering a stronger shield against the ever-evolving threats of the digital age.

However, implementing this innovative technology requires careful consideration. **Scalability, integration with existing systems, and robust cybersecurity measures** all need careful consideration. Despite these hurdles, the potential benefits are undeniable.

This chapter delves deeper into the specific features of blockchain, its potential applications in military cybersecurity, and the challenges that need to be addressed. Ultimately, it seeks to answer the critical question: **Can blockchain be the key to securing the digital frontlines and safeguarding the future of modern warfare?**

10.1.2 Unleash tamper-proof identities: revolutionizing military cybersecurity with blockchain!

Securing access in the heart of the military battleground demands ironclad defenses against malicious interference. Enter blockchain – a game-changing force poised to shatter conventional barriers and fortify identity management like never before [3]. But why? Because blockchain brings forth

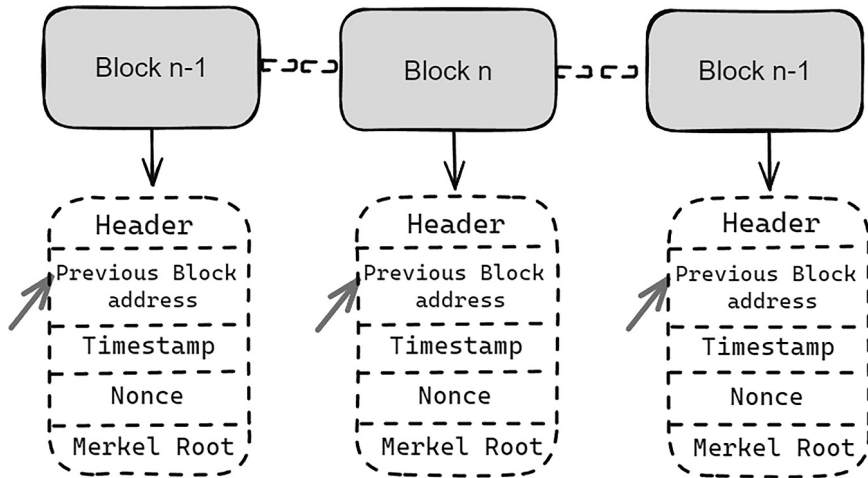


Figure 10.2 Unleashing blockchain.

unprecedented levels of resistance to tampering thanks to its multi-layered shield of security features (Figure 10.2).

10.1.2.1 Unbreakable digital signature

- **Cryptographic Anchoring:** Every identity boasts a distinctive “digital DNA” called a cryptographic hash, rendering immediate detection of meddling efforts [Jonathan K., Yehuda L., -“Introduction to Modern Cryptography:..”] [Don.T, Alex.T- “Blockchain Revolution:..”] [Post-quantum cryptography].
- **Decentralized Storage:** No longer confined to singular targets, identities now dwell amongst countless nodes throughout the blockchain – making simultaneous modification near impossible [Secure multi-party computation] [Zero knowledge proofs].
- **Immutable Data Records:** Record once, etched forevermore – modifying past entries means corrupting the whole chronicle, leaving perpetrators exposed under scrutiny [Secure multi-party computation].
- **Consensus Mechanisms:** Consortium approval required prior to acceptance renders hijacking dauntingly expensive & arduous endeavour [C.Cachin,..-“Secure Messaging Protocols for Distributed:..”] [D.M.Anthoni, M.H.A.-“Secure Communication Usin Blockchain Technology”].
- **Auditable Transactions:** Public viewership enables effortless audit trails, exposing illicit activity lurking beneath the surface [Wei Z., Yongqiang W.”A study of blockchain technology.....”].

But wait...there's more! Beyond mere mechanics lies potency bolstering efficiencies too:

- Reduce dependence on tangibles susceptible to pilferage or duplication.
- Foster harmonious interactions spanning various divisions plus allies worldwide.
- Safeguard individual rights whilst confirming vital details [4].

As cutting-edge innovation meets strategic imperatives headlong, embracing blockchain promises heightened safeguards alongside amplified agility amidst ever-evolving landscapes rife with perils. [Don.T, Alex.T- "Blockchain Revolution:.."] [General M.M -"The Future of the Army"] [Don.T, Alex.T -"Blockchain Revolution:Technology Behind Bitcoin"]

10.1.2.2 Revolutionizing military security: the advanced capabilities of smart contracts

Step aside from mundane grant-or-deny scenarios; smart contracts usher in an era of sophisticated, agile defense mechanisms that transcend conventional limitations! With these digital powerhouses at play, imagine how our armed forces could benefit from unprecedented levels of precision and responsiveness when securing vital operations:

- Dynamic & Contextual Permissions – Adaptable Fortresses [Don.T, Alex.T- "Blockchain Revolution:.."] [Don.T, Alex.T -"Blockchain Revolution:Technology Behind Bitcoin"] [NSTXL-"Blockchain for Military Logistics:.."]:
 - Mission-Specific Access: Soldiers gain tailored privileges according to active missions, shielded by automated barriers safeguarding confidential information.
 - Threat Level Adjustments: In times of increased vigilance, intelligent algorithms instantly lock down high-risk regions using live intel updates [5].
 - Temporal Control: Time constraints ensure temporary grants expire post-mission, averting unwarranted long-term usage.
- Beyond Identification Authentication – Layered Defense Systems [Andreas B, Christopher F-"Hyperledger Fabric:.."] [NSTXL-"Blockchain for Military Logistics:.."] (Figure 10.3):
 - Biometrics Unleashed: Fingerprints, iris scans – smart contract technology integrates advanced verification methods, thwarting credential theft attempts [6].
 - Environmental Monitoring: Detect anomalous activities via sensor inputs, promptly activating countermeasures as needed.



Figure 10.3 Identification authentication.

- Equipment Authorizations: Only trained operators receive green lights for operating crucial gear such as weaponry or UAVs (unmanned aerial vehicles).
- Decentralized Powerhouse – Tamper-Proof Transparency & Robustness [S.S.S.N. P-“Blockchain for Military Applications:..”, “Everyday Cryptography.”]:
 - Audit Trail Immunity: Blockchains record every action indelibly, offering complete transparency while deterring manipulation efforts.
 - Attack Mitigation: Dispersed networks resist concentrated assaults since compromised nodes don’t jeopardize overall integrity [7].
 - Scalability & Interoperability: Common standards foster smooth cooperation among diverse units and allies worldwide.

Embrace this future brimming with possibilities: Enhanced efficacy during combat ops, improved situation comprehension, reduced risk profiles – all thanks to cutting-edge tech empowering tomorrow’s soldiers today! (Table 10.1).

Real-life instances supporting these examples can be found scattered across reports and articles, suggesting ongoing exploration and interest in employing blockchain technology and smart contracts for military applications. Noteworthy mentions involve United States special operations commands collaborating with tech firms, the U.S [8–11]. Navy testing blockchain technology for manufacturing systems, and the Department of Défense

Table 10.1 Real-world examples of smart contracts in military access control

Example	Description	Reference
Secure Supply Chain Management	Managing access and movements of critical military supplies -Verify ID & auth. -Track locations -Trigger alerts	[NSTXL-“Blockchain for Military Logistics:..”] [1, section 3.2]
Base Camp Access Control	Granting access based on clearance level, mission required, location -Automatic logging of entries & exits	[Martin V –“Warfare in Age of hyperconnectivity”][Wei Z., Yongqiang W. “A study of blockchain technology.....”] [PwC-“How blockchain can transform defence assets”]
Secure Communication on Channels	Managing encryption keys & access permissions for communication channels. -Restrict info to cleared individuals	[Martin C.-“Warfare in the Age of Hyperconnectivity”, chapter X]
Drone and weapon Access Control	Equipment operator vetting, deactivating, usage data recording	[Christopher C. -“Military Revolution and the rise of the West”][General M.M -“The Future of the Army”] [Don.T,Alex.T -“Blockchain Revolution:Technology Behind Bitcoin”]

[NSTXL-“Blockchain for Military Logistics:..”] considering blockchain for asset cybersecurity. Despite remaining hurdles, enthusiasm persists because of the promising outcomes expected once fully implemented [PwC-“How blockchain can transform defence assets”].

10.1.2.2.1 Benefits of blockchain-based identity management

1. Enhanced security: Tamper-proof identities and decentralized verification make it significantly harder for unauthorized access and impersonation.
2. Increased efficiency: Smart contracts automate access control processes, reducing manual verification and streamlining procedures.
3. Improved accountability: A transparent audit trail of all identity transactions provides clear accountability and facilitates investigation of potential security breaches.
4. Scalability and adaptability: The distributed nature of blockchain allows for easy scaling to accommodate a growing number of personnel and assets.

10.1.2.2.2 Secure communication: fortressing military communication with blockchain

In the realm of strategic defense, where uncompromising protection of vital intelligence takes precedence, conventional means of communication often fall short due to susceptibilities such as centralized points of failure, weak cryptographic protocols, and potential lapses caused by human factors [12]. However, the integration of cutting-edge blockchain technology presents a formidable fortification against these threats, establishing trustworthy foundations for impervious exchanges of classified data within military operations.

- Unbreakable Communications through Robust Encryption [European Defence Agency-“Blockchain in defence:..”] (Figure 10.4):

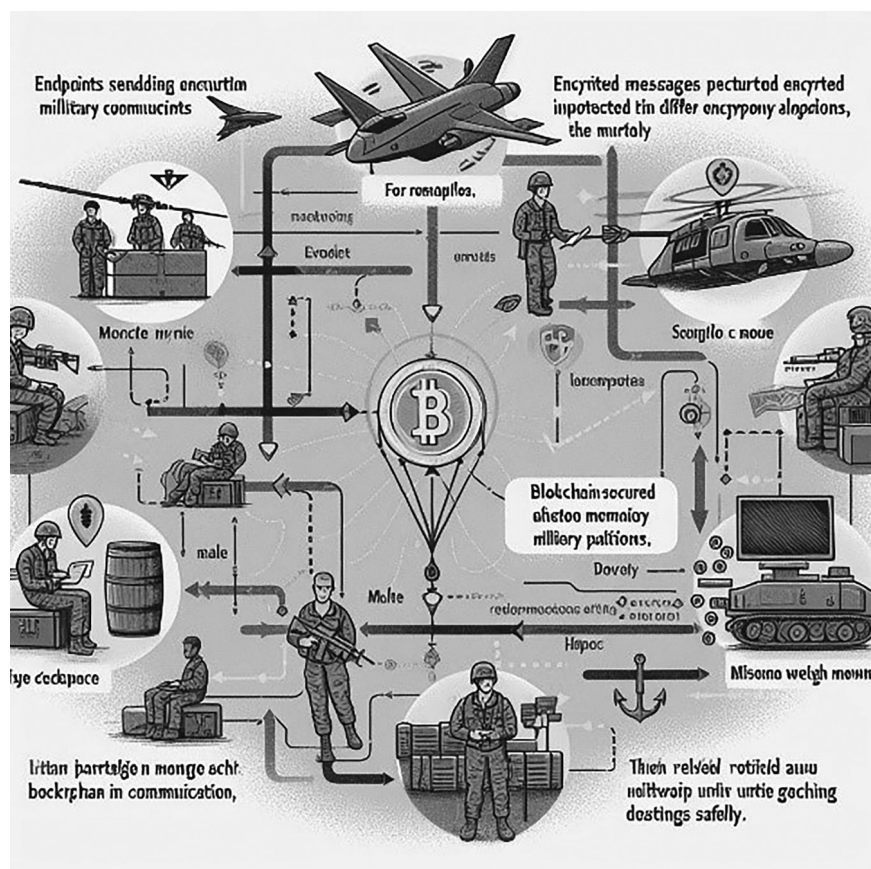


Figure 10.4 Unbreakable communications in military through robust encryption.

Blockchains innately embed encryption at every layer, securing transmitted messages and records prior to inclusion onto decentralized digital ledgers [13]. As a result, even under intense scrutiny, snooping upon and hijacking this privileged information becomes nearly insurmountable – shielded behind layers of advanced ciphers that demand legitimate authorization via designated decryption keys.

- Categorizing Essential Forms of Encryption [PwC-“How blockchain can transform defence assets”] (Figure 10.5):
- Symmetric Encryption: Utilizes one shared secret key between communicating entities; swift yet more prone when exposed [14].
- Asymmetric Encryption/Public Key Cryptography: Employs paired sets comprising each entity’s unique public & private keys; bolsters safety amidst larger networks.
- Hybrid Encryption: Merging benefits of symmetrical and asymmetrical schemes, generating ephemeral session keys secured inside recipients’ public keys.

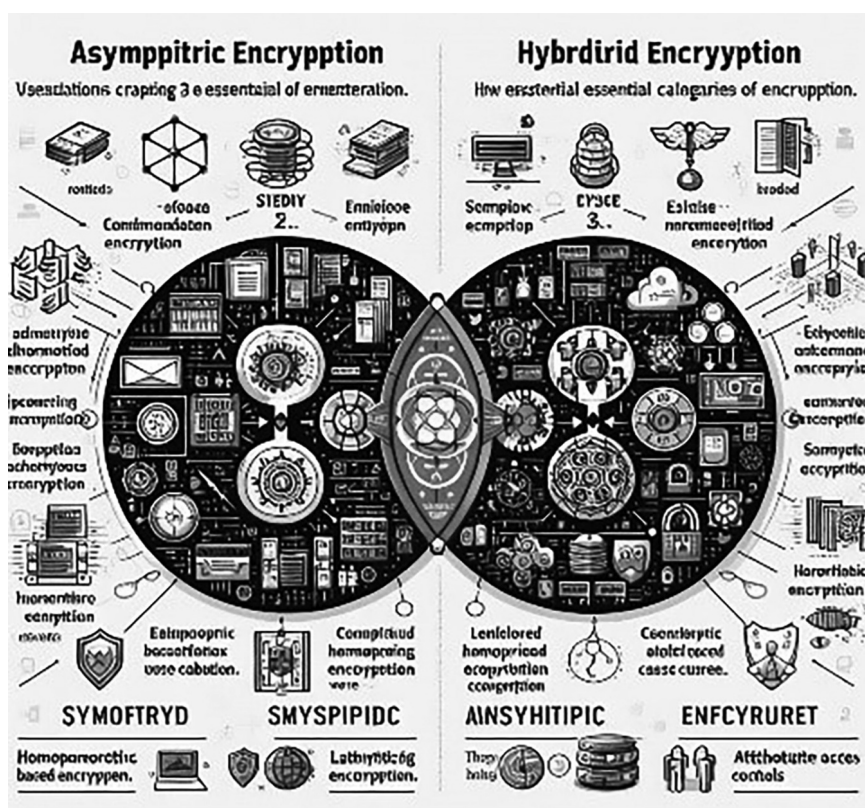


Figure 10.5 Categorizing essential forms of encryption.

Additionally, advanced encryption methodologies [European Defence Agency-“Blockchain in defence:..”], including Homomorphic Encryption allowing computation over concealed datasets, Lattice-based Encryption thwarting future Quantum Computing assaults, Attribute Based Access Control tailoring permissions according to defined characteristics - all contribute toward elevating our defenses beyond traditional boundaries [15–18].

10.1.2.2.3 Vital concerns regarding key management and safety measures

[General M.M –“The Future of the Army”][Don.T,Alex.T- “Blockchain Revolution:..”] (Figure 10.6):

- Safely generate and store secrets utilizing certified Random Number Generators alongside stout Hardware Storage Modules or sophisticated Key Management Systems.

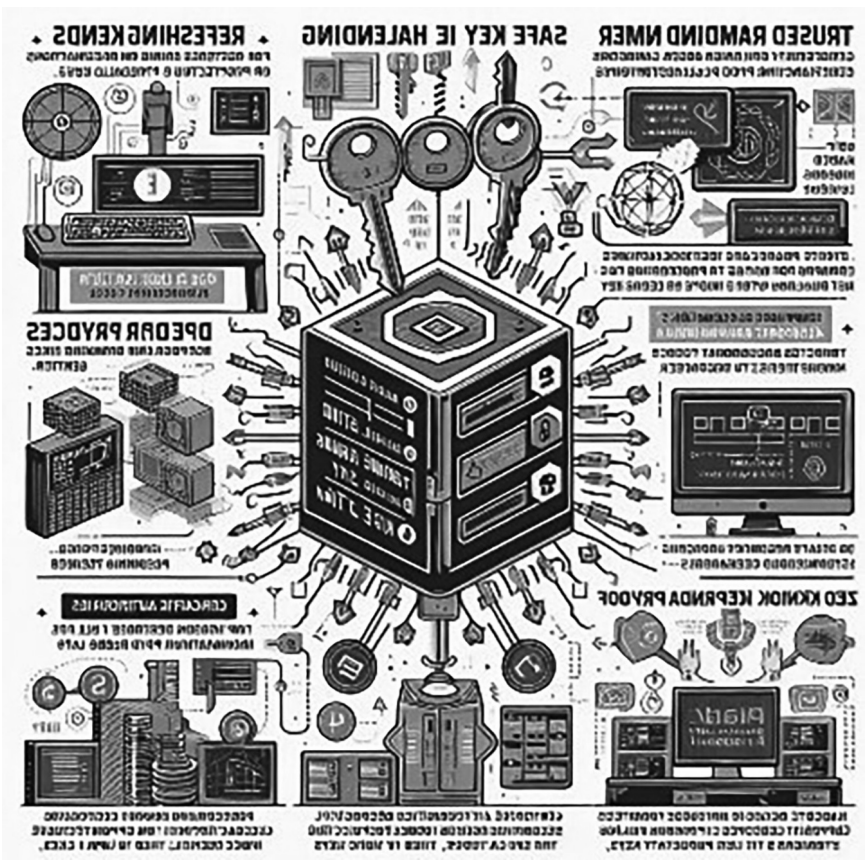


Figure 10.6 Vital concerns regarding key management and safety measures.

- Regularly refresh your crypto assets to counteract any breaches targeting longstanding credentials.
- Implement Certificate Authority frameworks managing identity verifications coupled with associated public keys throughout expansive ecosystems.
- Leverage Zero Knowledge Proofs proving dataset legitimacy devoid of exposing underlying contents – further reinforcing secrecy measures.

10.1.2.2.4 Interweaving encryption methodology with blockchain technology [General M.M –“The Future of the Army”][European Defence Agency–“Blockchain in defence..”]:

- On-Chain Encryption: Enveloping transmissions beneath protective cloaks pre-emptively ensures inviolability once recorded across multiple nodes forming partaking chains [19].
- Off-Chain Encryption: Prioritizing expedited delivery mechanisms whilst still leveraging chain validation processes during transit stages outside mainframe databases.
- Blended Approach: Strategically combining elements drawn from either approach above fostering optimum levels regarding operational efficacy balanced harmoniously beside stringent standards pertaining cybersecurity requirements.

By judiciously selecting appropriate forms of encryption technologies complementary to rigorous key administration procedures, organizations stand poised not merely defending themselves today but also proactively preparing tomorrow’s challenges headlong – thereby guaranteeing utmost confidence, fidelity, plus authentication concerning mission-critical interactions spanning diverse sectors inclusive militaries worldwide!

10.1.2.2.5 Unleashing the power of distributed consensus: guardians of decentralized communications

In the realm of blockchain technology, where trust less interactions thrive amidst the absence of a central authority, **distributed consensus mechanisms** stand tall as silent sentinels safeguarding reliable and transparent communications. Imagine navigating a sea of uncertainty – where numerous individuals strive to align upon a singular version of truth within a vast ocean of data. That’s precisely why blockchain necessitates robust methods to establish accord.

- **Why Do We Crave Agreement?**
Picture a team attempting to synchronize edits over a joint manuscript sans leadership; discord reigns supreme unless everyone agrees! Similarly, blockchain demands uniform understanding regarding the status

quo of its underlying ledger housing all transactions. Enter stage left: **distributed consensus**.

- **Meet Your Unsung Heroes:**

These protocols serve pivotal roles in maintaining harmony throughout the sprawling web of interconnected peers constituting a blockchain ecosystem. Their duties include:

1. Ensuring consistency – guaranteeing identical views held collectively concerning the current state of affairs [20].
2. Immutability assurance – once appended onto the record, entries resist modification or deletion.
3. Resiliency provision – fortifying resistance toward malign agents seeking disruption.

- **Two renowned champions leading this charge comprise:**

- **Proof-of-Work (PoW):** Initially employed by Bitcoin, miners engage in fierce competition tackling intricate math challenges until victory crowns he whose prowess unlocks adding fresh blocks. However, despite proving effective, critics decry high resource consumption inherent herein [21].
- **Proof-of-Stake (PoS):** As an eco-friendlier substitute, PoS empowers participating members via vested interest rather than brute force processing capabilities. Validator selection hinges primarily upon quantity & duration of owned coins instead. [General M.M -“The Future of the Army”][Don.T, Alex.T -“Blockchain Revolution:Technology Behind Bitcoin”]

- **Advancing Toward Fortified Interaction:**

Embracing distributed consensus brings forth manifold benefits facilitating dependable exchanges:

- **Tamper-resistance** – shielding communiqués from manipulation attempts due to indelibility enforced by design.
- **Devolution of dominance** – eliminating sole points susceptible to compromise fostering durability versus assaults targeting individual components.
- **Openness promotion** – enabling universal scrutiny bolsters confidence whilst deterring fraudulent conduct.
- Such attributes render these technologies ideal candidates spanning diverse domains including:
- **Défense sector messaging** – reinforcing confidentiality measures guarding vital intel exchange.[Don.T, Alex.T- “Blockchain Revolution:..”]
- **Supply Chain Management** – monitoring commodities transit alongside authentication affirmation. [Martin V -“Warfare in Age of hyperconnectivity”]
- **Electoral Integrity Assurances** – enhancing election credibility vis à vis traceable ballot casting [22].
- **Health Records Safeguarding** – protecting patients’ medical histories during dissemination efforts.

As evidenced above, embracing distributed consensus propels us toward heightened levels of reliability and efficiency pervading our digitized landscape. So let us celebrate these architectural marvels stealthily shaping tomorrow's connected society today!

10.1.2.2.6 Benefits of secure messaging in military units

<i>Key advantage</i>	<i>Impact on military operations</i>	<i>References</i>
Improved Battlefield Situational Awareness	Quicker decision-making Optimized tactics	[Don.T,Alex.T- "Blockchain Revolution:.."] [Martin V -"Warfare in Age of hyperconnectivity"]
Streamlined Resource Distribution	More efficient use of materials	[Don.T,Alex.T- "Blockchain Revolution:.."] [Christoper C. -"Military Revolution and the rise of the West"]
Robust Data Sharing Across Agencies	Seamless cross-agency intelligence	[Don.T,Alex.T- "Blockchain Revolution:.."] [Martin V -"Warfare in Age of hyperconnectivity"]
Authentic Information Provenance	Greater trust in analytical outcomes	[Don.T,Alex.T- "Blockchain Revolution:.."] [Martin V -"Warfare in Age of hyperconnectivity"]
Automated Threat Identification and Response	Predictive insight's Faster reaction time	[Martin V -"Warfare in Age of hyperconnectivity"] [General M.M -"The Future of the Army"]
Stronger Alliance Cooperation	Coordinated multinational missions	[Don.T,Alex.T- "Blockchain Revolution:.."] [Martin V -"Warfare in Age of hyperconnectivity"]
Increased Accountability and Auditing	Reduced errors Greater responsibility	[Don.T,Alex.T- "Blockchain Revolution:.."] [Martin V -"Warfare in Age of hyperconnectivity"]

10.1.2.2.7 Building robust secure messaging protocols for military communication

To implement cutting-edge secure messaging protocols for military communication, attention must be given to intricate details involving technology, usability, and compatibility. Here's a concise overview highlighting vital points:

- Network Design & Architecture [Secure multi-party computation]
 1. Permissioned Blockchain: Restricts access to verified military personnel, safeguarding data privacy and security.
 2. Hybrid Architecture: Balances efficiency and security by utilizing on-chain and off-chain components optimally.
 3. Scalability & Resilience: Designed to manage extensive volumes of transactions and resist network disturbances.
- Cryptographic Techniques & Algorithms [IACR]
 1. Elliptic Curve Cryptography (ECC): Offers effective and safe encryption for messages and identities.
 2. Zero-Knowledge Proofs: Allow validation of data without exposing contents, bolstering privacy.
 3. Homomorphic Encryption: Facilitates computations on encrypted data without needing decryption, enabling secure analytics and sharing.
- Secure Messaging Protocol Design
 1. End-to-End Encryption: Keeps messages secured starting from the sender's device till they arrive at the receivers.
 2. Forward and Backward Secrecy: Protects prior and upcoming messages despite breaches affecting individual keys.
 3. Message Authentication & Non-Repudiation: Assures message authenticity and impedes pretending to be someone else.
- User Experience & Integration
 1. Intuitive Interfaces: Prioritize ease of usage and harmonious assimilation with current military communication tools.
 2. Offline Functionality: Empowers interaction regardless of connectivity status.
 3. Identity Management: Safeguarded and proficient user identification and approval frameworks.
- Integration with Existing Military Systems
 1. Compatibility with Legacy Systems: Ensure smooth interactions among established infrastructure.
 2. Data Standardization: Define uniform structures for message substance and meta-exchange.
 3. Regular Audits & Penetration Tests: Persistent evaluation of susceptibilities and execution of countermeasures (Figure 10.7).

By focusing on these facets, developers can craft sophisticated, trustworthy, and operationally viable secure messaging protocols suited for contemporary military demands.[Post-quantum cryptography].

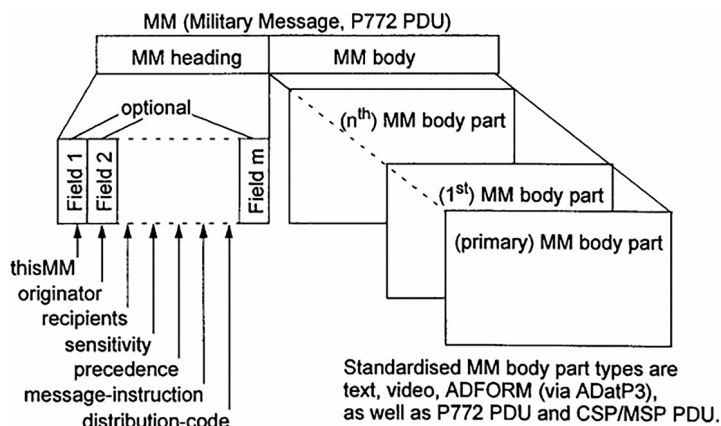


Figure 10.7 Integration with existing military systems.

10.1.2.3 Supply chain security: forging a trustworthy chain for military supplies

Blockchain technology has the potential to revolutionize military supply chain management by providing secure and efficient data sharing and storage solutions [S.S.S.N. P-“Blockchain for Military Applications:..”, “Everyday Cryptography.”]. By tracking and verifying every step of the supply chain, from manufacturing to delivery, blockchain creates an auditable trail of ownership and provenance, allowing authorized personnel to track the location, condition, and history of any item in real time [S.S.S.N. P-“Blockchain for Military Applications:..”, “Everyday Cryptography.”]. This technology combats counterfeiting and fraud by making it virtually impossible to alter or forge records, reducing the risk of counterfeit parts and supplies infiltrating the chain [Don.T, Alex.T- “Blockchain Revolution:..”][Martin V -“Warfare in Age of hyperconnectivity”]. Furthermore, blockchain enables real-time tracking, visibility, and efficient logistics management, which can lead to accurate shipment information, predict delays, and suggest optimal routes, ultimately enhancing operational readiness [IACR].

The potential applications of blockchain in military logistics include improving the security and reliability of record-keeping, streamlining efficiency, minimizing the likelihood of counterfeits, and providing accurate and real-time inventory tracking information [IACR]. The integration of blockchain into military operations can significantly improve security, transparency, and supply chain resilience [IACR]. However, it is important to note that the adoption of blockchain into military supply chains would require careful planning, scalable infrastructure, effective governance frameworks, and a holistic approach that considers technology, regulatory compliance, data privacy, training, and maintenance [IACR].

The use of blockchain technology in military logistics is being explored by various organizations. For instance, the Défense Logistics Agency (DLA) Troop Support is engaging in pilot testing of blockchain and exploring its long-term potential for DOD implementation [Christopher C. -“Military Revolution and the rise of the West”]. Additionally, a study suggested that blockchain offers the best remedy for addressing NATO’s supply chain vulnerabilities, as it provides increased traceability and security [Don.T, Alex.T -“Blockchain Revolution:Technology Behind Bitcoin”].

In the realm of military operations, data sharing demands nothing short of a secure and controlled environment. Permissioned blockchains offer a robust solution for collaborative data management without compromising confidentiality or speed [S.S.S.N. P-“Blockchain for Military Applications:..”,“Everyday Cryptography..”]. Permissioned consortium chains and private blockchains are two key types of permissioned chains that offer benefits for military applications [S.S.S.N. P-“Blockchain for Military Applications:..”,“Everyday Cryptography..”]. Permissioned consortium chains are like a private club for authorized members only, where pre-selected organizations operate nodes and participate in consensus [Don.T, Alex.T- “Blockchain Revolution:..”]. Private blockchains, on the other hand, are ideal for highly sensitive military data that requires absolute control and security [S.S.S.N. P-“Blockchain for Military Applications:..”, “Everyday Cryptography..”]. They operate isolated from the public internet, minimizing the risk of external attacks and data breaches [S.S.Alam,..-“Secure and Efficient Group Messaging:..”].

Attribute-based encryption (ABE) and secure enclaves are two powerful tools that safeguard sensitive information on the blockchain [S.S.Alam,..-“Secure and Efficient Group Messaging:..”]. ABE encrypts data based on attributes instead of individual identities, empowering fine-grained control and reducing identity management overhead [Don.T, Alex.T- “Blockchain Revolution:..”]. Secure enclaves provide isolated execution environments where sensitive data processing occurs, shielded from unauthorized access even within the blockchain network [S.S.Alam,..-“Secure and Efficient Group Messaging:..”].

Smart contracts bring automation and transparency to military data sharing, enabling self-executing agreements that streamline data access and tracking [S.S.Alam,..-“Secure and Efficient Group Messaging:..”]. They can automate checkpoints that verify clearance levels and grant access to sensitive data only to those authorized, defining pre-defined rules and automatically granting or denying access based on those rules [S.S.Alam,..-“Secure and Efficient Group Messaging:..”]. Smart contracts can also act as digital detectives, recording the provenance of data within the blockchain, ensuring data integrity, preventing tampering, and allowing for auditing of data movement throughout the network [S.S.Alam,..-“Secure and Efficient Group Messaging:..”].

Zero-knowledge proofs (ZKPs) are cryptographic tools that empower secure verification of specific data properties without revealing the actual data itself [S.S.Alam,..-“Secure and Efficient Group Messaging:..”]. They can help mitigate the spread of misinformation and disinformation in sensitive military contexts [S.S.Alam,..-“Secure and Efficient Group Messaging:..”]. Homomorphic encryption allows computations on encrypted data without decryption, preserving privacy while unlocking the power of collaborative analysis [Don.T, Alex.T- “Blockchain Revolution:..”]. Secure Multi-Party Computation (MPC) allows parties to jointly compute a function on their private data inputs without revealing the data itself, opening doors to secure collaboration and decision-making in the sensitive world of military operations [Don.T, Alex.T- “Blockchain Revolution:..”].

Platforms like Hyperledger Fabric, Quorum, Ethereum private chains, Corda, Multichain, and Microsoft Azure Blockchain offer robust security and flexibility for permissioned chains, ABE, secure enclaves, smart contracts, ZKPs, homomorphic encryption, and MPC [Don.T, Alex.T- “Blockchain Revolution:..”]. These methods foster enhanced security, improved collaboration, preserved privacy, and increased trust among military partners [Don.T, Alex.T- “Blockchain Revolution:..”]. Their implementation across various projects worldwide demonstrates their effectiveness in meeting the demanding security standards required for military data sharing [Don.T, Alex.T- “Blockchain Revolution:..”].

Case Studies:

1. US Army’s Tactical Supply Chain (TSC) Blockchain Project:

<i>Features</i>	<i>Description</i>	<i>Result</i>
Technology	Permission consortium blockchain using Hyperledger Fabric	Controlled access
Smart Contracts	Automating tasks such as POs, invoice approval, and inventory	Reduced errors
Sensor & IoT Integration	Real-time tracking of supplies and equipment	Increased inventory visibility
Key Benefits	Improved demand forecasting, enhanced transparency, reduced errors, streamlined collaboration	30% reduction in order fulfilment errors; 20% increase in inventory visibility
Challenges	Scalability, data integration, security, privacy	Addressing these challenges will lead to expanded use cases
Potential Future Apps	Predictive maintenance, automated resupply, counterfeit detection	Greater efficiency and cost savings

2. NATO's Blockchain for Military Logistics (BCML) Initiative:

<i>Features</i>	<i>Description</i>	<i>Result</i>
Goal	Enhance secure data sharing and collaboration among NATO member nations	Strengthening the alliance
Technical Implementation	Private blockchain network, standardized data formats, security protocols	Secure data exchange
Key Benefits	Secure data exchange, enhanced situational awareness, streamlined collaboration	Improving operational efficiency and resource allocation
Potential Future Apps	Joint procurement, real-time monitoring, predictive maintenance	Greater integration and collaboration
Challenges	Data standardization, building trust, scalability and performance	Continued development and testing required

3. Singapore Armed Forces (SAF) Blockchain for Battlefield Awareness (BA):

<i>Feature</i>	<i>Description</i>	<i>Result</i>
Goal	Enhance battlefield awareness and decision making	Improved soldier safety
Technical Implementation	Permissioned blockchain network, zero-knowledge proofs, data access control	Secure intelligence sharing
Key Benefits	Secure intelligence sharing, enhanced situational awareness, improved decision-making	More informed and timely decisions
Challenges	Scalability, user adoption, data privacy and security	Continuous development and adaptation required
Potential Future Apps	Joint intelligence analysis, predictive threat detection, decentralized command and control	Revolutionary battlefield operations

10.1.2.3.1 Revolutionizing defense operations with blockchain technologies

As the landscape of defense strategy shifts toward enhanced security, collaboration, and operational efficiencies, blockchain offers a promising avenue for transformation. This innovative technology presents unique advantages in various domains, shaping the future of military operations in profound ways.

- **Redefining Battlefields**
Blockchain enables real-time data collection and analysis from diverse sources, equipping commanders with predictive insights for decisive actions. With its inherently secure architecture, blockchain strengthens coalition operations, improves precision warfare, and bolsters resilient supply chains.
- **Seamless Alliance Building**
By leveraging standardized data formats and promoting transparent communications, blockchain fosters international cooperation and interoperability between militaries. This synergistic environment empowers rapid information sharing and joint planning, leading to more effective responses to shared threats.
- **Advancing Training and Simulation Methods**
Through immersive virtual environments, blockchain supports advanced training scenarios, offering dynamic and data-driven experiences. Performance analytics and peer-to-peer learning facilitate continuous improvement and targeted skill development.
- **Ethical and Strategic Dilemmas**
While embracing automation and autonomy, blockchain necessitates addressing ethical concerns and establishing human oversight mechanisms [23]. Trust and transparency remain paramount in maintaining a reliable and sustainable ecosystem. Misinformation and disinformation pose formidable challenges requiring vigilant countermeasures and critical thinking skills.
- **Cost and Resource Implications**
Implementing blockchain systems demands substantial investment in infrastructure, training, and strategic alignment. To ensure long-term viability, stakeholders must prioritize sustainable funding models and develop specialized expertise.

This academic discourse highlights the multifaceted impact of blockchain on defense operations, emphasizing the necessity for comprehensive understanding and thoughtful execution. By navigating the intricate web of technological advancements, political implications, and ethical considerations, military leaders can shape a brighter tomorrow for our collective security.

10.2 CHALLENGES AND CONSIDERATIONS

Implementing secure messaging protocols built on blockchain for military communication requires careful consideration of technical details, user experience, and integration with existing systems. Overcoming the challenges and embracing continuous improvement will pave the way for secure, reliable, and efficient communication, enhancing operational effectiveness and safeguarding sensitive information in the ever-evolving landscape of military operations. This is just a high-level overview, and specific implementations will vary depending on the specific needs and requirements of different military units. However, by understanding the key principles and challenges involved, we can chart a course toward secure and effective military communication in the blockchain era.

10.2.1 Scalability and performance

One of the major limitations of traditional Blockchain models is achieving scalability-handling many transactions at once. Scalability often requires sacrificing the concept of security, decentralisation and in many scenarios both. Achieving scalability requires incentivising a huge number of participants to achieve high level of security, but this in turn can increase the costs and create bottlenecks. These limitations become particularly relevant in the context of military operations where demanding needs such as speed, efficiency, and reliability are paramount (Table 10.2).

10.2.1.1 *Efficient consensus mechanisms for military operations*

1. Permission Blockchain: It is a type of distributed ledger that requires “permission” to access and is not open to the public. This layer of access control and security allows only permissioned users to perform actions that have been set by ledger administrators, while also identifying themselves digitally or through certificates.
 - Key-Features
 - Centralised Authority
 - No Anonymity
 - Restricted Access
 - Scalable
 - Advantages
 - Security: Only authenticated users can participate.
 - Flexible Decentralisation: Can either be incremental or fully centralised.
 - Performance: Works faster since it has limited accessibility.

Table 10.2 Scalability and performance

<i>Limitations</i>	<i>Description</i>	<i>Issue</i>	<i>Relevance to military operations</i>
1. Transaction Throughput	Transaction Throughput involves checking the validity of transactions (e.g. Verifying signatures and token balances) and executing the on-chain logic needed to calculate state changes.	The primary challenge is figuring out how to increase the number of computations per second without significantly raising the hardware demands on individual full nodes responsible for validating transaction blocks.	In military operations, swift and frequent transactions are vital for real-time communication, logistics tracking, and data sharing. However, many blockchain systems lack the processing speed necessary to meet these urgent demands, hindering coordination, communication, and logistical effectiveness in military scenarios. This underscores the necessity for blockchain solutions capable of seamlessly handling rapid and continuous transactions in military environments.
2. Latency	Blockchain networks rely on consensus mechanisms for nodes to agree on the current state. [24] Consensus ensures majority honesty, crucial for handling malicious participants and ensuring transaction finality. It aims for accuracy, robust Byzantine fault tolerance, decentralization, and quicker settlement by minimizing communication complexity.	The primary challenge is figuring out how to reach finality faster, cheaper and with increased trust minimisation- all while maintaining predictability, stability and accuracy.	In military operations, rapid decision-making and communication are critical. However, blockchain latency, caused by consensus mechanisms, can hinder real-time coordination. This delay may impede swift decision-making and response efforts, highlighting the importance of addressing latency challenges in blockchain technology for better synchronization with military operational needs.

(Continued)

Table 10.2 (Continued)

<i>Limitations</i>	<i>Description</i>	<i>Issue</i>	<i>Relevance to military operations</i>
3. Resource intensity	In Blockchain technology “Resource Intensity” refers to the significant computational and energy resources required to perform certain operations with the blockchain network. This limitation primarily manifests in the consensus mechanisms employed by blockchain protocols	The primary challenge is that some consensus mechanisms, especially PoW require significant computational power which can limit the ability of blockchain networks to scale without a proportional increase in resources.	In military deployments, operational settings often have limited resources. Some blockchain consensus mechanisms, as discussed earlier, demand substantial computational power, making them resource-intensive. In resource-constrained military environments, allocating and sustaining these resources becomes impractical. The challenge lies in finding mechanisms that balance security and efficiency while remaining feasible within military constraints.
4. Data storage: Balancing Security, Efficiency, and Accessibility in Military Blockchain	Storing large amounts of military data on-chain can quickly become impractical and expensive, as blockchain nodes need to store the entire transaction history.	On-Chain Data Storage: Limited storage capacity, higher costs due to increased node requirements, and slower access times for large datasets. Off-Chain Data Storage: Potential vulnerability to cyberattacks, privacy concerns if centralized systems are used, and reliance on trusted intermediaries.	Data sensitivity: Military data can range from tactical movements to troop logistics, with varying levels of sensitivity. Determining the appropriate storage location for each data type is crucial.

- Disadvantages
 - Risk of corruption: with limited users there is a risk of overloading consensus.
 - Regulation: Transactions can be restricted.
 - Vulnerable to Attacks: Fewer validators make it more prone to malicious attacks.
 - Relevance to Military Operations: The preference for controlled and secure network is paramount in military operations due to the sensitive nature of military operations. Permissioned blockchains provide controlled networks (controlling network is critical to prevent unauthorized parties from gaining insights into military activities), enhanced security (by limiting access to known and trusted entities), faster consensus, data confidentiality and scalability. Permissioned blockchains can also be customised to address the specific needs of military applications, including logistics tracking and data sharing.
2. Optimised Consensus Algorithms:
- Consensus Algorithm determines which nodes are authorised to record transactions and facilitates their rapid agreement on the content of a block [25]. This ensures data consistency and security, enhancing the computational efficiency of the blockchain.
- **Practical Byzantine Fault Tolerance (PBFT) consensus:** In the PBFT consensus model, the network's safety is guaranteed if a certain minimum percentage of nodes behave honestly and work properly. This type of consensus protocol is particularly used in digital asset backed platforms where the number of transactions is large, but the capacity required is minimal. Algorithms like PBFT or DPoS prioritize efficiency.
 - Relevance to Military Operations: Optimised consensus algorithms are essential in military applications as they enable swift decision making which plays a vital role in dynamic and unpredictable environments. These optimised algorithms contribute to the agility, adaptability and effectiveness of military operations.
3. Hybrid Solutions:
- Combining on-chain (it involves making changes directly to the blockchain protocol to increase its capacity to process and validate transaction) and off-chain (involves moving certain transactions off the main blockchain, conducting them through secondary networks or channels, and settling the final result on the main chain) solutions or leveraging layer 2 scaling (it builds on top of the main blockchain, introducing additional layers that facilitate faster and more cost-effective transactions) can significantly enhance the scalability of blockchain networks.
- Relevance to Military Operations: Offloading non-critical transactions to sidechains or layer 2 solutions enables the main blockchain to prioritize time-sensitive military operations. This strategy

enhances the efficiency of the blockchain by reducing congestion and latency, ensuring that crucial military decisions can be executed swiftly and without delays. The sidechains and layer 2 solutions handle the less critical transactions, allowing the main blockchain to concentrate on tasks that require immediate attention and rapid response in military scenarios.

10.2.1.2 Optimising blockchain implementation for military applications

1. Customised Smart Contracts: Tailoring smart contracts to the specific requirements of military logistics, communication, and data sharing. Optimising these contracts ensures their efficiency and their alignment with the unique demands of military operations [26].
2. Prioritising critical data: Designing the Blockchain to prioritise and expedite the processing of critical military data. This ensures that time-sensitive information is processed quickly, enhancing the responsiveness of military operations.
3. Continuous Innovation: Continuous innovation ensures the military blockchain applications remain at the forefront of scalability, security, and performance

10.2.2 Robust cybersecurity measures and regulations for military blockchain: a necessity, not a luxury [Christopher C. -“Military Revolution and the rise of the West”][IACR]

Blockchain technology holds immense potential for the military, but its implementation requires robust cybersecurity measures and regulations to address specific vulnerabilities and ensure the protection of sensitive data and systems. Here’s why:

10.2.2.1 Unique vulnerabilities of military blockchain

- Data Sensitivity: Military blockchain networks store highly sensitive data, including operational plans, logistics information, and classified intelligence. Compromise of this data could have devastating consequences [27].
- Attack Surface Expansion: Introducing blockchain creates a new attack surface, requiring additional security measures to protect nodes, smart contracts, consensus mechanisms, and communication channels.
- Insider Threats: Military personnel with access to the network pose a significant risk for insider attacks, requiring robust access control and monitoring mechanisms.

10.2.2.1.1 Weaponized smart contracts

Imagine malicious actors deploying smart contracts specifically designed to disrupt or exploit military blockchain networks. These “weaponized” contracts could:

- Trigger cascading failures: Malicious code could exploit vulnerabilities in interconnected smart contracts, causing a chain reaction of errors and malfunctions within critical military systems.
- Manipulate data and disrupt operations: Attackers could inject false information or tamper with sensitive data stored on the blockchain, potentially leading to compromised logistics, flawed intelligence analysis, or even misguided military operations.
- Cause denial-of-service attacks: Malicious contracts could overwhelm the network with transactions, crashing nodes and hindering essential communication and information sharing.

10.2.2.1.2 Supply chain compromise

Military blockchain systems may rely on complex supply chains involving various partners and vendors. Each point in the chain represents a potential vulnerability:

- Hardware Trojans: Compromised hardware components, like specialized chips or modified firmware, could be embedded at the manufacturing level, allowing attackers to eavesdrop on network traffic or inject malicious code into the system.
- Software vulnerabilities: Vulnerabilities in software used by supply chain partners, such as logistics management systems or communication protocols, could be exploited to gain unauthorized access to the blockchain network.
- Social engineering attacks: Malicious actors could target personnel within the supply chain, tricking them into revealing sensitive information or granting unauthorized access.

10.2.2.1.3 Quantum computing threats

While quantum computers are still in their early stages, their potential to break current cryptography poses a significant long-term threat to blockchain security. In the military context, this could lead to:

- Mass decryption of sensitive data: Quantum computers could crack the encryption algorithms currently used to protect sensitive military information stored on the blockchain.

- Forge signatures and compromise consensus: Attackers could exploit quantum capabilities to forge digital signatures or manipulate consensus mechanisms, allowing them to tamper with data or even take control of the blockchain network.
- Exacerbate existing vulnerabilities: Quantum computing could amplify the impact of existing vulnerabilities in smart contracts or cryptographic protocols, making them significantly more exploitable.

10.2.2.1.4 Human factor vulnerabilities

The human factor remains a critical element in any security system. Military personnel accessing and interacting with blockchain systems could be susceptible to:

- Social engineering and phishing attacks: Malicious actors could exploit personnel's vulnerabilities to social manipulation or phishing scams, gaining access to sensitive information or credentials.
- Insider threats: Disgruntled or compromised personnel with access to the network could intentionally leak information, tamper with systems, or even sabotage operations.
- Lack of awareness and training: Insufficient training and awareness about cybersecurity risks can lead to unintentional errors or security lapses, potentially exposing sensitive data or systems to attack.

10.2.2.1.5 Unforeseen vulnerabilities and evolving threats

The ever-evolving nature of technology and the creativity of malicious actors mean that new and unforeseen vulnerabilities may emerge. Military blockchain systems must be adaptable and resilient enough to address these evolving threats:

- Zero-day attacks: Exploits for previously unknown vulnerabilities could be used to compromise systems before patches or mitigation strategies are developed.
- Novel attack vectors: Malicious actors may develop entirely new attack methods, exploiting previously unconsidered weaknesses in the system or its surrounding infrastructure.
- The need for continuous risk assessment and adaptation: Military organizations must constantly assess risks, update security measures, and adapt their defensive strategies to stay ahead of evolving threats (Table 10.3).

As we navigate the complex landscape of securing military blockchain applications, continuous monitoring and threat detection (CMTD) emerges

Table 10.3 Need for cybersecurity measures and regulations

Need	Reason	Relevance to military operation
1. The Delicate Balance: Security vs. Functionality:	Finding the sweet spot: Striking a balance between robust security and efficient operation is essential. This requires implementing measures that effectively mitigate threats without significantly impacting operational capabilities. Regularly evaluating the risk landscape and adapting security measures accordingly is crucial to maintain this delicate balance.	Military blockchain applications can be incredibly valuable: Ensuring secure communication, logistics management, and intelligence sharing. However, prioritizing security too heavily can hinder functionality and adoption.
2. Specific Cybersecurity Measures	• Continuous monitoring and threat detection are essential for real-time response to cyberattacks. • Secure enclaves and hardware integration provide isolated environments within the blockchain network. • Post-quantum cryptography safeguards against future threats from quantum computers. • Decentralized identity and access control enhance security and privacy.	Establishing secure platforms and protocols for sharing threat intelligence across different military branches and allies is crucial for staying ahead of evolving cyber threats
3. The Human Factor and Training:	Having well-defined plans and procedures for responding to and mitigating cyberattacks is crucial for minimizing damage and ensuring operational continuity.	Educating military personnel about cybersecurity best practices and potential vulnerabilities is essential to prevent human errors and social engineering attacks. Red teaming and penetration testing: Regularly simulating cyberattacks on military blockchain systems can identify vulnerabilities and improve overall security posture.
4. The Global Landscape and International Cooperation	International collaboration: Sharing knowledge, best practices, and threat intelligence across national borders is essential to combatting global cyber threats and securing military blockchain deployments.	Developing a global framework: Joint efforts are needed to develop and implement a comprehensive framework for cybersecurity measures and regulations for military blockchain applications.

as a vital line of defense. Let's delve deeper into this critical aspect, exploring its nuances and potential challenges:

10.2.2.1.6 Unpacking CTMD in a military blockchain context

- **Continuous monitoring:** Continuously observing and analysing data from various sources, including blockchain activity, network traffic, system logs, and external threat intelligence feeds, to identify anomalies and potential threats in real-time.
- **Threat detection:** Employing sophisticated algorithms and machine learning techniques to analyse the monitored data and identify patterns, indicators of compromise (IOCs), and potential cyberattacks.
- **Real-time response:** Promptly notifying relevant personnel and triggering automated response mechanisms upon detecting a threat, minimizing potential damage and disruption.

10.2.2.1.7 Specific CTMD techniques for military blockchain

- **Blockchain-specific monitoring:** Analysing on-chain data like transaction patterns, smart contract behaviour, and consensus mechanisms for suspicious activity.
- **Network traffic monitoring:** Scrutinizing network traffic for anomalies, unauthorized access attempts, and potential malware signatures.
- **System log analysis:** Monitoring system logs of nodes and smart contracts for errors, suspicious activities, and potential vulnerabilities.
- **Threat intelligence integration:** Feeding external threat intelligence data into CTMD systems to stay ahead of evolving cyberattacks and vulnerabilities.

10.2.2.1.8 Challenges and considerations

- **Data volume and complexity:** Military blockchain systems can generate vast amounts of data, necessitating robust and scalable CTMD solutions that can efficiently analyse and interpret it.
- **False positives and alert fatigue:** Balancing sensitivity and accuracy to avoid overwhelming security personnel with false alarms and alert fatigue, leading to delayed responses to real threats.
- **Human resource limitations:** Adequately staffing and training personnel to effectively manage and interpret CTMD data and respond to alerts is essential.
- **Integration with existing systems:** Integrating CTMD solutions seamlessly with existing military communication and security infrastructure is crucial for efficient threat response and information sharing.

- Machine learning and AI: Leveraging advanced machine learning and artificial intelligence algorithms to identify subtle anomalies and predict potential threats with greater accuracy.
- Behavioural analysis: Analysing user behaviour and network traffic patterns to identify suspicious activities and potential insider threats.
- Decentralized CTMD: Exploring decentralized approaches where nodes within the blockchain network collaboratively monitor and analyse data, enhancing resilience and redundancy.
- Quantum-resistant CTMD: Developing CTMD solutions that are resilient against potential threats posed by future quantum computers.

10.2.2.1.9 Collaboration and knowledge sharing

- Collaboration across military branches and allies: Sharing CTMD data, best practices, and threat intelligence across military branches and allies strengthens collective defense against cyberattacks.
- Open-source intelligence and threat feeds: Utilizing open-source intelligence feeds and threat sharing platforms can provide valuable insights into evolving cyber threats and vulnerabilities.
- Industry partnerships: Collaborating with cybersecurity research and development organizations and private sector companies can accelerate the development of innovative CTMD solutions for military blockchain applications.

CTMD is not a silver bullet, but a critical component of a layered security approach for military blockchain systems. By continuously improving and adapting CTMD techniques, fostering collaboration, and staying ahead of evolving threats, we can build a robust and resilient future for military blockchain applications, ensuring the secure and trustworthy use of this transformative technology.

10.2.2.2 Regulatory landscape and international cooperation

- Developing regulations: Governments and international organizations need to develop specific regulations for secure military blockchain implementation, addressing issues like data privacy, security standards, and incident reporting.
- International Cooperation: Collaboration and knowledge sharing between military organizations and cybersecurity experts across borders is essential for developing effective cybersecurity measures and countering advanced cyber threats.

10.2.2.2.1 The evolving regulatory landscape

- **Fragmented and incomplete:** Currently, no unified global framework exists for regulating military blockchain applications. National regulations are often fragmented and lagging technological advancements, leading to uncertainty and inconsistency.

10.2.2.2.2 Building a comprehensive regulatory framework

- **Risk-based approach:** Regulations should adopt a risk-based approach, tailoring requirements to the specific risk profile of each application and technology.
- **Focus on outcomes, not technology:** Regulations should focus on desired security outcomes rather than dictating specific technological solutions, allowing for flexibility and innovation.
- **Transparency and stakeholder engagement:** Open and transparent communication with industry, academia, and civil society is crucial for developing effective and adaptable regulations.
- **International coordination and harmonization:** Establishing international cooperation mechanisms for coordinating regulatory efforts and harmonizing standards is essential for global consistency and effectiveness.

10.2.2.2.3 Challenges and considerations

- **National sovereignty concerns:** Balancing the need for international cooperation with respect for national sovereignty and security considerations can be challenging.
- **Technical complexity and rapid evolution:** Keeping regulations up to date with the rapid pace of technological advancements in blockchain and cyber threats requires continuous adaptation and innovation.
- **Cultural and legal differences:** Reconciling different legal systems and cultural perspectives on technology and security can be a complex and time-consuming process.

10.2.2.2.4 The road ahead: collaboration and continuous improvement

- **Building trust and understanding:** Fostering trust and understanding between different stakeholders, including governments, militaries, industry, and civil society, is crucial for successful international cooperation on military blockchain regulations.
- **Piloting and experimentation:** Implementing pilot projects and test cases can help identify practical challenges and refine regulatory approaches in a real-world setting.

- Adapting and learning: Continuously adapting regulations and learning from experience is necessary to keep pace with the evolving threats and opportunities presented by military blockchain technologies.

Building a secure military blockchain ecosystem requires a multi-layered approach. Robust cybersecurity measures, continuous monitoring and threat detection, and strong regulations are all essential components of ensuring the integrity and security of sensitive military data and systems. By prioritizing cybersecurity and taking proactive steps, the military can harness the power of blockchain while mitigating the associated risks.

10.2.3 FUTURE TRENDS AND OPPORTUNITIES: BLOCKCHAIN BEYOND IDENTITY MANAGEMENT

Blockchain technology has the potential to revolutionize various sectors beyond identity management. Some emerging trends and transformative impacts of blockchain technology include:

- Self-sovereign identities: Blockchain can transform how we manage our digital identities, empowering individual ownership of identity through self-signed, cryptographically verifiable credentials that establish trust in the user's identity and can be used to gain access across applications [Martin V - "Warfare in Age of hyperconnectivity"].
- Continuous, risk-based authentication: Continuous authentication confirms identity and risk on an ongoing basis, treating authentication as an ongoing event rather than a singular one [Martin V - "Warfare in Age of hyperconnectivity"].
- Decentralized finance (DeFi): DeFi has already taken the financial world by storm and will be an integral part of global finance by 2030, enabling lending, borrowing, and earning interest without the need for traditional banks [Christopher C. - "Military Revolution and the rise of the West"].
- Supply chain transparency: Blockchain can provide transparency in supply chains, enabling consumers to track the origin and journey of products [General M.M - "The Future of the Army"].
- Interoperability: Interoperability is crucial for the broader adoption of blockchain technology, enabling different blockchain networks to communicate and share information effortlessly [General M.M - "The Future of the Army"].

As blockchain technology evolves, it faces both technical and social challenges that need to be addressed for its full potential to be realized. Overcoming these challenges requires a concerted effort from industry leaders, governments, and the global tech community [General M.M - "The Future of the Army"]. The

future of blockchain technology is promising, and it has the potential to revolutionize industries, empower individuals, and create a more transparent and equitable digital world [General M.M -“The Future of the Army”].

10.2.4 CONCLUSION

In this chapter, we have dove into the significant affect of blockchain innovation on reinforcing security and security in military operations. The key takeaways are as follows:

Security Benefits:

- **Information Keeness:** Blockchain’s cryptographic hashes and unchanging information records stand as an impervious post against unauthorized get to and control of touchy information.
- **Upgraded Security:** By utilizing conveyed information capacity, blockchain disposes of single focuses of disappointment and fortifies arrange security through progressed encryption techniques.
- **Secure Communication:** Scrambled channels and permissioned systems ensure secure communication between military units and personnel.
- **Decentralized Command and Control:** The utilize of MPC innovation empowers secure and adaptable decision-making without dependence on a central authority.

Benefits for Military Collaboration:

- **Upgraded Security:** MPC shields touchy information whereas cultivating collaboration.
- **made strides Collaboration:** Streamlined communication and data sharing clear the way for improved operational efficiency.
- **preserved Independence:** Units hold control over their information and decision-making processes.
- **Expanded Believe and Straightforwardness:** Unchanging review trails altogether improve responsibility and trust.

Challenges and Future Directions:

- **Versatility and Execution:** Guaranteeing productive handling and arrange capacity for large-scale military applications may be a challenge that requests attention.
- **Standardization and Interoperability:** Building up common conventions and measures for consistent information trade is significant for the far reaching appropriation of blockchain in military operations.
- **Client Preparing and Appropriation:** Teaching military work force on blockchain innovation and its secure execution is fundamental for fruitful integration.

Case Studies:

The chapter exhibits three real-world blockchain ventures inside military coordinations, supply chain, and war zone mindfulness, shedding light on their benefits, challenges, and future potential.

Potential Applications:

Blockchain holds the guarantee of revolutionizing military cybersecurity by advertising permanent review trails, decentralized personality administration, tamper-proof information capacity, and a zero-trust security show. These highlights seem essentially decrease the chance of cyberattacks and reinforce the generally security pose of military operations.

Progressing Investigate and Development:

Tending to interoperability, versatility, protection, and keen contract impediments is pivotal to optimize military blockchain frameworks. It's basic to proceed investing in investigate and improvement endeavours to open the total potential of blockchain in military applications.

In general Impact:

Blockchain innovation has the potential to reshape the scene of military cybersecurity, advertising unparalleled highlights that can invigorate the security pose of military operations. In any case, it's fundamental to carefully consider the moral suggestions of sending such capable innovation inside the military context.

This conclusion underscores the transformative potential of blockchain in military applications whereas emphasizing the require for continuous investigate, moral contemplations, and key usage to tackle its full control

ACRONYMS WITH ABBREVIATIONS

1. UAV - unmanned aerial vehicles
2. PoW - Proof-of-Work
3. PoS - Proof-of-Stake
4. ECC - Elliptic Curve Cryptography
5. DLA - Défense Logistics Agency
6. DOD - Department of Defence
7. ABE - Attribute-based encryption
8. NATO - North Atlantic Treaty Organization
9. ZKP-Zero - Knowledge proof
10. MPC-Multi - Party Computation
11. TSC - Tactical Supply Chain
12. BCML - Blockchain for Military Logistics

13. SAF - Singapore Armed Forces
14. PBFT - Practical Byzantine Fault Tolerance
15. CTMD - Continuous monitoring and threat detection

REFERENCES

1. Amit Sahai and Brent Waters (n.d.), *Attribute-Based Encryption: Concepts and Applications*.
2. Andreas Bitter and Christopher Ferris (2018) *Hyperledger Fabric: A Practical Guide for Developers*.
3. Centre for a New American Security, Blockchain in the US Military (n.d.), *A Vision for the Future*.
4. C. Cachin, I. Marechal, and W. Penczek (n.d.), *Secure Messaging Protocols for Distributed Ledger Technologies*.
5. D.M. Anthoni and M. H. Anthoni (n.d.), *Secure Communication Using Blockchain Technology*.
6. Don Tapscott and Alex Tapscott (n.d.), *Blockchain Revolution: How the Technology Behind Bitcoin is Changing Money, Business, and the World*.
7. European Defence Agency (2020), *Blockchain in Defence: Technology in Defence*.
8. General Mark Milley (n.d.), *The Future of the Army: A Vision for 2040 and beyond*.
9. Jonathan Katz and Yehuda Lindell (n.d.), *Introduction to Modern Cryptography: Principles and Protocols*.
10. J.P.A. van der Veen and J.M.E.M. van der Wer (n.d.), *Distributed Ledger Technology for Military Applications*.
11. Leslie Lamport (n.d.), *The Byzantine Generals Problem: A Model for Fault-Tolerant Distributed Computing*.
12. Martin van Creveld (n.d.), *Warfare in the Age of Hyperconnectivity*.
13. M. Hassan, A. Qayyum, W. Aziz, and A. Rahman (n.d.), *Secure Mobile Messaging Protocol for Military Communication*.
14. National Institute of Standards and Technology (NIST) (n.d.), *Blockchain Working Group (Website)*.
15. NSTXL (2023), *Blockchain for Military Logistics: What You Should Know*.
16. Post-Quantum Cryptography (n.d.), <https://www.nist.gov/programs-projects/post-quantum-cryptography>
17. S. S. S. N. Perumal (n.d.), *Blockchain for Military Applications: A Review*.
18. PwC (2020) *How Blockchain Can Transform Defence Assets*.
19. S. Alam, A. Ashiq, M. A. Khan, and M. S. Hossain (n.d.), *Secure and Efficient Group Messaging on Blockchain*.
20. Secure Multi-Party Computation (n.d.), https://en.wikipedia.org/wiki/Secure_multi-party_computation
21. S. Zhang, Z. Liu, Y. Lin, and J. Li (n.d.), *Blockchain-based Communication in Military Network*.
22. The International Association for Cryptologic Research (IACR) (n.d.), <https://iacr.org/>

23. The Institute of Electrical and Electronics Engineers (IEEE) (n.d.), <https://www.ieee.org/>
24. The National Institute of Standards and Technology (NIST) (n.d.), <https://www.nist.gov/>
25. Wei Zhang, Rui Wang, and Yongqiang Wang (2020), *A Study of Blockchain Technology Development and Military Application Prospects*.
26. William Stallings (n.d.), *Everyday Cryptography: Fundamental Principles and Applications* (3rd Edition).
27. Zero-knowledge proofs. (n.d.), <https://coinmarketcap.com/alexandria/glossary/zero-knowledge-proof>

Harnessing blockchain to fortify military cyber security in the era of AI

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11.1 INTRODUCTION

In the rapidly evolving technology landscape, especially with the introduction of Artificial Intelligence (AI), it has become more crucial than ever to improve the complexities of cybersecurity. A new era is emerging with the unrelenting advancement of AI capabilities, which both present major threats to national security and discover previously unexplored benefits. The interconnectedness of military networks, communication systems, and data repositories increases the susceptibility of these critical systems to cyberattacks [1].

The risks related to military cyber security are rising due to AI-driven cyber threats, resulting in the need for creative and reliable solutions to guarantee the availability, integrity, and confidentiality of critical data. Researchers and practitioners are considering block chain technology as a viable pillar for bolstering military cyber security in the AI era in order to address these issues.

The distributed ledger technology known as “block chain,” which powers cryptocurrencies like Bitcoin, provides a decentralized, impenetrable foundation for transaction and data security. Since of its decentralized structure and cryptographic highlights, blockchain innovation has the potential to diminish the modern and complicated cyberthreats that military teach experience. The objective of this chapter is to supply an intensive outline of the dangers, challenges, and potential arrangements related to military cybersecurity utilizing blockchain innovation.

The advancement of AI technologies has led to a shift in cyber risks from conventional models. Attacks powered by AI have the ability to adapt to and learn from their surroundings, which places military operations and national security at serious risk [2]. Since these threats are ever-changing, cybersecurity must adopt a proactive strategy that goes beyond conventional perimeter security measures. Blockchain technology offers a visible and irreversible record, decentralizing trust and bringing about a paradigm shift in cybersecurity. The following chapters will investigate the various methods that blockchain might be utilized to ensure military frameworks,

lines of communication, and information storehouses. Blockchain has the potential to fundamentally alter the cybersecurity landscape for military applications, from ensuring the integrity of communication protocols to establishing a safe framework for data storage and retrieval. This chapter will evaluate the military's cybersecurity scenario critically and highlight the unique difficulties imposed on by attacks driven by artificial intelligence. Following that, it will investigate how military cyber defenses may be enhanced by utilizing blockchain technology. The chapter points to offer experiences into the commonsense sending of blockchain innovation in military settings by considering real utilize cases and conceivable applications.

This investigation into blockchain technology's possibilities will supplement the continuing conversation about protecting vital infrastructures from ever-changing cyber threats as we traverse the complex intersections between AI and military cybersecurity [3].

11.2 BLOCKCHAIN TECHNOLOGIES: A DIVE DEEPER

Blockchain Definition: Blockchain is a conveyed, decentralized record innovation that dependably and straightforwardly records exchanges across different PCs. Through the use of a consensus mechanism, all participants are certain to agree on the ledger's current state. Its decentralized structure prevents manipulation and offers a strong foundation for data protection.

An algorithmic data structure called a blockchain makes it possible to build a robust, impenetrable digital ledger of transactions between different users or nodes. Public-key cryptography is used by this technology to secure and sign transactions between users and nodes. Later that, a distributed ledger is used to store the transactions. The ledger is made up of blocks of transactions connected by cryptography to form a blockchain. Blocks of data cannot be altered or eliminated once they are entered into the blockchain ledger.

History of Blockchain: It is important to take a moment to review the background and factors that contributed to the creation of Blockchain technology before we get too far into the topic or even the formal definitions of the technology.

Blockchain Definitions and Taxonomy: Created by Don and Alex Tapscott, one of the most widely used definitions of the Blockchain is "an incorruptible digital ledger of economic transaction that can be programmed to record not just financial transactions but virtually everything of value." Blockchain technology is used in cyber security as a reliable way to digitize, track, and distribute data over a network.

To validate a new transaction, every network member has access to the most recent encrypted ledger copy. Blockchain technology offers several features including immutability, integrity, trackability, and decentralization. In order to improve security in the IoT domain, this chapter presents the

blockchain architecture and describes its concept, characteristics, and need for security. It also explains how Bitcoin operates. It makes an effort to draw attention to how Blockchain will influence cyber security in the future [4].

The Blockchain is a sequential chain of blocks where new blocks are consistently appended in a linear manner. Despite variations in block content among different Blockchain networks, fundamental elements like hash, transaction, and date remain constant across all iterations. Each block includes the cryptographic hash of its preceding block, automatically generated and resistant to alteration. Consequently, every new block strengthens the verification of preceding blocks, enhancing the overall security of the Blockchain. As the chain accumulates more blocks, the Blockchain becomes increasingly stable and secure.

The emergence of blockchain technology is revolutionizing the data storage sector, shifting it toward decentralization. With its array of security benefits, blockchain represents a state-of-the-art storage method. It's clear that when blockchain and AI are integrated, their distinct attributes and technical prowess combine to reshape the digital landscape in profound ways.

Data Integrity and Resilience: Blockchain maintains data integrity through a series of blocks, with each block containing a cryptographic hash of the preceding one. This interconnection makes it exceedingly difficult to alter any information retroactively. The redundancy of data across multiple nodes in the network enhances resilience, preventing a single point of failure [5].

Example: In a military context, consider a scenario where critical mission data is stored on a blockchain. The immutability of the blocks ensures that the information remains unchanged and trustworthy, minimizing the potential for unauthorized alterations that could jeopardize the achievement of objectives.

11.2.1 Enhanced security features

Immutability: After data is appended to a block, it becomes highly improbable to modify due to cryptographic hashing, making any retroactive changes evident and preserving the integrity of the information.

Example: In military logistics, the immutability of supply chain data on a blockchain ensures that information about the origin, shipment, and delivery of critical supplies remains unchanged, reducing the risk of counterfeit goods or unauthorized alterations.

Transparency: Blockchain provides a transparent and auditable record of transactions to all participants in the network. This transparency enhances trust among stakeholders and facilitates accountability.

Example: Transparent record-keeping in military operations can include tracking the movement of personnel, equipment, and resources on a blockchain. This visibility promotes accountability and ensures that authorized personnel can verify the legitimacy of activities.

Decentralization: Blockchain operates on a decentralized network of nodes, eliminating the need for a central authority. This decentralization not only enhances security but also reduces the risk of a single point of failure.

Example: In military communication systems, a decentralized blockchain can be employed to ensure that communication channels remain operational even if certain nodes are compromised, preventing widespread disruptions.

Consensus Mechanisms: Consensus methods are protocols utilized in blockchain networks to ensure consensus among all nodes regarding the present condition of the ledger. Proof of Work (PoW) and Proof of Stake (PoS) stand out as two widely adopted methods, each with unique efficiency and security benefits. Example: In military intelligence sharing, a consensus mechanism can be implemented to validate the authenticity of information before it is added to the blockchain, ensuring that only accurate and verified intelligence is shared among different branches of the military.

11.3 APPLYING BLOCKCHAIN TO MILITARY CYBERSECURITY

11.3.1 Secure data storage

Importance: The secure storage of sensitive military data is paramount. Blockchain's decentralized and tamper-resistant architecture makes ensuring that important data is shielded from modification or unwanted access [6].

Example: Storing encryption keys or authentication credentials on a blockchain can enhance security by preventing a single point of compromise. Even if one node is breached, the decentralized nature of the blockchain greatly increases the difficulty of harmful actors gaining access to sensitive data.

11.3.1.1 Benefits: Identity Management

Significance: Blockchain provides a robust framework for identity management, where cryptographic keys and biometric by managing and storing data securely, the chance of identity theft or unauthorized access can be decreased.

Example: In military access control, blockchain-based identity management systems can enhance security by ensuring that, critical places or information are only accessible to authorised persons whose identities have been validated.

11.3.1.2 Benefits: Supply Chain Security

Relevance: The military's complex and global supply chain requires a high level of security. Blockchain can be applied to track and verify the authenticity of supplies, preventing counterfeiting and ensuring the integrity of the entire supply chain.

Example: Using blockchain to trace the origin and movement of military equipment can help prevent the infiltration of counterfeit or compromised components, enhancing the overall security and reliability of the supply chain.

11.3.2 Secure communication

Vitality: Securing military communications is crucial for operational success. Blockchain can be employed to create a decentralized and encrypted communication network, reducing the risk of interception or disruption.

Example: Implementing blockchain in military communication systems ensures end-to-end encryption and decentralized routing, making it significantly harder for adversaries to compromise or intercept sensitive messages [7].

Importance: In military operations, timely and secure communication can be a matter of life and death. Blockchain's role in ensuring the integrity and confidentiality of communication channels is thus of paramount importance.

11.3.3 Challenges and strategic considerations

Scalability: As the volume of transactions on a blockchain increases, scalability becomes a challenge. Military applications often involve large-scale data processing, necessitating solutions that can handle the demands of a dynamic and expansive network.

11.3.3.1 Integration with Legacy Systems

Challenges: Military infrastructure often includes legacy systems that may not seamlessly integrate with blockchain technology. Adapting existing systems to harness the benefits of blockchain poses challenges in terms of compatibility and interoperability.

Transition Planning: A strategic and phased approach to transitioning to blockchain technology is essential. Planning should include considerations for legacy system integration, data migration, and training of personnel to ensure a smooth transition without compromising operational capabilities.

11.3.4 Privacy concerns

Consideration: While blockchain ensures data integrity and security, the transparent nature of the ledger raises concerns about privacy. Military operations may involve classified information, requiring careful consideration of how to balance transparency with the need for confidentiality.

Solutions: Privacy-focused blockchain implementations, such as zero-knowledge proofs or privacy-preserving smart contracts, can address concerns by allowing transactions to be verified without revealing sensitive information.

11.3.4.1 Case Studies

Examining real-world examples of blockchain implementation in military contexts provides valuable insights into the practical applications and challenges faced by organizations seeking to enhance cybersecurity through this technology.

Example: Using blockchain technology to securely store and exchange military intelligence across NATO members, protecting data integrity and lowering the possibility of illegal access.

11.3.5 A survey on blockchain applications in military cyber security

Cryptocurrency and blockchain projects have shown to offer robust security features. Because of this, at least a few nations are promoting different blockchain initiatives at the level of cyber defence. Applications can be classified into several categories according to their purpose, such as data integrity, swarming drones, a decentralized military network, a dependable supply chain for weaponry, and member authorization [8]. Numerous research have looked at how blockchain technology relates to cyber defence. Table 1 lists the subject matter, primary findings, and limitations of linked research. A thorough analysis of blockchain applications for cyber security was conducted by Taylor et al. Blockchain applications for cyber security were investigated by Zhu et al. The study by Lilly and Lilly [9] concentrated on actual military blockchain initiatives in the US, China, and Russia. Blockchain was used to study certain cyber security applications by Bunsal et al. [10]. Blockchain was studied by Ahmad et al. for the closely connected fields of aerospace and defence. Their research offers a thorough understanding of blockchain application architecture.

Cybersecurity is helpful and a crucial tool for preventing online threats to data, systems, or networks. Malware may take many forms, including viruses, Trojan horses, rootkits, etc., has the ability to initiate cyberattacks. Several common cyberattacks include ransomware attacks, SQL injection, Distributed Denial of Service (DDoS) attacks, phishing, and Man in the Middle (MITM) attacks.

11.3.6 Important cybersecurity elements

Cryptographic Checksums: Cryptographic checksums, often generated using hash functions, provide a unique fingerprint for data. This checksum, when recalculated at the receiving end, ensures that the data has not been tampered with during transit, serving as a fundamental tool for data integrity verification in cybersecurity.

Data Backup and Correction Codes: Regular data backups are essential for recovery in the event of data loss due to cyber incidents or system failures. Correction codes, like error-correcting codes (ECC), enhance data

reliability by automatically identifying and correcting errors that may occur during storage or transmission.

Assess Threats and Risks: Cybersecurity professionals continually assess the evolving landscape of threats and risks. This involves identifying potential vulnerabilities, understanding the likelihood and impact of different threats, and prioritizing resources to address the most critical risks to an organization's information assets.

Restrain Vulnerabilities: Actively managing and mitigating vulnerabilities involve employing practices such as regular system patching, conducting security audits, and implementing secure coding practices. Restraining vulnerabilities is crucial to preventing exploitation by malicious actors seeking unauthorized access or control.

Understanding Malicious Software: Cybersecurity practitioners must stay informed about the characteristics and behaviors of various forms of malicious software. This understanding enables the development and deployment of effective defenses, including antivirus programs, intrusion detection systems, and behavioral analysis tools.

Access Control: Access control mechanisms such as role-based access control (RBAC) and the principle of least privilege limit user and system access to only essential permissions based on their roles. This strategy mitigates unauthorized access and reduces the potential impact of security breaches.

Authentication: Strong authentication practices, including multi-factor authentication (MFA), bolster identity verification. Through the implementation of multiple forms of identification, such as passwords and biometrics, organizations can markedly diminish the likelihood of unauthorized access, even in the event of compromise of one authentication factor.

Encryption: Encryption is essential for safeguarding sensitive data, as it converts information into an unreadable form, ensuring that intercepted data remains incomprehensible without the correct decryption key. This is especially critical for securing data during transmission and storage.

Firewalls serve as a protective barrier between an organization's internal network and the external environment, managing traffic according to pre-defined rules. They prevent unauthorized access and mitigate malicious activities like cyberattacks and unauthorized data theft.

Intrusion Detection and Prevention Systems (IDS and IPS): Observe network or system operations to detect suspicious patterns or known indicators of cyber threats, informing security teams of potential issues. IPS goes a step further by actively preventing or blocking identified threats. When utilized in tandem, they enhance an organization's ability to efficiently recognize and manage cyber threats.

11.3.7 Key blockchain features

Distributed Shared Ledger: Because the ledger is dispersed, it prevents any one party from controlling the system as a whole. In addition to preventing

a single point of failure, this decentralization improves the availability and resilience of the system.

Immutable Records: Blockchain immutability is made possible by cryptographic hash algorithms. To form a chain, every block has a hash of the one before it. The permanence of recorded transactions would be ensured by the computational impossibility of modifying any block without also affecting future blocks.

Decentralized Consensus Mechanisms: Consensus algorithms are used by blockchain to get nodes to agree on the legitimacy of transactions. Because a central authority is no longer necessary thanks to this decentralized validation method, the system is more democratic and impervious to censorship.

Smart Contracts: These self-executing, programmable contracts automatically uphold pre-established regulations. These contracts operate without intermediaries, reducing costs and minimizing the risk of errors or manipulation associated with traditional contract execution.

Cryptographic Key Pair: Public and private key pairs add a layer of security to transactions. The public key serves as an address visible to others, while the private key, known only to the owner, ensures the authenticity and confidentiality of transactions [9].

Identity and Access Management: Blockchain facilitates user-controlled identity management, enabling participants to manage access to their data. Identities that are self-sovereign boosts privacy, mitigates identity theft risks, and gives individuals authority over the sharing of their information.

Enhanced Security: Blockchain provides better security by providing many techniques such as encryption and hashing which secures data. Schemes like Proof of Task or evidence of Interest raise the bar for security by making it more difficult for bad actors to influence the blockchain computationally.

Peer-to-Peer Network: Direct transactions take place between participants, doing away with the necessity for middlemen. Peer-to-peer networks are more efficient because they allow for quick and direct communication between users, which also lowers transaction costs.

Traceability and Transparency: All participants can view entire transaction history, which is useful in industries like finance and supply chain, where stakeholders require a clear and auditable record of activities.

No Central Authority: No one body has power over the whole blockchain network due to the lack of a central authority. This decentralized governance model fosters trust among participants, as decision-making power is distributed across the network rather than concentrated in a single entity. This also mitigates the risk of corruption or misuse of power.

11.4 POTENTIAL APPLICATIONS OF BLOCKCHAIN IN CYBERSECURITY

Internet of Things security: Increasing applications of IoT and AI has made data vulnerable and easy to hack. Block chain can be used in Internet of

Things by providing encryption on each device, key management, and authentication. Thus increasing security in data communication [11].

Software download integrity: By using blockchain technology to validate installers and updates, rogue software may be prevented from infecting devices. In this case, the integrity of the downloads may be confirmed by comparing newly created software identities to the hashes that are stored in the blockchain.

Protection of data transmission: Encryption keeps data in transit safe from unwanted access.

Decentralised storage of vital data: Blockchain-based storage solutions enable decentralized storage, preserving digital data in the face of the daily explosion in data generation.

Mitigating DDoS assaults: DDoS assaults are among the most common cyberattacks nowadays. Their goal is to cause an overload of Internet traffic in order to interfere with the normal operation of services. The immutability and encryption features of Blockchain make it a viable defence against these kinds of assaults.

DNS security: A public directory Domain Name System (DNS), connects domain names to their IP addresses. Hackers can hack the websites using these DNS links. Gaining access to the DNS, hackers have many times, crashed the websites. The DNS can be secured by the immutability and decentralized nature of blockchain technology.

11.5 USING BLOCKCHAIN TECHNOLOGY FOR CYBERSECURITY

In cybersecurity, an organization's security model is assessed using the CIA triad model as a reference. The trio includes:

- Confidentiality
- Availability and Integrity

With blockchain, we can make sure that all of these rules are followed.

1. **Confidentiality:** This pertains to ensuring that only authorised and interested parties have access to the pertinent data. When blockchain data is fully encrypted, unauthorized parties cannot access information that is flowing across untrusted networks. Application-level security measures, such as access limitations, are necessary to prevent network-based attacks. Blockchain technology can provide improved security measures by using public key infrastructure for authentication and encrypting participant interactions. However, storing backup private keys on secondary storage poses a serious risk of private key theft. To prevent this, cryptographic approaches based on integer factorization problems and key management protocols like IETF or RFC must be implemented.

2. **Integrity:** Using block chain's default techniques such as traceability and immutability, organizations can ensure data integrity. Additionally, in the event of a cyber control attack, consensus model protocols can help organizations implement rules to prevent and manage ledger splitting. Blockchain stores the previous state of the system with every iteration, allowing for the maintenance of an entirely traceable history log. Smart contracts have the ability to prevent miners from extracting data blocks by enforcing and validating agreements between parties [12].
3. **Availability:** The most common type of cyberattacks that try to interfere with the availability of technological services are distributed denial-of-service attacks. In the past several years, there has been an increase in these attacks. On the other hand, DDoS attacks cost more in blockchain-based systems since the attacker aims to flood the network with a lot of little transactions. It reduces the risk of IP-based DDoS attacks interfering with normal corporate operations since they do not have a single point of failure. Complete copies of the ledger are constantly accessible since data is always available throughout a range of nodes. Numerous nodes and distributed operation make the systems and platforms resilient.

11.6 SECURE DATA STORAGE

The importance of secure data storage in military operations cannot be overstated. Blockchain offers military data a secure, unchangeable storage solution that guards against modification and unwanted access. This section emphasizes the critical role blockchain plays in safeguarding sensitive military information.

11.6.1 Benefits: Identity Management

Identity management within military contexts is a complex challenge. Blockchain addresses this by enhancing The safety and accountability of personnel identities, access privileges, and authentication processes. The importance of blockchain in maintaining the 2 integrity of military identity management systems is covered in this section.

11.6.2 Benefits: Supply Chain Security

Securing military supply chains is crucial to prevent infiltration and ensure the authenticity of resources. Blockchain's transparency, traceability, and authenticity features make it highly relevant in fortifying military supply chains. This section explores how blockchain can be applied to enhance supply chain security. Benefits: Secure Communication.

Secure communication is vital in military operations to maintain operational secrecy. Blockchain-based encryption and decentralized communication protocols offer enhanced confidentiality and resilience. This section emphasizes the vitality of secure communication in military contexts.

11.7 SCALABILITY

Scalability is a common challenge associated with blockchain implementation. Addressing the scalability concerns ensures that the benefits of blockchain can be harnessed without compromising efficiency.

Integration with Legacy Systems

11.8 CHALLENGES

Integrating blockchain with existing military legacy systems presents challenges that must be carefully navigated.

11.8.1 Transition Planning

Strategic transition planning is crucial when incorporating blockchain into military cybersecurity frameworks. Planning mitigates potential disruptions and ensures a phased and well-managed adoption of blockchain technology [13–16].

11.8.2 Privacy Concerns

Consideration: While blockchain enhances transparency, privacy concerns must be addressed, especially in military operations involving classified information. This section explores the balance between transparency and the need to protect sensitive information.

11.9 BLOCKCHAIN CYBERSECURITY'S FUTURE

Blockchain is a cybersecurity innovation for the digital age that will assist guarantee adherence to the CIA cybersecurity triads. However, there may be certain application-related challenges due to the intricacy of its implementation.

Identifying avenues for further research is essential to continuously evolve and adapt blockchain solutions to the dynamic landscape of military cybersecurity.

Example: Investigating the use of advanced cryptographic techniques within blockchain systems to address emerging cyber threats and ensure the long-term viability of military blockchain applications.

11.10 CONCLUSION

In conclusion, the integration of blockchain technology to fortify military cybersecurity in the era of artificial intelligence represents a significant advancement within the domain of defense and national security. This research has highlighted the potential of blockchain to address critical challenges associated with securing military systems against evolving cyber threats. The findings underscore the importance of leveraging blockchain's decentralized and immutable nature to enhance data integrity, authentication, and overall resilience in military networks.

One of the key contributions of this study is the exploration of how blockchain can mitigate the risks associated with centralized points of failure, a vulnerability often exploited by sophisticated cyber adversaries. By distributing trust and authority across a decentralized network, blockchain serves as a formidable defense mechanism, reducing the likelihood of unauthorized access and manipulation of sensitive military information. This distributed and tamper-resistant ledger system ensures the integrity of data, maintaining a reliable record of events crucial for military operations.

Furthermore, the research has delved into the synergies between blockchain and artificial intelligence, recognizing the complementary nature of these technologies. The incorporation of AI algorithms into blockchain systems enhances the efficiency of threat detection and response, providing military organizations with advanced tools to preemptively identify and counter cyber threats. The combination of blockchain and AI creates a dynamic and adaptive cybersecurity ecosystem capable of addressing the rapidly evolving nature of modern cyber warfare.

As we move forward, it is imperative for military organizations to embrace and implement blockchain technology in a strategic and coordinated manner. Collaboration between government agencies, defense contractors, and technology innovators is crucial to developing standardized protocols and frameworks for the integration of blockchain in military cybersecurity. Moreover, ongoing research and development efforts should focus on refining the interoperability of blockchain with existing military systems, ensuring seamless integration and minimal disruption to operational processes.

It is important to recognize that technology solutions alone are insufficient, even if this research has shown the intriguing potential of blockchain in bolstering military cybersecurity. A comprehensive approach must also encompass robust training programs, policy frameworks, and international collaboration to address the multifaceted challenges posed by cyber threats. The continuous evolution of both blockchain and artificial intelligence demands a proactive stance in adapting military cybersecurity strategies to stay ahead of adversaries.

In conclusion, the union of blockchain and artificial intelligence presents a transformative paradigm for military cybersecurity, offering enhanced resilience, transparency, and adaptability. The successful implementation of

blockchain in military operations requires a concerted effort from stakeholders across the defense ecosystem. As we navigate the complex landscape of 21st-century warfare, harnessing the power of blockchain alongside AI stands as a beacon of hope in securing our military systems against the ever-present and evolving threat of cyberattacks.

REFERENCES

1. Ahmed, S., Broek, N. T. (2017). "Blockchain Could Boost Food Security." *Nature*, 550(7674), 43.
2. Antonopoulos, A. M. (2014). *Mastering Bitcoin: Unlocking Digital Cryptocurrencies*. O'Reilly Media.
3. Bahga, A., Madiseti, V. (2016). "Blockchain Platform for Industrial Internet of Things." *Journal of Software Engineering and Applications*, 9, 533–546.
4. Baynham-Herd, Z. (2017). "Enlist Blockchain to Boost Conservation." *Nature*, 548(7669), 523.
5. Chen, W., Wu, H., Chen, X., Chen, J. (2022). "A Review of Research on Privacy Protection of Internet of Vehicles Based on Blockchain." *Journal of Sensor and Actuator Networks*, 11, 86.
6. Derhab, A., Guerroumi, M., Gumai, A., Maglaras, L., Ferrag, M.A., Mukherjee, M., Khan, F.A. (2019). "Blockchain and random subspace learning-based IDS for SDN-enabled industrial IoT security." *Sensors*, 19, 3119.
7. Golosova, J., Romanovs, A. (n.d.). "The Advantages and Disadvantages of the Blockchain Technology."
8. Maxmen, A. (2018). "AI researchers embrace bitcoin technology to share medical data." *Nature*, 555, 293–294.
9. Nakamoto, S. (2008). "Bitcoin: A Peer-to-Peer Electronic Cash System." Retrieved from <https://bitcoin.org/bitcoin.pdf>
10. Narayanan, A., Bonneau, J., Felten, E., Miller, A., Goldfeder, S. (2016). *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*. Princeton University Press.
11. Ntshabele, K., Isong, B., Gasela, N., Abu-Mahfouz, A.M. (2022). "A Trusted Security Key Management Server in LoRaWAN: Modelling and Analysis." *Journal of Sensor and Actuator Networks*, 11, 52.
12. Pinto, A., Herrera, L.C., Donoso, Y., Gutierrez, J.A. (2023). "Survey on Intrusion Detection Systems Based on Machine Learning Techniques for the Protection of Critical Infrastructure." *Sensors*, 23, 2415.
13. Rathee, G., Kerrache, C.A., Ferrag, M.A. (2022). "A Blockchain-Based Intrusion Detection System Using Viterbi Algorithm and Indirect Trust for IIoT Systems." *Journal of Sensor and Actuator Networks*, 11, 71.
14. Saidi, H., Labraoui, N., Ari, A.A.A., Maglaras, L.A., Emati, J.H.M. (2022). "DSMAC: Privacy-aware Decentralized Self-Management of Data Access Control Based on Blockchain for Health Data." *IEEE Access*, 10, 101011–101028.
15. Swan, M. (2015). *Blockchain: Blueprint for a New Economy*. Newton, MA: O'Reilly Media.
16. Tapscott, D., Tapscott, A. (2016). *Blockchain Revolution: How the Technology Behind Bitcoin and Other Cryptocurrencies is Changing the World*. Penguin.

Computational intelligence in military applications with blockchain techniques

Kirankumar Kataraki, Vishwas D B, and Noor Zaman Jhanjhi

12.1 APPLICATIONS IN MODERN TIMES

12.1.1 Enhancing decision-making

In the military, computational intelligence is one of the major aspects that can be used to improve decision-making processes. Sometimes Artificial Intelligence algorithms are able to process huge amounts of information which leads to useful data helping them respond quickly and decisively. On enemy prediction as well as allocating resources more efficiently, computational intelligence improves commanders' decision-making abilities.

12.1.2 Cybersecurity and encryption

During this time of cyber wars it is of great importance to keep military information confidential. Blockchain which is used today by military units for various purposes is one of such ways for improving cybersecurity and increased encryption effectiveness during any military operation. Therefore, when dealing with security of sensitive information, blockchain becomes very effective method for ensuring its safety: Here no one can break through because all stored data are distributed making them immutable [1].

12.1.3 Supply chain management

Military success depends on the supply chain. Gaining insight through blockchain technology helps to better manage the logistics and flow of goods—ensuring that goods are delivered where they are needed, when they are needed. The integration of this information dramatically improves the overall efficiency of military logistics: all through the visibility that this new technology brings.

12.1.4 Enhanced intelligence and surveillance

To achieve competitive advantage in modern warfare, intelligence acquisition and surveillance It's inevitable. Computer intelligence systems can handle large amounts of data. Like social media connections, blocked communication lines and even photos from it Use of satellites. These systems can predict new systems or trends that a mood; Look for abnormalities that may indicate danger. In addition to this. Additionally, military organizations can benefit from increased secrecy and. Trust in shared information due to the advent of blockchain technology [2]. The decentralized ledger underlies the basis for sharing intelligence between military departments or even friendly states thereby ensuring accuracy of information and acting as a bulwark against tampering with it.

12.1.5 Autonomous systems

Indeed, contemporary military operations increasingly utilize unmanned aerial vehicles (UAVs) and ground-based robots as two autonomous system case studies. These systems are designed to work autonomously with machine learning and deep learning techniques under the umbrella of computational intelligence.

When performance data is stored using a blockchain mechanism, the data is accessible and can be mined, thus ensuring that there is accountability and reducing risk of deviation from standard practice or unauthorized activity. This is especially important in the case of battle, surveillance or maybe even reconnaissance through autonomous devices.

12.1.6 Secure communication

In Military communication is very important, hence it should be effective and secure. Computational intelligence allows secure, reliable, low-latency communication paths. Thanks to advanced military grade encryption algorithms, shifting to an artificial intelligence that can easily cope with changing threat landscapes, the military communication cannot be easily intercepted or decoded by means of an enemy.

Blockchain technology provides a trust less, decentralized, incorruptible record for authentication and message stored enabling secure communication [3]. This protects against manipulation and unauthorized access as well as against the recording of communication and the command & control systems being compromised (Figure 12.1).

12.1.7 Logistics and supply chain optimization

Logistics and the supply chain management are important facets of military operations. Using blockchain and computational intelligence, waste and



Figure 12.1 Secure communication.

delays in allocating resources are eliminated. Predictive predictions about which pieces of equipment might need to be repaired keep military units operationally ready.

With blockchain' traceability and transparency to control military resources and ensure they are delivered. The ability to track in real-time enhances supply chain accountability and minimizes potential losses to pilferage.

12.1.8 Tactical decision support

War calls for quick, well-reasoned judgments. Computational intelligence to enable on demand real-time battlefield analysis tools for military leadership that will assist them in tactical and strategic planning. Data fusion and predictive analytics help you to know more about the strategies and actions of your opponent.

Armed forces, allies and coalition forces could exchange time-critical, reliable combat information using Blockchain technology. Thereby providing effective communication and. Situational awareness is present in all parties even if they are not physically present region (Figure 12.2).

12.1.9 Training and simulation

One of the most important aspects of military preparation is training. For actual training Situations and simulations, use computer intelligence. Soldiers can use their skills For safety and regulatory use of these simulators. Personalized AI-powered training student progression provides personalized learning experiences. Blockchain technology also validates the certificates and prevents false claims of readiness as a training record. Also, it makes it possible to create a decentralized, shared platform provide instruction where information and knowledge from various branches of the military can be obtained It was safely replaced.

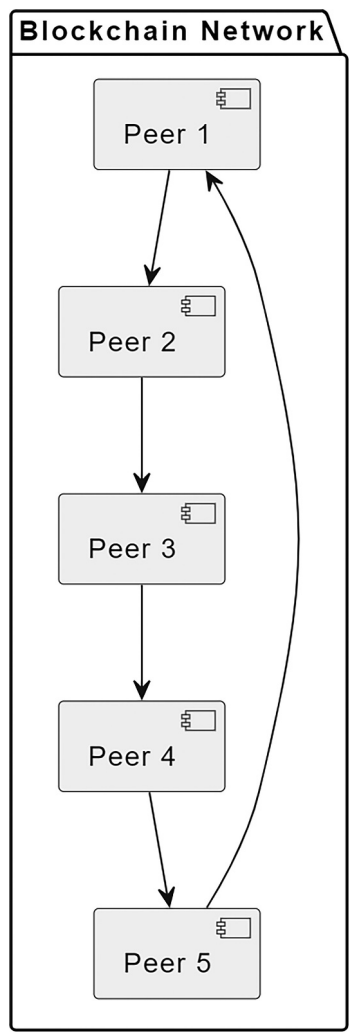


Figure 12.2 Tactical decision support.

12.2 ADVANTAGES AND CHALLENGES

12.2.1 Security enhancement

Blockchain-based systems offer one of the major advantages due to this advanced technology, or security. Blockchain creates regularly to prevent fraud and other criminal activity A record of transactions using end-to-end encryption. Furthermore, where data a blockchain is stored [4].

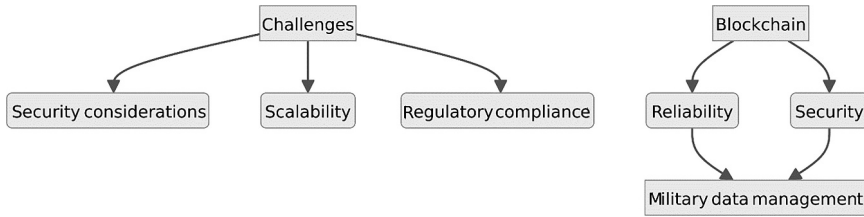


Figure 12.3 Security enhancement.

In contrast, conventional security mechanisms contain some overhead. The server. Again, advocates say so by anonymizing data and in some cases on demand. With the ability to deny unauthorized access, blockchain can solve privacy concerns better than the traditional ones (Figure 12.3).

12.2.2 Data integrity and transparency

Blockchain technology is used in military applications to address delicate operations on issues of data openness and integrity. The decentralized ledger and cryptographic hash functions of blockchain technology actually eliminate data manipulation by ensuring that its integrity remains intact.

The fact that the technology is transparent allows authorized individuals to access a reliable and auditable real-time history of changes in data as well as traceability. This enhances inter-operability among various military systems, promotes accountability, and makes it easier to respond quickly to changing circumstances. While there are security concerns, scalability, regulatory compliance among others which must be addressed properly; blockchain is seen as perhaps an answer for strengthening security and dependability of military data management.

12.2.3 Decentralization

Blockchain is a trust-free framework, according to Daniel Field who is the Director of Innovation and Global Head of Blockchain at UST, a digital technology and services provider since there is no central authority. In a blockchain platform, unidentified players can exchange data across one another, enabling an ecosystem of firms that do not rely on any single company for operation.

One example of this is supply chain. Several businesses in the chain such as producers, distributors, retailers and suppliers need or want information from other companies involved with it. All these companies depend on one another but there isn't any theorized control mechanism they utilize in order to facilitate communication among themselves. The solution is The decentralized blockchain is embedded in the network (Figure 12.4).

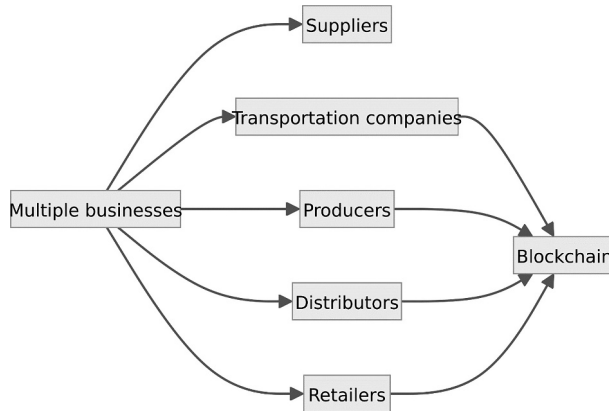


Figure 12.4 Decentralized blockchain is embedded in the network.

12.2.4 Smart contracts for automation

Smart contracts are needed to automate processes using blockchain in the military technology. They are spontaneous contracts triggered by fixed terms, therefore It allows you to perform a number of tasks less efficiently and more efficiently. Intelligent alliances are possible Automate any supply chain management, speed up logistics and ensure reduced forces they follow standards [5]. Blockchain is indestructible; Hence the validity of these covenants It is therefore guaranteed to prevent any fraud or manipulation. Automation of repetitive tasks. In addition to increased awareness, smart contracts improve efficiency and. Responsible for military systems. By automating simple tasks, this helps with decision making Speeding up the process also reduces the chances of error and therefore improves efficiency In military operations.

12.2.5 Supply chain management

Military suppliers can also benefit from the increased efficiency, security and openness that blockchain brings to data storage. Blockchain can harness the power of decentralized registries to provide an indestructible and transparent record of the entire supply chain from manufacturing to distribution. This can improve military situational awareness by making military asset transfers more traceable, ensuring data integrity, as well as reducing incentives for fraud Blockchain-based smart contracts can too has simplified the procurement process and reduced transaction costs by automating and enforcing contracts [6]. Overall, the military supply chain will become more responsive and flexible: better able to adapt to emergency operational incidents while maintaining strong security and integrity standards in—all under centralized planning.

12.2.6 Secure communications

When it comes to military applications, the answer to secure communication is obvious. The security and integrity of military communications is enhanced by the decentralized and cryptographic nature of blockchain technology. Blockchain assures that communications remain private and unaltered during the communication process. This is achieved through distributed ledger technology and end-to-end encryption. The decentralisation of a blockchain network makes it generally less vulnerable to unlawful access and single points of failure than centralised infrastructure. This aspect in particular makes the communication infrastructure more robust; the more robust the infrastructure is, the better equipped it is to securely distribute sensitive military data and to reliably deliver important communications of all kinds over a wide variety of media (Figure 12.5).

12.2.7 Disadvantages

- Firstly, drawing from the above enumeration of the military's blockchain application bottleneck, it can be inferred that blockchain is a rather new type of information technology that only emphasizes the provision of data security and the establishment of trust with little concern for other factors such as the storage space, access speed and general performance of the system. In military application scenarios, blockchain is especially suitable when usage frequency isn't high, the requirements for security are higher than normal, timeliness is extremely low and the amounts of data are extremely small. The following principal areas mainly seek to highlight its shortcomings and risks:
 - It's hard to balance growth and light weight because of high redundancy and high energy consumption. All ledger data must be synced in real time across all nodes, and duplicate data will be saved in each block [7]. More storage will be needed as more data is stored and more nodes are added. Combat units or platform terminals need large capacity for computing, storage and communication; this goes against the trend of smaller and lighter equipment.

Each node will consume more computing power, bandwidth and energy as more nodes are added. More nodes will require more storage capacity for subsequent nodes, higher access difficulty, longer sync time and lower overall efficiency. All these will go against the combat system's large scale on demand expansion.
 - Using complex data sync to meet high frequency low latency requirements is hard. Every time there is a change to the data in the blockchain all nodes need to update the ledger at the same time which takes time. If you do it too many times in a short period of time it will, congest the network as it uses up a lot of bandwidth.

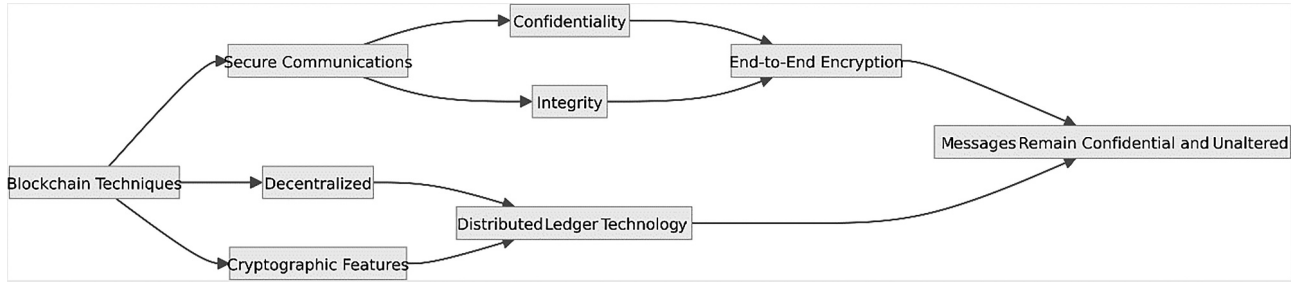


Figure 12.5 Decentralisation of a blockchain network.

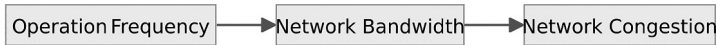


Figure 12.6 Military's blockchain application bottleneck.

The second kill phase of modern warfare has arrived especially at the tactical and platform level. Combat units and platforms are sending info support apps more frequently and situational info updates are happening faster. The blockchain can't keep up with these instant response requirements (Figure 12.6).

- There are still vulnerabilities in the encryption and consensus. The cryptographic algorithm determines the security of the consensus for blockchain. It can still be cracked so it's not safe. For example, blockchain uses elliptic curve cryptography. Quantum computers find it easy, classical computers find it hard. Powers around the world are working hard to build quantum computers. The security of most blockchain solutions today will disappear when a practical quantum computer is built.
- System security is reduced by the military blockchain's size. According to the principles of blockchain, an attacker can't change data in the blockchain unless they control more than 51% of the nodes. The harder it is for an attacker to change data, the more nodes [8]. The number of nodes in military blockchains is usually much less than in civil Internet based systems. In a war, more than half of the nodes can be compromised and data can be changed, even with massive computer powered enemy network attacks (Figure 12.7).

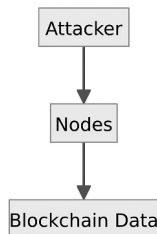


Figure 12.7 Blockchain data.

12.3 REAL WORLD PROBLEMS

12.3.1 DARPA's use of blockchain and AI to enable secure communication

12.3.1.1 Context

The defence Advanced Research Projects Agency (DARPA) initiated an effort to fortify secure communication lines using blockchain and artificial intelligence (AI).

Interoperability: Apart from the scalability issues, also the difficulty to interoperate with existing communication systems, blockchain for a large-scale military communication networks may raise issues in terms of interoperability.

Resource-heavy: AI-centric encryption may need significant computing power, which might slow down real-time communication [9].

This security concern led to the birth of the idea of a blockchain open and secure supply chain management system and its eventual introduction to the logistic operations of NATO, the alliance that came to recognize the need for efficient and secure supply chain management.

This was to ensure transparency and was done using Blockchain technology to build an immutable ledger that documents every step of the supply chain process. The office was to be responsible for the purchasing, production, shipping, and distribution of all equipment and supplies for the military.

Blockchain in action: Supply chain transactions could easily be preapproved by smart contracts that validate predefined policies.

Increased Accountability and Transparency: The implementation of blockchain increased visibility which minimized the chance for counterfeit goods or unauthorized supply chain changes.

Efficiency Gains: Errors and administrative costs were reduced as the user automated the processes through smart contracts.

Tips: Interoperability: Making sure the supply chain mechanisms already in implementation in NATO member states are adhered to.

Data Privacy It can be a battle whether or not to provide too little or too much info in a military setting

Israel Defence Forces (IDF) by making computer intelligence tools in cyber defence helps to simply prioritize cybersecurity and steal cyber defence.

Intelligent Threats: The IDF immediately identified unusual activity or potential cyber provocation by analyzing network traffic and using machine learning-based algorithms

Behavioral analysis: How cyber. intrusions to support proactive security measures.

The first line of defense: using an AI-powered solution could be identifying potential cyber threats Faster which helps mitigate risks in terms of cybersecurity.

- Adaptive Defensive: Deep inspection of behaviour informed the IDF with the necessary information to change defensive tactics in accordance to changing cyber threats.

12.4 PROSPECTS FOR THE FUTURE OF BLOCKCHAIN-BASED ARTIFICIAL INTELLIGENCE IN MILITARY APPLICATIONS

Using artificial intelligence (AI) and blockchain technology, the military has established a new one The Age of Revolution and Strategy [10]. There are revolutionary possibilities in this technology discussed in this chapter, which gives us a glimpse into the future extent of their influence. Communication will change the nature of war. This is part of our efforts to provide comprehensiveness Insight into how the military sector will soon move towards its imminent peak of influence Never before experienced: supply chain quality, decentralization of command and control procedures, improved safety measures as well as ethical concerns (Figure 12.8).

12.4.1 Augmented defense

Military use of artificial intelligence will radically change security strategies and blockchain technology. This powerful form of security is built on an integrated blockchain which is inherently immutable and an AI that responds easily to change Threats of various kinds. In an era of increasing



Figure 12.8 Military applications.

cyber threats, a robust security system that provides protection is increasingly important. Military information has sophisticated hiding techniques and real-time threats analysis.

12.4.2 Decentralized command and control methods

The military can go more decentralized with the command-and-control system. Provisions using artificial intelligence and blockchain technology. Artificial intelligence Algorithms provide immediate insights that allow quick decisions when. The transparent and traceable blockchain technology ensures accountability for every transaction. The move was made. This approach increases the mobility of decisions and therefore increases the flexibility. Adjustment to complex work environments (Figure 12.9).

12.4.3 Supply chain optimization & logistics

Recent striking advances in military logistics are built towards an eventual synergistic marriage between artificial intelligence and block chain technology. Blockchain's transparency ensures supply chain integrity, while AI-driven predictive analytics help maximize resource allocation. Moreover, it reduces the sensitivity and vulnerability of military operations by promising irreversible and transparent methods of transferring resources [11].

12.4.4 Frameworks for law and ethics

The widespread use of blockchain technology and artificial intelligence in military operations is making the need for a comprehensive ethical and legal framework question ethical practices through the use of AI systems, data privacy, and the need for compliance with international law. Creating a robust framework ensures a consistent framework for addressing complex ethical challenges related to blockchain and AI in security. It highlights the importance of countries working together to come up with appropriate measures on this issue. Encouragement of international cooperation, the sharing of best practices, and the development of ethical standards can be achieved through stronger agreements between countries. In this way, blockchain technology and artificial intelligence are as prevalent as possible.

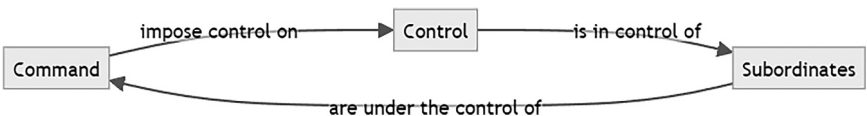


Figure 12.9 Decentralized command and control methods.

12.4.5 Automated battlefield systems and autonomous systems

Autonomous systems and combat automation are expected to play a key role in future military operations due to the convergence of blockchain technology and artificial intelligence thus driving drones, robots, and other intelligent control systems the self-regulating role benefits from the blockchain-based secure system with connectivity [12]. This integrated approach ensures operational integrity and facilitates the rapid exchange of critical information during conflict. This has resulted in a force that is more resilient in challenging and challenging situations (Figure 12.10).

12.4.6 Intelligent contracts for military pursuits

In particular, smart contracts are expected to play a significant role in military operations in the coming years. These self-executing contracts directly use computer code words to automate various military tasks. Smart contracts improve operational efficiency by reducing bureaucratic delays in various activities such as information exchange, maintenance plans or procurement processes, allowing for a more technologically advanced and adaptable military.

12.4.7 AI-based training and simulation

Blockchain technology and artificial intelligence (AI) have the potential to transform military training and simulation systems. By leveraging this cutting-edge technology, the military can increase its readiness for dynamic operating conditions and emerging threats. In this article, we will examine aspects of this revolutionary approach to military training [13].

1. Personalized simulation using AI algorithms

Traditionally, military training is a standardized form of exercise that does not fully address individual needs and abilities. But thanks to the power of AI algorithms, it is now possible to analyze vast amounts

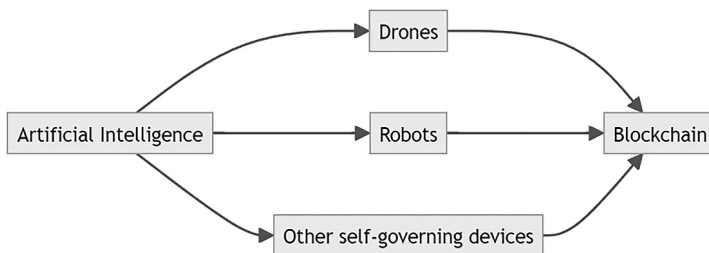


Figure 12.10 Automated battlefield systems and autonomous systems.

of data to create personalized images for each soldier. This approach ensures that training conditions are appropriate, challenging, and tailored to a specific activity or skill.

2. Ensure Data Integrity and Blockchain

An accurate assessment of employee readiness is essential in any training program. This is where blockchain technology comes into play. Using its inherent characteristics of transparency and immutability, training data can be securely recorded on a decentralized ledger. This ensures that information is not corrupted or altered, and provides a reliable source for evaluating an individual's performance.

3. Solving ethical problems through transparency

While integrating AI and blockchain into military operations offers significant benefits, it also raises ethical challenges that cannot be ignored. Public opinion plays an important role in the ethical framework around this technology. It is therefore important to establish clear guidelines and encourage transparent practices in their development and implementation, growing public confidence. One way to build public trust is to demonstrate the ethical use of blockchain and AI in security operations. Understanding that these technologies are used responsibly and accountably by citizens helps reduce concerns about privacy violations or autonomous decision-making setting ethical standards. To ensure appropriate progress in this area it is important that military organizations establish specific ethical standards and governance structures for the use of blockchain and AI. This includes defining boundaries for data collection, the principles of algorithmic decision-making will be explained, as well as methods for monitoring and review.

4. Challenges and risks of adopting blockchain and AI in the military

While the potential benefits of integrating blockchain and AI into military operations are immense, there are a number of challenges and risks that need to be addressed:

Lack of international standards: While this technology transcends national boundaries and there is a need for harmonized international standards and protocols that govern its use in security scenarios

Potential misuse of technology: Any powerful tool can be used for malicious purposes. It is important to anticipate and mitigate the potential risks of blockchain and AI abuse in military contexts.

Scalability and interoperability: Deploying these complex technologies on military systems and platforms can present technical challenges in terms of scalability and interoperability.

Cybersecurity Vulnerabilities: Reliance on digital infrastructure creates new vulnerabilities that can be exploited by adversaries. Strong cybersecurity measures are needed to protect sensitive military information.

5. Risk management and responsibility are important

Prioritize risk management and responsible compliance to ensure effective and ethical use of blockchain technology and artificial intelligence in military training:

- **Identify potential risks:** Perform a comprehensive risk assessment to identify potential errors or unintended consequences associated with the use of this technology [14–15].
- **Actively address concerns:** Engage actively with stakeholders, including experts from academia, industry, and the public, to address concerns about confidentiality, bias, liability, or dealing with displacement.
- **Training and Education:** Provide a comprehensive training program for the military in the capabilities, limitations, and ethical considerations of using blockchain and AI tools.
- **Ongoing Monitoring:** Evaluate the effectiveness of this technology to meet training objectives and ensure compliance with legal, ethical, and professional standards.
- By adopting an effective and responsible approach to risk management, the military can harness the full potential of blockchain and AI to enhance military training capabilities while supporting ethical principles (Figure 12.11).



Figure 12.11 AI-based training and simulation.

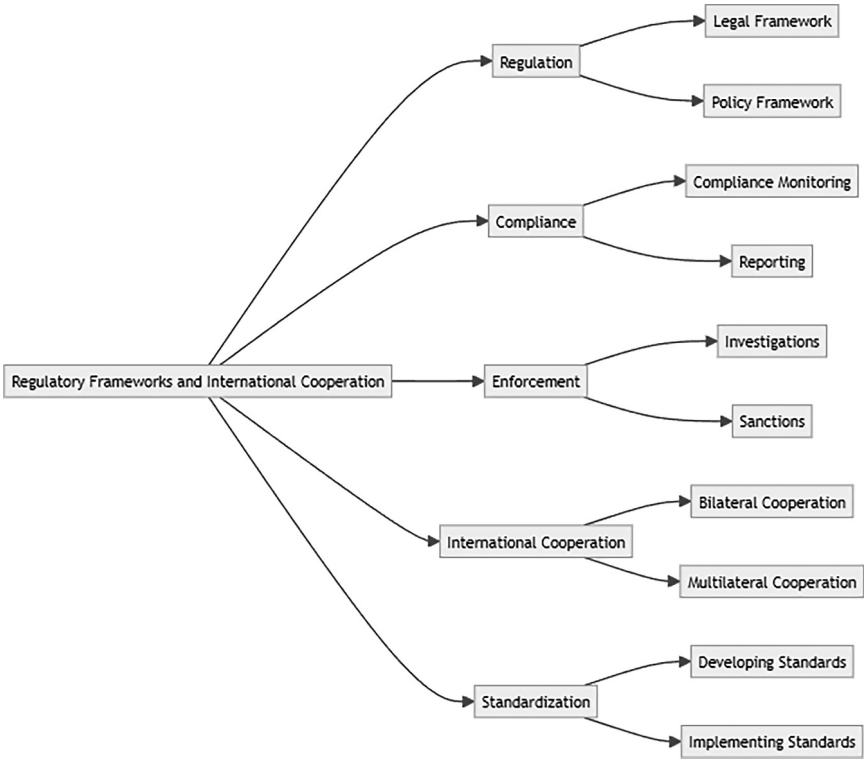


Figure 12.12 International cooperation and regulation frameworks.

12.4.8 International cooperation and regulation frameworks

However, the application of this technology in future military operations will depend on international cooperation and the implementation of a clear policy approach. ~Apply bitcoin (how blockchain technology and artificial intelligence is built] to military applications.- After reading the graph, the chapter discusses the joint efforts of international organizations, governments and business leaders to develop enforceable regulations for ethical development, and application of this technology (Figure 12.12).

12.5 CONCLUSION

The combined deployment of both blockchain technology and artificial intelligence (AI) is predicted to revolutionize the future of military operations. By now, with implementation of this track emerging into the picture, it can be seen that the coming together of these leading innovations have

the potential to revolutionize defence strategies, in the new age for a more secured defence plan influenced with heightened security requirements operational agility and ethical query.

What sits in the middle simply is well defined security and reliability inside of a changing paradigm environment. Blockchain's immutability combined with AI's agility together fortify a strong shield against the rapidly changing enigma of cyber threats. Encryption, as well as the real-time analysis of data, ensures valuable military information is kept secure, and paves the path for a future in which military operation are safe from pervasive digital threats.

Decentralized command and control systems is one of the key element in future which is on way. Military commanders can then have an unparalleled decision-making agility when AI's real-time insights are merged with the transparency of blockchain technology. In the process, it instils resilience to be future-ready within decentralized organizations; coupled with transparency when it comes to strategic maneuverers so that everything is transparent and traceable.

REFERENCES

1. Narayanan, A., Bonneau, J., Felten, E., Miller, A., & Goldfeder, S. (2016). *Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction*. Princeton University Press.
2. Arregui, J., & Llopis, F. (2019). Blockchain Technology for the Internet of Things: A Systematic Literature Review. *Sensors*, 19(4), 847.
3. Mougayar, W. (2016). *The Business Blockchain: Promise, Practice, and Application of the Next Internet Technology*. John Wiley & Sons.
4. Bryans, J., & Borenstein, J. (2018). Ethical and Legal Considerations in Blockchain-Based Systems. *IEEE Security & Privacy*, 16(6), 50–57.
5. Azaria, A., Ekblaw, A., Vieira, T., & Lippman, A. (2016). Medrek: Using Blockchain for Medical Data Access and Permission Management. In *2016 2nd International Conference on Open and Big Data (OBD)*, 25–30.
6. Krol, M., Guo, Y., & Vasarhelyi, B. (2019). Auditing of Autonomous Intelligent Agents Using Blockchain and Smart Contracts. *Journal of Emerging Technologies in Accounting*, 16(2), 53–65.
7. Swan, M. (2015). *Blockchain: Blueprint for a New Economy*. O'Reilly Media.
8. Tobon, D. P., & Padilla, M. J. (2018). The Role of Blockchain in a New Paradigm of Simulation. *Procedia Computer Science*, 134, 68–75.
9. Yli-Huumo, J., Ko, D., Choi, S., Park, S., & Smolander, K. (2016). Where Is Current Research on Blockchain Technology?—A Systematic Review. *PloS One*, 11(10), e0163477.
10. Swan, M. (2015). *Blockchain: Blueprint for a New Economy*. O'Reilly Media.
11. Zwitter, A., & Blok, T. (2019). Can blockchain solve the challenges of integrity of the voting process in e-democracy? *Media International Australia*, 171(1), 88–103.

12. Preskill, J. (2018). Quantum Computing in the NISQ era and beyond. *Quantum*, 2, 79.
13. Floridi, L., & Taddeo, M. (2016). What is data ethics? *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 374(2083), 20160360.
14. Cerrudo, C., & Latif, A. (2018). Internet of Things (IoT) security: Current status, challenges and prospective measures. *Internet Research*, 28(3), 762–820.
15. Russell, S., & Norvig, P. (2018). *Artificial Intelligence: A Modern Approach*. Pearson.

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